
Analytic Field Theory

— *EYNTKA* Series —

NATHANAEL CHWOJKO-SRAWLEY

June 30, 2026

Contents

Contents	3
A Review of Elementary Number Theory	i
0.1 Divisibility and the Euclidean Algorithm	i
0.2 Primes and the Fundamental Theorem of Arithmetic	vii
0.3 Congruences	xii
0.4 The Group $(\mathbb{Z}/m\mathbb{Z})^\times$ and Primitive Roots	xvii
0.5 Quadratic Residues and Reciprocity	xxiii
0.6 Asymptotic Notation	xxix
1 Arithmetic Functions and Summation Techniques	1
1.1 Arithmetic Functions and Multiplicativity	1
1.2 Dirichlet Convolution and Möbius Inversion	6
1.3 Average Orders and Summatory Functions	12
1.4 Partial Summation and Abel's Summation Formula	17
1.5 The Euler-Maclaurin Summation Formula	22
2 Dirichlet Series and Euler Products	31
2.1 Dirichlet Series: Convergence and Analyticity	31
2.2 The Ring Homomorphism: Convolution and Multiplication	37
2.3 Euler Products	42
2.4 Mellin Transforms and the Integral Representation of $\zeta(s)$	50
2.5 Perron's Formula	58

3	Elementary Results on Prime Distribution	67
3.1	Chebyshev's Functions	68
3.2	Chebyshev's Estimates	72
3.3	Mertens' Theorems	80
3.4	Bertrand's Postulate	90
4	The Riemann Zeta Function	97
4.1	Definition and the Euler Product	97
4.2	Meromorphic Continuation to $\operatorname{Re}(s) > 0$	101
4.3	First Arithmetic Consequences	104
4.4	Non-Vanishing on the Line $\operatorname{Re}(s) = 1$	110
	Index	115
5	The Prime Number Theorem	119
5.1	The Analytic Strategy	120
5.2	The Laplace Transform	122
5.3	Newman's Tauberian Theorem	126
5.4	Proof of the Prime Number Theorem	131
5.5	Error Terms and the Zero-Free Region	139

A Review of Elementary Number Theory

The machinery of analytic number theory rests on a foundation of elementary results about integers, primes, and congruences. This preliminary chapter collects the essential facts that the rest of the book takes for granted. Readers with a solid undergraduate number theory background will find this material familiar and may treat the chapter as a reference; those encountering some results for the first time should work through the proofs carefully, as the techniques—particularly the Euclidean algorithm, the Chinese Remainder Theorem, and quadratic reciprocity—reappear in more sophisticated forms throughout the book.

The progression is natural. Divisibility and the Euclidean algorithm provide the computational backbone. The Fundamental Theorem of Arithmetic establishes that primes are the multiplicative building blocks of $\mathbb{Z}$, a fact whose failure in more general rings drives much of algebraic number theory (see [14]). Congruences and the Chinese Remainder Theorem introduce the modular perspective that underlies Dirichlet characters and L -functions. The structure of the unit group $(\mathbb{Z}/m\mathbb{Z})^\times$ and the theory of quadratic residues prepare the ground for character theory and reciprocity laws. The chapter closes with a review of asymptotic notation, the language in which the main results of analytic number theory are expressed.

0.1 Divisibility and the Euclidean Algorithm

The integers carry two intertwined structures: addition and multiplication. The additive structure is simple— $\mathbb{Z}$ is a free abelian group of rank 1—but the multiplicative structure is the source of all the complexity that this book investigates. The first step toward understanding it is the divisibility relation, and the first computational tool is the Euclidean algorithm, which reduces questions about divisibility to iterated division with remainder.

The starting point is the division algorithm, which guarantees that long division in $\mathbb{Z}$ always terminates with a well-defined quotient and remainder.

Theorem 0.1.1: Division Algorithm

For any $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$, there exist unique integers q (the *quotient*) and r (the *remainder*) such that

$$a = bq + r, \quad 0 \leq r < b$$

Proof:

Step 1: Existence. Consider the set $S = \{a - bk \mid k \in \mathbb{Z}, a - bk \geq 0\}$. Since $b > 0$, choosing k sufficiently negative ensures that $a - bk > 0$, so S is nonempty. By the well-ordering principle, S has a least element $r = a - bq$ for some $q \in \mathbb{Z}$. By construction $r \geq 0$. If $r \geq b$, then $a - b(q + 1) = r - b \geq 0$ belongs to S and is strictly smaller than r , contradicting minimality. Hence $0 \leq r < b$.

Step 2: Uniqueness. Suppose $a = bq_1 + r_1 = bq_2 + r_2$ with $0 \leq r_1, r_2 < b$. Then $b(q_1 - q_2) = r_2 - r_1$, so b divides $r_2 - r_1$. Since $|r_2 - r_1| < b$, this forces $r_1 = r_2$ and consequently $q_1 = q_2$.

The division algorithm is the engine behind every computation in this section. Its role is analogous to that of the Euclidean metric in geometry: it provides a notion of “size” (the remainder r) that strictly decreases under a natural operation (repeated division), guaranteeing that iterative processes terminate. This principle—descent by remainder—also underlies the proof of unique factorization, which depends on it via Euclid’s lemma in the next section.

With division in hand, the notion of exact divisibility can be made precise.

Definition 0.1.2: Divisibility

Let $a, b \in \mathbb{Z}$. Then a *divides* b , written $a \mid b$, if there exists $k \in \mathbb{Z}$ such that $b = ak$. Equivalently, $a \mid b$ if and only if the remainder upon dividing b by a is zero. If a does not divide b , this is denoted $a \nmid b$.

The divisibility relation is a partial order on the positive integers: it is reflexive ($a \mid a$ since $a = a \cdot 1$), antisymmetric (if $a \mid b$ and $b \mid a$ with $a, b > 0$ then $a = b$), and transitive (if $a \mid b$ and $b \mid c$ then $a \mid c$, since $c = bk_1 = ak_2k_1$). It is also compatible with the ring operations of $\mathbb{Z}$ in the following sense.

Proposition 0.1.3: Linearity of Divisibility

If $d \mid a$ and $d \mid b$, then $d \mid (ax + by)$ for all $x, y \in \mathbb{Z}$.

Proof:

Write $a = dk_1$ and $b = dk_2$. Then $ax + by = d(k_1x + k_2y)$.

In the language of algebra, this says that the set of multiples of d forms an ideal $(d) = d\mathbb{Z}$ in $\mathbb{Z}$, and divisibility translates to ideal containment: $a \mid b$ if and only if $(b) \subseteq (a)$. The fact that *every* ideal of $\mathbb{Z}$ has this form— $\mathbb{Z}$ is a principal ideal domain—is the ring-theoretic content of Bézout’s identity, proved below. In more general number rings, ideals need not be principal, and the failure of this property is measured by the class group (see [14]).

The central notion built from divisibility is the greatest common divisor.

Definition 0.1.4: Greatest Common Divisor

For $a, b \in \mathbb{Z}$ not both zero, the *greatest common divisor* $\gcd(a, b)$ is the largest positive integer d such that $d|a$ and $d|b$. By convention, $\gcd(0, 0) = 0$. The integers a and b are *coprime* (or *relatively prime*) if $\gcd(a, b) = 1$.

The existence of $\gcd(a, b)$ is clear: the set of common divisors is nonempty (it contains 1 when a and b are not both zero) and bounded above by $\max(|a|, |b|)$, so a maximum exists. Before describing how to compute it, a few basic identities are worth recording.

Proposition 0.1.5: Properties of the GCD

For all $a, b \in \mathbb{Z}$ and $k \in \mathbb{Z} \setminus \{0\}$:

1. $\gcd(a, b) = \gcd(b, a) = \gcd(|a|, |b|)$.
2. $\gcd(a, 0) = |a|$.
3. $\gcd(ka, kb) = |k| \gcd(a, b)$.
4. If $a = bq + r$, then $\gcd(a, b) = \gcd(b, r)$.

Proof:

Parts (1)–(3) follow directly from the definition, since the set of common divisors is unchanged by reordering, by taking absolute values, or by scaling (respectively). For (4), proposition 0.1.3 gives the two containments: every common divisor of a and b divides $r = a - bq$, and every common divisor of b and r divides $a = bq + r$. Hence the sets of common divisors coincide, and their maxima are equal.

Part (4) is the identity that makes the Euclidean algorithm work. It reduces $\gcd(a, b)$ to $\gcd(b, r)$ where $0 \leq r < b$, so the arguments are strictly smaller. Iterating this reduction is the Euclidean algorithm.

Proposition 0.1.6: The Euclidean Algorithm

Let $a, b \in \mathbb{Z}$ with $b > 0$. Apply the division algorithm repeatedly:

$$\begin{aligned}
 a &= bq_0 + r_0, & 0 \leq r_0 < b, \\
 b &= r_0q_1 + r_1, & 0 \leq r_1 < r_0, \\
 r_0 &= r_1q_2 + r_2, & 0 \leq r_2 < r_1, \\
 &\vdots \\
 r_{n-2} &= r_{n-1}q_n + r_n, & 0 \leq r_n < r_{n-1}, \\
 r_{n-1} &= r_nq_{n+1}
 \end{aligned}$$

The process terminates after finitely many steps, and the last nonzero remainder is $\gcd(a, b) = r_n$.

Proof:

The sequence $b > r_0 > r_1 > \dots \geq 0$ is a strictly decreasing sequence of non-negative integers, so it must reach zero: there exists a smallest index n such that $r_{n+1} = 0$. At that step, $r_{n-1} = r_nq_{n+1}$, so $r_n \mid r_{n-1}$. Working backward through each line, r_n divides every preceding remainder and hence divides both b and a : it is a common divisor. Conversely, if d is any common divisor of a and b , the first equation gives $d \mid r_0$ (by proposition 0.1.3), the second gives $d \mid r_1$, and continuing inductively, $d \mid r_n$. Thus r_n is a common divisor that is divisible by every common divisor, which forces $r_n = \gcd(a, b)$.

The running time of the Euclidean algorithm is modest. After two consecutive division steps, the remainder drops by at least a factor of two: $r_{k+2} < r_k/2$. This gives an upper bound of $2\lfloor \log_2 b \rfloor + 1$ steps, making the algorithm logarithmic in the size of its input. A finer analysis shows that the worst-case inputs are consecutive Fibonacci numbers, where each quotient q_i equals 1 and the remainders decrease as slowly as possible.

Example 0.1.7: Computing a GCD

Take $a = 252$ and $b = 105$. The Euclidean algorithm proceeds:

$$\begin{aligned}
 252 &= 105 \cdot 2 + 42, \\
 105 &= 42 \cdot 2 + 21, \\
 42 &= 21 \cdot 2 + 0
 \end{aligned}$$

The last nonzero remainder is 21, so $\gcd(252, 105) = 21$. The algorithm terminates in three steps, reflecting the fact that $252/105$ has a short continued fraction expansion.

The Euclidean algorithm does more than compute the GCD—it simultaneously produces an explicit $\mathbb{Z}$ -linear combination. Tracing back through the successive divisions recovers the coefficients in what is arguably the single most useful identity in elementary number theory.

Theorem 0.1.8: Bézout's Identity

For any $a, b \in \mathbb{Z}$ not both zero, there exist $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a, b)$$

In particular, a and b are coprime if and only if 1 is a $\mathbb{Z}$ -linear combination of a and b .

This is a statement about the ideal structure of $\mathbb{Z}$: the ideal generated by a and b ,

$$(a, b) = \{ax + by \mid x, y \in \mathbb{Z}\},$$

is principal, generated by $\gcd(a, b)$. Every ideal of $\mathbb{Z}$ is principal—the ring $\mathbb{Z}$ is a PID. In rings of integers of number fields, this property can fail: in $\mathbb{Z}[\sqrt{-5}]$, for instance, the ideal $(2, 1 + \sqrt{-5})$ is not principal. The deviation from principality is measured by the ideal class group, one of the central objects studied in [14] and a key player in the analytic class number formula of ??.

Proof:

Define d to be the smallest positive element of $I = \{ax + by \mid x, y \in \mathbb{Z}\}$. This set contains positive elements (for instance, $a^2 + b^2$ if a and b are not both zero), so d exists by the well-ordering principle.

Step 1: d divides a and b . Apply the division algorithm to write $a = dq + r$ with $0 \leq r < d$. Since $d = ax_0 + by_0$ for some $x_0, y_0 \in \mathbb{Z}$, the remainder satisfies

$$r = a - dq = a(1 - qx_0) + b(-qy_0) \in I$$

If $r > 0$, then r is a positive element of I smaller than d , contradicting the choice of d . Hence $r = 0$, so $d \mid a$. The identical argument with b in place of a gives $d \mid b$.

Step 2: $d = \gcd(a, b)$. Since $d \mid a$ and $d \mid b$, the integer d is a common divisor. If c is any common divisor of a and b , then c divides every element of I (by proposition 0.1.3), and in particular $c \mid d$. Hence d is the greatest common divisor.

The proof is clean but non-constructive: it asserts existence via the well-ordering principle without producing x and y explicitly. To find them, one runs the Euclidean algorithm forward and then substitutes backward through the chain of divisions (see example 0.1.10 below).

A direct and frequently used consequence of Bézout's identity is the following cancellation property for coprime integers. It is the mechanism underlying unique factorization in the next section.

Corollary 0.1.9: Euclid's Lemma for Coprime Integers

If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.

Proof:

By theorem 0.1.8, there exist $x, y \in \mathbb{Z}$ with $ax + by = 1$. Multiplying through by c gives $acx + bcy = c$. Since $a \mid acx$ (trivially) and $a \mid bcy$ (by hypothesis), proposition 0.1.3 gives $a \mid c$.

The proof illustrates the standard technique: invoke Bézout to produce a linear combination equal to 1, then multiply by the desired target. This argument generalizes verbatim to any principal ideal domain, but fails in rings where Bézout's identity does not hold—such as $\mathbb{Z}[\sqrt{-5}]$, where $2 \mid (1 + \sqrt{-5})(1 - \sqrt{-5}) = 6$ but 2 divides neither factor.

Example 0.1.10: Back-Substitution for Bézout Coefficients

Continuing example 0.1.7, the Euclidean algorithm gave:

$$252 = 105 \cdot 2 + 42,$$

$$105 = 42 \cdot 2 + 21$$

From the second equation: $21 = 105 - 42 \cdot 2$. Substituting $42 = 252 - 105 \cdot 2$ from the first:

$$21 = 105 - (252 - 105 \cdot 2) \cdot 2 = 105 \cdot 5 + 252 \cdot (-2)$$

This gives $252 \cdot (-2) + 105 \cdot 5 = 21 = \gcd(252, 105)$, so the Bézout coefficients are $x = -2$ and $y = 5$. These are not unique: the general solution to $252x + 105y = 21$ is $(x, y) = (-2 + 5t, 5 - 12t)$ for $t \in \mathbb{Z}$, as the next result makes precise.

Bézout's identity settles the question of when a linear equation over $\mathbb{Z}$ has integer solutions.

Proposition 0.1.11: Linear Diophantine Equations

Let $a, b, c \in \mathbb{Z}$ with a and b not both zero, and set $d = \gcd(a, b)$. The equation

$$ax + by = c$$

has a solution $(x, y) \in \mathbb{Z}^2$ if and only if $d \mid c$. When solutions exist, they form the family

$$(x, y) = \left(x_0 + \frac{b}{d}t, y_0 - \frac{a}{d}t \right), \quad t \in \mathbb{Z},$$

where (x_0, y_0) is any particular solution.

Proof:

If (x, y) is a solution, then $d \mid a$ and $d \mid b$ give $d \mid (ax + by) = c$, establishing necessity. Conversely, if $d \mid c$, then theorem 0.1.8 provides $x'_0, y'_0 \in \mathbb{Z}$ with $ax'_0 + by'_0 = d$, and multiplying through by c/d gives the particular solution $(x_0, y_0) = ((c/d)x'_0, (c/d)y'_0)$.

For the general solution, note that (x, y) satisfies $ax + by = c$ if and only if $a(x - x_0) = -b(y - y_0)$. Dividing by d yields

$$\frac{a}{d}(x - x_0) = -\frac{b}{d}(y - y_0)$$

Since $\gcd(a/d, b/d) = 1$, corollary 0.1.9 gives $(b/d) \mid (x - x_0)$. Writing $x - x_0 = (b/d)t$ and substituting gives $y - y_0 = -(a/d)t$, as claimed.

The essential mechanism is the reduction to the coprime case: dividing by $d = \gcd(a, b)$ produces the coprimality condition $\gcd(a/d, b/d) = 1$, which then forces the structure of the solution set via

Bézout. This pattern—divide out the GCD, argue in the coprime case, and then scale back—recurs throughout number theory, from the analysis of congruences in the next section to the study of Dirichlet characters in ??.

Example 0.1.12: Solving a Linear Diophantine Equation

Consider $14x + 35y = 63$. Here $\gcd(14, 35) = 7$ and $7 \mid 63$, so solutions exist. Dividing through by 7 reduces the problem to $2x + 5y = 9$. By inspection, $(x_0, y_0) = (2, 1)$ satisfies $2 \cdot 2 + 5 \cdot 1 = 9$. The general solution to the original equation is

$$(x, y) = (2 + 5t, 1 - 2t), \quad t \in \mathbb{Z},$$

producing the solutions $\dots, (-3, 3), (2, 1), (7, -1), (12, -3), \dots$. For instance, the unique solution with $0 \leq x < 5$ is $(x, y) = (2, 1)$.

0.2 Primes and the Fundamental Theorem of Arithmetic

The Euclidean algorithm and Bézout’s identity treat all integers uniformly, without distinguishing between those that are “composed” and those that are not. This section introduces the multiplicative building blocks of $\mathbb{Z}$ —the prime numbers—and proves that every positive integer factors uniquely into primes. Unique factorization is the structural fact that makes the Euler product formula work (chapter 4): the identity $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ is, at its core, a restatement of the Fundamental Theorem of Arithmetic in the language of Dirichlet series.

Definition 0.2.1: Prime and Irreducible Elements

An integer $p > 1$ is *irreducible* if its only positive divisors are 1 and p . An integer $p > 1$ is *prime* if whenever $p \mid ab$, either $p \mid a$ or $p \mid b$.

In $\mathbb{Z}$, these two notions coincide—a fact that depends on Euclid’s lemma and therefore on the Euclidean algorithm. The distinction matters in more general rings. In $\mathbb{Z}[\sqrt{-5}]$, for instance, 2 is irreducible (it cannot be written as a product of two non-units) but not prime: $2 \mid (1 + \sqrt{-5})(1 - \sqrt{-5}) = 6$, yet 2 divides neither factor. The failure of “irreducible implies prime” is precisely the obstruction to unique factorization, and its resolution in number rings via the passage to ideals is a central theme of [14].

Proposition 0.2.2: In $\mathbb{Z}$, Prime and Irreducible Coincide

An integer $p > 1$ is prime if and only if it is irreducible.

Proof:

Prime implies irreducible. Suppose p is prime and $p = ab$ with $a, b > 0$. Then $p \mid ab$, so $p \mid a$ or $p \mid b$. If $p \mid a$, then since $a \leq p$ (as $b \geq 1$), this forces $a = p$ and $b = 1$. Similarly if $p \mid b$. Hence p is irreducible.

Irreducible implies prime. Suppose p is irreducible and $p \mid ab$. If $p \nmid a$, then the only positive common divisor of p and a is 1 (since p is irreducible and its only positive divisors are 1 and p , and $p \nmid a$ rules out p). Hence $\gcd(p, a) = 1$, and corollary 0.1.9 gives $p \mid b$.

The key step—the implication “irreducible implies prime”—relies entirely on corollary 0.1.9, which in turn depends on Bézout’s identity. This chain of dependencies is the mechanism by which the division algorithm guarantees unique factorization in $\mathbb{Z}$. In rings without a division algorithm (or more precisely, without the property that irreducibles are prime), unique factorization can fail.

The first result about the collection of all primes is among the oldest in mathematics.

Proposition 0.2.3: Infinitude of Primes

There are infinitely many primes.

Proof:

Suppose for contradiction that $p_1, p_2, \dots, p_n$ is a complete list of all primes. Consider the integer $N = p_1 p_2 \cdots p_n + 1$. Since $N > 1$, it has a prime divisor p . For each i , the prime p_i divides $p_1 \cdots p_n$ but does not divide N (since dividing N by p_i leaves remainder 1). Hence p is not among $p_1, \dots, p_n$, contradicting the assumption that the list was complete.

Euclid’s argument establishes the infinitude of primes but says nothing about how they are distributed among the integers. The quantitative study of this distribution—how many primes are there up to x , and how are they spaced?—is the central question of analytic number theory. The Prime Number Theorem (chapter 5) gives the asymptotic answer $\pi(x) \sim x/\log x$, but reaching it requires the full power of complex analysis applied to $\zeta(s)$.

The Fundamental Theorem of Arithmetic asserts that every integer greater than 1 can be written as a product of primes, and that this product is essentially unique.

Theorem 0.2.4: Fundamental Theorem of Arithmetic

Every integer $n > 1$ has a factorization

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

where $p_1 < p_2 < \cdots < p_k$ are primes and $a_1, \dots, a_k \geq 1$. This factorization is unique: if $n = q_1^{b_1} \cdots q_\ell^{b_\ell}$ is another such factorization with $q_1 < \cdots < q_\ell$, then $k = \ell$, $p_i = q_i$, and $a_i = b_i$ for all i .

Proof:

Step 1: Existence. Proceed by strong induction on n . If $n = 2$, then n is prime and $n = 2$ is itself a prime factorization. For $n > 2$: if n is prime, the factorization $n = n^1$ works. If n is not prime, then $n = ab$ for some $1 < a, b < n$, and by the inductive hypothesis both a and b have prime factorizations. Concatenating and reordering gives a prime factorization of n .

Step 2: Uniqueness. Suppose $p_1^{a_1} \cdots p_k^{a_k} = q_1^{b_1} \cdots q_\ell^{b_\ell}$. Then p_1 divides the right-hand side, so $p_1 \mid q_j^{b_j}$ for some j (by proposition 0.2.2, applied repeatedly). Since q_j is prime and $p_1 > 1$, this forces $p_1 = q_j$. As both lists are sorted in increasing order: if $p_1 < q_1$, then $p_1 \mid q_1^{b_1} \cdots q_\ell^{b_\ell}$ but $p_1 \neq q_j$ for any j (since $p_1 < q_1 \leq q_j$), a contradiction. Similarly $q_1 < p_1$ is impossible. Hence $p_1 = q_1$.

Dividing both sides by $p_1^{\min(a_1, b_1)}$ and applying the same argument inductively shows that the two factorizations agree term by term. In particular, $k = \ell$, $p_i = q_i$, and $a_i = b_i$ for all i .

The existence of a prime factorization uses only the well-ordering of $\mathbb{N}$ (via strong induction). The uniqueness, on the other hand, is the substantive content: it depends on the fact that primes in $\mathbb{Z}$ satisfy the cancellation property of proposition 0.2.2, which traces back through Euclid’s lemma to the division algorithm. Removing any link in this chain allows uniqueness to fail, as the example $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ in $\mathbb{Z}[\sqrt{-5}]$ demonstrates.

Example 0.2.5: Prime Factorizations

1. $360 = 2^3 \cdot 3^2 \cdot 5$. This is the unique way to write 360 as a product of prime powers (up to reordering).
2. $2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$. This is a *primorial*: the product of the first five primes. Primorials are squarefree integers (every prime appears with exponent exactly 1) and appear naturally in sieve arguments (??).
3. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$. The exponents can be computed directly from the Fundamental Theorem, but Legendre’s formula (proved at the end of this section) provides a much faster method.

The Fundamental Theorem is the reason why the study of multiplicative properties of $\mathbb{Z}$ reduces to the study of primes. Every question about divisibility, common factors, or the multiplicative structure of an integer can be answered by examining its prime factorization. This reductive principle has an analytic counterpart: the Euler product identity $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ converts a sum over all positive integers into a product over primes, and it is through this product that the analytic properties of $\zeta(s)$ yield information about the distribution of primes.

With unique factorization established, every positive integer n is completely described by its *prime signature*: the function $p \mapsto v_p(n)$ giving the exponent of each prime in the factorization of n .

Definition 0.2.6: The p -adic Valuation

For a prime p and a nonzero integer n , the *p -adic valuation* of n , denoted $v_p(n)$, is the largest integer $e \geq 0$ such that $p^e \mid n$. By convention, $v_p(0) = +\infty$. The Fundamental Theorem of Arithmetic gives

$$|n| = \prod_p p^{v_p(n)}$$

where the product is over all primes and is finite (since $v_p(n) = 0$ for all but finitely many p).

The p -adic valuation converts multiplicative questions about $\mathbb{Z}$ into additive questions about $\mathbb{Z}_{\geq 0}$: if $n = ab$ then $v_p(n) = v_p(a) + v_p(b)$ for every prime p . This decomposition—analyzing a global multiplicative problem one prime at a time—is the “local-to-global” philosophy that pervades number theory, from the Euler product for $\zeta(s)$ to the adelic methods of class field theory.

The valuation extends naturally to $\mathbb{Q}^\times$ by setting $v_p(a/b) = v_p(a) - v_p(b)$, giving a group homomorphism $v_p : \mathbb{Q}^\times \rightarrow \mathbb{Z}$. Every nonzero rational number r can then be written uniquely as $r = \pm \prod_p p^{v_p(r)}$, where the product is finite. The associated *p -adic absolute value* $|r|_p = p^{-v_p(r)}$ defines a metric on $\mathbb{Q}$ whose completion is the field $\mathbb{Q}_p$ of p -adic numbers (see [12]). The interplay

between all these completions—one for each prime and one for the ordinary absolute value—is a recurring theme beginning with the completed zeta function in ??.

Proposition 0.2.7: Divisibility via Valuations

For nonzero integers a and b :

1. $a|b$ if and only if $v_p(a) \leq v_p(b)$ for every prime p .
2. $\gcd(a, b) = \prod_p p^{\min(v_p(a), v_p(b))}$.
3. $\text{lcm}(a, b) = \prod_p p^{\max(v_p(a), v_p(b))}$.
4. $\gcd(a, b) \cdot \text{lcm}(a, b) = |ab|$.

Proof:

For (1): $a|b$ means $b = ak$ for some $k \in \mathbb{Z}$, which translates to $v_p(b) = v_p(a) + v_p(k) \geq v_p(a)$ for all p . The converse follows by writing $k = b/a$ and checking that $v_p(k) = v_p(b) - v_p(a) \geq 0$ for all p , so k is an integer. Parts (2) and (3) follow from (1): the greatest common divisor is the largest integer dividing both a and b , which means taking the minimum exponent at each prime; the least common multiple is the smallest integer divisible by both, which means taking the maximum. Part (4) follows from the identity $\min(x, y) + \max(x, y) = x + y$.

The valuation-based characterization of divisibility is the “right” perspective for many computations. For example, it makes the following formula for gcd immediate.

Example 0.2.8: GCD and LCM via Valuations

1. Take $a = 252 = 2^2 \cdot 3^2 \cdot 7$ and $b = 105 = 3 \cdot 5 \cdot 7$. Reading off:

$$\gcd(252, 105) = 2^0 \cdot 3^1 \cdot 5^0 \cdot 7^1 = 21, \quad \text{lcm}(252, 105) = 2^2 \cdot 3^2 \cdot 5 \cdot 7 = 1260$$

As a check: $\gcd \cdot \text{lcm} = 21 \cdot 1260 = 26460 = 252 \cdot 105 = |ab|$.

2. For $a = 360 = 2^3 \cdot 3^2 \cdot 5$ and $b = 750 = 2 \cdot 3 \cdot 5^3$:

$$\gcd(360, 750) = 2^1 \cdot 3^1 \cdot 5^1 = 30, \quad \text{lcm}(360, 750) = 2^3 \cdot 3^2 \cdot 5^3 = 9000$$

The Fundamental Theorem also provides a clean way to express many arithmetic functions. The number of positive divisors of $n = \prod p_i^{a_i}$ is $\prod (a_i + 1)$, since each divisor is obtained by independently choosing an exponent between 0 and a_i at each prime. The sum of divisors is $\prod (1 + p_i + p_i^2 + \cdots + p_i^{a_i}) = \prod \frac{p_i^{a_i+1} - 1}{p_i - 1}$. These formulas are simple consequences of unique factorization combined with the Chinese Remainder Theorem, and they generalize to the full family of multiplicative arithmetic functions studied in chapter 1.

Example 0.2.9: Divisor Count and Divisor Sum

1. The integer $n = 360 = 2^3 \cdot 3^2 \cdot 5$ has $(3 + 1)(2 + 1)(1 + 1) = 24$ positive divisors.

2. The sum of divisors of 360 is

$$\sigma(360) = \frac{2^4 - 1}{2 - 1} \cdot \frac{3^3 - 1}{3 - 1} \cdot \frac{5^2 - 1}{5 - 1} = 15 \cdot 13 \cdot 6 = 1170$$

3. For a prime p : the number of divisors of p^k is $k + 1$, and $\sigma(p^k) = \frac{p^{k+1} - 1}{p - 1}$. In particular, $\sigma(p) = p + 1$.

A useful bookkeeping device that appears throughout the book is the *canonical factorization notation*. For $n = \prod_{i=1}^k p_i^{a_i}$, a sum or product “over the prime factorization of n ” means over the pairs (p_i, a_i) . The notation $p^a \parallel n$ (read “ p^a exactly divides n ”) means $v_p(n) = a$, i.e., $p^a \mid n$ but $p^{a+1} \nmid n$.

Definition 0.2.10: Exact Divisibility

For a prime p and a positive integer n , the notation $p^a \parallel n$ means $v_p(n) = a$, that is, $p^a \mid n$ and $p^{a+1} \nmid n$.

To close this section, a formula that illustrates how the Fundamental Theorem interacts with factorials. It will be used in the proof of Chebyshev’s bounds in chapter 3.

Proposition 0.2.11: Legendre’s Formula

For any prime p and positive integer n :

$$v_p(n!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor$$

The sum is finite, as $\lfloor n/p^k \rfloor = 0$ for $p^k > n$.

Proof:

The exponent $v_p(n!)$ counts the total number of factors of p appearing in the product $1 \cdot 2 \cdots n$. Among the integers $1, 2, \dots, n$, there are exactly $\lfloor n/p \rfloor$ multiples of p , each contributing at least one factor. Of these, $\lfloor n/p^2 \rfloor$ are multiples of p^2 , each contributing a second factor. Continuing, $\lfloor n/p^k \rfloor$ multiples of p^k each contribute a k -th factor. Summing over all k counts each factor of p exactly once, giving the formula.

Example 0.2.12: Applying Legendre’s Formula

The exponent of 2 in $20!$ is

$$v_2(20!) = \left\lfloor \frac{20}{2} \right\rfloor + \left\lfloor \frac{20}{4} \right\rfloor + \left\lfloor \frac{20}{8} \right\rfloor + \left\lfloor \frac{20}{16} \right\rfloor = 10 + 5 + 2 + 1 = 18$$

Similarly, $v_3(20!) = 6 + 2 = 8$ and $v_5(20!) = 4$. In particular, $20!$ is divisible by $10^4 = 2^4 \cdot 5^4$ but not by 10^5 , so $20! = 2432902008176640000$ ends in exactly four trailing zeros.

An immediate consequence of Legendre’s formula is a clean expression for the p -adic valuation of binomial coefficients, which plays a role in Chebyshev’s estimates (chapter 3).

Corollary 0.2.13: Valuation of Binomial Coefficients

For a prime p and integers $0 \leq m \leq n$:

$$v_p \binom{n}{m} = \sum_{k=1}^{\infty} \left(\left\lfloor \frac{n}{p^k} \right\rfloor - \left\lfloor \frac{m}{p^k} \right\rfloor - \left\lfloor \frac{n-m}{p^k} \right\rfloor \right)$$

Each summand is either 0 or 1, and it equals 1 precisely when the addition $m + (n - m)$ produces a carry at the p^k -place in base p .

Proof:

Since $\binom{n}{m} = n!/(m!(n-m)!)$, the formula follows from Legendre's formula and the additivity of v_p :

$$v_p \binom{n}{m} = v_p(n!) - v_p(m!) - v_p((n-m)!) = \sum_{k \geq 1} \left(\left\lfloor \frac{n}{p^k} \right\rfloor - \left\lfloor \frac{m}{p^k} \right\rfloor - \left\lfloor \frac{n-m}{p^k} \right\rfloor \right)$$

For the “carry” interpretation: let $a = m$ and $b = n - m$, so $a + b = n$. The identity $\lfloor (a+b)/p^k \rfloor - \lfloor a/p^k \rfloor - \lfloor b/p^k \rfloor$ equals 1 when the addition of the base- p digits of a and b produces a carry into the p^k -position, and 0 otherwise. This is Kummer's theorem.

Example 0.2.14: Central Binomial Coefficient

Consider $\binom{10}{5} = 252$. In base 2: $5 = (101)_2$ and $5 + 5 = 10 = (1010)_2$. Adding $(101)_2 + (101)_2$, carries occur at positions 2^0 and 2^2 , so $v_2 \binom{10}{5} = 2$. Indeed, $252 = 2^2 \cdot 3^2 \cdot 7$. Similarly, in base 3: $5 = (12)_3$ and $5 + 5 = 10 = (101)_3$; adding $(12)_3 + (12)_3$ produces carries at 3^0 and 3^1 , giving $v_3 \binom{10}{5} = 2$.

0.3 Congruences

The previous sections studied the integers through their multiplicative structure: divisibility, primes, and factorization. Congruences introduce a complementary perspective by collapsing $\mathbb{Z}$ into finite quotient rings $\mathbb{Z}/m\mathbb{Z}$. This modular viewpoint is the natural setting for Dirichlet characters (??), and the Chinese Remainder Theorem—the structural result of this section—is the tool that decomposes these characters into local components, one for each prime power dividing the modulus.

Definition 0.3.1: Congruence

Let $m \in \mathbb{Z}_{>0}$. Two integers a and b are *congruent modulo m* , written $a \equiv b \pmod{m}$, if $m \mid (a - b)$. Equivalently, a and b leave the same remainder upon division by m .

Congruence modulo m is an equivalence relation on $\mathbb{Z}$, partitioning it into m equivalence classes called *residue classes*. The set of residue classes forms a ring under the inherited addition and multiplication:

Definition 0.3.2: The Ring $\mathbb{Z}/m\mathbb{Z}$

The *residue class ring modulo m* is the quotient ring

$$\mathbb{Z}/m\mathbb{Z} = \{0, 1, 2, \dots, m-1\}$$

with addition and multiplication performed modulo m . The canonical projection $\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$ sends each integer to its residue class.

The ring $\mathbb{Z}/m\mathbb{Z}$ is the simplest example of a quotient ring, and its structure—which residue classes are units, which are zero divisors, and how the ring decomposes—is controlled entirely by the prime factorization of m . The key structural result is the Chinese Remainder Theorem, which says that if m factors into coprime parts, then $\mathbb{Z}/m\mathbb{Z}$ splits as a product of smaller rings.

Before stating it, the following criterion for invertibility modulo m is needed.

Proposition 0.3.3: Units of $\mathbb{Z}/m\mathbb{Z}$

An element $a \in \mathbb{Z}/m\mathbb{Z}$ is a unit (i.e., has a multiplicative inverse modulo m) if and only if $\gcd(a, m) = 1$. When this holds, the inverse can be computed via the Euclidean algorithm: if $ax + my = 1$ (from theorem 0.1.8), then $a^{-1} \equiv x \pmod{m}$.

Proof:

If $\gcd(a, m) = 1$, then theorem 0.1.8 gives $ax + my = 1$ for some $x, y \in \mathbb{Z}$, which reads $ax \equiv 1 \pmod{m}$, so x is the inverse of a . Conversely, if $ax \equiv 1 \pmod{m}$ for some x , then $ax = 1 + km$ for some k , giving $ax - km = 1$. By proposition 0.1.3, any common divisor of a and m divides 1, so $\gcd(a, m) = 1$.

The group of units $(\mathbb{Z}/m\mathbb{Z})^\times = \{a \in \mathbb{Z}/m\mathbb{Z} \mid \gcd(a, m) = 1\}$ has order $\varphi(m)$, where φ is the Euler totient function. This group is the domain on which Dirichlet characters live, and its structure will be analyzed in detail in the next section.

An element $a \in \mathbb{Z}/m\mathbb{Z}$ with $a \neq 0$ and $\gcd(a, m) > 1$ is a *zero divisor*: writing $d = \gcd(a, m)$ with $1 < d < m$, the product $a \cdot (m/d) \equiv 0 \pmod{m}$ is zero even though neither factor is. Thus $\mathbb{Z}/m\mathbb{Z}$ is an integral domain if and only if m is prime, in which case every nonzero element is a unit and $\mathbb{Z}/m\mathbb{Z} = \mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$ is a field with p elements. This is the finite field that appears throughout the book, from character sums (??) to the local factors of L -functions.

Example 0.3.4: Units and Zero Divisors in $\mathbb{Z}/12\mathbb{Z}$

The elements of $\mathbb{Z}/12\mathbb{Z}$ partition as follows:

1. *Units* ($\gcd(a, 12) = 1$): $\{1, 5, 7, 11\}$. Each has a multiplicative inverse: $5 \cdot 5 = 25 \equiv 1$, $7 \cdot 7 = 49 \equiv 1$, $11 \cdot 11 = 121 \equiv 1$. So $5^{-1} = 5$, $7^{-1} = 7$, $11^{-1} = 11$ in $\mathbb{Z}/12\mathbb{Z}$.
2. *Zero divisors* ($1 < \gcd(a, 12) < 12$, $a \neq 0$): $\{2, 3, 4, 6, 8, 9, 10\}$. For instance, $3 \cdot 4 = 12 \equiv 0$, $6 \cdot 2 = 12 \equiv 0$.
3. *The zero element*: $\{0\}$.

Definition 0.3.5: Euler Totient Function

For $m \in \mathbb{Z}_{>0}$, the *Euler totient function* $\varphi(m)$ is the number of integers a with $1 \leq a \leq m$ and $\gcd(a, m) = 1$. Equivalently, $\varphi(m) = |(\mathbb{Z}/m\mathbb{Z})^\times|$.

Example 0.3.6: Small Values of φ

1. $\varphi(1) = 1$, since $\gcd(1, 1) = 1$.
2. $\varphi(p) = p - 1$ for any prime p , since every integer from 1 to $p - 1$ is coprime to p .
3. $\varphi(12) = |\{1, 5, 7, 11\}| = 4$, as these are the residues coprime to 12.
4. $\varphi(p^k) = p^k - p^{k-1} = p^{k-1}(p - 1)$ for a prime power: among $1, \dots, p^k$, the multiples of p are $p, 2p, \dots, p^{k-1} \cdot p$, and there are p^{k-1} of them.

The multiplicativity of φ is a consequence of the Chinese Remainder Theorem, stated next.

Theorem 0.3.7: Chinese Remainder Theorem

Let $m_1, m_2, \dots, m_k$ be pairwise coprime positive integers, and set $M = m_1 m_2 \cdots m_k$. Then the map

$$\mathbb{Z}/M\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}/m_1\mathbb{Z} \times \mathbb{Z}/m_2\mathbb{Z} \times \cdots \times \mathbb{Z}/m_k\mathbb{Z}, \quad a \mapsto (a \bmod m_1, a \bmod m_2, \dots, a \bmod m_k)$$

is a ring isomorphism.

The CRT says that specifying a residue class modulo M is the same as independently specifying a residue class modulo each m_i . This is the arithmetic analogue of the decomposition of a product of topological spaces into its factors, and it is the mechanism that allows global questions about $\mathbb{Z}/M\mathbb{Z}$ to be reduced to local questions about each $\mathbb{Z}/m_i\mathbb{Z}$.

Proof:

Step 1: The map is a ring homomorphism. The reduction map $a \mapsto (a \bmod m_1, \dots, a \bmod m_k)$ preserves addition and multiplication, since these operations in each factor are performed modulo m_i .

Step 2: Injectivity. If $a \equiv 0 \pmod{m_i}$ for all i , then $m_i \mid a$ for each i . Since the m_i are pairwise coprime, $M = m_1 \cdots m_k$ divides a (apply corollary 0.1.9 inductively: $m_1 \mid a$ and $\gcd(m_1, m_2) = 1$ together with $m_2 \mid a$ give $m_1 m_2 \mid a$; continuing gives $M \mid a$). Hence the kernel is trivial.

Step 3: Surjectivity. Both sides have the same cardinality $M = m_1 \cdots m_k$, so injectivity implies surjectivity.

Alternatively, surjectivity can be shown constructively. For each i , define $M_i = M/m_i$ and note that $\gcd(M_i, m_i) = 1$ (since the m_j for $j \neq i$ are all coprime to m_i). By theorem 0.1.8, there

exists y_i with $M_i y_i \equiv 1 \pmod{m_i}$. Given a target $(a_1, \dots, a_k)$, the integer

$$a = \sum_{i=1}^k a_i M_i y_i$$

satisfies $a \equiv a_i \pmod{m_i}$ for each i , since $M_j y_j \equiv 0 \pmod{m_i}$ when $j \neq i$ (as $m_i \mid M_j$) and $M_i y_i \equiv 1 \pmod{m_i}$.

The constructive proof is useful in practice: it gives an explicit formula for “gluing” local data into a global solution. The abstract proof via cardinality counting is cleaner but less informative.

Example 0.3.8: Solving a System of Congruences

Solve the system $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$. Here $M = 105$, $M_1 = 35$, $M_2 = 21$, $M_3 = 15$. The inverses are: $35y_1 \equiv 1 \pmod{3}$, giving $y_1 = 2$ (since $35 \cdot 2 = 70 \equiv 1$); $21y_2 \equiv 1 \pmod{5}$, giving $y_2 = 1$ (since $21 \equiv 1$); $15y_3 \equiv 1 \pmod{7}$, giving $y_3 = 1$ (since $15 \equiv 1$). Then

$$x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 140 + 63 + 30 = 233 \equiv 23 \pmod{105}$$

One verifies: $23 = 7 \cdot 3 + 2 \equiv 2 \pmod{3}$, $23 = 4 \cdot 5 + 3 \equiv 3 \pmod{5}$, $23 = 3 \cdot 7 + 2 \equiv 2 \pmod{7}$.

An immediate corollary of the CRT is the multiplicativity of the Euler totient.

Corollary 0.3.9: Multiplicativity of φ

If $\gcd(m, n) = 1$, then $\varphi(mn) = \varphi(m)\varphi(n)$.

Proof:

By theorem 0.3.7, $(\mathbb{Z}/mn\mathbb{Z})^\times \cong (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$, since the isomorphism $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ restricts to an isomorphism on unit groups (an element is a unit in the product if and only if each component is a unit). Taking cardinalities: $\varphi(mn) = \varphi(m)\varphi(n)$.

Combined with the prime power formula $\varphi(p^k) = p^{k-1}(p-1)$ from example 0.3.6, this gives a complete formula for φ .

Proposition 0.3.10: Formula for the Euler Totient

For any $n = p_1^{a_1} \cdots p_k^{a_k} > 1$:

$$\varphi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right) = \prod_{i=1}^k p_i^{a_i-1} (p_i - 1)$$

Proof:

By corollary 0.3.9 and induction, $\varphi(n) = \prod_{i=1}^k \varphi(p_i^{a_i}) = \prod_{i=1}^k p_i^{a_i-1} (p_i - 1)$. Factoring out $p_i^{a_i}$ from each term gives the product formula $\varphi(n) = n \prod_{p \mid n} (1 - 1/p)$.

Example 0.3.11: Computing φ

$$\varphi(360) = \varphi(2^3 \cdot 3^2 \cdot 5) = 360 \cdot (1 - \frac{1}{2})(1 - \frac{1}{3})(1 - \frac{1}{5}) = 360 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5} = 96.$$

The section closes with the two classical results that connect the Euler totient to modular exponentiation. These are the finite-group-theoretic facts that underlie the convergence arguments for Dirichlet series twisted by characters.

Theorem 0.3.12: Euler's Theorem

If $\gcd(a, m) = 1$, then $a^{\varphi(m)} \equiv 1 \pmod{m}$.

Proof:

The residue class of a is an element of the group $(\mathbb{Z}/m\mathbb{Z})^\times$, which has order $\varphi(m)$. By Lagrange's theorem (see [11]), the order of every element divides the order of the group. Hence $a^{\varphi(m)} \equiv 1 \pmod{m}$.

Euler's theorem is a statement about the group $(\mathbb{Z}/m\mathbb{Z})^\times$: every element has finite order dividing $\varphi(m)$. The special case $m = p$ prime gives the result traditionally attributed to Fermat.

Corollary 0.3.13: Fermat's Little Theorem

If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$. Equivalently, $a^p \equiv a \pmod{p}$ for all $a \in \mathbb{Z}$.

Proof:

Since $\varphi(p) = p - 1$, the first statement is a special case of theorem 0.3.12. For the second: if $p \mid a$ then $a^p \equiv 0 \equiv a \pmod{p}$; if $p \nmid a$ then $a^p = a \cdot a^{p-1} \equiv a \cdot 1 = a \pmod{p}$.

Another classical characterization of primes via modular arithmetic is Wilson's theorem.

Proposition 0.3.14: Wilson's Theorem

An integer $n \geq 2$ is prime if and only if $(n - 1)! \equiv -1 \pmod{n}$.

Proof:

If n is composite, say $n = ab$ with $1 < a \leq b < n$, then a appears as a factor in $(n - 1)!$. If $a \neq b$, both appear, and $n = ab$ divides $(n - 1)!$, so $(n - 1)! \equiv 0 \not\equiv -1 \pmod{n}$. If $a = b$, then $n = a^2 \geq 4$ (since $a \geq 2$), and both a and $2a$ appear in $\{1, \dots, n - 1\}$ when $n > 4$; the case $n = 4$ is verified directly: $3! = 6 \equiv 2 \not\equiv -1 \pmod{4}$.

If $n = p$ is prime, then $\mathbb{F}_p^\times = \{1, 2, \dots, p - 1\}$ is a group of order $p - 1$. Each element a pairs with its inverse a^{-1} . The elements satisfying $a = a^{-1}$, i.e., $a^2 \equiv 1 \pmod{p}$, are exactly $a \equiv \pm 1 \pmod{p}$ (since $p \mid (a - 1)(a + 1)$ forces $p \mid (a - 1)$ or $p \mid (a + 1)$). Hence in the product $1 \cdot 2 \cdots (p - 1)$, all elements except 1 and $p - 1$ cancel in pairs, leaving $(p - 1)! \equiv 1 \cdot (p - 1) = p - 1 \equiv -1 \pmod{p}$.

Wilson's theorem gives an exact criterion for primality, but it is computationally impractical since computing $(n-1)!$ modulo n requires $\Theta(n)$ multiplications. Its theoretical importance lies in providing explicit generators of ideals: the identity $(p-1)! + 1 \equiv 0 \pmod{p}$ is used in the proof of quadratic reciprocity (section 0.5).

Fermat's little theorem has a useful contrapositive: if $a^{p-1} \not\equiv 1 \pmod{p}$, then p is not prime. This is the basis of primality testing algorithms. The converse fails: there exist composite numbers n (called *Carmichael numbers*) such that $a^{n-1} \equiv 1 \pmod{n}$ for all a coprime to n . The smallest is $n = 561 = 3 \cdot 11 \cdot 17$, which satisfies this property because $(p-1) \mid 560$ for each prime factor $p \in \{3, 11, 17\}$ (since $560 = 2 \cdot 280 = 10 \cdot 56 = 16 \cdot 35$), and the CRT then guarantees $a^{560} \equiv 1 \pmod{561}$ for all a with $\gcd(a, 561) = 1$. This criterion—known as *Korselt's criterion*—states that n is a Carmichael number if and only if n is squarefree and $(p-1) \mid (n-1)$ for every prime p dividing n .

Example 0.3.15: Modular Exponentiation via Euler's Theorem

To compute $7^{222} \pmod{10}$, note that $\varphi(10) = 4$ and $\gcd(7, 10) = 1$, so $7^4 \equiv 1 \pmod{10}$ by theorem 0.3.12. Since $222 = 4 \cdot 55 + 2$:

$$7^{222} = (7^4)^{55} \cdot 7^2 \equiv 1^{55} \cdot 49 \equiv 9 \pmod{10}$$

The last digit of 7^{222} is therefore 9. More generally, the last digit of 7^n cycles with period 4: 7, 9, 3, 1, 7, 9, 3, 1, ... —a consequence of 7 having order exactly 4 in $(\mathbb{Z}/10\mathbb{Z})^\times$.

A complementary identity that counts divisors weighted by the totient is the following.

Proposition 0.3.16: Totient Summation Identity

For every positive integer n :

$$\sum_{d \mid n} \varphi(d) = n$$

Proof:

Partition the set $\{1, 2, \dots, n\}$ according to $\gcd(k, n)$. For each divisor d of n , the integers $k \in \{1, \dots, n\}$ with $\gcd(k, n) = d$ are exactly those of the form $k = d\ell$ where $1 \leq \ell \leq n/d$ and $\gcd(\ell, n/d) = 1$. There are $\varphi(n/d)$ such integers. Summing over all divisors d of n gives $n = \sum_{d \mid n} \varphi(n/d)$. Since $d \mapsto n/d$ is a bijection on the divisors of n , this equals $\sum_{d \mid n} \varphi(d)$.

This identity will reappear in chapter 1 as the Dirichlet convolution identity $\varphi * \mathbf{1} = \text{id}$, where $\mathbf{1}$ is the constant function 1 and id is the identity function $n \mapsto n$. Möbius inversion then recovers φ from this relation.

0.4 The Group $(\mathbb{Z}/m\mathbb{Z})^\times$ and Primitive Roots

Euler's theorem (theorem 0.3.12) guarantees that every element of $(\mathbb{Z}/m\mathbb{Z})^\times$ has finite multiplicative order dividing $\varphi(m)$. A natural question is whether this bound is achieved: does $(\mathbb{Z}/m\mathbb{Z})^\times$ always contain an element of order exactly $\varphi(m)$? Equivalently, is $(\mathbb{Z}/m\mathbb{Z})^\times$ cyclic? The answer is no in general— $(\mathbb{Z}/8\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ has exponent 2, so no element has order $\varphi(8) = 4$ —but a precise

characterization exists. Understanding the group structure of $(\mathbb{Z}/m\mathbb{Z})^\times$ is essential for the theory of Dirichlet characters in ??, since the characters of a finite abelian group are determined by its invariant factor decomposition.

Definition 0.4.1: Multiplicative Order

For $a \in (\mathbb{Z}/m\mathbb{Z})^\times$, the *multiplicative order* of a modulo m , denoted $\text{ord}_m(a)$, is the smallest positive integer k such that $a^k \equiv 1 \pmod{m}$.

By theorem 0.3.12, $\text{ord}_m(a)$ exists and divides $\varphi(m)$. The order is the period of the cyclic subgroup $\langle a \rangle = \{1, a, a^2, \dots, a^{k-1}\}$ inside $(\mathbb{Z}/m\mathbb{Z})^\times$, and the standard theory of cyclic groups (see [11]) gives the following.

Proposition 0.4.2: Basic Properties of Multiplicative Order

Let $a \in (\mathbb{Z}/m\mathbb{Z})^\times$ with $\text{ord}_m(a) = k$. Then:

1. $a^n \equiv 1 \pmod{m}$ if and only if $k \mid n$.
2. $\text{ord}_m(a^j) = k / \gcd(j, k)$.
3. The number of elements of order k in $(\mathbb{Z}/m\mathbb{Z})^\times$ is either 0 or $\varphi(k)$.

Proof:

For (1): if $k \mid n$ then $a^n = (a^k)^{n/k} \equiv 1$. Conversely, write $n = kq + r$ with $0 \leq r < k$; then $a^r = a^n \cdot (a^k)^{-q} \equiv 1$, and minimality of k forces $r = 0$.

For (2): let $d = \gcd(j, k)$. Then $(a^j)^{k/d} = (a^k)^{j/d} \equiv 1$, so $\text{ord}_m(a^j)$ divides k/d . If $(a^j)^n \equiv 1$ then $k \mid jn$ by (1), hence $(k/d) \mid (j/d)n$. Since $\gcd(j/d, k/d) = 1$, corollary 0.1.9 gives $(k/d) \mid n$. Hence $\text{ord}_m(a^j) = k/d$.

For (3): if some element a has order k , then $\langle a \rangle$ contains exactly one element of each order dividing k , and by (2) the number of generators of $\langle a \rangle$ (i.e., elements of order k in $\langle a \rangle$) is $\varphi(k)$. To see that no element outside $\langle a \rangle$ has order k : the polynomial $x^k - 1$ has at most k roots in $\mathbb{Z}/p\mathbb{Z}$ (since $\mathbb{Z}/p\mathbb{Z}$ is a field), and the k elements of $\langle a \rangle$ already exhaust them. For composite m , the argument requires the structure theorem below.

Part (3) uses a critical property of fields: a polynomial of degree k over a field has at most k roots. This fails over $\mathbb{Z}/m\mathbb{Z}$ when m is composite—for example, $x^2 - 1 \equiv 0 \pmod{8}$ has four solutions $\{1, 3, 5, 7\}$ —and this failure is precisely why the counting argument in the existence theorem below works only for prime moduli.

Definition 0.4.3: Primitive Root

A *primitive root* modulo m is an element $g \in (\mathbb{Z}/m\mathbb{Z})^\times$ of order $\varphi(m)$, i.e., a generator of $(\mathbb{Z}/m\mathbb{Z})^\times$. Equivalently, $(\mathbb{Z}/m\mathbb{Z})^\times$ is cyclic if and only if a primitive root exists.

When a primitive root g exists, every unit $a \in (\mathbb{Z}/m\mathbb{Z})^\times$ can be written as $a \equiv g^k \pmod{m}$ for a

unique $k \in \{0, 1, \dots, \varphi(m) - 1\}$. This integer k is the *discrete logarithm* of a to base g , and the map $a \mapsto k$ is an isomorphism $(\mathbb{Z}/m\mathbb{Z})^\times \xrightarrow{\sim} \mathbb{Z}/\varphi(m)\mathbb{Z}$ that converts multiplication to addition.

Example 0.4.4: The Discrete Logarithm Table Modulo 13

The group $(\mathbb{Z}/13\mathbb{Z})^\times$ has order 12. Testing $g = 2$: the successive powers are

$$\begin{aligned} 2^1 &= 2, & 2^2 &= 4, & 2^3 &= 8, & 2^4 &= 3, & 2^5 &= 6, & 2^6 &= 12, \\ 2^7 &= 11, & 2^8 &= 9, & 2^9 &= 5, & 2^{10} &= 10, & 2^{11} &= 7, & 2^{12} &= 1, \end{aligned}$$

confirming that 2 is a primitive root. The discrete logarithm $\log_2(a) \bmod 12$ converts multiplication in $(\mathbb{Z}/13\mathbb{Z})^\times$ to addition in $\mathbb{Z}/12\mathbb{Z}$. For instance, $5 \cdot 8 = 40 \equiv 1 \pmod{13}$, which corresponds to $\log_2(5) + \log_2(8) = 9 + 3 = 12 \equiv 0 \pmod{12}$. This conversion is the mechanism behind index calculus methods for computing in $(\mathbb{Z}/p\mathbb{Z})^\times$.

The following theorem characterizes exactly when primitive roots exist.

Theorem 0.4.5: Existence of Primitive Roots

The group $(\mathbb{Z}/m\mathbb{Z})^\times$ is cyclic (i.e., a primitive root modulo m exists) if and only if $m \in \{1, 2, 4, p^k, 2p^k\}$ where p is an odd prime and $k \geq 1$.

The proof has two parts: showing cyclicity for odd prime powers (the hard direction), and showing non-cyclicity for all other moduli.

Proof:

The cases $m = 1$ and $m = 2$ are trivial ($(\mathbb{Z}/1\mathbb{Z})^\times$ and $(\mathbb{Z}/2\mathbb{Z})^\times$ are both trivial). The case $m = 4$ is checked directly: $(\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\}$ is cyclic of order 2, with primitive root 3.

Step 1: $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic for p an odd prime. The group $(\mathbb{Z}/p\mathbb{Z})^\times$ has order $p-1$. For each divisor d of $p-1$, let $\psi(d)$ denote the number of elements of order d . Every element has some order dividing $p-1$, so $\sum_{d|(p-1)} \psi(d) = p-1$. By proposition 0.4.2(3), $\psi(d) \in \{0, \varphi(d)\}$ for each d . Since $\sum_{d|(p-1)} \varphi(d) = p-1$ (by proposition 0.3.16), and $\psi(d) \leq \varphi(d)$ for all d , the only possibility is $\psi(d) = \varphi(d)$ for every d . In particular, $\psi(p-1) = \varphi(p-1) \geq 1$, so an element of order $p-1$ exists.

Step 2: $(\mathbb{Z}/p^k\mathbb{Z})^\times$ is cyclic for p odd and $k \geq 2$. Let g be a primitive root modulo p . The group $(\mathbb{Z}/p^k\mathbb{Z})^\times$ has order $\varphi(p^k) = p^{k-1}(p-1)$. The reduction map $(\mathbb{Z}/p^k\mathbb{Z})^\times \rightarrow (\mathbb{Z}/p\mathbb{Z})^\times$ is surjective with kernel of order p^{k-1} , so $\text{ord}_{p^k}(g) = p^j(p-1)$ for some $0 \leq j \leq k-1$. It suffices to show $j = k-1$.

The key observation is: if $h \equiv 1 + tp \pmod{p^2}$ with $p \nmid t$, then $\text{ord}_{p^k}(h) = p^{k-1}$. To see this, use induction. The binomial theorem gives

$$h^p = (1 + tp)^p = 1 + tp^2 + \binom{p}{2} t^2 p^2 + \dots \equiv 1 + t' p^2 \pmod{p^3}$$

where $t' \equiv t \pmod{p}$ (since $p \mid \binom{p}{2}$ for $p \geq 3$), so $p \nmid t'$. Inductively, $h^{p^{j-1}} \equiv 1 + t_j p^j \pmod{p^{j+1}}$ with $p \nmid t_j$ for each j . In particular, $h^{p^{k-2}} \not\equiv 1 \pmod{p^k}$ while $h^{p^{k-1}} \equiv 1 \pmod{p^k}$, giving

$$\text{ord}_{p^k}(h) = p^{k-1}.$$

Applying this to $h = g^{p-1}$: if $g^{p-1} \not\equiv 1 \pmod{p^2}$, write $g^{p-1} = 1 + tp$ with $p \nmid t$, and the claim gives $\text{ord}_{p^k}(g^{p-1}) = p^{k-1}$. Then $\text{ord}_{p^k}(g) = p^{k-1}(p-1) = \varphi(p^k)$, so g is a primitive root modulo p^k . If instead $g^{p-1} \equiv 1 \pmod{p^2}$, replace g by $g' = g + p$ (which reduces to g modulo p , hence is still a primitive root modulo p). The binomial expansion gives $(g')^{p-1} \equiv g^{p-1} + (p-1)g^{p-2}p \equiv 1 - g^{p-2}p \pmod{p^2}$. Since $p \nmid g^{p-2}$, this shows $(g')^{p-1} \not\equiv 1 \pmod{p^2}$, so the argument applies to g' .

Step 3: $(\mathbb{Z}/2p^k\mathbb{Z})^\times$ is cyclic. By the Chinese Remainder Theorem (theorem 0.3.7), $(\mathbb{Z}/2p^k\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z})^\times \times (\mathbb{Z}/p^k\mathbb{Z})^\times \cong \{1\} \times (\mathbb{Z}/p^k\mathbb{Z})^\times$, which is cyclic.

Step 4: Non-cyclicity for other moduli. The remaining cases are handled by proposition 0.4.8 and proposition 0.4.9 below: $(\mathbb{Z}/2^k\mathbb{Z})^\times$ is not cyclic for $k \geq 3$, and the CRT decomposition for m with two or more distinct odd prime factors always produces a non-cyclic product (since each factor $(\mathbb{Z}/p_i^{a_i}\mathbb{Z})^\times$ has even order).

The proof that $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic uses a counting argument that avoids constructing a primitive root explicitly. Finding a primitive root modulo a given prime is a computational problem with no known polynomial-time deterministic algorithm, though efficient probabilistic methods exist.

Example 0.4.6: Primitive Roots for Small Primes

1. Modulo 7: $(\mathbb{Z}/7\mathbb{Z})^\times$ has order 6. Testing $g = 3$: the powers $3^1 = 3, 3^2 = 2, 3^3 = 6, 3^4 = 4, 3^5 = 5, 3^6 = 1$ exhaust all of $(\mathbb{Z}/7\mathbb{Z})^\times$, so 3 is a primitive root. The other primitive roots are the elements of order 6: by proposition 0.4.2(2), these are 3^j with $\gcd(j, 6) = 1$, giving $3^1 = 3$ and $3^5 = 5$. Hence $(\mathbb{Z}/7\mathbb{Z})^\times$ has $\varphi(6) = 2$ primitive roots.
2. Modulo 11: $\varphi(11) = 10$. Testing $g = 2$: $2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 5, 2^5 = 10, 2^6 = 9, 2^7 = 7, 2^8 = 3, 2^9 = 6, 2^{10} = 1$. So 2 is a primitive root modulo 11. There are $\varphi(10) = 4$ primitive roots: $2, 2^3 = 8, 2^7 = 7, 2^9 = 6$.

A fixed integer $a > 1$ that is not a perfect square is a primitive root modulo infinitely many primes—this is a conjecture of Artin (1927), still open in general. Hooley proved in 1967 that the conjecture follows from the Generalized Riemann Hypothesis. For instance, 2 is conjectured to be a primitive root modulo a positive proportion (approximately 37.4%) of all primes. The unconditional result closest to Artin's conjecture, due to Gupta-Murty and Heath-Brown, shows that at least one of 2, 3, or 5 is a primitive root modulo infinitely many primes.

Primitive roots also arise in a familiar context: the decimal expansion of $1/p$ for a prime $p \neq 2, 5$ has period exactly $\text{ord}_p(10)$. When 10 is a primitive root modulo p , the period is $p-1$, the maximum possible. For example, $1/7 = 0.\overline{142857}$ has period 6 = $\varphi(7)$ because $\text{ord}_7(10) = 6$ (i.e., 10 is a primitive root modulo 7), while $1/13 = 0.\overline{076923}$ also has maximal period 12 = $\varphi(13)$. On the other hand, $1/11 = 0.\overline{09}$ has period 2 because $\text{ord}_{11}(10) = 2$.

Example 0.4.7: Lifting a Primitive Root from p to p^2

Take $p = 7$ and $g = 3$, a primitive root modulo 7. Is 3 a primitive root modulo 49? The group $(\mathbb{Z}/49\mathbb{Z})^\times$ has order $\varphi(49) = 42 = 6 \cdot 7$, so the question reduces to checking whether $3^6 \equiv 1 \pmod{49}$. Computing: $3^6 = 729 = 14 \cdot 49 + 43$, so $3^6 \equiv 43 \equiv 1 + 42 \equiv 1 + 6 \cdot 7 \pmod{49}$. Since $3^6 \not\equiv 1 \pmod{49}$ and the coefficient 6 satisfies $7 \nmid 6$, the lifting argument from Step 2 applies: 3 is a primitive root modulo 49, and modulo 7^k for all $k \geq 1$.

On the other hand, take $p = 29$ and $g = 14$, a primitive root modulo 29. Computing $14^{28} \pmod{29^2}$: since $14^{28} \equiv 1 \pmod{29}$ (Fermat), write $14^{28} = 1 + 29t$. If $29 \nmid t$, then 14 does not lift, and one must replace it by $14 + 29 = 43$. In practice, such “exceptional” primitive roots are rare; for most primes p and most primitive roots g modulo p , the condition $g^{p-1} \not\equiv 1 \pmod{p^2}$ holds.

For moduli that do *not* admit primitive roots, the structure of $(\mathbb{Z}/m\mathbb{Z})^\times$ is determined by the following two results.

Proposition 0.4.8: Structure of $(\mathbb{Z}/2^k\mathbb{Z})^\times$

For $k \geq 3$: $(\mathbb{Z}/2^k\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{k-2}\mathbb{Z}$, generated by -1 and 5 .

Proof:

The group $(\mathbb{Z}/2^k\mathbb{Z})^\times$ has order $\varphi(2^k) = 2^{k-1}$. The element -1 has order 2, and 5 has order 2^{k-2} : to verify the latter, write $5 = 1 + 4$ and use induction. If $5^{2^{j-1}} \equiv 1 + c_j \cdot 2^{j+1} \pmod{2^{j+2}}$ with c_j odd, then

$$5^{2^j} = (5^{2^{j-1}})^2 \equiv 1 + c_j 2^{j+2} + c_j^2 2^{2j+2} \equiv 1 + c_j 2^{j+2} \pmod{2^{j+3}}$$

with c_j still odd (since only the linear term contributes modulo 2^{j+3} when $2j+2 \geq j+3$, i.e., $j \geq 1$). The base case $j = 0$: $5 = 1 + 1 \cdot 2^2$ with $c_0 = 1$. Inductively, $5^{2^{k-3}} \equiv 1 + c_{k-3} \cdot 2^{k-1} \not\equiv 1 \pmod{2^k}$, while $5^{2^{k-2}} \equiv 1 \pmod{2^k}$. Hence $\text{ord}_{2^k}(5) = 2^{k-2}$.

Since $5^j \equiv 1 \pmod{4}$ for all j (as $5 \equiv 1 \pmod{4}$) while $-1 \equiv 3 \pmod{4}$, no power of 5 equals -1 . Hence $\langle -1 \rangle \cap \langle 5 \rangle = \{1\}$, and the product $\langle -1 \rangle \times \langle 5 \rangle$ has order $2 \cdot 2^{k-2} = 2^{k-1} = |(\mathbb{Z}/2^k\mathbb{Z})^\times|$, giving the isomorphism.

Proposition 0.4.9: Structure of $(\mathbb{Z}/m\mathbb{Z})^\times$ for General m

For $m = 2^a p_1^{a_1} \cdots p_r^{a_r}$ with $p_1, \dots, p_r$ distinct odd primes:

$$(\mathbb{Z}/m\mathbb{Z})^\times \cong (\mathbb{Z}/2^a\mathbb{Z})^\times \times \prod_{i=1}^r (\mathbb{Z}/p_i^{a_i}\mathbb{Z})^\times$$

Each odd prime-power factor is cyclic of order $p_i^{a_i-1}(p_i - 1)$. A primitive root modulo m exists if and only if the product is cyclic, which happens exactly when $m \in \{1, 2, 4, p^k, 2p^k\}$.

Proof:

The decomposition is the CRT applied to unit groups, as in corollary 0.3.9. A direct product of cyclic groups is cyclic if and only if their orders are pairwise coprime (see [11]). If m has two distinct odd prime factors p_1, p_2 , then $\varphi(p_1^{a_1})$ and $\varphi(p_2^{a_2})$ are both even, so the product is not cyclic. Similarly, if $m = 2^a p^k$ with $a \geq 2$, then both $\varphi(2^a)$ and $\varphi(p^k)$ are even, ruling out cyclicity. The surviving cases are $m \in \{1, 2, 4, p^k, 2p^k\}$.

Example 0.4.10: Non-Cyclic Unit Groups

1. $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$. Every element satisfies $a^2 \equiv 1 \pmod{8}$, so the group has exponent 2 and $(\mathbb{Z}/8\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, the Klein four-group. The generators of the two $\mathbb{Z}/2\mathbb{Z}$ factors are $-1 = 7$ and 5 : indeed $\{(\pm 1)^a \cdot 5^b : a \in \{0, 1\}, b \in \{0, 1\}\} = \{1, 7, 5, 3\}$ gives all four elements. No primitive root exists.
2. $(\mathbb{Z}/15\mathbb{Z})^\times \cong (\mathbb{Z}/3\mathbb{Z})^\times \times (\mathbb{Z}/5\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$. This group has order 8 but exponent $\text{lcm}(2, 4) = 4$, so it is not cyclic. The maximum order 4 is achieved by elements like 2 (since $2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16 \equiv 1 \pmod{15}$).
3. $(\mathbb{Z}/24\mathbb{Z})^\times \cong (\mathbb{Z}/8\mathbb{Z})^\times \times (\mathbb{Z}/3\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z})^2 \times \mathbb{Z}/2\mathbb{Z} \cong (\mathbb{Z}/2\mathbb{Z})^3$. This has order 8 and exponent 2: every element is its own inverse. There are $2^3 - 1 = 7$ elements of order 2, confirming that the group is as far from cyclic as possible.

The *exponent* of $(\mathbb{Z}/m\mathbb{Z})^\times$ —the least common multiple of the orders of all its elements—is captured by the Carmichael function.

Definition 0.4.11: Carmichael Function

The *Carmichael function* $\lambda(m)$ is the exponent of $(\mathbb{Z}/m\mathbb{Z})^\times$: the smallest positive integer e such that $a^e \equiv 1 \pmod{m}$ for all a with $\gcd(a, m) = 1$.

By the structure theorem, $\lambda(m) = \text{lcm}(\lambda(p_1^{a_1}), \dots, \lambda(p_r^{a_r}), \lambda(2^a))$, where $\lambda(p^a) = \varphi(p^a) = p^{a-1}(p-1)$ for odd primes, $\lambda(2) = 1$, $\lambda(4) = 2$, and $\lambda(2^a) = 2^{a-2}$ for $a \geq 3$. When $(\mathbb{Z}/m\mathbb{Z})^\times$ is cyclic, $\lambda(m) = \varphi(m)$; otherwise $\lambda(m)$ is a proper divisor of $\varphi(m)$.

The Carmichael function refines Euler's theorem: $a^{\lambda(m)} \equiv 1 \pmod{m}$ for all a coprime to m , and $\lambda(m)$ is the *smallest* universal exponent with this property. When $\lambda(m) < \varphi(m)$, Euler's theorem with the exponent $\varphi(m)$ is valid but not sharp.

Example 0.4.12: Carmichael Function Computations

1. $\lambda(7) = \varphi(7) = 6$, since $(\mathbb{Z}/7\mathbb{Z})^\times$ is cyclic. The primitive root 3 achieves this order.
2. $\lambda(8) = 2$, since every element of $(\mathbb{Z}/8\mathbb{Z})^\times$ squares to 1, while $\varphi(8) = 4$. Euler's theorem gives $a^4 \equiv 1 \pmod{8}$, but the Carmichael function shows $a^2 \equiv 1$ already suffices.
3. $\lambda(15) = \text{lcm}(\lambda(3), \lambda(5)) = \text{lcm}(2, 4) = 4$, while $\varphi(15) = 8$. The element 2 has order exactly 4.
4. $\lambda(100) = \text{lcm}(\lambda(4), \lambda(25)) = \text{lcm}(2, 20) = 20$, while $\varphi(100) = 40$. Every element coprime to 100 satisfies $a^{20} \equiv 1 \pmod{100}$.

The structure theorem has a direct consequence for character theory. The dual group $(\widehat{\mathbb{Z}/m\mathbb{Z}})^\times$ of group homomorphisms $(\mathbb{Z}/m\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ is (non-canonically) isomorphic to $(\mathbb{Z}/m\mathbb{Z})^\times$ itself (see [11]). When $(\mathbb{Z}/m\mathbb{Z})^\times$ is cyclic, every character is determined by the image of a single generator; when it is not, specifying a character requires choosing the image of each invariant factor independently. The Dirichlet characters of modulus m (??) are the extensions of these group characters to all of $\mathbb{Z}$, and the invariant factor decomposition determines how many characters of each order exist.

Example 0.4.13: Orders and Characters Modulo 12

The group $(\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\} \cong (\mathbb{Z}/4\mathbb{Z})^\times \times (\mathbb{Z}/3\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ is the Klein four-group. Every non-identity element has order 2:

$$5^2 = 25 \equiv 1, \quad 7^2 = 49 \equiv 1, \quad 11^2 = 121 \equiv 1 \pmod{12}$$

The dual group $(\widehat{\mathbb{Z}/12\mathbb{Z}})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ has four characters $\chi_0, \chi_1, \chi_2, \chi_3$, each taking values in $\{\pm 1\}$. The trivial character sends every unit to 1. The three nontrivial characters correspond to the three subgroups of index 2 in $(\mathbb{Z}/12\mathbb{Z})^\times$:

$$\chi_1(n) = \begin{cases} 1 & n \equiv 1, 5 \\ -1 & n \equiv 7, 11 \end{cases}, \quad \chi_2(n) = \begin{cases} 1 & n \equiv 1, 7 \\ -1 & n \equiv 5, 11 \end{cases}, \quad \chi_3(n) = \begin{cases} 1 & n \equiv 1, 11 \\ -1 & n \equiv 5, 7 \end{cases}$$

modulo 12. Each nontrivial character has order 2 in the dual group. These Dirichlet characters of modulus 12 control the distribution of primes in residue classes modulo 12, as developed in ??.

This example illustrates a general principle: the number of Dirichlet characters of modulus m equals $\varphi(m) = |(\mathbb{Z}/m\mathbb{Z})^\times|$, and the order structure of the character group mirrors that of $(\mathbb{Z}/m\mathbb{Z})^\times$. When the group is cyclic of order n , there is a character of every order dividing n ; when it decomposes as a product $\mathbb{Z}/d_1\mathbb{Z} \times \cdots \times \mathbb{Z}/d_r\mathbb{Z}$, the characters decompose correspondingly.

0.5 Quadratic Residues and Reciprocity

The previous section determined when $(\mathbb{Z}/m\mathbb{Z})^\times$ is cyclic. This section addresses a finer structural question: which elements of $(\mathbb{Z}/p\mathbb{Z})^\times$ are perfect squares? The answer is governed by the Legendre symbol, and the relationship between squareness modulo different primes is the content of the law of quadratic reciprocity—one of the central results of elementary number theory. Quadratic reciprocity has far-reaching generalizations: Artin reciprocity in class field theory ([7]) subsumes it as a special case, and the Langlands program seeks non-abelian analogues. In this book, quadratic characters reappear in the study of Dirichlet L -functions (??) and the factorization of Dedekind zeta functions of quadratic fields (??).

Definition 0.5.1: Quadratic Residue

Let p be an odd prime. An integer a with $p \nmid a$ is a *quadratic residue* modulo p if there exists $x \in \mathbb{Z}$ with $x^2 \equiv a \pmod{p}$. Otherwise, a is a *quadratic non-residue* modulo p .

Since the squaring map $x \mapsto x^2$ on $(\mathbb{Z}/p\mathbb{Z})^\times$ sends x and $-x$ to the same value, and $x \neq -x$ in $(\mathbb{Z}/p\mathbb{Z})^\times$ for p odd (as $2x \neq 0$), the image of squaring has exactly $(p-1)/2$ elements. Hence exactly half the nonzero residues modulo p are quadratic residues, and half are non-residues.

Example 0.5.2: Quadratic Residues Modulo Small Primes

1. Modulo 7: the squares are $1^2 = 1$, $2^2 = 4$, $3^2 = 2$, so the quadratic residues are $\{1, 2, 4\}$ and the non-residues are $\{3, 5, 6\}$.
2. Modulo 13: the squares are $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 3$, $5^2 = 12$, $6^2 = 10$, giving quadratic residues $\{1, 3, 4, 9, 10, 12\}$ and non-residues $\{2, 5, 6, 7, 8, 11\}$.

The Legendre symbol packages the quadratic residue information into a multiplicative function.

Definition 0.5.3: Legendre Symbol

For an odd prime p and an integer a , the *Legendre symbol* is

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p, \\ 0 & \text{if } p \mid a. \end{cases}$$

The Legendre symbol $\left(\frac{\cdot}{p}\right) : \mathbb{Z} \rightarrow \{-1, 0, 1\}$ is a Dirichlet character of modulus p (as defined in ??): it is totally multiplicative and p -periodic. This is a consequence of the following criterion.

Proposition 0.5.4: Euler's Criterion

For an odd prime p and a with $p \nmid a$:

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$$

Proof:

The group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$ (theorem 0.4.5). Let g be a primitive root and write $a \equiv g^k \pmod{p}$. Then $a^{(p-1)/2} \equiv g^{k(p-1)/2} \pmod{p}$. Since g has order $p-1$, the value $g^{k(p-1)/2}$ is 1 if $(p-1) \mid k(p-1)/2$, i.e., if k is even, and $g^{(p-1)/2} = -1$ if k is odd (as $g^{(p-1)/2}$ is a root of $x^2 - 1$ other than 1, hence equals -1).

Now, $a = g^k$ is a quadratic residue if and only if $a = (g^{k/2})^2$ for some integer $k/2$, i.e., if and only if k is even. Hence $\left(\frac{a}{p}\right) = 1$ when k is even and -1 when k is odd, matching $a^{(p-1)/2} \pmod{p}$.

Euler's criterion immediately implies the total multiplicativity of the Legendre symbol: $\left(\frac{ab}{p}\right) = (ab)^{(p-1)/2} = a^{(p-1)/2} \cdot b^{(p-1)/2} = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right) \pmod{p}$. Since both sides take values in $\{-1, 1\}$ and agree modulo $p > 2$, they are equal as integers. This multiplicativity reduces the computation of $\left(\frac{a}{p}\right)$ for general a to the case of prime arguments.

The Legendre symbol $\left(\frac{\cdot}{p}\right)$ is the unique nontrivial quadratic character of $(\mathbb{Z}/p\mathbb{Z})^\times$: it is the group homomorphism $(\mathbb{Z}/p\mathbb{Z})^\times \rightarrow \{\pm 1\}$ whose kernel is the subgroup of quadratic residues, an index-2

subgroup. In the language of ??, it is the unique primitive Dirichlet character of modulus p and order 2. The Dirichlet L -function $L(s, \left(\frac{\cdot}{p}\right))$ controls the distribution of quadratic residues and non-residues among the primes, and its non-vanishing at $s = 1$ is one of the key inputs for Dirichlet's theorem on primes in arithmetic progressions.

The values of $\left(\frac{-1}{p}\right)$ and $\left(\frac{2}{p}\right)$ can be determined directly from Euler's criterion.

Proposition 0.5.5: First Supplement: -1 as a Quadratic Residue

For an odd prime p :

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2} = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4}, \\ -1 & \text{if } p \equiv 3 \pmod{4}. \end{cases}$$

Proof:

By Euler's criterion, $\left(\frac{-1}{p}\right) \equiv (-1)^{(p-1)/2} \pmod{p}$. Since $(p-1)/2$ is even when $p \equiv 1 \pmod{4}$ and odd when $p \equiv 3 \pmod{4}$, the result follows.

The first supplement has a clean interpretation: -1 is a square modulo p precisely when $(\mathbb{Z}/p\mathbb{Z})^\times$ has an element of order 4, which happens if and only if $4 \mid (p-1)$.

Proposition 0.5.6: Second Supplement: 2 as a Quadratic Residue

For an odd prime p :

$$\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8} = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod{8}, \\ -1 & \text{if } p \equiv \pm 3 \pmod{8}. \end{cases}$$

Proof:

The exponent $(p^2-1)/8 = (p-1)(p+1)/8$. When $p \equiv 1 \pmod{8}$, both $(p-1)/2$ and $(p+1)/2$ are even, so the product is divisible by 4 and $(p^2-1)/8$ is even. When $p \equiv 3 \pmod{8}$, then $(p-1)/2 = 1 + 4k$ for some k , and $(p+1)/4 = 1 + 2k$ is odd; working through the cases gives $(p^2-1)/8$ odd. The remaining cases $p \equiv 5, 7 \pmod{8}$ follow by symmetry. Alternatively, this can be deduced from Gauss's lemma (lemma 0.5.7 below) applied to $a = 2$.

Both supplements can also be proved using *Gauss sums*: the quantity $\tau(\chi) = \sum_{n=1}^{p-1} \left(\frac{n}{p}\right) e^{2\pi i n/p}$, which satisfies $\tau(\chi)^2 = (-1)^{(p-1)/2} p$. Gauss sums are the Fourier-analytic tool that connects quadratic residues to exponential sums, and they play a central role in the functional equations of Dirichlet L -functions (??).

The central tool for proving quadratic reciprocity is the following counting lemma.

Lemma 0.5.7: Gauss's Lemma

Let p be an odd prime and a an integer with $p \nmid a$. For each $k \in \{1, 2, \dots, (p-1)/2\}$, let r_k be the representative of $ka \pmod p$ in $\{-(p-1)/2, \dots, -1, 1, \dots, (p-1)/2\}$, and let n be the number of indices k for which $r_k < 0$. Then

$$\left(\frac{a}{p}\right) = (-1)^n$$

Proof:

The values $|r_1|, |r_2|, \dots, |r_{(p-1)/2}|$ are a permutation of $\{1, 2, \dots, (p-1)/2\}$: they are $(p-1)/2$ distinct elements of this set (distinct because $r_i = \pm r_j$ with $i \neq j$ would give $a(i \pm j) \equiv 0 \pmod p$, contradicting $p \nmid a$ and $0 < i \pm j < p$). Taking the product:

$$\prod_{k=1}^{(p-1)/2} r_k = (-1)^n \prod_{k=1}^{(p-1)/2} |r_k| = (-1)^n \prod_{k=1}^{(p-1)/2} k = (-1)^n \left(\frac{p-1}{2}\right)!$$

On the other hand, $r_k \equiv ka \pmod p$, so $\prod r_k \equiv a^{(p-1)/2} \prod_{k=1}^{(p-1)/2} k \pmod p$. Comparing with the line above and cancelling $((p-1)/2)!$ (which is nonzero modulo p) gives $a^{(p-1)/2} \equiv (-1)^n \pmod p$. The result follows from Euler's criterion.

Gauss's lemma converts the algebraic question “is a a square modulo p ?” into the combinatorial question “how many of the reductions $a, 2a, \dots, \frac{p-1}{2}a$ modulo p land in the upper half $\{(p+1)/2, \dots, p-1\}$?” This geometric reformulation is the key to Eisenstein's proof of quadratic reciprocity.

Example 0.5.8: Gauss's Lemma in Action

Take $a = 5$ and $p = 11$. The set $\{ka \pmod{11} : k = 1, \dots, 5\}$ is $\{5, 10, 4, 9, 3\}$. Representing these in $\{-5, \dots, -1, 1, \dots, 5\}$: $5 \mapsto 5, 10 \mapsto -1, 4 \mapsto 4, 9 \mapsto -2, 3 \mapsto 3$. The negative values are $\{-1, -2\}$, so $n = 2$ and $\left(\frac{5}{11}\right) = (-1)^2 = 1$. Indeed, $4^2 = 16 \equiv 5 \pmod{11}$, confirming that 5 is a quadratic residue.

Theorem 0.5.9: Quadratic Reciprocity

Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$$

Equivalently: $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$ unless $p \equiv q \equiv 3 \pmod{4}$, in which case $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$.

Quadratic reciprocity relates the solvability of $x^2 \equiv p \pmod q$ to that of $x^2 \equiv q \pmod p$ —two problems that, a priori, have nothing to do with each other. The sign $(-1)^{(p-1)(q-1)/4}$ is $+1$ unless both primes are congruent to 3 modulo 4, in which case the Legendre symbols “flip.” This single sign change is the entire content of the reciprocity law.

Proof:

By Gauss's lemma, $\left(\frac{q}{p}\right) = (-1)^{n_p}$ where $n_p = \#\{k \in \{1, \dots, (p-1)/2\} \mid kq \bmod p > p/2\}$. Since $kq \bmod p = kq - \lfloor kq/p \rfloor p$, the condition $kq \bmod p > p/2$ is equivalent to $kq - \lfloor kq/p \rfloor p > p/2$, which rearranges to $\lfloor kq/p \rfloor < kq/p - 1/2$. Counting more carefully:

$$n_p = \sum_{k=1}^{(p-1)/2} \left\lfloor \frac{kq}{p} \right\rfloor - 2 \sum_{k=1}^{(p-1)/2} \left\lfloor \frac{kq}{2p} \right\rfloor$$

Eisenstein's approach avoids this bookkeeping by counting lattice points. Consider the rectangle $R = (0, p/2) \times (0, q/2)$ in $\mathbb{R}^2$. The number of lattice points $(x, y) \in \mathbb{Z}^2$ with $0 < x < p/2$ and $0 < y < qx/p$ is $\sum_{x=1}^{(p-1)/2} \lfloor qx/p \rfloor$. Similarly, the number with $0 < y < q/2$ and $0 < x < py/q$ is $\sum_{y=1}^{(q-1)/2} \lfloor py/q \rfloor$. Since p and q are distinct odd primes, no lattice point (x, y) with $0 < x < p/2$ and $0 < y < q/2$ lies on the diagonal $qx = py$ (as this would require $p \mid x$ or $q \mid y$, contradicting the bounds). Hence every lattice point in $R \cap \mathbb{Z}^2$ lies strictly above or strictly below the diagonal, giving:

$$\sum_{x=1}^{(p-1)/2} \left\lfloor \frac{qx}{p} \right\rfloor + \sum_{y=1}^{(q-1)/2} \left\lfloor \frac{py}{q} \right\rfloor = \frac{p-1}{2} \cdot \frac{q-1}{2}$$

The left-hand side counts all lattice points in the open rectangle R : those below the diagonal contribute the first sum, and those above contribute the second. The total number of such points is $\frac{p-1}{2} \cdot \frac{q-1}{2}$.

By Gauss's lemma, one can verify that $(-1)^{n_p} = (-1)^{\sum_{x=1}^{(p-1)/2} \lfloor qx/p \rfloor}$ (since the parity of n_p matches that of this sum). Hence:

$$\left(\frac{q}{p}\right) \left(\frac{p}{q}\right) = (-1)^{n_p + n_q} = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$$

as claimed.

The lattice-point argument is due to Eisenstein and is among the most geometric of the many known proofs of quadratic reciprocity (over 240 proofs are known). The rectangle R and its diagonal will reappear—in a much more general form—in the lattice-point counting arguments used to prove the analytic class number formula in ??.

Example 0.5.10: Lattice Point Verification

Take $p = 5$ and $q = 7$. The rectangle $R = (0, 5/2) \times (0, 7/2)$ contains lattice points with $1 \leq x \leq 2$ and $1 \leq y \leq 3$. The diagonal is $y = (7/5)x$. Below the diagonal: $(1, 1)$ (since $1 < 7/5$); above the diagonal: $(1, 2)$, $(1, 3)$, $(2, 2)$, $(2, 3)$ (since $2 > 14/5$), and $(2, 1)$ lies below (since $1 < 14/5$). Counting: $\sum_{x=1}^2 \lfloor 7x/5 \rfloor = 1 + 2 = 3$ below, $\sum_{y=1}^3 \lfloor 5y/7 \rfloor = 0 + 1 + 2 = 3$ above, total $6 = 2 \cdot 3 = \frac{p-1}{2} \cdot \frac{q-1}{2}$. Since 6 is even, $\left(\frac{5}{7}\right) \left(\frac{7}{5}\right) = 1$. One checks: $\left(\frac{5}{7}\right) = 5^3 = 125 \equiv 6 \equiv -1 \pmod{7}$ and $\left(\frac{7}{5}\right) = 7^2 = 49 \equiv 4 \equiv -1 \pmod{5}$, confirming the product is 1.

Example 0.5.11: Using Quadratic Reciprocity

1. Is 21 a quadratic residue modulo 71? By multiplicativity, $\left(\frac{21}{71}\right) = \left(\frac{3}{71}\right) \left(\frac{7}{71}\right)$. For $\left(\frac{3}{71}\right)$: since $71 \equiv 3 \pmod{4}$ and $3 \equiv 3 \pmod{4}$, reciprocity gives $\left(\frac{3}{71}\right) = -\left(\frac{71}{3}\right) = -\left(\frac{2}{3}\right) = -(-1) = 1$. For $\left(\frac{7}{71}\right)$: since $71 \equiv 3 \pmod{4}$ and $7 \equiv 3 \pmod{4}$, reciprocity gives $\left(\frac{7}{71}\right) = -\left(\frac{71}{7}\right) = -\left(\frac{1}{7}\right) = -1$. Hence $\left(\frac{21}{71}\right) = 1 \cdot (-1) = -1$: the integer 21 is a non-residue modulo 71.
2. Is 5 a quadratic residue modulo 23? Since $5 \equiv 1 \pmod{4}$, reciprocity gives $\left(\frac{5}{23}\right) = \left(\frac{23}{5}\right) = \left(\frac{3}{5}\right)$. Since $5 \equiv 1 \pmod{4}$, applying reciprocity again: $\left(\frac{3}{5}\right) = \left(\frac{5}{3}\right) = \left(\frac{2}{3}\right) = -1$ (by the second supplement, since $3 \equiv 3 \pmod{8}$). Hence 5 is a non-residue modulo 23.

The Legendre symbol is defined only for prime moduli. The Jacobi symbol extends it to odd composite moduli and is the standard tool for efficient computation.

Definition 0.5.12: Jacobi Symbol

For an odd positive integer $n = p_1^{a_1} \cdots p_k^{a_k}$ (not necessarily prime) and any integer a , the *Jacobi symbol* is

$$\left(\frac{a}{n}\right) = \prod_{i=1}^k \left(\frac{a}{p_i}\right)^{a_i}$$

The Jacobi symbol inherits the multiplicativity properties of the Legendre symbol: $\left(\frac{ab}{n}\right) = \left(\frac{a}{n}\right) \left(\frac{b}{n}\right)$ and $\left(\frac{a}{mn}\right) = \left(\frac{a}{m}\right) \left(\frac{a}{n}\right)$. The reciprocity law, first supplement, and second supplement all extend to Jacobi symbols with identical formulas:

$$\left(\frac{m}{n}\right) \left(\frac{n}{m}\right) = (-1)^{\frac{m-1}{2} \cdot \frac{n-1}{2}}, \quad \left(\frac{-1}{n}\right) = (-1)^{(n-1)/2}, \quad \left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$$

for odd coprime $m, n > 0$. These follow from the corresponding Legendre symbol identities by multiplying over the prime factorizations.

A caution: $\left(\frac{a}{n}\right) = 1$ does *not* imply that a is a quadratic residue modulo n when n is composite. For example, $\left(\frac{2}{15}\right) = \left(\frac{2}{3}\right) \left(\frac{2}{5}\right) = (-1)(-1) = 1$, but $x^2 \equiv 2 \pmod{15}$ has no solution (since $x^2 \equiv 2 \pmod{3}$ has no solution). The Jacobi symbol detects quadratic residuosity only modulo each prime factor individually.

The computational value of the Jacobi symbol lies in its reciprocity law: evaluating $\left(\frac{a}{n}\right)$ reduces to repeated application of reciprocity and the supplements, without factoring n . The resulting algorithm runs in $O(\log^2 n)$ time, analogous to the Euclidean algorithm for GCD.

Example 0.5.13: Computing a Jacobi Symbol

Evaluate $\left(\frac{1001}{9907}\right)$. Neither number need be prime. Since $9907 \equiv 3 \pmod{4}$ and $1001 \equiv 1 \pmod{4}$, reciprocity gives $\left(\frac{1001}{9907}\right) = \left(\frac{9907}{1001}\right)$. Now $9907 = 9 \cdot 1001 + 898$, so $\left(\frac{9907}{1001}\right) = \left(\frac{898}{1001}\right) = \left(\frac{2}{1001}\right) \left(\frac{449}{1001}\right)$. Since $1001 \equiv 1 \pmod{8}$, the second supplement gives $\left(\frac{2}{1001}\right) = 1$. For $\left(\frac{449}{1001}\right)$: since $449 \equiv 1 \pmod{4}$, reciprocity gives $\left(\frac{449}{1001}\right) = \left(\frac{1001}{449}\right) = \left(\frac{103}{449}\right)$ (as $1001 = 2 \cdot 449 + 103$). Since $103 \equiv 3 \pmod{4}$ and $449 \equiv 1 \pmod{4}$, reciprocity gives $\left(\frac{103}{449}\right) = \left(\frac{449}{103}\right) = \left(\frac{37}{103}\right)$. Since $37 \equiv 1 \pmod{4}$: $\left(\frac{37}{103}\right) = \left(\frac{103}{37}\right) = \left(\frac{29}{37}\right)$. Since $29 \equiv 1 \pmod{4}$: $\left(\frac{29}{37}\right) = \left(\frac{37}{29}\right) = \left(\frac{8}{29}\right) = \left(\frac{2}{29}\right)^3 = (-1)^3 = -1$ (since $29 \equiv 5 \pmod{8}$). Hence $\left(\frac{1001}{9907}\right) = -1$.

Quadratic reciprocity is the first in a hierarchy of reciprocity laws. *Cubic reciprocity* governs when a is a perfect cube modulo p and is naturally formulated over the Eisenstein integers $\mathbb{Z}[\zeta_3]$; *quartic reciprocity* governs fourth powers and lives over the Gaussian integers $\mathbb{Z}[i]$. These higher reciprocity laws were developed by Gauss, Jacobi, and Eisenstein in the early 19th century and are part of the motivation for studying number rings beyond $\mathbb{Z}$. The ultimate generalization—Artin reciprocity—unifies all abelian reciprocity laws into a single statement about the Galois group $\text{Gal}(K^{\text{ab}}/K)$ of the maximal abelian extension of a number field K , and is the subject of [7]. The quadratic case treated here, while elementary, already exhibits the key phenomenon: a local condition (squareness modulo p) is controlled by global arithmetic data (the residue classes of p and q modulo 4).

0.6 Asymptotic Notation

The results of analytic number theory are expressed in the language of asymptotics: statements about how functions behave as their argument grows without bound. The Prime Number Theorem, for instance, asserts that $\pi(x) \sim x/\log x$ —a precise claim about the ratio $\pi(x)/(x/\log x)$ tending to 1, not about the difference $\pi(x) - x/\log x$. This section collects the standard asymptotic notations and their basic properties, establishing the conventions used throughout the book.

Definition 0.6.1: Big- O and Little- o

Let f and g be functions defined on a neighborhood of ∞ (or of some limit point), with g eventually positive.

1. $f(x) = O(g(x))$ as $x \rightarrow \infty$ means there exist constants $C > 0$ and x_0 such that $|f(x)| \leq Cg(x)$ for all $x \geq x_0$. The notation $f \ll g$ is synonymous.
2. $f(x) = o(g(x))$ as $x \rightarrow \infty$ means $\lim_{x \rightarrow \infty} f(x)/g(x) = 0$, i.e., $|f(x)| \leq \varepsilon g(x)$ for any $\varepsilon > 0$ and all sufficiently large x .

The big- O notation asserts a bound up to an unspecified multiplicative constant; the little- o notation asserts that the bound can be made arbitrarily small. Every $o(g)$ is also $O(g)$, but not conversely: $x = O(x)$ but $x \neq o(x)$.

A common source of confusion is that the “=” in $f = O(g)$ is not symmetric: it means “ f belongs to the class of functions bounded by Cg ,” not that two specific quantities are equal. The notation is best read as “ f is at most of order g .” When multiple big- O terms appear in an equation, each is understood to represent a (possibly different) function satisfying the stated bound. For instance, $O(x) + O(x^2) = O(x^2)$ is a valid statement: any function bounded by Cx plus any function bounded by $C'x^2$ is bounded by $C''x^2$ for $C'' = C + C'$.

Definition 0.6.2: Vinogradov Notation

The notation $f \ll g$ means $f = O(g)$, and $f \gg g$ means $g = O(f)$. If both $f \ll g$ and $f \gg g$ hold, this is written $f \asymp g$ (read “ f is of the same order as g ”).

The Vinogradov notation $\ll$ is standard in analytic number theory and avoids some of the awkwardness of the big- O symbol in chains of inequalities. For example, $\sum_{n \leq x} d(n) \ll x \log x$ is cleaner than $\sum_{n \leq x} d(n) = O(x \log x)$ when embedded in a longer computation. Both notations are used freely in

this book; the choice in any given instance is dictated by readability.

A critical convention: the implied constant in $f \ll g$ or $f = O(g)$ may depend on parameters that are considered fixed. When the dependence matters, it is indicated by a subscript: $f \ll_\varepsilon g$ means the implied constant depends on ε , and $f = O_k(g)$ means it depends on k . Failing to track such dependencies is a common source of error in analytic arguments, particularly when a parameter later varies (e.g., when $\varepsilon \rightarrow 0$ or $k \rightarrow \infty$).

Definition 0.6.3: Asymptotic Equivalence

Two functions f and g , both eventually nonzero, are *asymptotically equivalent*, written $f(x) \sim g(x)$, if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1$$

Equivalently, $f \sim g$ if and only if $f = g + o(g)$, i.e., $f(x) = g(x)(1 + o(1))$.

Asymptotic equivalence is the relation used to state the main theorems of analytic number theory. The Prime Number Theorem says $\pi(x) \sim x/\log x$; Mertens' theorem says $\prod_{p \leq x} (1 - 1/p) \sim e^{-\gamma}/\log x$. These are statements about the *leading-order* behavior of the function, ignoring lower-order corrections. A stronger statement specifies the error: $\pi(x) = \text{Li}(x) + O(x \exp(-c\sqrt{\log x}))$ gives both the leading term and a bound on the remainder.

The relation $\sim$ is an equivalence relation on the set of eventually positive functions: it is reflexive ($f \sim f$), symmetric ($f \sim g$ implies $g \sim f$), and transitive ($f \sim g$ and $g \sim h$ imply $f \sim h$). It is also compatible with multiplication ($f_1 \sim g_1$ and $f_2 \sim g_2$ imply $f_1 f_2 \sim g_1 g_2$) but *not* with addition: $f_1 \sim g_1$ and $f_2 \sim g_2$ do not imply $f_1 + f_2 \sim g_1 + g_2$, since cancellation can occur. For instance, $(x+1) \sim x$ and $(-x) \sim (-x)$, but $(x+1) + (-x) = 1 \not\sim 0 = x + (-x)$.

Two further notations occasionally appear in the literature.

Definition 0.6.4: Ω and Θ Notation

1. $f(x) = \Omega(g(x))$ means $f(x) \neq o(g(x))$, i.e., $|f(x)| \geq c g(x)$ for some $c > 0$ and infinitely many $x \rightarrow \infty$. This is the negation of $f = o(g)$: the function f is “at least of order g ” infinitely often.
2. $f(x) = \Theta(g(x))$ means $f \asymp g$, i.e., both $f = O(g)$ and $f = \Omega(g)$. This gives a tight two-sided bound: $c_1 g(x) \leq |f(x)| \leq c_2 g(x)$ for constants $0 < c_1 \leq c_2$ and all sufficiently large x .

The Ω notation is used in number theory to express *lower bounds* on oscillation. For example, the statement $\pi(x) - \text{Li}(x) = \Omega(x^{1/2} \log \log \log x / \log x)$ (a theorem of Littlewood) asserts that the error in the Prime Number Theorem is not just $O(x^{1/2})$ —it is at least as large as $x^{1/2} \log \log \log x / \log x$ infinitely often.

The following rules govern the manipulation of asymptotic expressions. They are used without comment throughout the book.

Proposition 0.6.5: Arithmetic of Asymptotic Notation

Let f, f_1, f_2, g, g_1, g_2 be functions with g, g_1, g_2 eventually positive. Then:

1. *Transitivity*: if $f = O(g)$ and $g = O(h)$, then $f = O(h)$.
2. *Sum rule*: $O(g_1) + O(g_2) = O(\max(g_1, g_2))$, and $O(g) + O(g) = O(g)$.
3. *Product rule*: $O(g_1) \cdot O(g_2) = O(g_1 g_2)$.
4. *Scalar rule*: for any constant $c \neq 0$, $c \cdot O(g) = O(g)$.
5. *Absorption*: if $g_1 = O(g_2)$, then $O(g_1) + O(g_2) = O(g_2)$.
6. *Little-o absorbs*: $o(g) + O(g) = O(g)$, and $o(g) \cdot O(h) = o(gh)$.

The same rules hold with $\ll$ in place of O , and analogous rules hold for o and $\sim$.

Proof:

All follow from the definition. For (1): $|f| \leq C_1 g$ and $g \leq C_2 h$ give $|f| \leq C_1 C_2 h$. For (2): if $|f_1| \leq C_1 g_1$ and $|f_2| \leq C_2 g_2$, then $|f_1 + f_2| \leq C_1 g_1 + C_2 g_2 \leq (C_1 + C_2) \max(g_1, g_2)$. The remaining parts are similar.

Asymptotic notation interacts well with summation and integration.

Proposition 0.6.6: Summation and Integration of O -Terms

1. If $f(n) = O(g(n))$ for all n , and g is non-negative, then $\sum_{n=1}^N f(n) = O\left(\sum_{n=1}^N g(n)\right)$.
2. If $f(t) = O(g(t))$ for $t \geq t_0$, and g is non-negative and integrable, then $\int_{t_0}^x f(t) dt = O\left(\int_{t_0}^x g(t) dt\right)$.

Proof:

For (1): $|f(n)| \leq C g(n)$ implies $|\sum f(n)| \leq \sum |f(n)| \leq C \sum g(n)$. Part (2) is identical with integrals replacing sums.

However, O -terms cannot be freely moved through nonlinear operations. For instance, $e^{O(1)}$ is *not* $O(1)$ —it denotes a function bounded between e^{-C} and e^C for some constant C , which is $\Theta(1)$. And $O(\log x)$ inside an exponent, as in $x^{O(1)} = e^{O(\log x)}$, denotes a function bounded by a power of x but with unspecified exponent.

The following table of growth rates, ordered from slowest to fastest, is a useful reference for comparing asymptotic estimates. Each function in the list is o of the next:

$$1 \ll \log \log x \ll \log x \ll (\log x)^A \ll x^\varepsilon \ll x^a \ll x^b \ll e^{x^c} \ll e^x \ll x! \ll x^x$$

where $0 < \varepsilon < a < b < 1 \leq c$ and $A > 0$. The gaps between consecutive entries in this hierarchy reflect qualitative differences in growth behavior. For example, the gap between $(\log x)^A$ and x^ε is

the reason that “logarithmic savings” in analytic number theory (gaining a factor of $(\log x)^4$ over a trivial bound) are fundamentally different from “power savings” (gaining x^ε). The Prime Number Theorem provides logarithmic savings over the trivial bound $\pi(x) \leq x$; the Riemann Hypothesis would upgrade this to a power saving.

Example 0.6.7: Manipulating Asymptotic Expressions

Suppose one has established that $f(x) = x^2 + 3x \log x + O(x)$ and wishes to estimate $f(x)/x^2$. Using the rules:

$$\frac{f(x)}{x^2} = 1 + \frac{3 \log x}{x} + O\left(\frac{1}{x}\right) = 1 + O\left(\frac{\log x}{x}\right)$$

since $\frac{3 \log x}{x} = O\left(\frac{\log x}{x}\right)$ and $\frac{1}{x} = O\left(\frac{\log x}{x}\right)$ (because $1 \leq \log x$ for $x \geq e$). Hence $f(x) \sim x^2$. The intermediate expression $1 + \frac{3 \log x}{x} + O(1/x)$ is more precise: it gives the leading term 1 and the next correction $3 \log x/x$, which tends to 0 but more slowly than $1/x$.

A common pitfall in asymptotic arguments is the hidden dependence of implied constants on parameters. Consider the estimate $\sum_{n \leq x} n^{-s} = O_s(1)$ for fixed $\operatorname{Re}(s) > 1$. This is valid for each fixed s , but the implied constant blows up as $s \rightarrow 1^+$ (since the sum diverges at $s = 1$). Any argument that takes $s \rightarrow 1^+$ *after* using this estimate produces an incorrect bound. The safest practice is to make all parameter dependencies explicit at the point of first use, as in $\sum_{n \leq x} n^{-\sigma} \leq \frac{x^{1-\sigma}}{1-\sigma} + \zeta(\sigma)$ for $\sigma > 1$, which exhibits the blowup at $\sigma = 1$ explicitly through the factor $1/(1-\sigma)$.

Example 0.6.8: Standard Asymptotic Estimates

The following estimates appear repeatedly throughout the book.

1. *Harmonic series:* $\sum_{n=1}^N \frac{1}{n} = \log N + \gamma + O(1/N)$, where $\gamma = 0.5772\dots$ is the Euler-Mascheroni constant. The error $O(1/N)$ can be refined to an asymptotic expansion in powers of $1/N$ using the Euler-Maclaurin formula (chapter 1).
2. *Geometric series:* for $|r| < 1$, $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$, and the partial sum satisfies $\sum_{n=0}^N r^n = \frac{1}{1-r} + O(r^{N+1})$.
3. *Logarithmic growth:* $\log x = o(x^\varepsilon)$ for every $\varepsilon > 0$. Equivalently, $\log x \ll_\varepsilon x^\varepsilon$. This slow growth of the logarithm is the reason that the Prime Number Theorem $\pi(x) \sim x/\log x$ says primes are “barely” infinite: they thin out, but only logarithmically.
4. *Stirling:* $\log(n!) = n \log n - n + \frac{1}{2} \log(2\pi n) + O(1/n)$. In cruder form, $n! \sim \sqrt{2\pi n} (n/e)^n$. This is the asymptotic tool for estimating binomial coefficients and factorials.
5. *Partial summation prototype:* $\sum_{n \leq x} \frac{1}{n^s} = \frac{x^{1-s}}{1-s} + \zeta(s) + O(x^{-\operatorname{Re}(s)})$ for $\operatorname{Re}(s) > 0$, $s \neq 1$. This is proved in chapter 2 using partial summation.

The interaction between O , o , and $\sim$ deserves emphasis, as it governs the precision of every asymptotic statement in the book. Saying $f = O(g)$ gives an upper bound but no information about the actual size of f : the function f could be much smaller than g , or even identically zero. Saying $f \sim g$ pins down the leading behavior precisely, but says nothing about the error $f - g$ beyond it being $o(g)$. Giving an explicit error term, as in $f = g + O(h)$ with $h = o(g)$, is the strongest type of asymptotic statement and is the form in which the main theorems (PNT, class number formula, Chebotarev) are most useful.

Example 0.6.9: Distinguishing Asymptotic Statements

Consider the three statements about $\pi(x)$:

1. $\pi(x) = O(x/\log x)$: there is a constant C such that $\pi(x) \leq Cx/\log x$ for large x . This is a consequence of Chebyshev's estimates (chapter 3) and is proved without complex analysis.
2. $\pi(x) \sim x/\log x$: the ratio $\pi(x)/(x/\log x) \rightarrow 1$. This is the Prime Number Theorem (chapter 5), and its proof requires the non-vanishing of $\zeta(s)$ on $\operatorname{Re}(s) = 1$.
3. $\pi(x) = \operatorname{Li}(x) + O(x \exp(-c\sqrt{\log x}))$: the error between $\pi(x)$ and $\operatorname{Li}(x)$ is bounded by $x \exp(-c\sqrt{\log x})$, which is $o(x/(\log x)^n)$ for every n but *not* $O(x^{1-\varepsilon})$ for any $\varepsilon > 0$. This sharper form requires a quantitative zero-free region for $\zeta(s)$.

Each successive statement is strictly more informative than the previous one, and each requires strictly more analytic input to prove. This hierarchy—from order of magnitude, to leading term, to explicit error—recurs throughout the book.

The distinction between $\sim$ and explicit error terms has practical consequences. The two approximations $\pi(x) \sim x/\log x$ and $\pi(x) \sim \operatorname{Li}(x) = \int_2^x dt/\log t$ are both valid forms of the Prime Number Theorem, since $\operatorname{Li}(x) \sim x/\log x$. However, $\operatorname{Li}(x)$ is a significantly better numerical approximation: at $x = 10^9$, the prime counting function satisfies $\pi(10^9) = 50,847,534$, while $10^9/\log(10^9) \approx 48,254,942$ (an error of about 5%) and $\operatorname{Li}(10^9) \approx 50,849,235$ (an error of about 0.003%). The asymptotic relation $\sim$ treats both approximations as equally valid, but the explicit error terms distinguish them.

Example 0.6.10: Growth Rate Comparisons

1. $n^{100} = o(1.01^n)$: any fixed polynomial is eventually dominated by any exponential with base greater than 1. This follows from $\lim_{n \rightarrow \infty} n^{100}/1.01^n = 0$, proved by taking logarithms: $100 \log n - n \log 1.01 \rightarrow -\infty$.
2. $(\log n)^{1000} = o(n^{0.001})$: any fixed power of $\log n$ is eventually dominated by any positive power of n . At $n = 10^6$, however, $(\log n)^{1000} \approx (13.8)^{1000}$ vastly exceeds $n^{0.001} \approx 1.014$. The crossover point is astronomically large, illustrating that asymptotic statements say nothing about the rate of convergence.
3. $n! \gg (n/e)^n$: Stirling's approximation gives $n! \sim \sqrt{2\pi n} (n/e)^n$, so $n! \asymp n^{1/2} (n/e)^n$. The factor $\sqrt{2\pi n}$ is asymptotically negligible compared to $(n/e)^n$ but is needed for precise estimates of binomial coefficients.

Example 0.6.11: Summing with Asymptotic Notation

Estimate $\sum_{n=1}^N \sqrt{n}$. Since $\sqrt{n}$ is monotonically increasing, comparison with the integral gives

$$\int_1^N \sqrt{t} \, dt \leq \sum_{n=1}^N \sqrt{n} \leq \sqrt{N} + \int_1^N \sqrt{t} \, dt$$

Computing $\int_1^N \sqrt{t} \, dt = \frac{2}{3}N^{3/2} - \frac{2}{3}$, this yields

$$\sum_{n=1}^N \sqrt{n} = \frac{2}{3}N^{3/2} + O(N^{1/2})$$

The Euler-Maclaurin formula (chapter 1) refines this to $\sum_{n=1}^N \sqrt{n} = \frac{2}{3}N^{3/2} + \frac{1}{2}N^{1/2} + \frac{1}{24}N^{-1/2} + O(N^{-3/2})$, but the leading term $\frac{2}{3}N^{3/2}$ is often sufficient.

Finally, a convention on logarithms: throughout this book, $\log$ denotes the natural logarithm $\ln = \log_e$. This is standard in analytic number theory (and in most of mathematics outside of computer science). Iterated logarithms are written $\log \log x = \log(\log x)$, and occasionally $\log_k x$ denotes the k -fold iterated logarithm.

Example 0.6.12: Asymptotic Expansion of a Sum

Estimate $S(N) = \sum_{n=1}^N \frac{\log n}{n}$. Using the integral comparison $\sum_{n=1}^N f(n) = \int_1^N f(t) \, dt + O(f(N))$ for monotone f (a rough form of Abel summation, proved in chapter 1):

$$S(N) = \int_1^N \frac{\log t}{t} \, dt + O\left(\frac{\log N}{N}\right) = \frac{(\log N)^2}{2} + O\left(\frac{\log N}{N}\right)$$

This gives $S(N) \sim \frac{1}{2}(\log N)^2$. For $N = 10^6$: the exact value is $S(10^6) \approx 95.38$, while $\frac{1}{2}(\log 10^6)^2 = \frac{1}{2}(6 \log 10)^2 \approx 95.40$ —close agreement, as expected from the small error term $O(\log N/N) \approx 10^{-5}$.

The notations introduced in this section form the grammar in which the results of this book are stated. The reader should be comfortable manipulating O , o , $\sim$, and $\ll$ fluently before proceeding, as every chapter from this point forward uses them without further comment.

Exercise 0.6.1

1. Use the Euclidean algorithm to compute $\gcd(1001, 385)$, and find integers x, y such that $1001x + 385y = \gcd(1001, 385)$.
2. Find all integer solutions (x, y) to $91x + 65y = 13$.
3. Let p be a prime and n a positive integer. Prove that

$$v_p(n!) = \frac{n - s_p(n)}{p - 1}$$

where $s_p(n)$ denotes the sum of the digits of n in base p .

4. Find the smallest positive integer x satisfying $x \equiv 3 \pmod{7}$, $x \equiv 5 \pmod{11}$, and $x \equiv 2 \pmod{13}$.
5.
 1. Find the order of 2 modulo 31.
 2. Determine whether 2 is a primitive root modulo 31.
 3. List all primitive roots modulo 13.
6. Compute $\lambda(360)$, where λ is the Carmichael function. Find an element of $(\mathbb{Z}/360\mathbb{Z})^\times$ achieving this order.

7. Evaluate the following Legendre and Jacobi symbols:

1. $\left(\frac{-3}{17}\right)$
2. $\left(\frac{11}{41}\right)$
3. $\left(\frac{1009}{2027}\right)$ (both are prime)

8. Using Wilson's theorem, prove that if p is a prime with $p \equiv 1 \pmod{4}$, then $x = \left(\frac{p-1}{2}\right)!$ satisfies $x^2 \equiv -1 \pmod{p}$.

9. Let p be an odd prime. Prove that the product of all quadratic residues modulo p in $\{1, 2, \dots, p-1\}$ is congruent to $(-1)^{(p+1)/2} \pmod{p}$.

10. 1. Show that $\sum_{n=1}^N \frac{1}{n^2} = \frac{\pi^2}{6} + O\left(\frac{1}{N}\right)$.
 2. Show that $\sum_{p \leq N} \frac{1}{p^2} = O(1)$, where the sum is over primes.
 3. Prove that $\prod_{n=1}^N \left(1 + \frac{1}{n}\right) \sim \frac{N}{e^\gamma} e^{1/N} \dots$ is false, and find the correct asymptotic: $\prod_{n=1}^N \left(1 + \frac{1}{n}\right) = N + 1$.

11. Prove that for any fixed $m \geq 1$:

$$\frac{\#\{n \leq x : \gcd(n, m) = 1\}}{x} = \frac{\varphi(m)}{m} + O\left(\frac{1}{x}\right)$$

Interpret this as saying that the “probability” that a random integer is coprime to m is $\varphi(m)/m = \prod_{p|m} (1 - 1/p)$.

12. Let n be an odd positive integer with $n > 1$. Prove that $\left(\frac{-1}{n}\right) = (-1)^{(n-1)/2}$ for the Jacobi symbol, without assuming n is prime.

Arithmetic Functions and Summation Techniques

The preliminary chapter studied individual integers: their divisibility, their prime factorizations, their residue classes. Analytic number theory asks a different kind of question. Rather than examining a single integer n , it attaches a numerical invariant $f(n)$ to every positive integer and studies how these values behave in aggregate—how $\sum_{n \leq x} f(n)$ grows as $x \rightarrow \infty$, what the “average” value of f is, and how the values fluctuate around that average. This shift from the particular to the statistical is the defining move of the subject.

This chapter builds the toolkit for that program. The first two sections are algebraic: they introduce the arithmetic functions that encode number-theoretic data and the Dirichlet convolution that organizes them into a ring. The last three sections are analytic: they develop the summation techniques—the hyperbola method, Abel’s formula, and the Euler-Maclaurin formula—that convert discrete sums into continuous integrals amenable to the methods of analysis. Every later chapter in this book depends on this material.

1.1 Arithmetic Functions and Multiplicativity

The starting point is to formalize the idea of a “number-theoretic quantity attached to each positive integer.” The divisor count $d(n)$, the Euler totient $\varphi(n)$, and the von Mangoldt function $\Lambda(n)$ are all examples of the same type of object: a function from the positive integers to the complex numbers. The structure that makes these functions tractable is *multiplicativity*—the property that $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$ —which reduces the study of f on all positive integers to its behavior at prime powers.

Definition 1.1.1: Arithmetic Function

An *arithmetic function* is a function $f : \mathbb{Z}_{>0} \rightarrow \mathbb{C}$.

No regularity conditions are imposed: an arithmetic function is simply a sequence of complex numbers indexed by the positive integers. The interest lies in the additional structure that specific arithmetic functions possess.

Definition 1.1.2: Multiplicative, Totally Multiplicative, and Additive Functions

An arithmetic function f is:

1. *Multiplicative* if $f(1) = 1$ and $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$.
2. *Totally multiplicative* if $f(1) = 1$ and $f(mn) = f(m)f(n)$ for all $m, n \geq 1$.
3. *Additive* if $f(mn) = f(m) + f(n)$ whenever $\gcd(m, n) = 1$.
4. *Totally additive* if $f(mn) = f(m) + f(n)$ for all $m, n \geq 1$.

The condition $f(1) = 1$ for multiplicative functions is a normalization: setting $m = n = 1$ in $f(mn) = f(m)f(n)$ gives $f(1) = f(1)^2$, so $f(1) \in \{0, 1\}$, and the convention excludes the zero function. The logarithm function $\log : \mathbb{Z}_{>0} \rightarrow \mathbb{R}$ is totally additive.

The fundamental observation is that multiplicativity reduces global information to local information at each prime.

Proposition 1.1.3: Determination by Prime Powers

A multiplicative function f is completely determined by its values at prime powers $\{f(p^k) : p \text{ prime}, k \geq 1\}$. Specifically, for $n = p_1^{a_1} \cdots p_r^{a_r}$:

$$f(n) = \prod_{i=1}^r f(p_i^{a_i})$$

If f is totally multiplicative, then $f(p^k) = f(p)^k$, so f is determined by its values at primes.

Proof:

Since $\gcd(p_i^{a_i}, p_j^{a_j}) = 1$ for $i \neq j$, repeated application of the multiplicativity condition gives $f(n) = f(p_1^{a_1}) \cdots f(p_r^{a_r})$. For totally multiplicative f : $f(p^k) = f(p \cdot p^{k-1}) = f(p)f(p^{k-1})$, and induction gives $f(p^k) = f(p)^k$.

This reduction to prime powers is the arithmetic analogue of the Chinese Remainder Theorem (theorem 0.3.7), which decomposes $\mathbb{Z}/n\mathbb{Z}$ into a product of rings $\mathbb{Z}/p_i^{a_i}\mathbb{Z}$. In both cases, the global object splits into independent local pieces, one for each prime. This “local-to-global” decomposition is the structural principle that, in the language of Dirichlet series (chapter 2), becomes the Euler product.

The rest of this section introduces the arithmetic functions that appear most frequently throughout the book. Each is defined, its values at prime powers are recorded, and a small table of values is

given for orientation.

Definition 1.1.4: The Möbius Function

The *Möbius function* $\mu : \mathbb{Z}_{>0} \rightarrow \{-1, 0, 1\}$ is defined by

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1 p_2 \cdots p_k \text{ is a product of } k \text{ distinct primes,} \\ 0 & \text{if } p^2 | n \text{ for some prime } p. \end{cases}$$

Equivalently, μ is the multiplicative function with $\mu(p) = -1$ and $\mu(p^k) = 0$ for $k \geq 2$.

The Möbius function detects squarefreeness: $\mu(n) \neq 0$ if and only if n is squarefree. Its deeper role as the identity element for Möbius inversion, or equivalently as the Dirichlet inverse of the constant function $\mathbf{1}$, is the subject of the next section. The alternative sign is for the inclusion-exclusion principle: for a finite set with k elements, the inclusion-exclusion formula alternates signs $(-1)^k$, and μ does the same over the prime factors of n .

Definition 1.1.5: Divisor Functions

For $k \in \mathbb{C}$, the k -th *divisor function* is

$$\sigma_k(n) = \sum_{d|n} d^k$$

The two most important special cases are the *number-of-divisors function* $d(n) = \sigma_0(n) = \sum_{d|n} 1$ and the *sum-of-divisors function* $\sigma(n) = \sigma_1(n) = \sum_{d|n} d$.

Each σ_k is multiplicative (but not totally multiplicative for $k \neq 0$). At a prime power:

$$\sigma_k(p^a) = 1 + p^k + p^{2k} + \cdots + p^{ak} = \frac{p^{(a+1)k} - 1}{p^k - 1}$$

for $k \neq 0$, and $d(p^a) = a + 1$. The multiplicativity of σ_k follows from the fact that the divisors of mn (with $\gcd(m, n) = 1$) are exactly the products $d_1 d_2$ where $d_1 | m$ and $d_2 | n$, so the sum factors.

Definition 1.1.6: The Liouville Function

The *Liouville function* $\lambda : \mathbb{Z}_{>0} \rightarrow \{-1, 1\}$ is defined by $\lambda(n) = (-1)^{\Omega(n)}$, where $\Omega(n) = \sum_{p^k || n} k$ is the number of prime factors of n counted with multiplicity.

The Liouville function is *totally* multiplicative: $\lambda(mn) = (-1)^{\Omega(mn)} = (-1)^{\Omega(m) + \Omega(n)} = \lambda(m)\lambda(n)$ for all m, n . At primes, $\lambda(p) = -1$, and $\lambda(p^k) = (-1)^k$. The Liouville function is a “signed” version of the constant function $\mathbf{1}$: it assigns $+1$ to integers with an even number of prime factors and -1 to those with an odd number. The relationship between λ and μ is that $\mu(n) = \lambda(n)$ when n is squarefree, and $\mu(n) = 0$ otherwise (so λ is a “squarefree-ified” extension of μ).

The next two functions count prime factors.

Definition 1.1.7: Prime Factor Counting Functions

For $n \geq 1$:

1. $\omega(n) = \sum_{p|n} 1$ is the number of *distinct* prime factors of n .
2. $\Omega(n) = \sum_{p^k || n} k$ is the number of prime factors of n *counted with multiplicity*.

By convention, $\omega(1) = \Omega(1) = 0$.

The function ω is additive (but not totally additive), and Ω is totally additive. For a prime power p^a : $\omega(p^a) = 1$ and $\Omega(p^a) = a$. The distinction between ω and Ω matters for non-squarefree integers: $\omega(12) = 2$ (the primes 2 and 3) while $\Omega(12) = 3$ (two factors of 2 and one of 3).

The final function in the catalogue is different in character from the preceding ones: it is not multiplicative, and it is supported on prime powers rather than on all positive integers. It is, however, the single most important arithmetic function for the study of prime distribution.

Definition 1.1.8: The Von Mangoldt Function

The *von Mangoldt function* $\Lambda : \mathbb{Z}_{>0} \rightarrow \mathbb{R}_{\geq 0}$ is defined by

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k \text{ for some prime } p \text{ and } k \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

The von Mangoldt function is not multiplicative: $\Lambda(2) = \log 2$ and $\Lambda(3) = \log 3$, but $\Lambda(6) = 0 \neq \Lambda(2)\Lambda(3)$. Its importance comes from the identity that links it to the natural logarithm.

Proposition 1.1.9: Von Mangoldt Identity

For all $n \geq 1$:

$$\sum_{d|n} \Lambda(d) = \log n$$

Proof:

Write $n = p_1^{a_1} \cdots p_r^{a_r}$. The divisors d of n with $\Lambda(d) \neq 0$ are exactly the prime powers p_i^j with $1 \leq j \leq a_i$ and $1 \leq i \leq r$. Hence:

$$\sum_{d|n} \Lambda(d) = \sum_{i=1}^r \sum_{j=1}^{a_i} \log p_i = \sum_{i=1}^r a_i \log p_i = \log \left(\prod_{i=1}^r p_i^{a_i} \right) = \log n$$

This identity says that $\log$ is the summatory function of Λ over divisors. In the language of Dirichlet convolution (introduced in the next section), it reads $\Lambda * \mathbf{1} = \log$, which is one of the key identities in analytic number theory. The von Mangoldt function appears as the coefficient of the logarithmic

derivative of $\zeta(s)$: the identity

$$-\zeta'(s)/\zeta(s) = \sum_{n=1}^{\infty} \Lambda(n)n^{-s}$$

(chapter 4) is the Dirichlet series translation of the von Mangoldt identity, and it is through this series that the zeros of $\zeta(s)$ control the distribution of primes.

The Chebyshev functions, which are the summatory functions of Λ and $\log p$ respectively, will be central in chapter 3:

$$\psi(x) = \sum_{n \leq x} \Lambda(n), \quad \vartheta(x) = \sum_{p \leq x} \log p$$

The Prime Number Theorem (chapter 5) is equivalent to $\psi(x) \sim x$.

Example 1.1.10: Verifying the Von Mangoldt Identity

Take $n = 12 = 2^2 \cdot 3$. The divisors of 12 are $\{1, 2, 3, 4, 6, 12\}$. Among these, the prime powers are $2, 3, 4 = 2^2$, giving:

$$\sum_{d|12} \Lambda(d) = \Lambda(1) + \Lambda(2) + \Lambda(3) + \Lambda(4) + \Lambda(6) + \Lambda(12) = 0 + \log 2 + \log 3 + \log 2 + 0 + 0 = 2 \log 2 + \log 3 = \log 12$$

The non-prime-power divisors 1, 6, and 12 contribute nothing, while $4 = 2^2$ contributes $\log 2$ (not $\log 4$). This is the mechanism by which Λ “distributes” the logarithm of n across its prime-power divisors, assigning weight $\log p$ to each prime p and each power of p dividing n .

The following table collects the values of all the arithmetic functions introduced in this section for $n = 1, \dots, 12$, providing a concrete reference.

Example 1.1.11: Table of Arithmetic Functions

n	1	2	3	4	5	6	7	8	9	10	11	12
$\mu(n)$	1	-1	-1	0	-1	1	-1	0	0	1	-1	0
$\varphi(n)$	1	1	2	2	4	2	6	4	6	4	10	4
$d(n)$	1	2	2	3	2	4	2	4	3	4	2	6
$\sigma(n)$	1	3	4	7	6	12	8	15	13	18	12	28
$\lambda(n)$	1	-1	-1	1	-1	1	-1	-1	1	1	-1	-1
$\omega(n)$	0	1	1	1	1	2	1	1	1	2	1	2
$\Omega(n)$	0	1	1	2	1	2	1	3	2	2	1	3
$\Lambda(n)$	0	$\log 2$	$\log 3$	$\log 2$	$\log 5$	0	$\log 7$	$\log 2$	$\log 3$	0	$\log 11$	0

The patterns are visible: μ vanishes on non-squarefree integers (4, 8, 9, 12); $d(n)$ is large when n has many factorizations (e.g., $d(12) = 6$); Λ is nonzero only at prime powers (2, 3, 4, 5, 7, 8, 9, 11).

Example 1.1.12: Computing Arithmetic Functions from Prime Factorizations

1. Let $n = 2520 = 2^3 \cdot 3^2 \cdot 5 \cdot 7$. Then:

$$d(2520) = (3+1)(2+1)(1+1)(1+1) = 48,$$

$$\sigma(2520) = \frac{2^4-1}{1} \cdot \frac{3^3-1}{2} \cdot \frac{5^2-1}{4} \cdot \frac{7^2-1}{6} = 15 \cdot 13 \cdot 6 \cdot 8 = 9360,$$

$$\mu(2520) = 0 \quad (\text{since } 2^2 | 2520),$$

$$\lambda(2520) = (-1)^{3+2+1+1} = (-1)^7 = -1$$

2. Let $n = 30 = 2 \cdot 3 \cdot 5$. This is squarefree, so $\mu(30) = (-1)^3 = -1$ and $\lambda(30) = -1$ as well. The divisor count is $d(30) = 2^3 = 8$, and $\sigma(30) = 3 \cdot 4 \cdot 6 = 72$. Note $\sigma(30)/30 = 72/30 = 12/5 > 2$, so 30 is an *abundant* number.

The multiplicativity of each function in the catalogue is summarized as follows: μ , φ , σ_k , d , σ are multiplicative; λ is totally multiplicative; ω is additive; Ω is totally additive; Λ and $\log$ are none of the above. Several relationships between these functions are already visible. The Möbius function μ and the Liouville function λ agree on squarefree integers and differ elsewhere. The identity $|\mu(n)| = 1$ if and only if n is squarefree can be expressed as $|\mu| = \mu^2$, and $\sum_{d|n} |\mu(d)| = 2^{\omega(n)}$ counts the squarefree divisors of n . The von Mangoldt identity $\sum_{d|n} \Lambda(d) = \log n$ expresses $\log$ as a “divisor sum” of Λ . All such identities acquire a unified algebraic meaning once the notion of Dirichlet convolution is in place.

1.2 Dirichlet Convolution and Möbius Inversion

The arithmetic functions of the previous section satisfy many identities: $\sum_{d|n} \mu(d) = [n=1]$, $\sum_{d|n} \Lambda(d) = \log n$, $\sum_{d|n} \varphi(d) = n$, and so forth. These identities share a common structure, i.e. a sum over the divisors of n , with the summand depending on both d and n/d . This structure has a name: *Dirichlet convolution*. Equipping the set of arithmetic functions with Dirichlet convolution as multiplication turns it into a commutative ring, and the divisor-sum identities become algebraic statements in this ring:

Definition 1.2.1: Dirichlet Convolution

The *Dirichlet convolution* of two arithmetic functions f and g is the arithmetic function $f * g$ defined by

$$(f * g)(n) = \sum_{d|n} f(d) g(n/d) = \sum_{ab=n} f(a) g(b)$$

The two expressions are equivalent: summing over divisors d of n with complementary divisor n/d is the same as summing over ordered factorizations $n = ab$. The second form makes the commutativity $f * g = g * f$ evident since $ab = ba$.

Several of the identities from section 1.1 are now recognizable as convolution statements:

1. The von Mangoldt identity $\sum_{d|n} \Lambda(d) = \log n$ is $\Lambda * \mathbf{1} = \log$ where $\mathbf{1}$ denotes the constant function $\mathbf{1}(n) = 1$.
2. The totient summation $\sum_{d|n} \varphi(d) = n$ is $\varphi * \mathbf{1} = \text{id}$ where $\text{id}(n) = n$.
3. The divisor function $d(n) = \sum_{d|n} 1$ is $d = \mathbf{1} * \mathbf{1}$ and the sum-of-divisors function $\sigma(n) = \sum_{d|n} d$ is $\sigma = \text{id} * \mathbf{1}$.

Theorem 1.2.2: The Ring of Arithmetic Functions

The set $\mathcal{A}$ of all arithmetic functions, equipped with pointwise addition and Dirichlet convolution as multiplication, forms a commutative ring with identity. The additive identity is the zero function, and the multiplicative identity is

$$\varepsilon(n) = [n = 1] = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1. \end{cases}$$

Proof:

1. *Commutativity.* Commutativity of $*$ follows from the symmetry of the second form: $(f * g)(n) = \sum_{ab=n} f(a)g(b) = \sum_{ab=n} g(a)f(b) = (g * f)(n)$.

2. *Associativity.* For any three arithmetic functions f, g, h :

$$((f * g) * h)(n) = \sum_{de=n} (f * g)(d) h(e) = \sum_{de=n} \sum_{ab=d} f(a)g(b)h(e) = \sum_{abe=n} f(a)g(b)h(e)$$

The final expression is symmetric in the grouping, so $(f * (g * h))(n)$ gives the same triple sum.

3. *Identity.* $(f * \varepsilon)(n) = \sum_{d|n} f(d)\varepsilon(n/d) = f(n)\varepsilon(1) = f(n)$, since $\varepsilon(n/d) = 0$ unless $d = n$.

4. *Distributivity.* $(f * (g + h))(n) = \sum_{d|n} f(d)(g + h)(n/d) = \sum_{d|n} f(d)g(n/d) + \sum_{d|n} f(d)h(n/d) = (f * g)(n) + (f * h)(n)$.

The ring $(\mathcal{A}, +, *)$ is a commutative ring with identity but not an integral domain: for instance, if f is supported on $\{1\}$ and g is supported on $\{1\}$ with $f(1) = g(1) = 0$, then $f * g = 0$. The unit group of $\mathcal{A}$ (the set of arithmetic functions possessing a convolution inverse) is characterized by a simple condition:

Proposition 1.2.3: Units of the Convolution Ring

An arithmetic function f has a Dirichlet convolution inverse f^{-1} (satisfying $f * f^{-1} = \varepsilon$) if and only if $f(1) \neq 0$. When it exists, the inverse is unique and is given recursively by

$$f^{-1}(1) = \frac{1}{f(1)}, \quad f^{-1}(n) = -\frac{1}{f(1)} \sum_{\substack{d|n \\ d < n}} f(n/d) f^{-1}(d) \quad \text{for } n > 1$$

Proof:

The condition $(f * f^{-1})(1) = f(1)f^{-1}(1) = 1$ forces $f(1) \neq 0$ and $f^{-1}(1) = 1/f(1)$. For $n > 1$, the condition $(f * f^{-1})(n) = 0$ reads

$$\sum_{d|n} f(n/d) f^{-1}(d) = 0,$$

and isolating the $d = n$ term gives $f(1)f^{-1}(n) = -\sum_{d|n, d < n} f(n/d)f^{-1}(d)$, which determines $f^{-1}(n)$ recursively from the values $f^{-1}(d)$ for $d < n$.

The recursive formula computes f^{-1} one value at a time in increasing order: $f^{-1}(1)$, then $f^{-1}(2)$, then $f^{-1}(3)$, and so on, each depending only on previously computed values. This is effective for small arguments but does not directly give a closed-form expression.

Example 1.2.4: Computing a Dirichlet Inverse

Compute id^{-1} using the recursive formula. At $n = 1$: $\text{id}^{-1}(1) = 1$. At $n = 2$: $\text{id}^{-1}(2) = -(1/1) \cdot \text{id}(2) \cdot \text{id}^{-1}(1) = -2$. At $n = 3$: $\text{id}^{-1}(3) = -3$. At $n = 4$, the divisors less than 4 are 1 and 2:

$$\text{id}^{-1}(4) = -(\text{id}(4) \cdot \text{id}^{-1}(1) + \text{id}(2) \cdot \text{id}^{-1}(2)) = -(4 \cdot 1 + 2 \cdot (-2)) = 0$$

At $n = 6$: $\text{id}^{-1}(6) = -(6 \cdot 1 + 3 \cdot (-2) + 2 \cdot (-3)) = -(6 - 6 - 6) = 6$. The pattern is $\text{id}^{-1}(n) = n\mu(n)$: the function vanishes at non-squarefree integers, equals $(-1)^{\omega(n)}n$ at squarefree integers, and is multiplicative with $\text{id}^{-1}(p) = -p$ and $\text{id}^{-1}(p^k) = 0$ for $k \geq 2$.

For multiplicative functions, a cleaner description is available.

Proposition 1.2.5: Multiplicative Functions Form a Subgroup

If f and g are multiplicative, then $f * g$ is multiplicative. If f is multiplicative, then f^{-1} exists (since $f(1) = 1$) and is also multiplicative.

Proof:

Let $\gcd(m, n) = 1$. The divisors of mn are exactly the products d_1d_2 where $d_1|m$ and $d_2|n$

(since $\gcd(m, n) = 1$). Hence:

$$\begin{aligned} (f * g)(mn) &= \sum_{d|mn} f(d) g(mn/d) = \sum_{d_1|m} \sum_{d_2|n} f(d_1 d_2) g\left(\frac{m}{d_1} \cdot \frac{n}{d_2}\right) \\ &= \sum_{d_1|m} \sum_{d_2|n} f(d_1) f(d_2) g(m/d_1) g(n/d_2) = (f * g)(m) \cdot (f * g)(n) \end{aligned}$$

Since $(f * g)(1) = f(1)g(1) = 1$, the function $f * g$ is multiplicative.

For the inverse: if f is multiplicative, then $f(1) = 1 \neq 0$, so f^{-1} exists. Multiplicativity of f^{-1} can be verified by induction on mn using the recursive formula: the key observation is that for $\gcd(m, n) = 1$, the divisors of mn factor as products of divisors of m and n , so the recursion for $f^{-1}(mn)$ splits into independent recursions for $f^{-1}(m)$ and $f^{-1}(n)$.

The multiplicative arithmetic functions thus form a group under Dirichlet convolution, with identity ε . This group structure is the reason that identities among multiplicative functions can be verified by checking them at prime powers.

The most important Dirichlet inverse is that of the constant function $\mathbf{1}$.

Proposition 1.2.6: The Möbius Function as a Dirichlet Inverse

$\mu * \mathbf{1} = \varepsilon$. That is, μ is the Dirichlet inverse of $\mathbf{1}$:

$$\sum_{d|n} \mu(d) = [n = 1] = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n > 1. \end{cases}$$

Proof:

Both μ and $\mathbf{1}$ are multiplicative, so $\mu * \mathbf{1}$ is multiplicative by proposition 1.2.5. It suffices to check the identity at prime powers. For p^k with $k \geq 1$:

$$\sum_{d|p^k} \mu(d) = \mu(1) + \mu(p) + \mu(p^2) + \cdots + \mu(p^k) = 1 + (-1) + 0 + \cdots + 0 = 0$$

At $n = 1$: $(\mu * \mathbf{1})(1) = \mu(1) \cdot 1 = 1$. Hence $\mu * \mathbf{1} = \varepsilon$.

This identity is the algebraic content of inclusion-exclusion. For a squarefree integer $n = p_1 \cdots p_r$, the divisors are the 2^r products of subsets of $\{p_1, \dots, p_r\}$, and $\sum_{d|n} \mu(d) = \sum_{k=0}^r \binom{r}{k} (-1)^k = (1-1)^r = 0$.

With the Möbius function identified as the inverse of $\mathbf{1}$, the central inversion formula follows immediately.

Theorem 1.2.7: Möbius Inversion Formula

Let f and g be arithmetic functions. Then:

$$g = f * \mathbf{1} \quad \text{if and only if} \quad f = g * \mu$$

In explicit notation: $g(n) = \sum_{d|n} f(d)$ for all n if and only if $f(n) = \sum_{d|n} \mu(d) g(n/d)$ for all n .

Proof:

If $g = f * \mathbf{1}$, then $g * \mu = (f * \mathbf{1}) * \mu = f * (\mathbf{1} * \mu) = f * \varepsilon = f$, using associativity and $\mathbf{1} * \mu = \varepsilon$ (proposition 1.2.6). Conversely, if $f = g * \mu$, then $f * \mathbf{1} = (g * \mu) * \mathbf{1} = g * (\mu * \mathbf{1}) = g * \varepsilon = g$.

Thus, Möbius inversion is the tool for “solving” a divisor-sum relation: if g is defined as a sum of f over divisors, then f can be recovered from g by convolving with μ .

Example 1.2.8: Recovering φ and Λ via Möbius Inversion

1. The identity $\varphi * \mathbf{1} = \text{id}$ (proposition 0.3.16) inverts to $\varphi = \text{id} * \mu$. In explicit form:

$$\varphi(n) = \sum_{d|n} \mu(d) \cdot \frac{n}{d} = n \sum_{d|n} \frac{\mu(d)}{d} = n \prod_{p|n} \left(1 - \frac{1}{p}\right),$$

where the last step uses multiplicativity: for $n = \prod p_i^{a_i}$, the sum $\sum_{d|n} \mu(d)/d$ factors over primes as $\prod_{p|n} (1 + \mu(p)/p) = \prod_{p|n} (1 - 1/p)$. This recovers the formula from proposition 0.3.10.

2. The von Mangoldt identity $\Lambda * \mathbf{1} = \log$ (proposition 1.1.9) inverts to $\Lambda = \log * \mu$, i.e.,

$$\Lambda(n) = \sum_{d|n} \mu(d) \log(n/d) = - \sum_{d|n} \mu(d) \log d$$

The second equality uses $\sum_{d|n} \mu(d) \log(n/d) = \log n \sum_{d|n} \mu(d) - \sum_{d|n} \mu(d) \log d = - \sum_{d|n} \mu(d) \log d$, since $\sum_{d|n} \mu(d) = 0$ for $n > 1$ (and the case $n = 1$ gives $\Lambda(1) = 0$ directly).

Example 1.2.9: Counting Squarefree Integers

Let $Q(x) = \#\{n \leq x : n \text{ is squarefree}\} = \sum_{n \leq x} |\mu(n)|$. Since $|\mu(n)| = \sum_{d^2|n} \mu(d)$ (an identity verifiable at prime powers: both sides equal 1 if n is squarefree and 0 otherwise), the count can be rewritten as:

$$Q(x) = \sum_{n \leq x} \sum_{d^2|n} \mu(d) = \sum_{d \leq \sqrt{x}} \mu(d) \left\lfloor \frac{x}{d^2} \right\rfloor = \sum_{d \leq \sqrt{x}} \mu(d) \left(\frac{x}{d^2} + O(1) \right) = x \sum_{d \leq \sqrt{x}} \frac{\mu(d)}{d^2} + O(\sqrt{x})$$

The sum $\sum_{d=1}^{\infty} \mu(d)/d^2$ converges to $1/\zeta(2) = 6/\pi^2$ (as the Euler product gives $\sum \mu(d)/d^s = 1/\zeta(s)$,

a fact established rigorously in chapter 2). Accepting this for now:

$$Q(x) = \frac{6}{\pi^2}x + O(\sqrt{x})$$

Hence the density of squarefree integers is $6/\pi^2 \approx 0.6079$: about 61% of all positive integers are squarefree.

The remaining identities from the outline of section 1.1 are now provable in the convolution ring. A systematic collection is recorded here for later reference.

Proposition 1.2.10: Key Convolution Identities

The following identities hold in $(\mathcal{A}, +, *)$:

1. $\mu * \mathbf{1} = \varepsilon$ (Möbius is the inverse of $\mathbf{1}$).
2. $\varphi * \mathbf{1} = \text{id}$, equivalently $\varphi = \text{id} * \mu$.
3. $\Lambda * \mathbf{1} = \log$, equivalently $\Lambda = \log * \mu$.
4. $d = \mathbf{1} * \mathbf{1}$ (the divisor function is the self-convolution of $\mathbf{1}$).
5. $\sigma = \text{id} * \mathbf{1}$.
6. $\sigma_k = \text{id}^k * \mathbf{1}$, where $\text{id}^k(n) = n^k$.
7. $|\mu| * |\mu| = 2^\omega$, where $2^\omega(n) = 2^{\omega(n)}$.
8. $\lambda * |\mu| = \varepsilon$ (λ is the inverse of $|\mu|$).
9. $\lambda * \mathbf{1} = \mathbf{1}_\square$, where $\mathbf{1}_\square(n) = [n \text{ is a perfect square}]$.

Proof:

Identities (1) - (6) have been established above or follow directly from definitions. For the remaining:

7. Both sides are multiplicative (the left by proposition 1.2.5; the right since ω is additive gives $2^{\omega(mn)} = 2^{\omega(m)}2^{\omega(n)}$ for $\gcd(m, n) = 1$). At p^k : $(|\mu| * |\mu|)(p^k) = \sum_{j=0}^k |\mu(p^j)| |\mu(p^{k-j})| = |\mu(1)| |\mu(p^k)| + |\mu(p)| |\mu(p^{k-1})| + \dots$. The nonzero terms are those with $j \in \{0, 1\}$ and $k-j \in \{0, 1\}$, so only $j = 0, k-j \leq 1$ and $j = 1, k-j \leq 1$ contribute. For $k = 0$: the sum is 1. For $k = 1$: $|\mu(1)| |\mu(p)| + |\mu(p)| |\mu(1)| = 2$. For $k \geq 2$: $|\mu(1)| |\mu(p^k)| + |\mu(p)| |\mu(p^{k-1})| = 0$. Hence $(|\mu| * |\mu|)(p^k) = 2$ for $k \geq 1$ and 1 for $k = 0$, matching $2^{\omega(p^k)}$.
8. At p^k : $(\lambda * |\mu|)(p^k) = \sum_{j=0}^k \lambda(p^j) |\mu(p^{k-j})| = (-1)^0 \cdot |\mu(p^k)| + (-1)^1 \cdot |\mu(p^{k-1})| + \dots$. For $k = 0$: the sum is 1. For $k = 1$: $|\mu(p)| + (-1) |\mu(1)| = 1 - 1 = 0$. For $k \geq 2$: $|\mu(p^k)| + (-1) |\mu(p^{k-1})| + |\mu(p^{k-2})| + \dots$; the only nonzero terms are $j = k-1$ (giving $(-1)^{k-1}$) and $j = k$ (giving $(-1)^k$), which sum to 0. Hence $\lambda * |\mu| = \varepsilon$.

$$9. \text{ At } p^k: (\lambda * \mathbf{1})(p^k) = \sum_{j=0}^k (-1)^j = \begin{cases} 1 & k \text{ even,} \\ 0 & k \text{ odd} \end{cases}, \text{ which is } [p^k \text{ is a perfect square}].$$

Identity (9) deserves a moment of interpretation: the summatory function $\sum_{d|n} \lambda(d)$ is 1 if n is a perfect square and 0 otherwise. This identity connects the Liouville function to the distribution of perfect squares, and its Dirichlet series translation $\zeta(2s)/\zeta(s) = \sum \lambda(n)n^{-s}$ will appear in chapter 2.

1.3 Average Orders and Summatory Functions

The convolution identities of the previous section describe exact relationships among arithmetic functions at individual integers. The present section asks a different question: how do these functions behave *on average*? The divisor function $d(n)$ fluctuates wildly, for example $d(p) = 2$ for primes, while $d(n)$ can be arbitrarily large for highly composite n . But its average over $\{1, \dots, N\}$ grows smoothly like $\log N$. Extracting this smooth average from the pointwise behavior is the first analytic problem of the book, and the main tool is the *Dirichlet hyperbola method*: a technique for evaluating $\sum_{n \leq x} (f * g)(n)$ by splitting the double sum along the hyperbola $ab = n$.

Definition 1.3.1: Summatory Function and Average Order

For an arithmetic function f , the *summatory function* is

$$M_f(x) = \sum_{n \leq x} f(n)$$

The function $g(x)$ is an *average order* of f if $M_f(x) \sim g(x)$ as $x \rightarrow \infty$, i.e., $\sum_{n \leq x} f(n)/g(x) \rightarrow 1$.

The summatory function converts the discrete sequence $(f(n))$ into a step function of a continuous variable x . The average order captures the typical size of $f(n)$: if $g(x)$ is an average order of f , then $f(n)$ is “typically” of size $g(n)/n$. The fundamental technique for estimating summatory functions of convolutions is the following:

Theorem 1.3.2: Dirichlet’s Hyperbola Method

Let f and g be arithmetic functions, and set $h = f * g$. For any $x \geq 1$ and any y with $1 \leq y \leq x$:

$$\sum_{n \leq x} h(n) = \sum_{a \leq y} f(a) G\left(\frac{x}{a}\right) + \sum_{b \leq x/y} g(b) F\left(\frac{x}{b}\right) - F(y) G\left(\frac{x}{y}\right)$$

where $F(t) = \sum_{n \leq t} f(n)$ and $G(t) = \sum_{n \leq t} g(n)$.

The idea is to write $\sum_{n \leq x} h(n) = \sum_{ab \leq x} f(a)g(b)$ and split the region $\{(a, b) : ab \leq x\}$ into two parts along the line $a = y$. The first sum counts pairs with $a \leq y$, the second counts pairs with $b \leq x/y$ (equivalently $a > y$), and the subtracted term corrects for the overlap where both $a \leq y$ and $b \leq x/y$.

Proof:

Start from $h = f * g$:

$$\sum_{n \leq x} h(n) = \sum_{\substack{a, b \geq 1 \\ ab \leq x}} f(a)g(b)$$

Step 1: Partition the summation region. Split $\{(a, b) \in \mathbb{Z}_{>0}^2 : ab \leq x\}$ into: $S_1 = \{a \leq y\}$ and $S_2 = \{a > y, b \leq x/y\}$. Every pair (a, b) with $ab \leq x$ belongs to at least one of these sets (if $a > y$, then $b \leq x/a < x/y$), and the overlap is $S_1 \cap S_2 = \{a \leq y, b \leq x/y\}$. By inclusion-exclusion:

$$\sum_{ab \leq x} f(a)g(b) = \sum_{\substack{a \leq y \\ ab \leq x}} f(a)g(b) + \sum_{\substack{a > y \\ b \leq x/y}} f(a)g(b) - \sum_{\substack{a \leq y \\ b \leq x/y}} f(a)g(b) + \sum_{\substack{a \leq y \\ b \leq x/y}} f(a)g(b)$$

Step 2: Evaluate each piece. The first sum: for fixed $a \leq y$, the condition $ab \leq x$ means $b \leq x/a$, so $\sum_{\substack{a \leq y \\ ab \leq x}} f(a)g(b) = \sum_{a \leq y} f(a)G(x/a)$. The second sum plus the overlap: $\sum_{b \leq x/y} \sum_{a \leq x/b} f(a)g(b) = \sum_{b \leq x/y} g(b)F(x/b)$. The overlap alone: $\sum_{a \leq y} f(a) \sum_{b \leq x/y} g(b) = F(y)G(x/y)$. Combining gives the stated formula.

The optimal choice of the splitting parameter y depends on the specific functions f and g ; the symmetric choice $y = \sqrt{x}$ is the most common and typically minimizes the error term. The power of the method lies in reducing the estimation of $\sum h(n)$ to the estimation of F and G separately—a significant simplification when f and g are simpler than h .

Geometrically, the lattice points $(a, b) \in \mathbb{Z}_{>0}^2$ with $ab \leq x$ lie below the hyperbola $ab = x$ in the first quadrant. The horizontal line $a = y$ divides this region into an upper-left piece (where the sum over a is short and G handles the sum over b) and a lower-right piece (where the sum over b is short and F handles the sum over a). The rectangle $\{a \leq y, b \leq x/y\}$ is the overlap region. This geometric picture—counting lattice points under a hyperbola—is a prototype for the lattice-point counting arguments that appear in the proof of the analytic class number formula (??).

The first and most classical application is to the divisor function $d = \mathbf{1} * \mathbf{1}$.

Theorem 1.3.3: Average Order of the Divisor Function

$$\sum_{n \leq x} d(n) = x \log x + (2\gamma - 1)x + O(\sqrt{x})$$

where $\gamma = 0.5772\dots$ is the Euler-Mascheroni constant. In particular, the average order of $d(n)$ is $\log n$.

Proof:

Since $d = \mathbf{1} * \mathbf{1}$, apply theorem 1.3.2 with $f = g = \mathbf{1}$ and $y = \sqrt{x}$. The summatory function of $\mathbf{1}$ is $F(t) = G(t) = [t] = t + O(1)$. By symmetry, the first two sums in the hyperbola formula are equal, giving:

$$\sum_{n \leq x} d(n) = 2 \sum_{a \leq \sqrt{x}} \left\lfloor \frac{x}{a} \right\rfloor - [\sqrt{x}]^2$$

Using $\lfloor x/a \rfloor = x/a + O(1)$:

$$2 \sum_{a \leq \sqrt{x}} \left\lfloor \frac{x}{a} \right\rfloor = 2x \sum_{a \leq \sqrt{x}} \frac{1}{a} + O(\sqrt{x})$$

The harmonic sum satisfies $\sum_{a \leq \sqrt{x}} 1/a = \frac{1}{2} \log x + \gamma + O(1/\sqrt{x})$, so:

$$2x \sum_{a \leq \sqrt{x}} \frac{1}{a} = x \log x + 2\gamma x + O(\sqrt{x})$$

For the subtracted term: $\lfloor \sqrt{x} \rfloor^2 = (\sqrt{x} + O(1))^2 = x + O(\sqrt{x})$. Combining:

$$\sum_{n \leq x} d(n) = x \log x + 2\gamma x + O(\sqrt{x}) - x - O(\sqrt{x}) = x \log x + (2\gamma - 1)x + O(\sqrt{x})$$

1.3.4 Remark: The Dirichlet Divisor Problem The error term $O(\sqrt{x})$ in theorem 1.3.3 is not optimal. Let $\Delta(x) = \sum_{n \leq x} d(n) - x \log x - (2\gamma - 1)x$ denote the error. The best known upper bound as of writing this book, due to Huxley (2003), is $\Delta(x) = O(x^{131/416+\varepsilon})$, where $131/416 \approx 0.3149$. Hardy proved $\Delta(x) = \Omega(x^{1/4})$, establishing that the error is at least of order $x^{1/4}$ infinitely often. The conjectured optimal exponent is $1/4$, and determining the true order of $\Delta(x)$ (the *Dirichlet divisor problem*) remains open.

Example 1.3.5: Numerical Verification of the Divisor Sum

For $x = 100$: the exact value is $\sum_{n=1}^{100} d(n) = 482$. The asymptotic formula gives $100 \log 100 + (2\gamma - 1) \cdot 100 \approx 460.5 + 15.4 = 476.0$, with error $482 - 476 = 6$. The bound $O(\sqrt{100}) = O(10)$ is consistent. At $x = 10,000$: the exact sum is 93,668, and the formula gives $10,000 \log(10,000) + (2\gamma - 1) \cdot 10,000 \approx 92,103 + 1,544 = 93,648$, with error $20 = O(\sqrt{10,000}) = O(100)$.

For comparison, the summatory function of μ exhibits much more complicated behavior. For future reference, we box the following:

Definition 1.3.6: The Mertens Function

The *Mertens function* is $M(x) = \sum_{n \leq x} \mu(n)$.

The identity $\mu * \mathbf{1} = \varepsilon$ (proposition 1.2.6) implies $\sum_{d \leq x} M(x/d) = 1$, a self-referencing identity that constrains the growth of M . The Mertens function oscillates heavily and grows much more slowly than x : the bound $M(x) = o(x)$ is equivalent to the Prime Number Theorem (chapter 5), and the Riemann Hypothesis is equivalent to $M(x) = O(x^{1/2+\varepsilon})$ for every $\varepsilon > 0$. Mertens conjectured in 1897 that $|M(x)| < \sqrt{x}$ for all $x \geq 1$; this was disproved by Odlyzko and te Riele in 1985, though no explicit counterexample is known.

The rest of this section is dedicated to computing average orders of various arithmetic functions:

Theorem 1.3.7: Average Order of the Sum-of-Divisors Function

$$\sum_{n \leq x} \sigma(n) = \frac{\pi^2}{12} x^2 + O(x \log x)$$

Proof:

Since $\sigma = \text{id} * \mathbf{1}$, the double-sum representation gives $\sum_{n \leq x} \sigma(n) = \sum_{ab \leq x} a$. Summing over b first:

$$\sum_{n \leq x} \sigma(n) = \sum_{b \leq x} \sum_{a \leq x/b} a = \sum_{b \leq x} \frac{1}{2} \left\lfloor \frac{x}{b} \right\rfloor \left(\left\lfloor \frac{x}{b} \right\rfloor + 1 \right)$$

Since $\lfloor t \rfloor (\lfloor t \rfloor + 1)/2 = t^2/2 + O(t)$:

$$\sum_{b \leq x} \left(\frac{x^2}{2b^2} + O\left(\frac{x}{b}\right) \right) = \frac{x^2}{2} \sum_{b \leq x} \frac{1}{b^2} + O(x \log x) = \frac{x^2}{2} \left(\frac{\pi^2}{6} + O(1/x) \right) + O(x \log x) = \frac{\pi^2}{12} x^2 + O(x \log x)$$

A numerical check: $\sum_{n=1}^{100} \sigma(n) = 8299$, while $\frac{\pi^2}{12} \cdot 100^2 \approx 8225$, giving an error of $74 = O(100 \log 100) = O(461)$.

Theorem 1.3.8: Average Order of the Euler Totient

$$\sum_{n \leq x} \varphi(n) = \frac{3}{\pi^2} x^2 + O(x \log x)$$

The leading coefficient $3/\pi^2 = 1/(2\zeta(2))$ reflects the density of coprime pairs: if two integers are chosen uniformly at random from $\{1, \dots, N\}$, the probability that they are coprime tends to $6/\pi^2$ as $N \rightarrow \infty$.

Proof:

By Möbius inversion (theorem 1.2.7), $\varphi(n) = \sum_{d|n} \mu(d)(n/d)$. Summing over $n \leq x$ and interchanging the order:

$$\sum_{n \leq x} \varphi(n) = \sum_{d \leq x} \mu(d) \sum_{\substack{n \leq x \\ d|n}} \frac{n}{d}$$

Writing $n = dm$ with $m \leq x/d$:

$$\sum_{\substack{n \leq x \\ d|n}} \frac{n}{d} = \sum_{m \leq x/d} m = \frac{1}{2} \left\lfloor \frac{x}{d} \right\rfloor \left(\left\lfloor \frac{x}{d} \right\rfloor + 1 \right) = \frac{x^2}{2d^2} + O\left(\frac{x}{d}\right)$$

Substituting:

$$\sum_{n \leq x} \varphi(n) = \frac{x^2}{2} \sum_{d \leq x} \frac{\mu(d)}{d^2} + O\left(x \sum_{d \leq x} \frac{1}{d}\right) = \frac{x^2}{2} \left(\frac{6}{\pi^2} + O(1/x) \right) + O(x \log x) = \frac{3}{\pi^2} x^2 + O(x \log x)$$

The identity $\sum_{d=1}^{\infty} \mu(d)/d^2 = 1/\zeta(2) = 6/\pi^2$ is the Euler product $\prod_p (1 - p^{-2})$; a rigorous proof appears in chapter 2.

The appearance of $6/\pi^2$ in the totient average has a probabilistic interpretation. The number of pairs (a, b) with $1 \leq a, b \leq N$ and $\gcd(a, b) = 1$ is $\sum_{a=1}^N \sum_{b=1}^N [\gcd(a, b) = 1] = 2 \sum_{n=1}^N \varphi(n) - 1$ (the term -1 accounts for the pair $(1, 1)$ counted once). The density of coprime pairs is therefore

$$\frac{2 \sum_{n=1}^N \varphi(n) - 1}{N^2} = \frac{6/\pi^2 \cdot N^2 + O(N \log N)}{N^2} \rightarrow \frac{6}{\pi^2} \approx 0.6079$$

Concretely, if a and b are chosen uniformly at random from $\{1, \dots, N\}$, the probability that $\gcd(a, b) = 1$ approaches $6/\pi^2$. Each prime p contributes independently: the probability that $p \nmid \gcd(a, b)$ is $1 - 1/p^2$, and the product $\prod_p (1 - 1/p^2) = 6/\pi^2$ reflects this independence.

The three estimates $\sum d(n) \sim x \log x$, $\sum \sigma(n) \sim \frac{\pi^2}{12} x^2$, and $\sum \varphi(n) \sim \frac{3}{\pi^2} x^2$ illustrate a pattern: the average order of a multiplicative function defined as a convolution $f * \mathbf{1}$ is controlled by the summatory behavior of f and the value $\zeta(2) = \pi^2/6$. The Dirichlet series framework of chapter 2 makes this pattern systematic.

The additive function $\omega(n)$ (number of distinct prime divisors) has a subtler average order.

Proposition 1.3.9: Average Order of $\omega(n)$

$$\sum_{n \leq x} \omega(n) = x \log \log x + Bx + O\left(\frac{x}{\log x}\right)$$

where $B = \gamma + \sum_p (\log(1 - 1/p) + 1/p) = 0.2614\dots$ is related to Mertens' constant. In particular, the average order of $\omega(n)$ is $\log \log n$.

The proof uses the representation $\omega(n) = \sum_p |n|_p$, which after interchanging the order of summation gives $\sum_{n \leq x} \omega(n) = \sum_{p \leq x} \lfloor x/p \rfloor = x \sum_{p \leq x} 1/p + O(\pi(x))$, and the estimate $\sum_{p \leq x} 1/p = \log \log x + M + O(1/\log x)$ (Mertens' second theorem, chapter 3) completes the argument. The full proof is given after Mertens' theorems are established; an alternative proof using the Euler product of $\zeta(s)$ appears in chapter 3 as well.

The average $\log \log n$ is strikingly small: a “typical” integer near 10^9 has only about $\log \log(10^9) \approx 3.0$ distinct prime factors. The celebrated Erdős-Kac theorem (1940) refines this by showing that $(\omega(n) - \log \log n)/\sqrt{\log \log n}$ is approximately normally distributed—a Gaussian central limit theorem for a number-theoretic function. This theorem lies outside the scope of this book, but the average order $\log \log n$ established here is the mean of that Gaussian.

Example 1.3.10: Summatory Functions and the Squarefree Indicator

The squarefree indicator function $|\mu(n)|$ (which is 1 if n is squarefree, 0 otherwise) has summatory function $Q(x) = \sum_{n \leq x} |\mu(n)|$. In example 1.2.9, the estimate $Q(x) = \frac{6}{\pi^2} x + O(\sqrt{x})$ was derived using the identity $|\mu(\tilde{n})| = \sum_{d^2 | n} \mu(d)$. This can also be obtained from the convolution $|\mu| * \mathbf{1}_{\square} = \mathbf{1}$ (where $\mathbf{1}_{\square}(n) = [n \text{ is a perfect square}]$), which inverts to $|\mu| = \mu * \mathbf{1}_{\square}^{-1}$. The approach via the identity $|\mu(n)| = \sum_{d^2 | n} \mu(d)$ is more direct and avoids computing the inverse of $\mathbf{1}_{\square}$.

Example 1.3.11: Comparing Average Orders

The average orders established in this section are:

$$d(n) \sim \log n, \quad \sigma(n) \sim \frac{\pi^2}{6}n, \quad \varphi(n) \sim \frac{6}{\pi^2}n, \quad \omega(n) \sim \log \log n$$

The first and fourth grow logarithmically (or doubly logarithmically), while σ and φ grow linearly. The product $\sigma(n)\varphi(n) \sim n^2$ on average, reflecting the identity $\varphi(n)\sigma(n)/n^2 = \prod_{p|n} (1 - 1/p^2)$, which is bounded between $6/\pi^2$ and 1 for all n .

Example 1.3.12: Average Behavior of $\varphi(n)/n$

The estimate $\sum_{n \leq x} \varphi(n) = \frac{3}{\pi^2}x^2 + O(x \log x)$ implies that the average value of $\varphi(n)/n$ over $\{1, \dots, N\}$ tends to $6/\pi^2 \approx 0.6079$. This is the same constant as the density of squarefree integers (example 1.2.9): both equal $1/\zeta(2)$. The connection is structural—the “probability” that two integers are coprime and the density of squarefree integers are both governed by $\zeta(2)^{-1} = \prod_p (1 - p^{-2})$.

1.4 Partial Summation and Abel's Summation Formula

The previous section estimated summatory functions by exploiting the convolution structure of specific arithmetic functions. This section introduces a technique that applies to *any* weighted sum $\sum a_n f(n)$, regardless of whether the coefficients a_n have multiplicative structure: *Abel's summation formula*, the discrete analogue of integration by parts. The formula converts a sum $\sum a_n f(n)$ into an integral involving the summatory function $A(t) = \sum_{n \leq t} a_n$ and the derivative f' , thereby transferring information about the partial sums of (a_n) into analytic information about the weighted sum. It is arguably the single most frequently used technical tool in this book.

Theorem 1.4.1: Abel's Summation Formula

Let $(a_n)_{n \geq 1}$ be a sequence of complex numbers, and let $f : [1, \infty) \rightarrow \mathbb{C}$ be a continuously differentiable function. Define $A(t) = \sum_{n \leq t} a_n$. Then for any $x \geq 1$:

$$\sum_{n \leq x} a_n f(n) = A(x)f(x) - \int_1^x A(t) f'(t) dt$$

The formula is the exact analogue of integration by parts: $\int u dv = uv - \int v du$, with $u = f$, $dv = dA$ (the Stieltjes measure associated to the step function A), $v = A$, and $du = f' dt$. The “boundary term” $A(x)f(x)$ corresponds to $[uv]$, and the integral $\int A(t)f'(t) dt$ corresponds to $\int v du$. The lower limit of integration is 1 rather than 0 because $A(t) = 0$ for $t < 1$.

Proof:

For $x \geq 1$, write $N = \lfloor x \rfloor$ and split the sum into a telescoping form. For each n with $1 \leq n \leq N$:

$$a_n = A(n) - A(n-1)$$

Hence:

$$\begin{aligned}
 \sum_{n=1}^N a_n f(n) &= \sum_{n=1}^N (A(n) - A(n-1))f(n) \\
 &= \sum_{n=1}^N A(n)f(n) - \sum_{n=1}^N A(n-1)f(n) \\
 &= \sum_{n=1}^N A(n)f(n) - \sum_{n=0}^{N-1} A(n)f(n+1) \\
 &= A(N)f(N) + \sum_{n=1}^{N-1} A(n)(f(n) - f(n+1)) - A(0)f(1)
 \end{aligned}$$

Since $A(0) = 0$ and $f(n) - f(n+1) = -\int_n^{n+1} f'(t) dt$:

$$\sum_{n=1}^N a_n f(n) = A(N)f(N) - \sum_{n=1}^{N-1} A(n) \int_n^{n+1} f'(t) dt$$

On the interval $[n, n+1)$, the function $A(t)$ is constant with value $A(n)$. Hence $\sum_{n=1}^{N-1} A(n) \int_n^{n+1} f'(t) dt = \int_1^N A(t)f'(t) dt$. Finally, since $A(t) = A(N)$ for $t \in [N, x]$:

$$A(x)f(x) - \int_1^x A(t)f'(t) dt = A(N)f(x) - \int_1^N A(t)f'(t) dt - A(N) \int_N^x f'(t) dt = A(N)f(N) - \int_1^N A(t)f'(t) dt$$

using $f(x) - \int_N^x f'(t) dt = f(N)$. This matches the expression above.

The power of Abel's formula lies in the asymmetry of its inputs and outputs. The input is a *discrete* sum $\sum a_n f(n)$, which may be difficult to estimate directly. The output replaces this with a *continuous* integral $\int A(t)f'(t) dt$ plus a boundary term, and the integral is often much easier to handle, especially when $A(t)$ has a known asymptotic and $f'(t)$ is smooth and decaying.

The formula can also be written in Stieltjes integral notation as

$$\sum_{n \leq x} a_n f(n) = \int_1^x f(t) dA(t)$$

where $dA(t)$ is the Stieltjes measure that places mass a_n at each integer n . Integration by parts for Stieltjes integrals then gives $\int f dA = [fA] - \int A df = A(x)f(x) - \int_1^x A(t)f'(t) dt$, recovering theorem 1.4.1. This perspective will return when studying the Mellin and Laplace transforms that appear in chapter 5 and ??.

A useful variant handles the “other direction”: given knowledge of a weighted sum, deduce information about the partial sums.

Corollary 1.4.2: Summatory Function Estimate from Weighted Sums

If $\sum_{n \leq x} a_n f(n) = G(x) + O(E(x))$ and f is positive, monotonically decreasing, and continuously differentiable, then

$$A(x) = \frac{G(x)}{f(x)} + \frac{1}{f(x)} \int_1^x A(t) f'(t) dt + O\left(\frac{E(x)}{f(x)}\right)$$

When $A(t)$ appears on both sides, this is an implicit equation that can sometimes be solved by iteration.

Proof:

Rearranging theorem 1.4.1: $A(x) = \frac{1}{f(x)} \left(\sum_{n \leq x} a_n f(n) + \int_1^x A(t) f'(t) dt \right)$. Substituting the hypothesis gives the result.

This corollary is used in chapter 3 to pass from the estimate $\sum_{p \leq x} \log p/p = \log x + O(1)$ (Mertens' first theorem) to $\sum_{p \leq x} 1/p = \log \log x + O(1)$ (Mertens' second theorem).

Corollary 1.4.3: Summation by Parts for Convergent Series

Under the hypotheses of theorem 1.4.1, if $f(x) \rightarrow 0$ as $x \rightarrow \infty$ and the integral $\int_1^\infty A(t) f'(t) dt$ converges absolutely, then:

$$\sum_{n=1}^{\infty} a_n f(n) = - \int_1^{\infty} A(t) f'(t) dt$$

Proof:

Take $x \rightarrow \infty$ in theorem 1.4.1. The boundary term $A(x)f(x)$ tends to zero provided $A(x)f(x) \rightarrow 0$, which holds when $A(x) = O(x^c)$ for some c and $f(x) = o(x^{-c})$. The integral converges by hypothesis.

The first application recovers the harmonic sum asymptotic, providing an alternative to the integral comparison used in section 0.6.

Example 1.4.4: The Harmonic Sum via Abel Summation

Take $a_n = 1$ and $f(t) = 1/t$, so $A(t) = [t] = t - \{t\}$ and $f'(t) = -1/t^2$. Abel's formula gives:

$$\sum_{n \leq x} \frac{1}{n} = \frac{[x]}{x} + \int_1^x \frac{[t]}{t^2} dt = 1 + O(1/x) + \int_1^x \frac{t - \{t\}}{t^2} dt = 1 + \int_1^x \frac{dt}{t} - \int_1^x \frac{\{t\}}{t^2} dt + O(1/x)$$

The first integral is $\log x$. The second integral converges as $x \rightarrow \infty$ (since $0 \leq \{t\} < 1$ and $\int_1^\infty t^{-2} dt < \infty$) to a value $\int_1^\infty \{t\}/t^2 dt = 1 - \gamma$, where γ is the Euler-Mascheroni constant. Hence:

$$\sum_{n \leq x} \frac{1}{n} = \log x + \gamma + O(1/x)$$

The error $O(1/x)$ comes from the boundary term $[x]/x = 1 - \{x\}/x$ and the tail of $\int_x^\infty \{t\}/t^2 dt =$

$O(1/x)$.

Example 1.4.5: Weighted Divisor Sum

Take $a_n = d(n)$ and $f(t) = 1/t$. By theorem 1.3.3, $A(t) = \sum_{n \leq t} d(n) = t \log t + (2\gamma - 1)t + O(\sqrt{t})$. Abel's formula gives:

$$\begin{aligned} \sum_{n \leq x} \frac{d(n)}{n} &= \frac{A(x)}{x} + \int_1^x \frac{A(t)}{t^2} dt \\ &= \log x + (2\gamma - 1) + O(x^{-1/2}) + \int_1^x \frac{t \log t + (2\gamma - 1)t + O(\sqrt{t})}{t^2} dt \\ &= \log x + (2\gamma - 1) + \int_1^x \frac{\log t}{t} dt + (2\gamma - 1) \int_1^x \frac{dt}{t} + O(1) \\ &= \log x + (2\gamma - 1) + \frac{(\log x)^2}{2} + (2\gamma - 1) \log x + O(1) \end{aligned}$$

Hence $\sum_{n \leq x} d(n)/n = \frac{1}{2}(\log x)^2 + 2\gamma \log x + O(1)$. This estimate is used in chapter 2 to analyze the Dirichlet series $\sum d(n)n^{-s} = \zeta(s)^2$.

The next application previews the connection between partial summation and Dirichlet series that will be developed in chapter 2.

Proposition 1.4.6: Partial Sums of n^{-s}

For $s \in \mathbb{C}$ with $\operatorname{Re}(s) > 0$ and $s \neq 1$:

$$\sum_{n \leq x} \frac{1}{n^s} = \frac{x^{1-s}}{1-s} + \zeta(s) + O(x^{-\operatorname{Re}(s)})$$

where $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ for $\operatorname{Re}(s) > 1$, and the formula provides the analytic continuation of the right-hand side to $\operatorname{Re}(s) > 0$.

Proof:

Apply theorem 1.4.1 with $a_n = 1$ and $f(t) = t^{-s}$. Then $A(t) = [t] = t - \{t\}$ and $f'(t) = -st^{-s-1}$:

$$\sum_{n \leq x} n^{-s} = \frac{[x]}{x^s} + s \int_1^x \frac{[t]}{t^{s+1}} dt = \frac{x - \{x\}}{x^s} + s \int_1^x \frac{t - \{t\}}{t^{s+1}} dt$$

Separating the main term from the fractional part:

$$= x^{1-s} + s \int_1^x t^{-s} dt - s \int_1^x \frac{\{t\}}{t^{s+1}} dt + O(x^{-\operatorname{Re}(s)})$$

The first integral evaluates to $\frac{x^{1-s}-1}{1-s}$ for $s \neq 1$. The second integral converges absolutely for $\operatorname{Re}(s) > 0$ (since $|\{t\}/t^{s+1}| \leq t^{-\operatorname{Re}(s)-1}$), and its value from 1 to ∞ defines a constant depending

on s . Combining:

$$\sum_{n \leq x} n^{-s} = x^{1-s} + \frac{s}{1-s}(x^{1-s} - 1) - s \int_1^x \frac{\{t\}}{t^{s+1}} dt + O(x^{-\operatorname{Re}(s)}) = \frac{x^{1-s}}{1-s} + C(s) + O(x^{-\operatorname{Re}(s)})$$

where $C(s) = \frac{s}{s-1} - s \int_1^\infty \{t\} t^{-s-1} dt$. For $\operatorname{Re}(s) > 1$, taking $x \rightarrow \infty$ shows $C(s) = \zeta(s)$, and the formula extends $C(s)$ to $\operatorname{Re}(s) > 0$ by the convergence of the integral.

This result simultaneously provides the meromorphic continuation of $\zeta(s)$ to $\operatorname{Re}(s) > 0$ (with a simple pole at $s = 1$) and a useful asymptotic for the partial sums $\sum_{n \leq x} n^{-s}$. The full meromorphic continuation to all of $\mathbb{C}$ requires the functional equation (??).

Example 1.4.7: Estimating $\sum_{p \leq x} 1/p$ from $\pi(x)$

Abel summation converts knowledge of $\pi(x) = \sum_{p \leq x} 1$ (the prime counting function) into an estimate for $\sum_{p \leq x} 1/p$. Taking $a_p = 1$ for primes p and $a_n = 0$ otherwise, so $A(t) = \pi(t)$, and $f(t) = 1/t$:

$$\sum_{p \leq x} \frac{1}{p} = \frac{\pi(x)}{x} + \int_2^x \frac{\pi(t)}{t^2} dt$$

If the Prime Number Theorem $\pi(t) \sim t/\log t$ is assumed, then $\pi(x)/x \sim 1/\log x \rightarrow 0$ and

$$\int_2^x \frac{\pi(t)}{t^2} dt \sim \int_2^x \frac{dt}{t \log t} = \log \log x - \log \log 2$$

Hence $\sum_{p \leq x} 1/p \sim \log \log x$, recovering Mertens' second theorem (up to the precise constant). The rigorous version of this argument, using Chebyshev's bounds instead of PNT, appears in chapter 3.

This example illustrates a general principle: Abel summation lets the estimation of $\sum f(p)$ over primes be reduced to the estimation of $\pi(x)$, and conversely. The relationship is bidirectional: knowledge of $\pi(x)$ yields information about $\sum 1/p$, and the divergence of $\sum 1/p$ (proved in chapter 4 via the Euler product) gives a lower bound on $\pi(x)$.

Example 1.4.8: Partial Summation with the Totient Function

To estimate $\sum_{n \leq x} \varphi(n)/n^2$, apply Abel summation with $a_n = \varphi(n)$ and $f(t) = 1/t^2$. By theorem 1.3.8, $A(t) = \sum_{n \leq t} \varphi(n) = \frac{3}{\pi^2} t^2 + O(t \log t)$. Then:

$$\begin{aligned} \sum_{n \leq x} \frac{\varphi(n)}{n^2} &= \frac{A(x)}{x^2} + 2 \int_1^x \frac{A(t)}{t^3} dt = \frac{3}{\pi^2} + O\left(\frac{\log x}{x}\right) + 2 \int_1^x \frac{3t^2/\pi^2 + O(t \log t)}{t^3} dt \\ &= \frac{3}{\pi^2} + \frac{6}{\pi^2} \int_1^x \frac{dt}{t} + O\left(\int_1^x \frac{\log t}{t^2} dt\right) + O\left(\frac{\log x}{x}\right) \\ &= \frac{3}{\pi^2} + \frac{6}{\pi^2} \log x + O(1) \end{aligned}$$

The dominant term is $\frac{6}{\pi^2} \log x$, confirming that $\sum \varphi(n)/n^2$ diverges logarithmically.

Abel summation also provides a useful criterion for the convergence of series with oscillating terms.

Corollary 1.4.9: Abel's Test for Dirichlet Series

Let (a_n) be a sequence with bounded partial sums: $|A(N)| \leq M$ for all N . If $f : [1, \infty) \rightarrow \mathbb{R}$ is monotonically decreasing with $f(t) \rightarrow 0$ as $t \rightarrow \infty$, then $\sum_{n=1}^{\infty} a_n f(n)$ converges, and

$$\left| \sum_{n>N} a_n f(n) \right| \leq 2M f(N+1)$$

Proof:

By theorem 1.4.1 applied to the tail sum:

$$\sum_{N < n \leq x} a_n f(n) = A'(x)f(x) - A'(N)f(N+1) - \int_{N+1}^x A'(t)f'(t) dt$$

where $A'(t) = \sum_{N < n \leq t} a_n = A(t) - A(N)$, so $|A'(t)| \leq 2M$. Since f is decreasing, $f' \leq 0$, and:

$$\left| \int_{N+1}^x A'(t)f'(t) dt \right| \leq 2M \int_{N+1}^x |f'(t)| dt = 2M(f(N+1) - f(x))$$

The boundary terms satisfy $|A'(x)f(x)| \leq 2Mf(x) \rightarrow 0$ and $|A'(N)f(N+1)| = 0$. Taking $x \rightarrow \infty$: $|\sum_{n>N} a_n f(n)| \leq 2Mf(N+1)$.

This criterion applies, for example, to $\sum \mu(n)/n^s$ for $\operatorname{Re}(s) > 0$: the partial sums $\sum_{n \leq N} \mu(n) = M(N)$ satisfy $M(N) = o(N)$ (equivalent to PNT, as noted in section 1.3), so the series converges for $\operatorname{Re}(s) > 0$ once PNT is established. Before PNT, convergence is known only for $\operatorname{Re}(s) > 1$ from absolute convergence.

1.5 The Euler-Maclaurin Summation Formula

Abel's summation formula from the previous section converts a sum into an integral plus a boundary term, with an error controlled by the fractional part of the summation variable. For many applications, a single error term is sufficient. But when high-precision asymptotic expansions are needed (expansions of the form $\sum_{n \leq N} f(n) = I(N) + c_1 g_1(N) + c_2 g_2(N) + \dots$ with explicit coefficients) Abel summation isn't enough. The *Euler-Maclaurin formula* remedies this by providing an arbitrarily accurate comparison between $\sum f(n)$ and $\int f(t) dt$, with error terms involving successively higher derivatives of f weighted by *Bernoulli numbers*. These numbers, appearing repeatedly throughout analytic number theory, are introduced first:

Definition 1.5.1: Bernoulli Numbers

The *Bernoulli numbers* $B_k \in \mathbb{Q}$ for $k \geq 0$ are defined by the generating function

$$\frac{t}{e^t - 1} = \sum_{k=0}^{\infty} \frac{B_k}{k!} t^k$$

valid for $|t| < 2\pi$.

Equating coefficients with the Taylor expansion of $t/(e^t - 1)$ gives a recursive formula: multiplying both sides by $(e^t - 1)/t = \sum_{j=0}^{\infty} t^j/(j+1)!$ and identifying coefficients yields $B_0 = 1$ and, for $n \geq 1$,

$$\sum_{k=0}^n \binom{n+1}{k} B_k = 0$$

This recursion computes the Bernoulli numbers one at a time. The first several values are:

$$B_0 = 1, \quad B_1 = -\frac{1}{2}, \quad B_2 = \frac{1}{6}, \quad B_3 = 0, \quad B_4 = -\frac{1}{30}, \quad B_5 = 0, \quad B_6 = \frac{1}{42}, \quad B_8 = -\frac{1}{30}, \quad B_{10} = \frac{5}{66}$$

A fundamental property is the vanishing of odd-indexed Bernoulli numbers beyond B_1 .

Proposition 1.5.2: Vanishing of Odd Bernoulli Numbers

For $k \geq 1$, $B_{2k+1} = 0$.

Proof:

Consider the function $G(t) = t/(e^t - 1) + t/2$. Then

$$G(t) = \sum_{k=0}^{\infty} \frac{B_k}{k!} t^k + \frac{t}{2} = 1 + \sum_{k=2}^{\infty} \frac{B_k}{k!} t^k,$$

using $B_0 = 1$ and $B_1 = -1/2$. Direct computation gives $G(-t) = -t/(e^{-t} - 1) - t/2 = t \cdot e^t/(e^t - 1) - t/2 = t(1 + 1/(e^t - 1)) - t/2 = t/(e^t - 1) + t/2 = G(t)$. Hence G is an even function of t , so all coefficients of odd powers of t in $G(t)$ vanish. This forces $B_{2k+1} = 0$ for $k \geq 1$.

The Bernoulli numbers grow rapidly in absolute value: $|B_{2k}| \sim 2(2k)!/(2\pi)^{2k}$ as $k \rightarrow \infty$. This growth means that the Euler-Maclaurin expansion is typically an *asymptotic* series rather than a convergent one—truncating at a well-chosen index gives an excellent approximation, but including too many terms causes divergence.

Definition 1.5.3: Bernoulli Polynomials

The *Bernoulli polynomials* $B_k(x)$ are defined by

$$B_k(x) = \sum_{j=0}^k \binom{k}{j} B_j x^{k-j},$$

or equivalently by the generating function $\frac{te^{xt}}{e^t-1} = \sum_{k=0}^{\infty} \frac{B_k(x)}{k!} t^k$ for $|t| < 2\pi$.

The Bernoulli polynomials satisfy several identities that make them the right objects for Euler-Maclaurin. The first few are $B_0(x) = 1$, $B_1(x) = x - 1/2$, $B_2(x) = x^2 - x + 1/6$, $B_3(x) = x^3 - 3x^2/2 + x/2$, and $B_4(x) = x^4 - 2x^3 + x^2 - 1/30$. Note $B_k(0) = B_k$ for all k .

Proposition 1.5.4: Properties of Bernoulli Polynomials

For $k \geq 1$:

1. $B'_k(x) = k B_{k-1}(x)$.
2. $\int_0^1 B_k(x) dx = 0$.
3. $B_k(x+1) - B_k(x) = kx^{k-1}$.
4. $B_k(1-x) = (-1)^k B_k(x)$.

Proof:

For (1): differentiate the generating function with respect to x :

$$\frac{t^2 e^{xt}}{e^t - 1} = \sum_{k=0}^{\infty} \frac{B'_k(x)}{k!} t^k$$

The left side equals $t \cdot \frac{te^{xt}}{e^t-1} = \sum_{k=0}^{\infty} \frac{B_k(x)}{k!} t^{k+1} = \sum_{k=1}^{\infty} \frac{k B_{k-1}(x)}{k!} t^k$. Matching coefficients gives $B'_k(x) = k B_{k-1}(x)$ for $k \geq 1$.

For (2): integrate the generating function from $x = 0$ to $x = 1$:

$$\int_0^1 \frac{te^{xt}}{e^t-1} dx = \frac{e^t-1}{e^t-1} = 1$$

Equating this with $\sum_{k=0}^{\infty} \frac{1}{k!} \int_0^1 B_k(x) dx \cdot t^k$ forces the $k = 0$ coefficient to be 1 and all higher coefficients to vanish: $\int_0^1 B_k(x) dx = 0$ for $k \geq 1$.

For (3): the generating function satisfies $\frac{te^{(x+1)t}}{e^t-1} - \frac{te^{xt}}{e^t-1} = te^{xt}$. Matching coefficients of t^k : $\frac{B_k(x+1) - B_k(x)}{k!} = \frac{x^{k-1}}{(k-1)!}$ for $k \geq 1$, giving $B_k(x+1) - B_k(x) = kx^{k-1}$.

For (4): the generating function satisfies $\frac{te^{(1-x)t}}{e^t-1} = \frac{te^{-xt}e^t}{e^t-1}$. Using the identity $\frac{e^t}{e^t-1} = \frac{-1}{e^{-t}-1} \cdot (-1)$,

this equals $\frac{-te^{-xt}}{e^{-t}-1}$, which is the generating function of $B_k(x)$ evaluated at $-t$. Matching coefficients of t^k on both sides: $B_k(1-x)/k! = (-1)^k B_k(x)/k!$, giving $B_k(1-x) = (-1)^k B_k(x)$.

Property (3) is the key identity for Euler-Maclaurin: it expresses the power kx^{k-1} as the discrete difference $B_k(x+1) - B_k(x)$, which allows telescoping sums to be converted into integrals of Bernoulli polynomials. Property (4) gives the reflection symmetry: combined with (1), it shows that $B_{2k+1}(1/2) = 0$ for $k \geq 0$, generalizing the vanishing of odd Bernoulli numbers.

Definition 1.5.5: Periodic Bernoulli Functions

For $k \geq 0$, the *periodic Bernoulli function* $\tilde{B}_k : \mathbb{R} \rightarrow \mathbb{R}$ is

$$\tilde{B}_k(x) = B_k(\{x\})$$

where $\{x\} = x - \lfloor x \rfloor$ is the fractional part of x . The function $\tilde{B}_k$ is periodic with period 1.

For $k = 0$, $\tilde{B}_0(x) = 1$ is constant. For $k = 1$, $\tilde{B}_1(x) = \{x\} - 1/2$, which is the sawtooth function (discontinuous at integers). For $k \geq 2$, $\tilde{B}_k$ is continuous everywhere because $B_k(0) = B_k(1) = B_k$ for $k \geq 2$ (from property (3): $B_k(1) - B_k(0) = k \cdot 0^{k-1} = 0$).

With these tools in place, the Euler-Maclaurin formula follows from repeated integration by parts.

Theorem 1.5.6: Euler-Maclaurin Summation Formula

Let $f \in C^{2p}[a, b]$ with $a, b \in \mathbb{Z}$ and $p \geq 1$. Then

$$\sum_{n=a}^b f(n) = \int_a^b f(t) dt + \frac{f(a) + f(b)}{2} + \sum_{k=1}^p \frac{B_{2k}}{(2k)!} (f^{(2k-1)}(b) - f^{(2k-1)}(a)) + R_p$$

where the remainder term is

$$R_p = - \int_a^b \frac{\tilde{B}_{2p}(t)}{(2p)!} f^{(2p)}(t) dt$$

Proof:

Step 1: The base case. On each unit interval $[n, n+1]$, use $\tilde{B}_1(t) = t - n - 1/2$ for $t \in [n, n+1]$ and note that $\tilde{B}'_1(t) = 1$ on this interval. Then:

$$\int_n^{n+1} f(t) dt = \int_n^{n+1} f(t) \cdot 1 dt = \int_n^{n+1} f(t) d\tilde{B}_1(t)$$

Integration by parts gives

$$\int_n^{n+1} f(t) d\tilde{B}_1(t) = [f(t)\tilde{B}_1(t)]_n^{n+1} - \int_n^{n+1} \tilde{B}_1(t) f'(t) dt$$

Since $\tilde{B}_1(n^+) = -1/2$ and $\tilde{B}_1((n+1)^-) = 1/2$:

$$\int_n^{n+1} f(t) dt = \frac{f(n) + f(n+1)}{2} - \int_n^{n+1} \tilde{B}_1(t) f'(t) dt$$

Summing over $n = a, a + 1, \dots, b - 1$:

$$\int_a^b f(t) dt = \sum_{n=a}^{b-1} \frac{f(n) + f(n+1)}{2} - \int_a^b \tilde{B}_1(t) f'(t) dt = \sum_{n=a}^b f(n) - \frac{f(a) + f(b)}{2} - \int_a^b \tilde{B}_1(t) f'(t) dt$$

Rearranging:

$$\sum_{n=a}^b f(n) = \int_a^b f(t) dt + \frac{f(a) + f(b)}{2} + \int_a^b \tilde{B}_1(t) f'(t) dt$$

This is the formula for $p = 0$ (no Bernoulli correction terms), with remainder $\int \tilde{B}_1 f'$.

Step 2: Iteration. For the remainder integral, apply integration by parts using $\tilde{B}'_{k+1}(t) = (k+1)\tilde{B}_k(t)$ for $t \notin \mathbb{Z}$ (from proposition 1.5.4(1)). Then

$$\int_a^b \tilde{B}_k(t) f^{(k)}(t) dt = \left[\frac{\tilde{B}_{k+1}(t)}{k+1} f^{(k)}(t) \right]_a^b - \int_a^b \frac{\tilde{B}_{k+1}(t)}{k+1} f^{(k+1)}(t) dt$$

At the endpoints $a, b \in \mathbb{Z}$: $\tilde{B}_{k+1}(a) = \tilde{B}_{k+1}(b) = B_{k+1}(0) = B_{k+1}$. Hence

$$\int_a^b \tilde{B}_k(t) f^{(k)}(t) dt = \frac{B_{k+1}}{k+1} (f^{(k)}(b) - f^{(k)}(a)) - \int_a^b \frac{\tilde{B}_{k+1}(t)}{k+1} f^{(k+1)}(t) dt$$

Dividing by $k!$: $\int_a^b \frac{\tilde{B}_k(t)}{k!} f^{(k)}(t) dt = \frac{B_{k+1}}{(k+1)!} (f^{(k)}(b) - f^{(k)}(a)) - \int_a^b \frac{\tilde{B}_{k+1}(t)}{(k+1)!} f^{(k+1)}(t) dt$.

Step 3: Apply iteratively. Starting from $\int \tilde{B}_1 f'$ and applying the above recursion $2p - 1$ times, the odd-indexed Bernoulli numbers $B_3, B_5, \dots$ vanish, contributing nothing to the boundary terms. Only the terms with even index $2k$ survive, giving the stated formula with remainder $R_p = - \int_a^b \frac{\tilde{B}_{2p}(t)}{(2p)!} f^{(2p)}(t) dt$.

The power of the Euler-Maclaurin formula is that it gives an exact identity for any p , not just an asymptotic approximation. The practical question is: how should p be chosen? For smooth decaying f , the remainder R_p is typically small for moderate p but grows rapidly as $p \rightarrow \infty$ (reflecting the rapid growth of $|B_{2p}|$). The optimal p depends on the specific f and is generally the value that minimizes $|R_p|$.

The first major application is to Stirling's formula.

Theorem 1.5.7: Stirling's Formula

For $x > 0$ as $x \rightarrow \infty$:

$$\log \Gamma(x+1) = x \log x - x + \frac{1}{2} \log(2\pi x) + \sum_{k=1}^{p-1} \frac{B_{2k}}{2k(2k-1)} \cdot \frac{1}{x^{2k-1}} + O_p(x^{-2p+1})$$

In particular, the leading-order form is $\log(n!) = n \log n - n + \frac{1}{2} \log(2\pi n) + O(1/n)$, or equivalently $n! \sim \sqrt{2\pi n}(n/e)^n$.

Proof:

Apply theorem 1.5.6 to $f(t) = \log t$ on $[1, N]$ with N a positive integer. The derivatives are $f'(t) = 1/t$, $f''(t) = -1/t^2$, and generally $f^{(k)}(t) = (-1)^{k-1}(k-1)!/t^k$. The integral is $\int_1^N \log t \, dt = N \log N - N + 1$. The boundary terms and Bernoulli corrections give:

$$\sum_{n=1}^N \log n = N \log N - N + 1 + \frac{\log N}{2} + \sum_{k=1}^p \frac{B_{2k}}{(2k)!} (f^{(2k-1)}(N) - f^{(2k-1)}(1)) + R_p$$

With $f^{(2k-1)}(t) = (2k-2)!/t^{2k-1}$:

$$\frac{B_{2k}}{(2k)!} \cdot (2k-2)! \left(\frac{1}{N^{2k-1}} - 1 \right) = \frac{B_{2k}}{2k(2k-1)} \left(\frac{1}{N^{2k-1}} - 1 \right)$$

The sum $\log(N!) = \sum_{n=1}^N \log n$ on the left, and letting $N \rightarrow \infty$ with the correction terms at infinity vanishing, identifies the constant as $\frac{1}{2} \log(2\pi)$. The details of the constant determination use Wallis's product or the duplication formula for Γ ; the result is as stated.

Example 1.5.8: Numerical Stirling

For $n = 10$: the exact value is $10! = 3,628,800$. The leading Stirling approximation $\sqrt{20\pi}(10/e)^{10} \approx 3,598,696$ gives relative error about 0.83%. Including the $B_2/(2x) = 1/(12x)$ correction gives $\sqrt{20\pi}(10/e)^{10} \exp(1/120) \approx 3,628,810$, matching the exact value to 5 significant figures.

A second application gives an asymptotic expansion for the partial sums of n^{-s} refined beyond proposition 1.4.6.

Proposition 1.5.9: Asymptotic Expansion of $\sum n^{-s}$

For $s \in \mathbb{C}$ with $\operatorname{Re}(s) > 0$ and $s \neq 1$, and any integer $p \geq 1$:

$$\sum_{n=1}^N n^{-s} = \frac{N^{1-s}}{1-s} + \zeta(s) + \frac{N^{-s}}{2} + \sum_{k=1}^{p-1} \frac{B_{2k}}{(2k)!} s(s+1) \cdots (s+2k-2) \cdot N^{-s-2k+1} + O_{p,s}(N^{-\operatorname{Re}(s)-2p+1})$$

Proof:

Apply theorem 1.5.6 to $f(t) = t^{-s}$ on $[1, N]$. The integral is $\int_1^N t^{-s} dt = \frac{N^{1-s} - 1}{1-s}$. Derivatives: $f^{(k)}(t) = (-1)^k s(s+1) \cdots (s+k-1) t^{-s-k}$. The boundary terms $(f(1) + f(N))/2 = (1 + N^{-s})/2$ and the Bernoulli corrections combine to give the stated expansion. The constant $\zeta(s)$ emerges from the value at $N = 1$ when $\operatorname{Re}(s) > 1$, and the expansion extends the right-hand side to $\operatorname{Re}(s) > 1 - 2p$ as an analytic function of s .

1.5.10 Foreshadowing: Bernoulli Numbers and Special Values of $\zeta(s)$ The proposition above suggests the following identity. Setting $s = -n$ (a negative integer) formally in the expansion and taking $N = 1$, the boundary and Bernoulli terms combine to yield

$$\zeta(-n) = -\frac{B_{n+1}}{n+1} \quad \text{for } n \geq 1$$

This is not rigorous here: the meromorphic continuation of $\zeta(s)$ to all of $\mathbb{C}$ has not yet been established, and the formal manipulation of the expansion at $s = -n$ requires justification. Both issues are resolved in ??, where the functional equation of $\zeta(s)$ provides the continuation and confirms the identity. In particular: $\zeta(0) = -1/2$, $\zeta(-1) = -1/12$, $\zeta(-3) = 1/120$, and $\zeta(-2k) = 0$ for $k \geq 1$ (the trivial zeros). The famous “identity” $1 + 2 + 3 + \cdots = -1/12$ is a shorthand for $\zeta(-1) = -1/12$.

Example 1.5.11: High-Precision Computation of the Harmonic Sum

To compute $H_{100} = \sum_{n=1}^{100} 1/n$ to high precision, apply theorem 1.5.6 with $f(t) = 1/t$ on $[1, 100]$ and $p = 3$. With $f^{(2k-1)}(t) = -(2k-1)!/t^{2k}$, the Bernoulli term for index $2k$ contributes

$$\frac{B_{2k}}{(2k)!} (f^{(2k-1)}(100) - f^{(2k-1)}(1)) = \frac{B_{2k}}{2k} \left(1 - \frac{1}{100^{2k}} \right)$$

The formula gives:

$$H_{100} = \log 100 + \frac{1 + 1/100}{2} + \frac{1/6}{2} \left(1 - \frac{1}{100^2} \right) + \frac{-1/30}{4} \left(1 - \frac{1}{100^4} \right) + \frac{1/42}{6} \left(1 - \frac{1}{100^6} \right) + R_3$$

Numerically: $\log 100 \approx 4.60517$, boundary = 0.505, $k = 1$ term ≈ 0.08333 , $k = 2$ term ≈ -0.00833 , $k = 3$ term ≈ 0.00397 . The sum is approximately 5.18914, matching the exact value $H_{100} \approx 5.18738$ to within 0.002. The discrepancy is the remainder R_3 , which is of order $|B_6|/(6 \cdot 100^0) \approx 0.004$ (the worst contribution being at the lower endpoint $t = 1$, where the derivatives of $1/t$ are largest).

Exercise 1.5.1

1. For each of the following arithmetic functions, determine whether it is multiplicative, totally multiplicative, additive, totally additive, or none of these. Justify each answer.
 1. $f(n) = n^2$
 2. $f(n) = \log n$
 3. $f(n) = \sigma(n)/n$

$$4. f(n) = \sigma_2(n) = \sum_{d|n} d^2$$

2.
 1. Compute $\Lambda(n)$ for $n = 1, 2, \dots, 12$.
 2. Verify the von Mangoldt identity $\sum_{d|n} \Lambda(d) = \log n$ for $n = 12$ and $n = 18$.
 3. Show that $\Lambda(n) \leq \log n$ for all $n \geq 1$, with equality if and only if n is prime.
3. Prove that σ_k is multiplicative for every $k \in \mathbb{C}$, and derive the formula

$$\sigma_k(p^a) = \frac{p^{(a+1)k} - 1}{p^k - 1}$$

for $k \neq 0$ and any prime p . State the corresponding formula for $k = 0$.

4. Prove that the Liouville function λ is totally multiplicative, and show that $\lambda(n)\mu(n) = \mu(n)^2$ for all $n \geq 1$. Deduce that λ and μ agree on squarefree integers.
5. Compute the Dirichlet convolution $(f * g)(12)$ for:
 1. $f = g = \mathbf{1}$.
 2. $f = \mu, g = \text{id}$.
 3. $f = \lambda, g = \mathbf{1}$.

Compare each result with the value predicted by the convolution identities in proposition 1.2.10.

6. Using the recursive formula for Dirichlet inverses, compute $\sigma^{-1}(n)$ for $n = 1, 2, 3, 4, 6$.
7. Prove the *multiplicative* version of Möbius inversion: if $F(n) = \prod_{d|n} f(d)$ for all $n \geq 1$, then $f(n) = \prod_{d|n} F(d)^{\mu(n/d)}$.
8. Prove the identity $d * \varphi = \sigma$. [Hint: use $d = \mathbf{1} * \mathbf{1}$ and $\varphi = \text{id} * \mu$.]
9. Let f be a multiplicative function. Show that f^{-1} (the Dirichlet inverse of f) is also multiplicative, and prove that for any prime p ,

$$f^{-1}(p^a) = \sum_{k=1}^a (-1)^k \sum_{\substack{a_1 + \dots + a_k = a \\ a_i \geq 1}} f(p^{a_1}) \cdots f(p^{a_k})$$

10. Using the hyperbola method, provide an independent derivation of the estimate $\sum_{n \leq x} d(n) = x \log x + O(x)$. Then verify numerically for $x = 20$: compute the exact sum $\sum_{n=1}^{20} d(n)$ and compare with $20 \log 20 + (2\gamma - 1) \cdot 20$.
11. Let $Q(x) = \sum_{n \leq x} |\mu(n)|$ count the squarefree integers up to x . Using the hyperbola method applied to the identity $|\mu(n)| = \sum_{d^2|n} \mu(d)$, give a self-contained proof that $Q(x) = 6x/\pi^2 + O(\sqrt{x})$.
12. Prove that

$$\sum_{\substack{n \leq x \\ n \text{ squarefree}}} \frac{1}{n} = \frac{6}{\pi^2} \log x + C + O(x^{-1/2})$$

for a constant C . [Hint: use Abel summation with $a_n = |\mu(n)|$ and $f(t) = 1/t$, then apply exercise 1.5.1 (11).]

13. Prove the trivial bound $|M(x)| = |\sum_{n \leq x} \mu(n)| \leq x$ for all $x \geq 1$. Then, using the identity $\sum_{d \leq x} M(x/d) = 1$, derive the recursion-based estimate $M(x) = 1 - \sum_{2 \leq d \leq x} M(x/d)$ and explain why this does not immediately yield $M(x) = o(x)$.
14. Using Abel summation and the estimate $\sum_{n \leq x} \sigma(n) = \frac{\pi^2}{12}x^2 + O(x \log x)$, prove that

$$\sum_{n \leq x} \frac{\sigma(n)}{n} = \frac{\pi^2}{6}x + O((\log x)^2)$$

15. Using Abel summation and $\sum_{n \leq x} \varphi(n) = \frac{3}{\pi^2}x^2 + O(x \log x)$, show that

$$\sum_{n \leq x} \frac{\varphi(n)}{n^2} = \frac{6}{\pi^2} \log x + C + O\left(\frac{\log x}{x}\right)$$

for a constant C . (An explicit formula for C is not required.)

16. Let a_n satisfy $|A(N)| = |\sum_{n \leq N} a_n| \leq M$ for all N . Use Abel summation to prove that $\sum_{n=1}^{\infty} a_n/n^s$ converges for every $s \in \mathbb{C}$ with $\operatorname{Re}(s) > 0$, and the convergence is uniform on compact subsets of $\{\operatorname{Re}(s) > 0\}$.
17. Using Abel summation with $a_n = 1$ and $f(t) = (\log t)/t$, prove that

$$\sum_{n \leq x} \frac{\log n}{n} = \frac{(\log x)^2}{2} + C_1 + O\left(\frac{\log x}{x}\right)$$

for a constant C_1 . (An explicit value for C_1 is not required.)

18. Using the recursion $\sum_{k=0}^n \binom{n+1}{k} B_k = 0$ for $n \geq 1$, compute B_6 , B_8 , and B_{10} directly from the values $B_0 = 1$, $B_1 = -1/2$, $B_2 = 1/6$, $B_4 = -1/30$, and the vanishing of odd-indexed B_k for $k \geq 3$.
19.
 1. Compute the Bernoulli polynomials $B_2(x)$, $B_3(x)$, and $B_4(x)$ from the formula $B_k(x) = \sum_{j=0}^k \binom{k}{j} B_j x^{k-j}$.
 2. Verify $B_k(1) = B_k(0)$ for $k \in \{2, 3, 4\}$.
 3. Use the telescoping identity $B_3(x+1) - B_3(x) = 3x^2$ to derive the closed form $\sum_{j=0}^{N-1} j^2 = N(N-1)(2N-1)/6$.
20. Apply the Euler-Maclaurin formula to $f(t) = 1/t^2$ on $[1, N]$ with $p = 2$ (two Bernoulli correction terms). Extract the asymptotic expansion for $\sum_{n=1}^N 1/n^2$ and identify the N -independent constant as $\zeta(2) = \pi^2/6$.
21. The functional equation of $\zeta(s)$ implies $\zeta(2k) = (-1)^{k+1} \frac{(2\pi)^{2k} B_{2k}}{2(2k)!}$ for $k \geq 1$ (this will be proved in ??). Verify this identity for $k = 1$ (giving $\zeta(2) = \pi^2/6$) and $k = 2$ (giving $\zeta(4) = \pi^4/90$). What does the formula give for $k = 3$?

Dirichlet Series and Euler Products

Chapter 1 organized the arithmetic functions into a ring under Dirichlet convolution and extracted asymptotic information about their summatory functions through summation techniques. The present chapter introduces the analytic objects that encode the same information in a fundamentally different way: *Dirichlet series*. The Dirichlet series $D(f, s) = \sum_{n=1}^{\infty} f(n)/n^s$ associates to each arithmetic function f an analytic function of a complex variable s , and this association converts Dirichlet convolution into pointwise multiplication, multiplicativity into an Euler product over primes, and the summatory function into a contour integral. The poles of the Dirichlet series control the main terms in asymptotic estimates; its zeros govern the oscillatory corrections; and its analytic continuation beyond the region of absolute convergence is the mechanism by which the distribution of primes becomes accessible to the methods of complex analysis.

This chapter builds the dictionary between the algebraic side (arithmetic functions, convolution, multiplicativity) and the analytic side (Dirichlet series, products, Mellin transforms). Sections 2.1-2.3 establish the algebraic-analytic correspondence: convergence, the multiplication theorem, and Euler products. Section 2.4 develops the Mellin transform as a self-contained tool, and Section 2.5 proves Perron's formula, which inverts the Dirichlet series back into information about partial sums. Together, these tools are the analytic engine of the book.

2.1 Dirichlet Series: Convergence and Analyticity

An arithmetic function f is a sequence of complex numbers; its Dirichlet series is the analytic function obtained by summing $f(n)/n^s$ over all positive integers. The first question is: for which values of the complex variable s does the series converge? The answer has a clean geometric structure: there is a

critical vertical line in the complex plane, to the right of which the series converges and to the left of which it diverges. This *abscissa of convergence* is the Dirichlet series analogue of the radius of convergence for a power series, and it is determined by the growth rate of the partial sums of the coefficients.

Definition 2.1.1: Dirichlet Series

A *Dirichlet series* is a series of the form

$$D(f, s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s}$$

where $f : \mathbb{Z}_{>0} \rightarrow \mathbb{C}$ is an arithmetic function and $s \in \mathbb{C}$.

The variable s is conventionally written as $s = \sigma + it$ with $\sigma = \operatorname{Re}(s)$ and $t = \operatorname{Im}(s)$. Since $n^s = n^\sigma \cdot n^{it} = n^\sigma e^{it \log n}$, the factor $n^{-\sigma}$ controls convergence while n^{-it} contributes oscillation without affecting absolute convergence. This separation between the real and imaginary parts of s underlies the half-plane structure of the convergence region.

The fundamental convergence theorem is the following.

Theorem 2.1.2: Half-Plane of Convergence

Let $D(f, s) = \sum f(n)/n^s$ be a Dirichlet series. If $D(f, s_0)$ converges for some $s_0 \in \mathbb{C}$, then $D(f, s)$ converges for all s with $\operatorname{Re}(s) > \operatorname{Re}(s_0)$, and the convergence is uniform on compact subsets of this half-plane.

Proof:

Write $\sigma_0 = \operatorname{Re}(s_0)$ and fix s with $\sigma = \operatorname{Re}(s) > \sigma_0$. Set $a_n = f(n)/n^{s_0}$ and $g(t) = t^{s_0-s}$, so that $f(n)/n^s = a_n \cdot n^{s_0-s}$. By hypothesis, $A(N) = \sum_{n \leq N} a_n$ is bounded (as the partial sums of a convergent series). The function $g(t) = t^{s_0-s}$ has $|g(t)| = t^{\sigma_0-\sigma} \rightarrow 0$ as $t \rightarrow \infty$ (since $\sigma_0 - \sigma < 0$), and $g'(t) = (s_0 - s)t^{s_0-s-1}$.

By Abel summation (theorem 1.4.1):

$$\sum_{n=1}^N \frac{f(n)}{n^s} = \sum_{n=1}^N a_n \cdot n^{s_0-s} = A(N) \cdot N^{s_0-s} - \int_1^N A(t)(s_0 - s)t^{s_0-s-1} dt$$

As $N \rightarrow \infty$: $|A(N) \cdot N^{s_0-s}| \leq M \cdot N^{\sigma_0-\sigma} \rightarrow 0$, where $M = \sup_N |A(N)|$. The integral converges absolutely:

$$\int_1^\infty |A(t)| |s_0 - s| t^{\sigma_0-\sigma-1} dt \leq M |s_0 - s| \int_1^\infty t^{\sigma_0-\sigma-1} dt = \frac{M |s_0 - s|}{\sigma - \sigma_0} < \infty$$

Hence $D(f, s)$ converges.

For uniform convergence on a compact set $K \subset \{\operatorname{Re}(s) > \sigma_0\}$: let $\sigma_1 = \min_{s \in K} \operatorname{Re}(s) > \sigma_0$. The same bound gives $|\sum_{n > N} f(n)/n^s| \leq M |s_0 - s| N^{\sigma_0-\sigma_1} / (\sigma_1 - \sigma_0) + M N^{\sigma_0-\sigma_1}$, and $|s_0 - s|$ is bounded on K , so the tail tends to 0 uniformly.

This theorem shows that the region of convergence of a Dirichlet series is always a right half-plane (or the entire complex plane, or the empty set). The boundary of this half-plane defines the abscissa of convergence.

Definition 2.1.3: Abscissae of Convergence and Absolute Convergence

For a Dirichlet series $D(f, s)$:

1. The *abscissa of convergence* σ_c is the infimum of the set of $\sigma \in \mathbb{R}$ such that $D(f, \sigma + it)$ converges for all t . Equivalently, $D(f, s)$ converges for $\operatorname{Re}(s) > \sigma_c$ and diverges for $\operatorname{Re}(s) < \sigma_c$.
2. The *abscissa of absolute convergence* σ_a is the infimum of the set of σ such that $\sum |f(n)|/n^\sigma < \infty$.

By convention, $\sigma_c = -\infty$ if the series converges everywhere, $\sigma_c = +\infty$ if it converges nowhere, and similarly for σ_a .

By definition, $\sigma_c \leq \sigma_a$ (absolute convergence implies convergence). The gap between σ_c and σ_a reflects the amount of cancellation in the coefficients.

Proposition 2.1.4: Gap Between Abscissae

For any Dirichlet series: $\sigma_c \leq \sigma_a \leq \sigma_c + 1$. Both inequalities can be equalities.

Proof:

The inequality $\sigma_c \leq \sigma_a$ is immediate. For the upper bound: if $D(f, s_0)$ converges at s_0 with $\operatorname{Re}(s_0) = \sigma_0$, then $f(n)/n^{s_0} \rightarrow 0$, so $|f(n)| \leq Cn^{\sigma_0}$ for some constant C . Hence $\sum |f(n)|/n^\sigma \leq C \sum n^{\sigma_0 - \sigma}$, which converges for $\sigma > \sigma_0 + 1$. Taking the infimum over convergent s_0 gives $\sigma_a \leq \sigma_c + 1$.

Equality $\sigma_c = \sigma_a$ holds for $\zeta(s) = \sum 1/n^s$ (both equal 1, since the coefficients are non-negative). Equality $\sigma_a = \sigma_c + 1$ holds for the alternating series $\sum (-1)^{n+1}/n^s = (1 - 2^{1-s})\zeta(s)$: here $\sigma_c = 0$ (by Abel's test, since the partial sums of $(-1)^{n+1}$ are bounded), while $\sigma_a = 1$.

The gap between σ_c and σ_a measures the cancellation among the coefficients $f(n)$. For non-negative coefficients, there is no cancellation, and $\sigma_c = \sigma_a$. For coefficients with heavy sign alternation (like $(-1)^{n+1}$), the gap can reach its maximum value of 1. The intermediate case $0 < \sigma_a - \sigma_c < 1$ occurs for the Dirichlet series $\sum \mu(n)/n^s$: the cancellation in the Möbius function is sufficient to push σ_c below $\sigma_a = 1$, but determining σ_c precisely is equivalent to determining the supremum of the real parts of the zeros of $\zeta(s)$.

More generally, whenever the coefficients $f(n)$ are non-negative, the abscissa of convergence equals the abscissa of absolute convergence, and the Dirichlet series $D(f, s)$ has a singularity (non-removable) at the point $s = \sigma_a$ on the real axis. This is the Landau-Vivanti theorem for Dirichlet series.

Proposition 2.1.5: Landau's Theorem on Non-Negative Coefficients

If $f(n) \geq 0$ for all n , then $\sigma_c = \sigma_a$, and $D(f, s)$ has a singularity at $s = \sigma_a$.

Proof:

Since $f(n) \geq 0$: $|f(n)|/n^\sigma = f(n)/n^\sigma$, so $\sigma_c = \sigma_a$. For the singularity: suppose for contradiction that $D(f, s)$ extends analytically to a neighborhood of σ_a . Then the Taylor expansion of $D(f, s)$ at some point $s_1 > \sigma_a$ would converge in a disk centered at s_1 reaching past σ_a , giving

$$D(f, s) = \sum_{k=0}^{\infty} \frac{D^{(k)}(f, s_1)}{k!} (s - s_1)^k = \sum_{k=0}^{\infty} \frac{(s_1 - s)^k}{k!} \sum_{n=1}^{\infty} \frac{f(n)(\log n)^k}{n^{s_1}}$$

for $|s - s_1| < s_1 - \sigma_a + \varepsilon$. Since $f(n) \geq 0$, the double sum has non-negative terms, and Tonelli's theorem allows swapping the summations. At a real point $s_2 < \sigma_a$ inside the disk:

$$D(f, s_2) = \sum_{n=1}^{\infty} \frac{f(n)}{n^{s_1}} \sum_{k=0}^{\infty} \frac{(s_1 - s_2)^k (\log n)^k}{k!} = \sum_{n=1}^{\infty} \frac{f(n)}{n^{s_1}} n^{s_1 - s_2} = \sum_{n=1}^{\infty} \frac{f(n)}{n^{s_2}},$$

so the Dirichlet series converges at $s_2 < \sigma_a$, contradicting the definition of σ_a .

For $\zeta(s) = \sum 1/n^s$, this theorem guarantees a singularity at $s = 1$, consistent with the pole of $\zeta(s)$ at $s = 1$ established by the integral representation of section 1.4. For $\sum d(n)/n^s = \zeta(s)^2$, the theorem gives a singularity at $s = 1$ (a double pole). The theorem does not determine the *nature* of the singularity (pole, branch point, essential singularity), only its existence.

The abscissae can be computed from the coefficients via the following formulas.

Proposition 2.1.6: Formulas for the Abscissae

Let $A(x) = \sum_{n \leq x} f(n)$ and $A^*(x) = \sum_{n \leq x} |f(n)|$. Then:

$$\sigma_c = \limsup_{x \rightarrow \infty} \frac{\log |A(x)|}{\log x}, \quad \sigma_a = \limsup_{x \rightarrow \infty} \frac{\log A^*(x)}{\log x}$$

(With the convention that $\sigma_c = -\infty$ if $A(x)$ is eventually zero, etc.)

Proof:

For σ_a : if $\sigma > \limsup_{x \rightarrow \infty} \frac{\log A^*(x)}{\log x}$, then $A^*(x) \leq x^{\sigma - \varepsilon}$ for large x and some $\varepsilon > 0$. Abel summation applied to $\sum |f(n)|/n^\sigma$ with partial sums $A^*(x)$ gives convergence, since the integral $\int_1^\infty A^*(t)/t^{\sigma+1} dt \leq \int_1^\infty t^{-1-\varepsilon} dt < \infty$. Conversely, if $\sigma < \limsup \log A^*(x)/\log x$, then $A^*(x) \geq x^{\sigma + \varepsilon}$ for infinitely many x , and the series diverges. The proof for σ_c is analogous, replacing $A^*(x)$ with $A(x)$ and using Abel summation for conditional convergence.

The behavior at the boundary $\operatorname{Re}(s) = \sigma_c$ is delicate. The series may converge at every point of this line, at no point, or at some points but not others. For $\zeta(s)$, the boundary is $\operatorname{Re}(s) = 1$, and the series diverges at $s = 1$ but converges conditionally at every other point $1 + it$ with $t \neq 0$ (by Dirichlet's test, since the partial sums of n^{-it} are bounded for $t \neq 0$ and $1/n$ decreases to 0). For

the eta function $\eta(s)$, the boundary is $\operatorname{Re}(s) = 0$, and the series diverges at $s = 0$ by oscillation but converges at $s = it$ for $t \neq 0$.

A final remark on the strip $\sigma_c < \operatorname{Re}(s) < \sigma_a$ (when it is nonempty). In this region, the Dirichlet series converges but not absolutely. The convergence is conditional, and the series does not converge uniformly on all of $\{\operatorname{Re}(s) > \sigma_c\}$ (only on compact subsets). This strip is analogous to the annulus of conditional convergence for Laurent series. The arithmetic content of this strip is often significant: for $\sum \mu(n)/n^s$, the identity $1/\zeta(s) = \sum \mu(n)/n^s$ extends into the strip $\sigma_c < \operatorname{Re}(s) \leq 1$, and the non-vanishing of $\zeta(s)$ on the line $\operatorname{Re}(s) = 1$ is what makes the extension to $\sigma_c < 1$ possible.

A Dirichlet series is not just convergent in its half-plane—it is analytic there, and can be differentiated term by term.

Theorem 2.1.7: Analyticity and Term-by-Term Differentiation

A Dirichlet series $D(f, s) = \sum f(n)/n^s$ is an analytic function on $\{\operatorname{Re}(s) > \sigma_c\}$, and for all $k \geq 1$:

$$D^{(k)}(f, s) = \sum_{n=1}^{\infty} \frac{f(n)(-\log n)^k}{n^s}$$

in the same half-plane. This series also has abscissa of convergence σ_c .

Proof:

Fix a compact set $K \subset \{\operatorname{Re}(s) > \sigma_c\}$ with $\sigma_1 = \min_{s \in K} \operatorname{Re}(s) > \sigma_c$. Choose σ_0 with $\sigma_c < \sigma_0 < \sigma_1$. The partial sums $A(N) = \sum_{n \leq N} f(n)/n^{\sigma_0}$ are bounded for $\operatorname{Re}(s_0) = \sigma_0$, and by the proof of theorem 2.1.2, $D(f, s)$ converges uniformly on K . Since each term $f(n)/n^s$ is an entire function of s , and the series converges uniformly on compact subsets, the sum is analytic on $\{\operatorname{Re}(s) > \sigma_c\}$ (by Weierstrass's theorem on uniformly convergent series of analytic functions, [13]).

For differentiation: $\frac{d}{ds}(f(n)/n^s) = -f(n) \log n / n^s$. The series $\sum f(n) \log n / n^s$ has the same abscissa of convergence as $\sum f(n)/n^s$: the factor $\log n$ grows slower than any power n^ε , so for $\operatorname{Re}(s) > \sigma_c + \varepsilon$, the additional factor $\log n$ does not affect absolute convergence, and letting $\varepsilon \rightarrow 0$ shows the abscissa is unchanged. Uniform convergence on compacta allows exchanging the derivative and the sum.

The analyticity theorem has an important consequence: a Dirichlet series is determined by its coefficients.

Corollary 2.1.8: Uniqueness of Coefficients

If $D(f, s) = D(g, s)$ for all s in some half-plane $\{\operatorname{Re}(s) > \sigma_0\}$, then $f(n) = g(n)$ for all n .

Proof:

Set $h = f - g$, so $D(h, s) = 0$ for $\operatorname{Re}(s) > \sigma_0$. Restricting to real $s = \sigma > \sigma_0$: $D(h, \sigma) = \sum_{n=1}^{\infty} h(n)/n^\sigma = 0$. As $\sigma \rightarrow +\infty$, the terms $h(n)/n^\sigma$ decay at different rates (faster for larger n), and the dominant term is $h(1)$. Formally: dividing by $1^{-\sigma} = 1$ and taking $\sigma \rightarrow \infty$, the terms with $n \geq 2$ vanish, giving $h(1) = 0$. Now $0 = \sum_{n \geq 2} h(n)/n^\sigma$; multiplying by 2^σ and taking

$\sigma \rightarrow \infty$ shows $h(2) = 0$. Inductively, $h(n) = 0$ for all n .

This uniqueness theorem is the Dirichlet series analogue of the fact that a power series is determined by its coefficients. The proof exploits a feature specific to Dirichlet series: the “basis functions” n^{-s} decay at *different rates* as $\sigma \rightarrow \infty$, allowing the coefficients to be peeled off one at a time. For power series $\sum a_n z^n$, the analogous statement requires evaluating derivatives at $z = 0$; for Dirichlet series, the multiplicative structure of the basis $\{n^{-s}\}$ gives a simpler argument.

Example 2.1.9: Abscissae of Standard Dirichlet Series

1. $\zeta(s) = \sum 1/n^s$: since the coefficients are non-negative, proposition 2.1.5 gives $\sigma_c = \sigma_a$. The harmonic series $\sum 1/n$ diverges, so $\sigma_a \geq 1$. For $\sigma > 1$: $\sum 1/n^\sigma \leq 1 + \int_1^\infty t^{-\sigma} dt = \sigma/(\sigma - 1) < \infty$. Hence $\sigma_c = \sigma_a = 1$.
2. $\sum d(n)/n^s$: the divisor function satisfies $d(n) \geq 1$ for all n and $d(n) \leq C_\varepsilon n^\varepsilon$ for any $\varepsilon > 0$ (a proof appears in chapter 4), so $\sigma_a = 1$. The identity $\sum d(n)/n^s = \zeta(s)^2$ (proved in the next section) shows the series has a double pole at $s = 1$.
3. $\sum \Lambda(n)/n^s$: since $0 \leq \Lambda(n) \leq \log n \leq n^\varepsilon$ for any $\varepsilon > 0$, the abscissa $\sigma_a = 1$. By proposition 2.1.5, $\sigma_c = 1$ as well, and there is a singularity at $s = 1$. The identity $-\zeta'(s)/\zeta(s) = \sum \Lambda(n)/n^s$ (§2.2) identifies this singularity as a simple pole with residue 1.
4. $\sum \mu(n)/n^s$: since $|\mu(n)| \leq 1$, $\sigma_a = 1$. The exact value of σ_c is a major open problem. The bound $\sigma_c \leq 1$ follows from the Prime Number Theorem ($M(x) = o(x)$ implies convergence for $\operatorname{Re}(s) > 1$ by Abel summation). The Riemann Hypothesis is equivalent to $\sigma_c = 1/2$; the best unconditional result, $\sigma_c \leq 1$, does not pin down the exact value.

The term-by-term differentiation identity $D'(f, s) = D(f \cdot (-\log), s)$ has an important interpretation: differentiating a Dirichlet series with respect to s multiplies the coefficients by $-\log n$. This connects differentiation to the von Mangoldt function, since $-\zeta'(s) = \sum (\log n)/n^s$ is the Dirichlet series of $\log$, and dividing by $\zeta(s)$ gives $-\zeta'(s)/\zeta(s) = D(\Lambda, s)$ (the logarithmic derivative encodes the von Mangoldt function). More generally, for any Dirichlet series $D(f, s)$ with f multiplicative and $D(f, s) \neq 0$, the logarithmic derivative $-D'(f, s)/D(f, s)$ is the Dirichlet series of an arithmetic function whose values at prime powers are determined by those of f . This mechanism is developed systematically in the next section.

Example 2.1.10: A Dirichlet Series with $\sigma_c < \sigma_a$

The Dirichlet eta function $\eta(s) = \sum_{n=1}^\infty (-1)^{n+1}/n^s$ has $\sigma_a = 1$ and $\sigma_c = 0$. The partial sums $A(N) = \sum_{n=1}^N (-1)^{n+1}$ alternate between 1 and 0, so $|A(N)| \leq 1$ for all N , and Abel’s test (corollary 1.4.9) gives convergence for $\operatorname{Re}(s) > 0$. At $s = 0$: $\sum (-1)^{n+1} = 1 - 1 + 1 - \dots$ diverges by oscillation, confirming $\sigma_c = 0$.

The identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ is verified by writing $\eta(s) = \sum 1/n^s - 2 \sum 1/(2n)^s = \zeta(s) - 2^{1-s}\zeta(s)$. The factor $(1 - 2^{1-s})$ vanishes at $s = 1 + 2\pi i k / \log 2$ for $k \in \mathbb{Z}$, cancelling the pole of $\zeta(s)$ at $s = 1$ when $k = 0$. In particular, $\eta(1) = 1 - 1/2 + 1/3 - 1/4 + \dots = \log 2$, and this value can also be obtained as $\lim_{s \rightarrow 1} (1 - 2^{1-s})\zeta(s) = \log 2$ (using $\zeta(s) \sim 1/(s - 1)$ and $(1 - 2^{1-s}) \sim (s - 1) \log 2$ near $s = 1$).

The convergence theory of Dirichlet series should be compared with that of power series. A power

series $\sum a_n z^n$ converges in a *disk* $|z| < R$, determined by a single parameter R (the radius of convergence). A Dirichlet series $\sum a_n n^{-s}$ converges in a *half-plane* $\operatorname{Re}(s) > \sigma_c$, also determined by a single parameter. The disk is replaced by a half-plane because $|n^{-s}| = n^{-\sigma}$ depends only on $\operatorname{Re}(s)$, not on $\operatorname{Im}(s)$: changing the imaginary part of s rotates each term n^{-it} on the unit circle without affecting convergence. The analogy extends further: the Hadamard formula $1/R = \limsup |a_n|^{1/n}$ for the radius of convergence corresponds to the formula $\sigma_a = \limsup (\log A^*(x)) / \log x$ for the abscissa of absolute convergence.

Example 2.1.11: Dirichlet Series That Converge Everywhere or Nowhere

1. The Dirichlet series $D(s) = \sum_{n=1}^N a_n / n^s$ (a finite sum) is an entire function of s , with $\sigma_c = \sigma_a = -\infty$.
2. The series $\sum n! / n^s$ has $\sigma_c = \sigma_a = +\infty$: the terms $n! / n^s$ grow faster than any power of n , so the series diverges for all s .
3. The series $\sum 1 / (n^s \log^2 n)$ (starting from $n = 2$) converges absolutely for $\operatorname{Re}(s) \geq 1$ (since $\sum 1 / (n \log^2 n) < \infty$). Hence $\sigma_a \leq 1$. Since the coefficients are positive, $\sigma_c = \sigma_a$; and $\sum 1 / (n^\sigma \log^2 n)$ diverges for $\sigma < 1$ (by comparison with $\sum 1/n$), so $\sigma_a = 1$ exactly. This series converges on the *closed* half-plane $\operatorname{Re}(s) \geq 1$, unlike $\zeta(s)$ which diverges at $s = 1$.

2.2 The Ring Homomorphism: Convolution and Multiplication

The previous section developed Dirichlet series as analytic functions on a half-plane. The next structural fact is that the operation of *Dirichlet convolution* on arithmetic functions, introduced in section 1.2, corresponds to ordinary *multiplication* of the associated Dirichlet series. This single identity is the algebraic engine of analytic number theory: every standard convolution identity from chapter 1 translates into an identity among Dirichlet series, and the convolution ring $(\mathcal{A}, +, *)$ realizes concretely as a ring of analytic functions on a half-plane. The isomorphism is also the route by which the von Mangoldt function Λ enters the analytic theory through the logarithmic derivative $-\zeta'(s)/\zeta(s)$, the identity that drives the analytic proof of the Prime Number Theorem in chapter 5. The crux is the following.

Theorem 2.2.1: Dirichlet Multiplication

Let f, g be arithmetic functions, and let $\sigma^* = \max(\sigma_a(f), \sigma_a(g))$. For $\operatorname{Re}(s) > \sigma^*$, the series $D(f, s)$, $D(g, s)$, and $D(f * g, s)$ all converge absolutely, and

$$D(f * g, s) = D(f, s) \cdot D(g, s)$$

Proof:

Fix s with $\sigma = \operatorname{Re}(s) > \sigma^*$. By the definition of σ^* , both $\sum |f(m)|/m^\sigma$ and $\sum |g(k)|/k^\sigma$ converge.

Step 1: Form the absolutely convergent product series. The product

$$\left(\sum_{m=1}^{\infty} \frac{f(m)}{m^s} \right) \left(\sum_{k=1}^{\infty} \frac{g(k)}{k^s} \right) = \sum_{m=1}^{\infty} \sum_{k=1}^{\infty} \frac{f(m)g(k)}{(mk)^s}$$

converges absolutely, since the absolute value of the double sum is bounded by the product of the absolute series.

Step 2: Group terms by $n = mk$. For each $n \geq 1$, the pairs $(m, k) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$ with $mk = n$ are in bijection with the divisors d of n via $(m, k) = (d, n/d)$. Hence

$$\sum_{m,k} \frac{f(m)g(k)}{(mk)^s} = \sum_{n=1}^{\infty} \frac{1}{n^s} \sum_{d|n} f(d)g(n/d) = \sum_{n=1}^{\infty} \frac{(f * g)(n)}{n^s}$$

by the definition of Dirichlet convolution. The interchange of summation order is justified by the absolute convergence established in Step 1 (Tonelli's theorem applied to the counting measure on $\mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$).

*Step 3: Absolute convergence of $D(f * g, s)$.* The same argument with $|f|$ and $|g|$ in place of f and g gives

$$\sum_{n=1}^{\infty} \frac{|(f * g)(n)|}{n^\sigma} \leq \sum_{n=1}^{\infty} \frac{(|f| * |g|)(n)}{n^\sigma} = \left(\sum_{m=1}^{\infty} \frac{|f(m)|}{m^\sigma} \right) \left(\sum_{k=1}^{\infty} \frac{|g(k)|}{k^\sigma} \right) < \infty$$

completing the proof.

The proof says something more than the bare identity. The same rearrangement, applied to $|f|$ and $|g|$, also bounds the abscissa of absolute convergence of the convolution: $\sigma_a(f * g) \leq \max(\sigma_a(f), \sigma_a(g))$. Equality need not hold (cancellation in $f * g$ can shift the abscissa downward), but the inequality places $D(f * g, s)$ on a half-plane no smaller than the common half-plane of $D(f, s)$ and $D(g, s)$.

The theorem converts every convolution identity from section 1.2 into a Dirichlet series identity. The corollary collects the most-used cases.

Corollary 2.2.2: Standard Identities from Convolution

For $\operatorname{Re}(s)$ in the appropriate half-plane:

1. $\zeta(s)^2 = \sum_{n=1}^{\infty} \frac{d(n)}{n^s}$ for $\operatorname{Re}(s) > 1$, from $d = \mathbf{1} * \mathbf{1}$.
2. $\zeta(s) \zeta(s-k) = \sum_{n=1}^{\infty} \frac{\sigma_k(n)}{n^s}$ for $\operatorname{Re}(s) > \max(1, k+1)$, from $\sigma_k = \operatorname{id}_k * \mathbf{1}$, where $\operatorname{id}_k(n) = n^k$.
3. $\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}$ for $\operatorname{Re}(s) > 1$, from $\mu * \mathbf{1} = \varepsilon$.
4. $\sum_{n=1}^{\infty} \frac{\varphi(n)}{n^s} = \frac{\zeta(s-1)}{\zeta(s)}$ for $\operatorname{Re}(s) > 2$, from $\varphi * \mathbf{1} = \operatorname{id}$.
5. $\frac{\zeta(2s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\lambda(n)}{n^s}$ for $\operatorname{Re}(s) > 1$, from $\lambda * |\mu| = \varepsilon$ together with $\sum |\mu(n)|/n^s = \zeta(s)/\zeta(2s)$.

Proof:

Each identity follows from theorem 2.2.1 together with the elementary computations $D(\varepsilon, s) = 1$, $D(\mathbf{1}, s) = \zeta(s)$, and $D(\operatorname{id}_k, s) = \sum n^k/n^s = \zeta(s-k)$ on $\operatorname{Re}(s) > k+1$.

Identity (1): $D(d, s) = D(\mathbf{1} * \mathbf{1}, s) = \zeta(s) \zeta(s) = \zeta(s)^2$.

Identity (2): $D(\sigma_k, s) = D(\operatorname{id}_k * \mathbf{1}, s) = D(\operatorname{id}_k, s) D(\mathbf{1}, s) = \zeta(s-k) \zeta(s)$.

Identity (3): $D(\varepsilon, s) = 1 = D(\mu * \mathbf{1}, s) = D(\mu, s) \zeta(s)$, so $D(\mu, s) = 1/\zeta(s)$.

Identity (4): $D(\operatorname{id}, s) = \zeta(s-1) = D(\varphi * \mathbf{1}, s) = D(\varphi, s) \zeta(s)$, so $D(\varphi, s) = \zeta(s-1)/\zeta(s)$.

Identity (5): The squarefree indicator $|\mu|$ satisfies $|\mu| * \mathbf{1}_{\square} = \mathbf{1}$, where $\mathbf{1}_{\square}(n) = 1$ if n is a square and 0 otherwise. Hence $D(|\mu|, s) D(\mathbf{1}_{\square}, s) = \zeta(s)$. Since $D(\mathbf{1}_{\square}, s) = \sum_{m \geq 1} 1/(m^2)^s = \zeta(2s)$, this gives $D(|\mu|, s) = \zeta(s)/\zeta(2s)$. The Liouville identity $\lambda * |\mu| = \varepsilon$ (proposition 1.2.10) then gives $D(\lambda, s) \cdot \zeta(s)/\zeta(2s) = 1$, i.e., $D(\lambda, s) = \zeta(2s)/\zeta(s)$, as claimed.

These identities form the fundamental dictionary at the level of $\zeta(s)$. Identity (1) makes the divisor function $d(n)$ accessible to analytic methods through $\zeta(s)^2$, whose double pole at $s = 1$ is the analytic source of the leading $x \log x$ term in the asymptotic $\sum_{n \leq x} d(n) \sim x \log x$ from section 1.3. Identity (2) generalizes this to all σ_k . Identity (3) inverts the role of the Möbius function at the analytic level: the reciprocal of the zeta function is the Dirichlet series of μ , linking the analytic behavior of $\zeta(s)$ to summatory questions about $\mu(n)$ and ultimately to the Prime Number Theorem and the Riemann Hypothesis. Identity (4) gives the Dirichlet series of the Euler totient as a ratio of zeta functions; its

2.2. The Ring Homomorphism: Convolution and Multiplication Dirichlet Series and Euler Products

residue at $s = 2$ is $1/\zeta(2) = 6/\pi^2$, the asymptotic density of coprime pairs (section 1.3). Identity (5) is the Liouville series, whose ratio form $\zeta(2s)/\zeta(s)$ encodes the multiplicative structure of squarefree integers analytically.

The first identity admits a direct verification that makes the convolution-multiplication mechanism transparent.

Example 2.2.3: Verifying $\zeta(s)^2 = \sum d(n)/n^s$ by Squaring

Squaring the series for $\zeta(s)$ on $\operatorname{Re}(s) > 1$:

$$\zeta(s)^2 = \left(\sum_{m=1}^{\infty} \frac{1}{m^s} \right) \left(\sum_{k=1}^{\infty} \frac{1}{k^s} \right) = \sum_{m,k \geq 1} \frac{1}{(mk)^s} = \sum_{n=1}^{\infty} \frac{c(n)}{n^s}$$

where $c(n) = \#\{(m, k) \in \mathbb{Z}_{>0} \times \mathbb{Z}_{>0} : mk = n\}$. The pairs (m, k) with product n are in bijection with the divisors d of n via $(m, k) = (d, n/d)$, so $c(n) = d(n)$ exactly.

The first few coefficients confirm the count: $d(1) = 1$, $d(2) = d(3) = d(5) = d(7) = 2$, $d(4) = d(9) = 3$, $d(6) = d(8) = d(10) = 4$, $d(12) = 6$. Hence

$$\zeta(s)^2 = 1 + \frac{2}{2^s} + \frac{2}{3^s} + \frac{3}{4^s} + \frac{2}{5^s} + \frac{4}{6^s} + \frac{2}{7^s} + \frac{4}{8^s} + \frac{3}{9^s} + \frac{4}{10^s} + \cdots$$

The general theorem reduces this verification to the convolution identity $d = \mathbf{1} * \mathbf{1}$, sparing the explicit double-sum manipulation in every individual case.

The convolution-multiplication identity admits a more conceptual restatement.

Theorem 2.2.4: Ring Isomorphism: Arithmetic Functions and Dirichlet Series

Let $\mathcal{A}_a \subseteq \mathcal{A}$ denote the subring of arithmetic functions f with finite abscissa of absolute convergence $\sigma_a(f) < \infty$. The map

$$\Phi : \mathcal{A}_a \longrightarrow \{\text{Dirichlet series with finite } \sigma_a\}, \quad \Phi(f) = D(f, s)$$

is an injective ring homomorphism, where the target is regarded as the ring of analytic functions on common half-planes of absolute convergence with pointwise addition and multiplication. Under Φ :

1. $\varepsilon \mapsto 1$,
2. $\mathbf{1} \mapsto \zeta(s)$,
3. $\mu \mapsto 1/\zeta(s)$,
4. $\text{id} \mapsto \zeta(s-1)$.

Proof:

Linearity of Φ is immediate: $D(f + g, s) = D(f, s) + D(g, s)$ on the common half-plane of absolute convergence. Multiplicativity is theorem 2.2.1: $D(f * g, s) = D(f, s) D(g, s)$ on

$\operatorname{Re}(s) > \max(\sigma_a(f), \sigma_a(g))$. The unit ε maps to the constant function 1 since $D(\varepsilon, s) = \varepsilon(1) = 1$.

Injectivity is the content of corollary 2.1.8: if $D(f, s) = D(g, s)$ on a common half-plane, then $f(n) = g(n)$ for all n . The named correspondences come from direct computation: $D(\varepsilon, s) = 1$, $D(\mathbf{1}, s) = \zeta(s)$, $D(\operatorname{id}, s) = \zeta(s-1)$, and $D(\mu, s) = 1/\zeta(s)$ from corollary 2.2.2, completing the proof.

This restatement clarifies the structural picture. The formal-algebraic content of arithmetic functions (the convolution ring $\mathcal{A}$) and the analytic content of Dirichlet series (functions on a half-plane with a given abscissa of absolute convergence) are not just compatible but identical, up to the inclusion $\mathcal{A}_a \subseteq \mathcal{A}$. Algebraic identities like $\mu * \mathbf{1} = \varepsilon$ become analytic identities like $\zeta(s) \sum \mu(n)/n^s = 1$, and conversely, analytic computations involving products and quotients of Dirichlet series translate back into convolution identities at the level of coefficients. The ring structure on the analytic side imposes constraints that are invisible algebraically: for instance, the analytic identity $\zeta(s) (\sum \mu(n)/n^s) = 1$ on $\operatorname{Re}(s) > 1$ forces the strict inequality $\sigma_c(\mu) \leq 1$, since $1/\zeta(s)$ continues analytically to a neighborhood of every point of $\{\operatorname{Re}(s) = 1\} \setminus \{1\}$ (using the non-vanishing $\zeta(1+it) \neq 0$, established in chapter 4).

The first nontrivial consequence of the ring isomorphism is a non-vanishing statement.

Example 2.2.5: Non-Vanishing of $\zeta(s)$ for $\operatorname{Re}(s) > 1$

The Möbius identity $1/\zeta(s) = \sum \mu(n)/n^s$ on $\operatorname{Re}(s) > 1$ shows that $\zeta(s) \neq 0$ on this half-plane. If $\zeta(s_0) = 0$ for some s_0 with $\operatorname{Re}(s_0) > 1$, then $1/\zeta(s)$ would have a pole at s_0 , contradicting the fact that $\sum \mu(n)/n^{s_0}$ converges absolutely to a finite value at s_0 (since $\operatorname{Re}(s_0) > 1 = \sigma_a(\mu)$). The next section gives a more conceptual proof of this fact via the Euler product, which extends without change to any Dirichlet series with an Euler product. The harder boundary case $\operatorname{Re}(s) = 1$, where the Möbius series sits on the boundary of its half-plane of absolute convergence and the argument above breaks down, is treated in chapter 4.

The headline application of the ring isomorphism is the identity that connects $\zeta(s)$ to prime-counting.

Theorem 2.2.6: Logarithmic Derivative of $\zeta(s)$

For $\operatorname{Re}(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

where Λ is the von Mangoldt function (definition 1.1.8).

Proof:

The strategy combines two ingredients: the convolution identity $\Lambda * \mathbf{1} = \log$ from proposition 1.1.9, and term-by-term differentiation of $\zeta(s)$ from theorem 2.1.7.

Step 1: Differentiate $\zeta(s)$. By theorem 2.1.7, for $\operatorname{Re}(s) > 1$,

$$\zeta'(s) = - \sum_{n=1}^{\infty} \frac{\log n}{n^s}$$

Equivalently, $-\zeta'(s) = D(\log, s)$, the Dirichlet series of the function $n \mapsto \log n$.

Step 2: Apply the convolution identity. Under the ring isomorphism of theorem 2.2.4, the von Mangoldt identity $\Lambda * \mathbf{1} = \log$ becomes

$$D(\Lambda, s) \cdot \zeta(s) = D(\log, s) = -\zeta'(s)$$

on $\operatorname{Re}(s) > 1$. Absolute convergence of $D(\Lambda, s)$ on this half-plane follows from $0 \leq \Lambda(n) \leq \log n$ together with $\sigma_a(\log) = 1$ (since $\log n \leq n^\varepsilon$ for any $\varepsilon > 0$, so $\sum (\log n)/n^\sigma$ converges for every $\sigma > 1$).

Step 3: Divide. Since $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > 1$ (example 2.2.5), division by $\zeta(s)$ yields

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} = -\frac{\zeta'(s)}{\zeta(s)}$$

on $\operatorname{Re}(s) > 1$, completing the proof.

The identity $-\zeta'(s)/\zeta(s) = \sum \Lambda(n)/n^s$ is the single most important identity in elementary analytic number theory. Its left side is an analytic object: the logarithmic derivative of $\zeta(s)$, whose poles encode the analytic structure of $\zeta(s)$. Its right side is the Dirichlet series of the von Mangoldt function, whose summatory function $\psi(x) = \sum_{n \leq x} \Lambda(n)$ counts prime powers (weighted by $\log p$) and is the natural object whose asymptotics encode the Prime Number Theorem $\psi(x) \sim x$. The identity converts an analytic problem (locating the poles and zeros of $-\zeta'(s)/\zeta(s)$) into an arithmetic one (estimating $\psi(x)$). The simple pole of $\zeta(s)$ at $s = 1$, with residue 1, becomes a simple pole of $-\zeta'(s)/\zeta(s)$ at $s = 1$ with residue 1 (since $-\zeta'(s)/\zeta(s) \sim 1/(s-1)$ near $s = 1$); this residue is the analytic source of the leading term $\psi(x) \sim x$ in the Prime Number Theorem. The contributions of the zeros of $\zeta(s)$ inside the critical strip give the oscillatory error terms in the explicit formula. The whole strategy of the analytic proof of the Prime Number Theorem, developed in chapter 5, rests on the identity proved above.

The structural extension of theorem 2.2.6 to general multiplicative arithmetic functions is the following: for multiplicative f with $D(f, s) \neq 0$ in some half-plane, the logarithmic derivative $-D'(f, s)/D(f, s)$ is the Dirichlet series of an arithmetic function Λ_f supported on prime powers, generalizing the von Mangoldt function. The analytic structure of the logarithmic derivative is therefore controlled entirely by the values of f at prime powers, mirroring the determination-by-prime-powers principle for multiplicative arithmetic functions (proposition 1.1.3).

2.3 Euler Products

The ring isomorphism of theorem 2.2.4 converts Dirichlet convolution into multiplication of Dirichlet series, but its full force is felt only for *multiplicative* functions. For such functions the values $f(n)$ at all integers are reconstructed from the values $f(p^k)$ at prime powers (proposition 1.1.3), and this reconstruction lifts to the analytic level: the Dirichlet series $D(f, s)$ factors as a product of local pieces, one for each prime, with the local piece at p built entirely from $f(p), f(p^2), f(p^3), \dots$. The resulting factorization is the *Euler product*, and it is the analytic incarnation of unique factorization in $\mathbb{Z}$. Where theorem 2.2.1 expresses one global relation $D(f * g, s) = D(f, s)D(g, s)$, the Euler product expresses a relation between $D(f, s)$ and the infinitely many local factors at once.

The mechanism is the local-to-global principle of section 1.1 made analytic. A multiplicative function decouples across primes; absolute convergence of $D(f, s)$ lets the corresponding infinite sum be regrouped as an infinite product over primes without disturbing its value. The product form exposes structure that the series hides: it shows immediately that $D(f, s)$ cannot vanish in the region of absolute convergence (every factor is a nonzero number near 1), and its logarithm is a sum over prime powers that ties the analytic object directly to the primes. These two consequences, non-vanishing and the logarithmic expansion, are the threads that the study of prime distribution in chapter 3 and the theory of $\zeta(s)$ in chapter 4 will pull on.

Theorem 2.3.1: Euler Product for Multiplicative Functions

Let f be a multiplicative arithmetic function (definition 1.1.2), and let $\sigma_a(f)$ be its abscissa of absolute convergence (definition 2.1.3). For $\operatorname{Re}(s) > \sigma_a(f)$,

$$D(f, s) = \sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_p \left(\sum_{k=0}^{\infty} \frac{f(p^k)}{p^{ks}} \right)$$

where the product runs over all primes and converges absolutely. If f is totally multiplicative, the local factor at p is the geometric series

$$\sum_{k=0}^{\infty} \frac{f(p^k)}{p^{ks}} = \sum_{k=0}^{\infty} \left(\frac{f(p)}{p^s} \right)^k = \frac{1}{1 - f(p)p^{-s}}$$

Proof:

Fix s with $\sigma = \operatorname{Re}(s) > \sigma_a(f)$, so that $\sum_{n=1}^{\infty} |f(n)|n^{-\sigma} < \infty$. Write $f_p(s) = \sum_{k=0}^{\infty} f(p^k)p^{-ks}$ for the local factor at p .

Step 1: Each local factor converges absolutely. For a single prime p , the terms $f(p^k)p^{-ks}$ are a subsequence of the terms $f(n)n^{-s}$ (those indexed by $n = p^k$), so

$$\sum_{k=0}^{\infty} \left| \frac{f(p^k)}{p^{ks}} \right| = \sum_{k=0}^{\infty} \frac{|f(p^k)|}{p^{k\sigma}} \leq \sum_{n=1}^{\infty} \frac{|f(n)|}{n^{\sigma}} < \infty$$

Hence $f_p(s)$ is a well-defined complex number, and $f_p(s) = 1 + \sum_{k=1}^{\infty} f(p^k)p^{-ks}$ since $f(p^0) = f(1) = 1$ by multiplicativity.

Step 2: The finite product over small primes equals a partial sum. Fix $y \geq 2$ and let $p_1 < p_2 < \dots < p_r$ be the primes not exceeding y . Each $f_{p_i}(s)$ is an absolutely convergent series, so their finite product may be expanded by multiplying out and rearranging (a finite product of absolutely convergent series converges absolutely to the iterated sum):

$$\prod_{i=1}^r f_{p_i}(s) = \sum_{k_1=0}^{\infty} \dots \sum_{k_r=0}^{\infty} \frac{f(p_1^{k_1}) \dots f(p_r^{k_r})}{(p_1^{k_1} \dots p_r^{k_r})^s}$$

By multiplicativity of f , the numerator $f(p_1^{k_1}) \dots f(p_r^{k_r})$ equals $f(p_1^{k_1} \dots p_r^{k_r})$, since the prime powers $p_i^{k_i}$ are pairwise coprime. By unique factorization (??, section 0.2), the map $(k_1, \dots, k_r) \mapsto$

$n = p_1^{k_1} \cdots p_r^{k_r}$ is a bijection from r -tuples of non-negative integers onto the set

$$S_y = \{n \geq 1 : \text{every prime factor of } n \text{ is } \leq y\}$$

of y -smooth integers. Therefore

$$\prod_{i=1}^r f_{p_i}(s) = \sum_{n \in S_y} \frac{f(n)}{n^s}$$

the local factors at the primes $\leq y$ assemble exactly into the sum over integers built from those primes.

Step 3: Control the tail. Every integer $n \leq y$ has all prime factors $\leq y$, so $\{1, 2, \dots, \lfloor y \rfloor\} \subseteq S_y$. The difference between $D(f, s)$ and the partial product is supported on integers not in S_y , each of which exceeds y :

$$\left| D(f, s) - \prod_{i=1}^r f_{p_i}(s) \right| = \left| \sum_{n \notin S_y} \frac{f(n)}{n^s} \right| \leq \sum_{n \notin S_y} \frac{|f(n)|}{n^\sigma} \leq \sum_{n > y} \frac{|f(n)|}{n^\sigma}$$

The right-hand side is the tail of the convergent series $\sum_n |f(n)|n^{-\sigma}$, so it tends to 0 as $y \rightarrow \infty$. Letting $y \rightarrow \infty$ gives $\prod_p f_p(s) = D(f, s)$, with the product interpreted as the limit of its partial products.

Step 4: Absolute convergence of the infinite product. An infinite product $\prod_p (1 + a_p)$ converges absolutely precisely when $\sum_p |a_p| < \infty$ ([13]). Here $a_p = f_p(s) - 1 = \sum_{k=1}^{\infty} f(p^k)p^{-ks}$, and

$$\sum_p |a_p| = \sum_p \left| \sum_{k=1}^{\infty} \frac{f(p^k)}{p^{ks}} \right| \leq \sum_p \sum_{k=1}^{\infty} \frac{|f(p^k)|}{p^{k\sigma}} \leq \sum_{n=2}^{\infty} \frac{|f(n)|}{n^\sigma} < \infty$$

where the last inequality holds because each prime power p^k with $k \geq 1$ is a distinct integer ≥ 2 , so the double sum over (p, k) is a subseries of $\sum_{n \geq 2} |f(n)|n^{-\sigma}$. Thus the product converges absolutely, and in particular its value is independent of the order of the factors.

Step 5: The totally multiplicative case. If f is totally multiplicative, then $f(p^k) = f(p)^k$ by proposition 1.1.3. The local factor is the geometric series $\sum_{k \geq 0} (f(p)p^{-s})^k$ with ratio $f(p)p^{-s}$, and its absolute convergence (Step 1) forces $|f(p)p^{-s}| < 1$, since a geometric series whose ratio has modulus ≥ 1 has terms that do not tend to 0 and hence diverges. Summing the geometric series gives $f_p(s) = (1 - f(p)p^{-s})^{-1}$, completing the proof.

The Euler product is unique factorization read analytically. Step 2 is the heart of the matter: multiplying the local factors and expanding produces exactly one term for each way of choosing a prime-power part at each prime, and unique factorization guarantees that these choices correspond bijectively to integers. The hypothesis of absolute convergence is what permits the rearrangement; without it, the regrouping of the double series into a product would not be justified, and the value could change. The role of multiplicativity is equally precise: it is what makes the numerator $f(p_1^{k_1}) \cdots f(p_r^{k_r})$ collapse to the single value $f(n)$, so that the expanded product is genuinely the Dirichlet series and not some twisted variant. For totally multiplicative f , the further collapse

$f(p^k) = f(p)^k$ turns each local factor into a closed-form geometric series, and the Euler product becomes a product of simple rational functions of p^{-s} .

The first and defining instance is the zeta function itself, where $f = \mathbf{1}$ is the constant function 1, totally multiplicative with $\mathbf{1}(p) = 1$.

Corollary 2.3.2: The Euler Product for the Riemann Zeta Function

For $\operatorname{Re}(s) > 1$,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1 - p^{-s}}$$

the product converging absolutely.

Proof:

The constant function $\mathbf{1}$ is totally multiplicative with $\mathbf{1}(p) = 1$ and $\sigma_a(\mathbf{1}) = 1$ (example 2.1.9). Applying theorem 2.3.1 with $f = \mathbf{1}$, the local factor at p is the geometric series $\sum_{k=0}^{\infty} p^{-ks} = (1 - p^{-s})^{-1}$, so $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ for $\operatorname{Re}(s) > 1$, as required.

This identity, due to Euler, is the analytic statement of the fundamental theorem of arithmetic. Reading the product from right to left and expanding each factor as a geometric series, $\prod_p (1 + p^{-s} + p^{-2s} + \dots)$, the coefficient of n^{-s} in the expansion counts the number of ways to write n as a product of prime powers, which is exactly one. The equality with $\sum_n n^{-s}$ therefore encodes the fact that every integer factors into primes in exactly one way. The product representation also carries arithmetic information that the sum conceals. As $s \rightarrow 1^+$ along the real axis, $\zeta(s) \rightarrow +\infty$ because the harmonic series diverges; on the product side this forces $\prod_p (1 - p^{-1})^{-1}$ to diverge, which can happen only if there are infinitely many primes and $\sum_p p^{-1}$ diverges. Euler's original proof of the infinitude of primes is precisely this observation, and the quantitative form of the divergence is the subject of example 2.3.6 below and, in full, of the elementary theory of prime distribution in chapter 3 (a finite Euler product would give a finite limit, contradicting the pole). The analytic behavior of $\zeta(s)$ near $s = 1$, and throughout the critical strip, is developed in chapter 4.

The convolution identities of corollary 2.2.2 all have Euler product counterparts, obtained by computing one local factor and recognizing the resulting product. Carrying out these computations gives a second, independent route to those identities, one that never invokes the multiplication theorem but instead reads each global product off its local factors.

Example 2.3.3: Euler Products of Standard Dirichlet Series

Each of the following multiplicative functions has an Euler product computed from its values at prime powers. The global product is then identified with a ratio of zeta values, cross-checking corollary 2.2.2.

1. *The Möbius function μ (definition 1.1.4).* Here $\mu(p) = -1$ and $\mu(p^k) = 0$ for $k \geq 2$, so the local factor terminates:

$$\sum_{k=0}^{\infty} \frac{\mu(p^k)}{p^{ks}} = 1 + \frac{\mu(p)}{p^s} = 1 - \frac{1}{p^s}$$

Hence $D(\mu, s) = \prod_p (1 - p^{-s})$ for $\operatorname{Re}(s) > 1$. Comparing with corollary 2.3.2, this is exactly $\prod_p (1 - p^{-s})^{-1}$ inverted factor by factor, so $D(\mu, s) = 1/\zeta(s)$, matching corollary 2.2.2(3).

2. The divisor function $d = \mathbf{1} * \mathbf{1}$ (definition 1.1.5). Since $d(p^k) = k + 1$, the local factor is

$$\sum_{k=0}^{\infty} \frac{k+1}{p^{ks}} = \sum_{k=0}^{\infty} (k+1)x^k = \frac{1}{(1-x)^2}, \quad x = p^{-s}$$

using the standard power series $\sum_{k \geq 0} (k+1)x^k = (1-x)^{-2}$ for $|x| < 1$. Therefore $D(d, s) = \prod_p (1-p^{-s})^{-2} = \zeta(s)^2$ for $\operatorname{Re}(s) > 1$, matching corollary 2.2.2(1). The local factor $(1-p^{-s})^{-2}$ is the square of the local factor of $\zeta(s)$, the product form of $d = \mathbf{1} * \mathbf{1}$.

3. The Liouville function λ (definition 1.1.6). The function λ is totally multiplicative with $\lambda(p) = -1$, so theorem 2.3.1 gives the geometric local factor

$$\sum_{k=0}^{\infty} \frac{\lambda(p^k)}{p^{ks}} = \frac{1}{1 - \lambda(p)p^{-s}} = \frac{1}{1 + p^{-s}}$$

Hence $D(\lambda, s) = \prod_p (1 + p^{-s})^{-1}$ for $\operatorname{Re}(s) > 1$. To identify the global product, write $1 + p^{-s} = (1 - p^{-2s})/(1 - p^{-s})$, so that

$$\prod_p \frac{1}{1 + p^{-s}} = \prod_p \frac{1 - p^{-s}}{1 - p^{-2s}} = \frac{\prod_p (1 - p^{-s})}{\prod_p (1 - p^{-2s})} = \frac{1/\zeta(s)}{1/\zeta(2s)} = \frac{\zeta(2s)}{\zeta(s)}$$

where the last step applies corollary 2.3.2 at s and at $2s$ (both products converge absolutely for $\operatorname{Re}(s) > 1$). This matches corollary 2.2.2(5).

4. The Euler totient φ . The totient is multiplicative with $\varphi(p^k) = p^k - p^{k-1}$ for $k \geq 1$ and $\varphi(1) = 1$. The local factor is

$$1 + \sum_{k=1}^{\infty} \frac{p^k - p^{k-1}}{p^{ks}} = 1 + \sum_{k=1}^{\infty} p^{k(1-s)} - \sum_{k=1}^{\infty} p^{k-1-ks}$$

Both geometric series converge for $\operatorname{Re}(s) > 1$, where $|p^{1-s}| = p^{1-\sigma} < 1$. Summing them,

$$\sum_{k=1}^{\infty} p^{k(1-s)} = \frac{p^{1-s}}{1 - p^{1-s}}, \quad \sum_{k=1}^{\infty} p^{k-1-ks} = \frac{p^{-s}}{1 - p^{1-s}}$$

so the local factor is

$$1 + \frac{p^{1-s} - p^{-s}}{1 - p^{1-s}} = \frac{1 - p^{-s}}{1 - p^{1-s}}$$

The local factor is thus valid for $\operatorname{Re}(s) > 1$. Identifying the global product, however, requires corollary 2.3.2 at $s-1$: the factor $\prod_p (1 - p^{1-s}) = \prod_p (1 - p^{-(s-1)}) = 1/\zeta(s-1)$ converges only for $\operatorname{Re}(s-1) > 1$, so it is this $\zeta(s-1)$ factor, not the local series, that pins the region. Applying corollary 2.3.2 at s and at $s-1$,

$$D(\varphi, s) = \prod_p \frac{1 - p^{-s}}{1 - p^{1-s}} = \frac{1/\zeta(s)}{1/\zeta(s-1)} = \frac{\zeta(s-1)}{\zeta(s)}$$

for $\operatorname{Re}(s) > 2$, matching corollary 2.2.2(4).

In every case the local factor encodes the arithmetic of f at a single prime, and the shape of the global product records the convolution identity. The vanishing of $\mu(p^k)$ for $k \geq 2$ truncates the Möbius local factor to a polynomial $1 - p^{-s}$, the product reciprocal of $\zeta(s)$. The squaring $d = \mathbf{1} * \mathbf{1}$ appears as a squared local factor. The sign $\lambda(p) = -1$ flips the geometric series from $(1 - p^{-s})^{-1}$ to $(1 + p^{-s})^{-1}$, and the algebraic identity $(1 + x) = (1 - x^2)/(1 - x)$ converts this into the ratio $\zeta(2s)/\zeta(s)$ that isolates the squarefree structure. These computations are interchangeable with the convolution route of corollary 2.2.2. The Euler product route also exposes how the half-plane of validity arises: for the Liouville function the region $\operatorname{Re}(s) > 1$ is inherited directly from the local geometric series, whereas for the totient the local factors already converge on $\operatorname{Re}(s) > 1$ yet the global identity is pinned to $\operatorname{Re}(s) > 2$ by the $\zeta(s - 1)$ factor that appears upon recombination. The product form thus separates the local convergence of each factor from the global constraint imposed by the zeta values that assemble the answer.

A structural consequence of the product form, independent of any particular f , is that a Dirichlet series with an Euler product cannot vanish where the product converges. This is because a convergent infinite product of nonzero factors is nonzero, a criterion that has no counterpart on the series side.

Proposition 2.3.4: Non-Vanishing in the Half-Plane of Absolute Convergence

Let f be multiplicative, and let s be a point with $\operatorname{Re}(s) = \sigma$ at which

$$\sum_p \sum_{k=1}^{\infty} \frac{|f(p^k)|}{p^{k\sigma}} < \infty$$

and at which every local factor $f_p(s) = \sum_{k \geq 0} f(p^k)p^{-ks}$ is nonzero. Then the Euler product converges to a nonzero value, so $D(f, s) \neq 0$. In particular $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > 1$.

Proof:

The hypothesis $\sum_p \sum_{k \geq 1} |f(p^k)|p^{-k\sigma} < \infty$ is exactly the absolute convergence condition $\sum_p |f_p(s) - 1| < \infty$ from Step 4 of theorem 2.3.1, since $f_p(s) - 1 = \sum_{k \geq 1} f(p^k)p^{-ks}$ and $|f_p(s) - 1| \leq \sum_{k \geq 1} |f(p^k)|p^{-k\sigma}$. Absolute convergence of the product $\prod_p f_p(s)$ follows, and by theorem 2.3.1 its value is $D(f, s)$.

The criterion for a nonzero limit is the following: an absolutely convergent infinite product $\prod_p f_p(s)$ with every factor $f_p(s) \neq 0$ has a nonzero value ([13]). The reason is that absolute convergence means $f_p(s) \rightarrow 1$, so all but finitely many factors lie in the disk $|z - 1| < 1/2$ on which $\log$ is defined; the tail $\sum_p \log f_p(s)$ converges to a finite complex number L , the corresponding tail product converges to $e^L \neq 0$, and multiplying by the finitely many remaining nonzero factors keeps the value nonzero. Hence $D(f, s) = \prod_p f_p(s) \neq 0$.

For the zeta function, take $f = \mathbf{1}$ and $\operatorname{Re}(s) > 1$. The local factors $(1 - p^{-s})^{-1}$ are nonzero (a reciprocal is never zero), and the convergence hypothesis holds since $\sum_p \sum_{k \geq 1} p^{-k\sigma} \leq \sum_{n \geq 2} n^{-\sigma} < \infty$. Therefore $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > 1$, as we sought to show.

This proposition reproves the non-vanishing of example 2.2.5 from a structural vantage point. The earlier argument used the specific identity $1/\zeta(s) = \sum \mu(n)n^{-s}$ and observed that a zero of ζ would

produce a pole of an absolutely convergent series. The present argument is cleaner and more general: it uses only that $\zeta(s)$ has an Euler product with nonzero local factors, and it applies verbatim to any Dirichlet series possessing an Euler product, such as the Dirichlet L -functions whose non-vanishing on the line $\operatorname{Re}(s) = 1$ drives the theorem on primes in arithmetic progressions. The product form makes the non-vanishing transparent because the obstruction to a zero is local: a product vanishes only if a factor vanishes, and each local factor here is a value near 1. The difficult cases in the theory, such as the non-vanishing of $\zeta(1 + it)$ on the boundary line $\operatorname{Re}(s) = 1$, lie outside the region of absolute convergence, where this elementary argument no longer reaches; those are the province of chapter 4.

Taking the logarithm of an Euler product turns the product of local factors into a sum over primes, and expanding each logarithm as a power series produces a sum over prime powers. For totally multiplicative f the local factors are geometric, and the logarithm has an explicit expansion that connects the Dirichlet series directly to the prime powers.

Proposition 2.3.5: The Logarithm of an Euler Product

Let f be totally multiplicative with absolutely convergent Euler product on $\operatorname{Re}(s) > \sigma_a(f)$ and $|f(p)p^{-s}| < 1$ for every p . Then the double series

$$\log D(f, s) := \sum_p \sum_{k=1}^{\infty} \frac{f(p)^k}{k p^{ks}}$$

converges absolutely for $\operatorname{Re}(s) > \sigma_a(f)$ and defines a branch of the logarithm of $D(f, s)$ there: it is the analytic branch characterized by $\log D(f, s) \rightarrow 0$ as $\operatorname{Re}(s) \rightarrow +\infty$. This branch need not be the principal logarithm of $D(f, s)$; the two coincide precisely when the imaginary part of the series lies in $(-\pi, \pi]$, which holds whenever $\operatorname{Re}(s)$ is large enough. In particular, for $f = \mathbf{1}$ the series defines

$$\log \zeta(s) = \sum_p \sum_{k=1}^{\infty} \frac{1}{k p^{ks}}$$

for $\operatorname{Re}(s) > 1$, agreeing with the ordinary real logarithm on the real segment $s > 1$ (there $\zeta(s) > 1$). Differentiating term by term recovers

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

the identity of theorem 2.2.6.

Proof:

Step 1: A logarithm of the product as a sum of factor logarithms. By theorem 2.3.1 the local factor is $f_p(s) = (1 - f(p)p^{-s})^{-1}$, and the product $\prod_p f_p(s)$ converges absolutely. Throughout, $\log$ on a single factor denotes the principal branch. Each factor satisfies $|f(p)p^{-s}| < 1$, so $1 - f(p)p^{-s}$ lies in the open right half-plane $\operatorname{Re}(z) > 0$, away from the branch cut, and the principal expansion $-\log(1 - z) = \sum_{k \geq 1} z^k/k$ applies with $z = f(p)p^{-s}$. Setting

$$\log D(f, s) := \sum_p \log \frac{1}{1 - f(p)p^{-s}} = - \sum_p \log (1 - f(p)p^{-s}) = \sum_p \sum_{k=1}^{\infty} \frac{f(p)^k}{k p^{ks}}$$

gives the double series of the statement. The symbol $\log D(f, s)$ is *defined* by this sum; it is a logarithm of $D(f, s)$ because exponentiating term by term gives $\exp(\sum_p \log f_p(s)) = \prod_p f_p(s) = D(f, s)$ by theorem 2.3.1, the interchange of exp with the convergent sum being the continuity of exp applied to the partial sums. It need not be the *principal* logarithm of $D(f, s)$: the principal logarithm has imaginary part in $(-\pi, \pi]$, whereas the imaginary part $\sum_p \sum_{k \geq 1} \operatorname{Im}(f(p)^k p^{-ks})/k$ of the series is constrained only by absolute convergence and need not lie in $(-\pi, \pi]$; the two coincide precisely when it does. The double series converges absolutely: $\sum_p \sum_{k \geq 1} |f(p)|^k p^{-k\sigma}/k \leq \sum_p \sum_{k \geq 1} |f(p)p^{-s}|^k = \sum_p |f(p)p^{-s}|/(1 - |f(p)p^{-s}|)$. Absolute convergence of each local factor (Step 1 of theorem 2.3.1) forces $|f(p)p^{-s}| \rightarrow 0$, so $|f(p)p^{-s}| \leq 1/2$ for all but finitely many p , and for those p the summand is at most $2|f(p)p^{-s}|$. Since $\sum_p |f(p)p^{-s}|$ converges (its terms are a subseries of $\sum_n |f(n)|n^{-\sigma}$), the double series converges absolutely.

Step 2: Specialize to $\zeta(s)$. For $f = \mathbf{1}$, every $f(p) = 1$, so $f(p)^k = 1$, giving

$$\log \zeta(s) = \sum_p \sum_{k=1}^{\infty} \frac{1}{k p^{ks}}$$

for $\operatorname{Re}(s) > 1$.

Step 3: Differentiate term by term. Each summand $1/(k p^{ks}) = k^{-1} p^{-ks}$ is an entire function of s with derivative $-(\log p) p^{-ks}$ (the factor k cancels). The differentiated series is

$$\sum_p \sum_{k=1}^{\infty} \frac{-\log p}{p^{ks}}$$

which converges absolutely and uniformly on $\operatorname{Re}(s) \geq 1+\delta$ for each $\delta > 0$: there $\sum_p \sum_{k \geq 1} (\log p) p^{-k(1+\delta)} = \sum_p (\log p) p^{-(1+\delta)}/(1 - p^{-(1+\delta)})$, which is finite because $\log p \leq p^{\delta/2}$ for large p makes the series comparable to $\sum_p p^{-1-\delta/2}$. Local uniform convergence of the differentiated series, together with convergence of the original at one point, justifies differentiating $\log \zeta(s)$ term by term ([13]). The derivative of the left side is $\zeta'(s)/\zeta(s)$, so

$$\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k=1}^{\infty} \frac{-\log p}{p^{ks}}, \quad \text{hence} \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k=1}^{\infty} \frac{\log p}{p^{ks}}$$

Step 4: Recognize the von Mangoldt series. The pairs (p, k) with p prime and $k \geq 1$ are in bijection with the prime powers $n = p^k$, and the summand $\log p$ is exactly $\Lambda(p^k)$ by definition 1.1.8. For n not a prime power, $\Lambda(n) = 0$, so the sum over prime powers equals the sum over all n :

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k=1}^{\infty} \frac{\Lambda(p^k)}{p^{ks}} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

for $\operatorname{Re}(s) > 1$. This is the identity of theorem 2.2.6, completing the proof.

This gives a second derivation of the logarithmic-derivative formula $-\zeta'(s)/\zeta(s) = \sum \Lambda(n)n^{-s}$,

consistent with the convolution-based proof of theorem 2.2.6. The two proofs illuminate the identity from opposite sides. The convolution proof builds the formula from the identity $\Lambda * \mathbf{1} = \log$ at the level of arithmetic functions, transported through the ring isomorphism. The Euler product proof builds it from the factorization of $\zeta(s)$ over primes, with the von Mangoldt function emerging as the coefficient that records $\log p$ at each prime power p^k . The agreement of the two is a consistency check on the whole dictionary: the von Mangoldt function is at once the convolution inverse-image of $\log$ under $*$ and the logarithmic-derivative coefficient of the Euler product. The expansion $\log \zeta(s) = \sum_{p,k} 1/(k p^{ks})$ is itself the starting point for many estimates near $s = 1$, since the $k = 1$ term $\sum_p p^{-s}$ dominates and carries the divergence examined next.

Example 2.3.6: The Product Near $s = 1$

As $s \rightarrow 1^+$ along the real axis, $\zeta(s) \rightarrow +\infty$ because of the pole at $s = 1$, so $\log \zeta(s) \rightarrow +\infty$. In the expansion of proposition 2.3.5,

$$\log \zeta(s) = \sum_p \frac{1}{p^s} + \sum_p \sum_{k=2}^{\infty} \frac{1}{k p^{ks}}$$

the double tail with $k \geq 2$ stays bounded as $s \rightarrow 1^+$: it is dominated by $\sum_p \sum_{k \geq 2} p^{-k} = \sum_p p^{-2}/(1 - p^{-1}) \leq \sum_{n \geq 2} 1/(n(n-1)) = 1$ uniformly for $s \geq 1$. Since $\log \zeta(s) \rightarrow +\infty$ while this tail remains bounded, the prime sum must diverge:

$$\sum_p \frac{1}{p^s} \rightarrow +\infty \quad (s \rightarrow 1^+)$$

Letting $s \rightarrow 1^+$ shows that $\sum_p 1/p$ cannot converge, for if it did, $\sum_p p^{-s} \leq \sum_p p^{-1}$ would stay bounded. This foreshadows the divergence of $\sum_p 1/p$ and the infinitude of primes, treated quantitatively in chapter 3.

2.4 Mellin Transforms and the Integral Representation of $\zeta(s)$

The convergence theory and the convolution-multiplication dictionary of the previous sections treat a Dirichlet series as a formal-analytic object built from its coefficients. To extract asymptotic information from $\zeta(s)$ and its relatives, the series must be connected to integrals, where the machinery of complex analysis (contour shifting, residues, the calculus of poles) can be brought to bear. The device that makes this connection is the *Mellin transform*. Where the Fourier transform is adapted to the additive group $\mathbb{R}$ and the Laplace transform to the additive half-line, the Mellin transform is the integral transform adapted to the *multiplicative* group $(0, \infty)$: it converts dilation $t \mapsto at$ into a character a^{-s} , exactly as the Fourier transform converts translation into a character $e^{i\xi a}$. This is precisely the symmetry that a Dirichlet series $\sum f(n)n^{-s}$ respects, since each term n^{-s} is a multiplicative character of the variable n evaluated at the parameter s .

The payoff is twofold. First, the Mellin transform expresses the gamma function, the Riemann zeta function, and their products as integrals over $(0, \infty)$ whose integrands have manifest analytic structure. The integral representation $\Gamma(s)\zeta(s) = \int_0^\infty t^{s-1}/(e^t - 1) dt$ derived below is the starting point from which the meromorphic continuation and the functional equation of $\zeta(s)$ are obtained in

chapter 4 and ???. Second, the Mellin inversion formula runs the transform backward: it recovers an arithmetic sum from a contour integral of its Dirichlet series, and this is the analytic content of Perron's formula in section 2.5, the tool that converts analytic facts about $\zeta(s)$ into counting statements about primes. The gamma function will be developed in full in ???; here only the integral $\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt$ for $\operatorname{Re}(s) > 0$ is needed, and it appears below as the first example of a Mellin transform.

Definition 2.4.1: The Mellin Transform

Let $f: (0, \infty) \rightarrow \mathbb{C}$ be a function that is (Lebesgue) measurable on $(0, \infty)$. The *Mellin transform* of f is the function

$$(\mathcal{M}f)(s) = \int_0^\infty f(t) t^{s-1} dt$$

defined for those $s \in \mathbb{C}$ for which the integral converges absolutely. Suppose f satisfies the two-sided power bound

$$f(t) = O(t^{-a}) \text{ as } t \rightarrow 0^+, \quad f(t) = O(t^{-b}) \text{ as } t \rightarrow \infty$$

with $a < b$. Then the integral converges absolutely for every s with $a < \operatorname{Re}(s) < b$, and the open vertical strip

$$\{s \in \mathbb{C} \mid a < \operatorname{Re}(s) < b\}$$

is called the *fundamental strip* of f .

The fundamental strip is the multiplicative analogue of the half-plane of convergence for a Dirichlet series, and the reason it is a strip rather than a half-plane is the presence of *two* endpoints to control. Near $t = 0$ the factor t^{s-1} contributes $t^{\operatorname{Re}(s)-1}$, so the integral $\int_0^1 f(t) t^{s-1} dt$ converges when $\operatorname{Re}(s) - 1 > a - 1$, that is, when $\operatorname{Re}(s) > a$; the lower bound on $\operatorname{Re}(s)$ is forced by the behavior of f at the origin. Near $t = \infty$ the same factor must decay against the growth of f , and $\int_1^\infty f(t) t^{s-1} dt$ converges when $\operatorname{Re}(s) - 1 < b - 1$, that is, when $\operatorname{Re}(s) < b$; the upper bound is forced by the behavior of f at infinity. Only on the overlap $a < \operatorname{Re}(s) < b$ do both tails converge, which requires $a < b$. As with Dirichlet series, $|t^{s-1}| = t^{\operatorname{Re}(s)-1}$ depends only on the real part of s , so the region of absolute convergence is invariant under vertical translation, hence a vertical strip.

Example 2.4.2: The Gamma Function as a Mellin Transform

Take $f(t) = e^{-t}$. Its Mellin transform is, by definition, the integral that defines the gamma function:

$$(\mathcal{M}e^{-t})(s) = \int_0^\infty e^{-t} t^{s-1} dt = \Gamma(s)$$

The fundamental strip is the half-plane $\operatorname{Re}(s) > 0$, which is the degenerate case $a = 0$, $b = +\infty$ of definition 2.4.1. Convergence at both ends is checked directly. Near $t = 0^+$, the factor e^{-t} is bounded (indeed $e^{-t} \rightarrow 1$), so the integrand is comparable to $t^{\operatorname{Re}(s)-1}$, and

$$\int_0^1 e^{-t} t^{\sigma-1} dt \leq \int_0^1 t^{\sigma-1} dt = \frac{1}{\sigma}$$

with $\sigma = \operatorname{Re}(s)$, which is finite precisely when $\sigma > 0$. Near $t = \infty$, the exponential decay of e^{-t} dominates every power of t : for any $\sigma \in \mathbb{R}$ there is a constant C_σ with $t^{\sigma-1} \leq C_\sigma e^{t/2}$ for $t \geq 1$,

whence

$$\int_1^\infty e^{-t} t^{\sigma-1} dt \leq C_\sigma \int_1^\infty e^{-t/2} dt = 2C_\sigma e^{-1/2} < \infty$$

for every real σ . The upper endpoint of the strip is therefore $b = +\infty$, and the integral defines $\Gamma(s)$ as an analytic function on the entire half-plane $\operatorname{Re}(s) > 0$. The continuation of Γ to a meromorphic function on all of $\mathbb{C}$, with simple poles at the non-positive integers, together with the functional equation $\Gamma(s+1) = s\Gamma(s)$ and the reflection formula, is developed in ??; the present section uses only the integral on $\operatorname{Re}(s) > 0$.

The elementary properties of the transform are the precise statements of the slogans “dilation becomes a character” and “the Mellin transform is the Fourier transform in multiplicative coordinates.” These are the identities that will be applied repeatedly when manipulating the integrand of $\Gamma(s)\zeta(s)$.

Proposition 2.4.3: Elementary Properties of the Mellin Transform

Let $f: (0, \infty) \rightarrow \mathbb{C}$ have fundamental strip $a < \operatorname{Re}(s) < b$.

1. *Scaling.* For any real $\alpha > 0$, the function $t \mapsto f(\alpha t)$ has the same fundamental strip, and

$$(\mathcal{M}f(\alpha \cdot))(s) = \alpha^{-s}(\mathcal{M}f)(s)$$

2. *Shift by a power.* For any $\mu \in \mathbb{C}$, the function $t \mapsto t^\mu f(t)$ has fundamental strip $a - \operatorname{Re}(\mu) < \operatorname{Re}(s) < b - \operatorname{Re}(\mu)$, and

$$(\mathcal{M}t^\mu f(t))(s) = (\mathcal{M}f)(s + \mu)$$

3. *Reduction to a two-sided Laplace transform.* The substitution $t = e^{-x}$ identifies the Mellin transform with a two-sided Laplace transform. Writing $g(x) = f(e^{-x})$ for $x \in \mathbb{R}$,

$$(\mathcal{M}f)(s) = \int_{-\infty}^{\infty} g(x) e^{-sx} dx$$

for all s in the fundamental strip. In particular, on a vertical line $\operatorname{Re}(s) = c$ with $a < c < b$, setting $s = c + i\xi$ gives

$$(\mathcal{M}f)(c + i\xi) = \int_{-\infty}^{\infty} (e^{-cx} g(x)) e^{-i\xi x} dx$$

which is the Fourier transform of the function $x \mapsto e^{-cx} g(x) = e^{-cx} f(e^{-x})$.

Proof:

Throughout write $\sigma = \operatorname{Re}(s)$.

1. Fix $\alpha > 0$ and s with $a < \sigma < b$. In the integral $\int_0^\infty f(\alpha t) t^{s-1} dt$ substitute $u = \alpha t$, so $t = u/\alpha$, $dt = du/\alpha$, and $t^{s-1} = (u/\alpha)^{s-1} = \alpha^{1-s} u^{s-1}$. As t runs over $(0, \infty)$ so does u , and

$$\int_0^\infty f(\alpha t) t^{s-1} dt = \int_0^\infty f(u) \alpha^{1-s} u^{s-1} \frac{du}{\alpha} = \alpha^{-s} \int_0^\infty f(u) u^{s-1} du = \alpha^{-s} (\mathcal{M}f)(s)$$

The substitution is a C^1 diffeomorphism of $(0, \infty)$ with itself, so it preserves absolute convergence: the integral on the left converges absolutely if and only if the integral on the right does, which holds precisely for $a < \sigma < b$. Hence the fundamental strip is unchanged.

2. Fix $\mu \in \mathbb{C}$. The integrand of $\mathcal{M}(t^\mu f(t))$ at the parameter s is $t^\mu f(t) t^{s-1} = f(t) t^{(s+\mu)-1}$, which is exactly the integrand of $\mathcal{M}f$ at the parameter $s + \mu$. Therefore the integrals are equal whenever either converges:

$$(\mathcal{M} t^\mu f(t))(s) = \int_0^\infty f(t) t^{(s+\mu)-1} dt = (\mathcal{M}f)(s + \mu)$$

Absolute convergence of the right-hand integral holds for $a < \operatorname{Re}(s + \mu) < b$, that is, for $a - \operatorname{Re}(\mu) < \sigma < b - \operatorname{Re}(\mu)$, which is the asserted fundamental strip.

3. Substitute $t = e^{-x}$, so that $x = -\log t$, $dt = -e^{-x} dx$, and $t^{s-1} = e^{-(s-1)x}$. As t decreases from ∞ to 0 , the variable x increases from $-\infty$ to $+\infty$, so the limits reverse and absorb the sign of dt :

$$(\mathcal{M}f)(s) = \int_0^\infty f(t) t^{s-1} dt = \int_{-\infty}^\infty f(e^{-x}) e^{-(s-1)x} e^{-x} dx = \int_{-\infty}^\infty f(e^{-x}) e^{-sx} dx$$

With $g(x) = f(e^{-x})$ this is $\int_{-\infty}^\infty g(x) e^{-sx} dx$, the two-sided Laplace transform of g evaluated at s . The substitution $t = e^{-x}$ is again a C^1 diffeomorphism of $(0, \infty)$ onto $\mathbb{R}$, so absolute convergence is preserved and the identity holds throughout the fundamental strip. Writing $s = c + i\xi$ splits the exponential as $e^{-sx} = e^{-cx} e^{-i\xi x}$, and grouping the real exponential with g gives

$$(\mathcal{M}f)(c + i\xi) = \int_{-\infty}^\infty (e^{-cx} g(x)) e^{-i\xi x} dx$$

which is the Fourier transform of $x \mapsto e^{-cx} g(x)$ under the convention $\hat{h}(\xi) = \int_{-\infty}^\infty h(x) e^{-i\xi x} dx$, as we sought to show.

The third property is the structural heart of the transform. The multiplicative group $(0, \infty)$ is isomorphic, as a topological group, to the additive group $\mathbb{R}$ via the logarithm $t \mapsto -\log t = x$, and under this isomorphism Haar measure dt/t on $(0, \infty)$ becomes Lebesgue measure dx on $\mathbb{R}$. The Mellin transform is what the Fourier–Laplace transform on $\mathbb{R}$ becomes when transported back to $(0, \infty)$ through this isomorphism, with the complex parameter s playing the role of the (complexified) frequency variable. Every property of the Mellin transform is, after this change of variables, a property of the Fourier or Laplace transform; the scaling identity $(\mathcal{M}f(\alpha \cdot))(s) = \alpha^{-s} (\mathcal{M}f)(s)$, for instance, is the multiplicative form of the rule that translation by $\log \alpha$ in x multiplies the Fourier transform by a character.

This isomorphism does more than supply intuition: it converts the inversion problem for the Mellin transform into the inversion problem for the Fourier transform, which is settled by the classical Fourier inversion theorem. The result is the Mellin inversion formula.

Theorem 2.4.4: Mellin Inversion

Let $f: (0, \infty) \rightarrow \mathbb{C}$ be continuous, and let $c \in \mathbb{R}$ lie in the fundamental strip of f . Suppose that

1. the function $t \mapsto t^c f(t)$ is integrable on $(0, \infty)$ with respect to the Haar measure dt/t , that is, $\int_0^\infty |f(t)| t^c \frac{dt}{t} < \infty$; and
2. the function $\xi \mapsto (\mathcal{M}f)(c + i\xi)$ is integrable on $\mathbb{R}$, that is, $\int_{-\infty}^\infty |(\mathcal{M}f)(c + i\xi)| d\xi < \infty$.

Then for every $t > 0$,

$$f(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} (\mathcal{M}f)(s) t^{-s} ds$$

where the integral is taken over the vertical line $\operatorname{Re}(s) = c$.

Proof:

The proof transports the statement to $\mathbb{R}$ by the substitution $t = e^{-x}$ of proposition 2.4.3 and applies the Fourier inversion theorem there.

Step 1: Pass to the additive variable. Set $g(x) = f(e^{-x})$ for $x \in \mathbb{R}$ and define

$$G(x) = e^{-cx} g(x) = e^{-cx} f(e^{-x})$$

Since f is continuous on $(0, \infty)$ and $x \mapsto e^{-x}$ is continuous from $\mathbb{R}$ onto $(0, \infty)$, the function G is continuous on $\mathbb{R}$. Hypothesis (1) states that G is integrable: substituting $t = e^{-x}$ (so $dt/t = -dx$, and the orientation reversal cancels the sign) in the change-of-variables identity for the Haar measure gives

$$\int_{-\infty}^\infty |G(x)| dx = \int_{-\infty}^\infty |f(e^{-x})| e^{-cx} dx = \int_0^\infty |f(t)| t^c \frac{dt}{t} < \infty$$

so $G \in L^1(\mathbb{R})$.

Step 2: Identify the Fourier transform of G . By proposition 2.4.3(3), for $s = c + i\xi$,

$$\widehat{G}(\xi) = \int_{-\infty}^\infty G(x) e^{-i\xi x} dx = \int_{-\infty}^\infty e^{-cx} g(x) e^{-i\xi x} dx = (\mathcal{M}f)(c + i\xi)$$

under the angular-frequency convention $\widehat{G}(\xi) = \int_{-\infty}^\infty G(x) e^{-i\xi x} dx$, which is the convention fixed for the Mellin variable in proposition 2.4.3(3). The Fourier inversion theorem [5] is stated in the alternative normalization $\widehat{G}_{2\pi}(\eta) = \int_{-\infty}^\infty G(x) e^{-2\pi i \eta x} dx$, carrying the 2π in the exponent and no prefactor on inversion. The two transforms are affine reparametrizations of one another, $\widehat{G}_{2\pi}(\eta) = \widehat{G}(2\pi\eta)$, so $\widehat{G}_{2\pi} \in L^1(\mathbb{R})$ if and only if $\widehat{G} \in L^1(\mathbb{R})$. Hypothesis (2) is therefore exactly the statement that $\widehat{G} \in L^1(\mathbb{R})$, equivalently $\widehat{G}_{2\pi} \in L^1(\mathbb{R})$.

Step 3: Apply Fourier inversion. The function G is continuous and lies in $L^1(\mathbb{R})$ (Step 1), and its Fourier transform $\widehat{G}_{2\pi}$ lies in $L^1(\mathbb{R})$ (Step 2). These are precisely the hypotheses of the Fourier

inversion theorem [5], which in this normalization asserts that under them

$$G(x) = \int_{-\infty}^{\infty} \widehat{G}_{2\pi}(\eta) e^{2\pi i \eta x} d\eta \quad \text{for all } x \in \mathbb{R}$$

Substituting $\eta = \xi/(2\pi)$, so that $d\eta = d\xi/(2\pi)$, $\widehat{G}_{2\pi}(\eta) = \widehat{G}(2\pi\eta) = \widehat{G}(\xi)$, and $e^{2\pi i \eta x} = e^{i\xi x}$, transforms this into the angular-convention inversion formula

$$G(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{G}(\xi) e^{i\xi x} d\xi \quad \text{for all } x \in \mathbb{R}$$

in which the prefactor $1/(2\pi)$ is the Jacobian of the reparametrization. (The integrability of $\widehat{G}$ guarantees the right-hand integral converges absolutely, and continuity of G promotes the conclusion from “almost every x ” to “every x ”; see [6] for the measure-theoretic version and [5] for the statement used here.)

Step 4: Return to the multiplicative variable. Substitute $\widehat{G}(\xi) = (\mathcal{M}f)(c + i\xi)$ and $G(x) = e^{-cx} f(e^{-x})$ into the inversion formula of Step 3:

$$e^{-cx} f(e^{-x}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} (\mathcal{M}f)(c + i\xi) e^{i\xi x} d\xi$$

Multiplying through by e^{cx} moves it inside the integral as $e^{cx} e^{i\xi x} = e^{(c+i\xi)x}$:

$$f(e^{-x}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} (\mathcal{M}f)(c + i\xi) e^{(c+i\xi)x} d\xi$$

Now set $t = e^{-x}$, so $e^x = t^{-1}$ and $e^{(c+i\xi)x} = t^{-(c+i\xi)} = t^{-s}$ with $s = c + i\xi$. Along the vertical line $\operatorname{Re}(s) = c$ the variable s satisfies $ds = i d\xi$, so $d\xi = ds/i$, and as ξ runs from $-\infty$ to $+\infty$ the point s runs from $c - i\infty$ to $c + i\infty$. Therefore

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} (\mathcal{M}f)(c + i\xi) t^{-s} d\xi = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} (\mathcal{M}f)(s) t^{-s} ds$$

the factor $1/i$ converting $d\xi$ into ds and producing the $1/(2\pi i)$ normalization, completing the proof.

Two features of the inversion formula deserve comment. First, the integral runs up a vertical line $\operatorname{Re}(s) = c$ inside the fundamental strip, and the value of $f(t)$ is independent of which line c is chosen, since the left-hand side does not mention c ; this freedom to slide the contour within the strip is what makes the formula useful, because shifting the line of integration across a pole of $(\mathcal{M}f)(s)$ picks up a residue, and these residues are the main terms in asymptotic expansions. Second, the normalization $1/(2\pi i)$ together with the factor t^{-s} is exactly the data of a contour integral in the sense of [13]: the Mellin inversion formula presents $f(t)$ as a contour integral of its transform, and the entire apparatus of residue calculus applies to its evaluation. This is the precise sense in which the Mellin transform “converts an arithmetic sum into a contour integral,” and it is the mechanism underlying Perron’s formula in section 2.5, where $(\mathcal{M}f)(s)$ is replaced by a Dirichlet series and the contour integral recovers a summatory function.

The transform is now applied to the object of central interest. The Dirichlet series $\zeta(s) = \sum_{n \geq 1} n^{-s}$ of definition 2.1.1 converges on the half-plane $\operatorname{Re}(s) > 1$, where its abscissa of absolute convergence is $\sigma_a = 1$ (definition 2.1.3). Multiplying $\zeta(s)$ by $\Gamma(s)$ produces a single integral over $(0, \infty)$ whose integrand is an elementary function of t , with no reference to the individual terms n^{-s} .

Theorem 2.4.5: Integral Representation of the Riemann Zeta Function

For every s with $\operatorname{Re}(s) > 1$,

$$\Gamma(s) \zeta(s) = \int_0^\infty \frac{t^{s-1}}{e^t - 1} dt$$

the integral converging absolutely on this half-plane.

Proof:

Fix s with $\sigma = \operatorname{Re}(s) > 1$.

Step 1: Expand the integrand as a geometric series. For $t > 0$ one has $0 < e^{-t} < 1$, so the geometric series in e^{-t} converges:

$$\frac{1}{e^t - 1} = \frac{e^{-t}}{1 - e^{-t}} = \sum_{n=1}^{\infty} e^{-nt}$$

This is the sum of the geometric series with first term e^{-t} and ratio $e^{-t} \in (0, 1)$, valid for every $t > 0$. Multiplying by t^{s-1} gives the pointwise identity

$$\frac{t^{s-1}}{e^t - 1} = \sum_{n=1}^{\infty} t^{s-1} e^{-nt}$$

for each fixed $t > 0$.

Step 2: Evaluate the integral of a single term. For each $n \geq 1$, the substitution $u = nt$ in the gamma integral of example 2.4.2 gives, with $t = u/n$ and $dt = du/n$,

$$\int_0^\infty t^{s-1} e^{-nt} dt = \int_0^\infty \left(\frac{u}{n}\right)^{s-1} e^{-u} \frac{du}{n} = n^{-s} \int_0^\infty u^{s-1} e^{-u} du = \frac{\Gamma(s)}{n^s}$$

the integral $\int_0^\infty u^{s-1} e^{-u} du = \Gamma(s)$ converging absolutely because $\sigma > 1 > 0$ (example 2.4.2). In particular the same computation with s replaced by σ shows $\int_0^\infty t^{\sigma-1} e^{-nt} dt = \Gamma(\sigma) n^{-\sigma}$.

Step 3: Interchange summation and integration. The terms $t^{s-1} e^{-nt}$ are not all nonnegative (the factor t^{s-1} is complex when s is), so the interchange is justified on the absolute values, where Tonelli's theorem applies to the nonnegative integrand $(n, t) \mapsto |t^{s-1}| e^{-nt} = t^{\sigma-1} e^{-nt}$ over the product of counting measure on $\mathbb{Z}_{\geq 1}$ and Lebesgue measure on $(0, \infty)$. By Step 2 applied with σ in place of s ,

$$\sum_{n=1}^{\infty} \int_0^\infty t^{\sigma-1} e^{-nt} dt = \sum_{n=1}^{\infty} \frac{\Gamma(\sigma)}{n^\sigma} = \Gamma(\sigma) \zeta(\sigma) < \infty$$

the series converging because $\sigma > 1 = \sigma_a$ for ζ (definition 2.1.3). Tonelli's theorem [6] therefore guarantees that the function $(n, t) \mapsto t^{s-1} e^{-nt}$ is integrable for the product measure, and Fubini's

theorem [6] then permits the interchange of the sum and the integral for the (complex) integrand itself.

Step 4: Assemble. Using the pointwise identity of Step 1 and the justified interchange of Step 3,

$$\int_0^\infty \frac{t^{s-1}}{e^t - 1} dt = \int_0^\infty \sum_{n=1}^\infty t^{s-1} e^{-nt} dt = \sum_{n=1}^\infty \int_0^\infty t^{s-1} e^{-nt} dt = \sum_{n=1}^\infty \frac{\Gamma(s)}{n^s} = \Gamma(s) \zeta(s)$$

where the last equality factors the constant $\Gamma(s)$ out of the sum $\sum_n n^{-s} = \zeta(s)$, valid on $\operatorname{Re}(s) > 1$. The absolute convergence asserted in the statement is the finiteness computed in Step 3, completing the proof.

This integral representation is the analytic launch-point for everything that follows about $\zeta(s)$. The Dirichlet series $\sum n^{-s}$ converges only on $\operatorname{Re}(s) > 1$ and gives no direct access to the values of $\zeta(s)$ elsewhere; the integral $\int_0^\infty t^{s-1}/(e^t - 1) dt$, by contrast, has an integrand whose singularity structure can be analyzed by splitting the range at $t = 1$ and expanding $1/(e^t - 1)$ near the origin, where it behaves like $1/t$. That expansion separates a part that continues meromorphically to the whole plane from a part that is already entire, and this is exactly how the meromorphic continuation of $\zeta(s)$ to $\mathbb{C}$, with its single simple pole at $s = 1$, is produced in chapter 4. The same integral, after the contour is deformed into a Hankel loop around the positive real axis, yields the functional equation relating $\zeta(s)$ to $\zeta(1-s)$ in ???. The representation also exhibits the gamma factor $\Gamma(s)$ as the inseparable companion of $\zeta(s)$: the natural object is the product $\Gamma(s)\zeta(s)$, and it is this product, not $\zeta(s)$ alone, whose Mellin-integral form has elementary content. The appearance of $\Gamma(s)$ here is the first sign of the completed zeta function $\xi(s)$ that organizes the functional equation.

The alternating analogue of the representation replaces $e^t - 1$ by $e^t + 1$ and brings in the Dirichlet eta function, the alternating series studied in example 2.1.10. The computation is identical except for a sign that turns the geometric series into an alternating one, and the resulting integrand is better behaved at the origin.

Example 2.4.6: The Dirichlet Eta Function and the Alternating Representation

The Dirichlet eta function is $\eta(s) = \sum_{n=1}^\infty (-1)^{n+1} n^{-s}$, which by example 2.1.10 satisfies $\eta(s) = (1 - 2^{1-s})\zeta(s)$ on $\operatorname{Re}(s) > 1$. Its product with $\Gamma(s)$ admits the integral representation

$$(1 - 2^{1-s}) \zeta(s) \Gamma(s) = \eta(s) \Gamma(s) = \int_0^\infty \frac{t^{s-1}}{e^t + 1} dt \quad (\operatorname{Re}(s) > 1)$$

derived exactly as in theorem 2.4.5. For $t > 0$ the geometric series gives

$$\frac{1}{e^t + 1} = \frac{e^{-t}}{1 + e^{-t}} = \sum_{n=1}^\infty (-1)^{n+1} e^{-nt}$$

(the series with first term e^{-t} and ratio $-e^{-t}$, with $|-e^{-t}| < 1$). Multiplying by t^{s-1} and integrating term by term gives

$$\int_0^\infty \frac{t^{s-1}}{e^t + 1} dt = \sum_{n=1}^\infty (-1)^{n+1} \int_0^\infty t^{s-1} e^{-nt} dt = \sum_{n=1}^\infty (-1)^{n+1} \frac{\Gamma(s)}{n^s} = \Gamma(s) \eta(s)$$

using the single-term evaluation $\int_0^\infty t^{s-1} e^{-nt} dt = \Gamma(s) n^{-s}$ from Step 2 of theorem 2.4.5. The interchange of summation and integration is justified by absolute convergence: the absolute values

$\sum_n \int_0^\infty t^{\sigma-1} e^{-nt} dt = \Gamma(\sigma)\zeta(\sigma)$ are summable for $\sigma = \operatorname{Re}(s) > 1$, so the same Tonelli–Fubini argument as in theorem 2.4.5 applies ([6]).

The contrast with the $1/(e^t - 1)$ integrand is the point of the example. Near $t = 0^+$ one has $e^t + 1 \rightarrow 2$, so $1/(e^t + 1) \rightarrow 1/2$ is bounded, and the integrand $t^{s-1}/(e^t + 1)$ is comparable to $\frac{1}{2}t^{\sigma-1}$, which is integrable on $(0, 1]$ for every $\sigma > 0$, not only for $\sigma > 1$. At $t = \infty$ the integrand decays like $t^{\sigma-1}e^{-t}$ for every σ . Hence the integral $\int_0^\infty t^{s-1}/(e^t + 1) dt$ converges and defines an analytic function on the entire half-plane $\operatorname{Re}(s) > 0$, even though the series defining $\eta(s)$ was used only on $\operatorname{Re}(s) > 1$. Foreshadowing the continuation of chapter 4: this convergent integral provides the meromorphic continuation of $\eta(s)\Gamma(s)$ to $\operatorname{Re}(s) > 0$, and dividing by the elementary factor $(1 - 2^{1-s})\Gamma(s)$ continues $\zeta(s)$ past the line $\operatorname{Re}(s) = 1$. The pole of $\zeta(s)$ at $s = 1$ survives this division because $(1 - 2^{1-s})$ vanishes there, while the factor $1/(e^t + 1)$ removed the obstruction at the origin that the $1/(e^t - 1)$ integrand carried.

2.5 Perron's Formula

The transition from a Dirichlet series to the arithmetic information it encodes runs in two directions. Forming $F(s) = \sum_{n \geq 1} a_n n^{-s}$ from a sequence (a_n) packages the coefficients into an analytic function; the analyticity, the location of poles, and the residues of F are the analytic data. Perron's formula runs the construction backward: it recovers the partial sum $\sum_{n \leq x} a_n$ of the coefficients from a contour integral of F . The kernel that performs the recovery is x^s/s , integrated up a vertical line to the right of the region where F converges. The effect is that analytic facts about F , principally the behavior of F near its singularities, are converted into asymptotic statements about the coefficient sum, since shifting the contour of integration leftward across a pole of $F(s)x^s/s$ contributes a residue, and these residues are the main terms in the asymptotics.

The instance that organizes this chapter and the next is $F(s) = -\zeta'(s)/\zeta(s) = \sum_{n \geq 1} \Lambda(n)n^{-s}$ (theorem 2.2.6). Here the coefficient sum is the Chebyshev function $\psi(x) = \sum_{n \leq x} \Lambda(n)$, and Perron's formula expresses $\psi(x)$ as a contour integral of $-\zeta'(s)/\zeta(s) \cdot x^s/s$. The single pole of $\zeta(s)$ at $s = 1$ produces a pole of the integrand with residue x , which is the source of the leading term $\psi(x) \sim x$ of the Prime Number Theorem; the zeros of $\zeta(s)$ inside the critical strip produce the oscillatory corrections. Perron's formula is therefore the gateway through which the analytic study of $\zeta(s)$ in chapter 4 becomes the analytic proof of the Prime Number Theorem in chapter 5. The formula itself is the Mellin-type inversion of theorem 2.4.4 specialized to the kernel x^s/s : where Mellin inversion recovers a function from its transform along a vertical line, Perron inversion recovers a step function (the summatory function) from the Dirichlet series, and the discontinuity of the step function at each integer is exactly the jump produced by the kernel x^s/s .

The entire formula rests on a single contour integral, the value of the kernel x^s/s integrated up a vertical line. This integral is discontinuous in the ratio it depends on, jumping from 0 to 1 as the ratio crosses 1, and it is this discontinuity that reproduces the indicator $n \leq x$ inside the coefficient sum.

Lemma 2.5.1: The Discontinuous Integral

Let $c > 0$. For $y > 0$ define

$$I(y) = \frac{1}{2\pi i} \int_{(c)} \frac{y^s}{s} ds = \lim_{T \rightarrow \infty} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \frac{y^s}{s} ds$$

the integral taken up the vertical line $\operatorname{Re}(s) = c$. Then

$$I(y) = \begin{cases} 1 & y > 1 \\ \frac{1}{2} & y = 1 \\ 0 & 0 < y < 1 \end{cases}$$

Moreover, writing $I(y, T) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} y^s/s ds$ for the truncation to the segment $[c-iT, c+iT]$ and $\delta(y) = 1$ for $y > 1$, $\delta(y) = 0$ for $0 < y < 1$, the truncation satisfies, for every $T > 0$,

$$|I(y, T) - \delta(y)| \leq y^c \min\left(1, \frac{1}{\pi T |\log y|}\right) \quad (y \neq 1)$$

and, in the boundary case $y = 1$,

$$\left|I(1, T) - \frac{1}{2}\right| \leq \frac{c}{\pi T}$$

Proof:

Throughout, $s = \sigma + it$ and the residue theorem is applied in the form established in [13]. The integrand y^s/s is meromorphic on $\mathbb{C}$ with a single simple pole at $s = 0$, where its residue is $\lim_{s \rightarrow 0} s \cdot y^s/s = y^0 = 1$.

Step 1: The boundary case $y = 1$. Here $y^s = 1$, so the truncated integral is computed directly. Parametrizing $s = c + it$ with $t \in [-T, T]$ and $ds = i dt$,

$$I(1, T) = \frac{1}{2\pi i} \int_{-T}^T \frac{i dt}{c + it} = \frac{1}{2\pi} \int_{-T}^T \frac{c - it}{c^2 + t^2} dt$$

The imaginary part of the integrand, $-t/(c^2 + t^2)$, is odd in t and integrates to zero over the symmetric interval. The real part gives

$$I(1, T) = \frac{1}{2\pi} \int_{-T}^T \frac{c}{c^2 + t^2} dt = \frac{1}{2\pi} \cdot 2 \arctan \frac{T}{c} = \frac{1}{\pi} \arctan \frac{T}{c}$$

As $T \rightarrow \infty$ this tends to $\frac{1}{\pi} \cdot \frac{\pi}{2} = \frac{1}{2}$, establishing $I(1) = \frac{1}{2}$. For the error, $\frac{1}{2} - \frac{1}{\pi} \arctan(T/c) = \frac{1}{\pi}(\frac{\pi}{2} - \arctan(T/c)) = \frac{1}{\pi} \arctan(c/T)$, and since $\arctan u \leq u$ for $u \geq 0$,

$$\left|I(1, T) - \frac{1}{2}\right| = \frac{1}{\pi} \arctan \frac{c}{T} \leq \frac{c}{\pi T}$$

which is the boundary estimate.

Step 2: The sharp truncation bound for $y \neq 1$. Fix $y \neq 1$ and $T > 0$. The estimate to prove is $|I(y, T) - \delta(y)| \leq y^c / (\pi T |\log y|)$. The argument depends on the sign of $\log y$ through which direction the contour is closed.

Suppose first $y > 1$, so $\log y > 0$ and $|y^s| = y^\sigma$ decays as $\sigma \rightarrow -\infty$. For $U > 0$ consider the rectangle $\mathcal{R}_U$ with vertices $c - iT$, $c + iT$, $-U + iT$, $-U - iT$, traversed counterclockwise. It encloses the pole at $s = 0$, so by the residue theorem

$$\frac{1}{2\pi i} \oint_{\mathcal{R}_U} \frac{y^s}{s} ds = 1$$

The boundary $\partial\mathcal{R}_U$ consists of the right edge (the segment from $c - iT$ to $c + iT$, contributing $I(y, T)$), the left edge at $\operatorname{Re}(s) = -U$, and the two horizontal edges at $\operatorname{Im}(s) = \pm T$. On the left edge, $|y^s/s| \leq y^{-U}/U \rightarrow 0$ as $U \rightarrow \infty$ because $y > 1$, so its contribution vanishes in the limit. On each horizontal edge, $s = \sigma \pm iT$ with σ running from $-U$ to c , and $|s| \geq T$, so $|y^s/s| \leq y^\sigma/T$; integrating in σ over $(-\infty, c]$ bounds each horizontal contribution by

$$\frac{1}{2\pi} \int_{-\infty}^c \frac{y^\sigma}{T} d\sigma = \frac{1}{2\pi T} \cdot \frac{y^c}{\log y}$$

using $\int_{-\infty}^c y^\sigma d\sigma = y^c / \log y$ (valid since $\log y > 0$). Letting $U \rightarrow \infty$, the residue identity reads $1 = I(y, T) + (\text{two horizontal contributions})$, whence

$$|I(y, T) - 1| \leq 2 \cdot \frac{1}{2\pi T} \cdot \frac{y^c}{\log y} = \frac{y^c}{\pi T \log y} = \frac{y^c}{\pi T |\log y|}$$

since $\log y = |\log y|$ here. This is the claim for $y > 1$.

Now suppose $0 < y < 1$, so $\log y < 0$ and $|y^s| = y^\sigma$ decays as $\sigma \rightarrow +\infty$. The same argument is run with the rectangle closed to the right: vertices $c - iT$, $c + iT$, $U + iT$, $U - iT$. This rectangle does not contain $s = 0$, so the closed integral is zero. The right edge at $\operatorname{Re}(s) = U$ contributes at most $y^U/U \rightarrow 0$. The horizontal edges, with σ running from c to U , are bounded by $\frac{1}{2\pi T} \int_c^\infty y^\sigma d\sigma = \frac{1}{2\pi T} \cdot y^c / |\log y|$ each, using $\int_c^\infty y^\sigma d\sigma = -y^c / \log y = y^c / |\log y|$. Hence

$$|I(y, T) - 0| \leq \frac{y^c}{\pi T |\log y|}$$

completing the sharp bound for both signs.

Step 3: The trivial bound $|I(y, T) - \delta(y)| \leq y^c$ for $y \neq 1$. The sharp bound degrades as $|\log y| \rightarrow 0$, so a second estimate independent of $|\log y|$ is needed to control y near 1. This is obtained by deforming the vertical segment to a circular arc on which $\operatorname{Re}(s) \geq c$ (for $y < 1$) or which encircles the pole (for $y > 1$). Set $R = \sqrt{c^2 + T^2}$, so that the endpoints $c \pm iT$ lie on the circle $|s| = R$, and write $\theta_0 = \arctan(T/c) \in (0, \pi/2)$, the argument of $c + iT$.

Suppose $0 < y < 1$. Let Γ be the arc of $|s| = R$ from $c + iT$ to $c - iT$ passing through $s = R$ on the right, that is, $s = R e^{i\theta}$ with θ decreasing from θ_0 to $-\theta_0$. The arc Γ and the segment from $c - iT$ to $c + iT$ together bound a region lying in $\operatorname{Re}(s) \geq c > 0$, which does not contain the pole $s = 0$; since the segment and Γ share endpoints, Cauchy's theorem gives $I(y, T) = \frac{1}{2\pi i} \int_\Gamma y^s/s ds$. On Γ , $\operatorname{Re}(s) = R \cos \theta$ with $|\theta| \leq \theta_0$, and $\cos \theta \geq \cos \theta_0 = c/R$, so $R \cos \theta \geq c$; because $0 < y < 1$, this forces $|y^s| = y^{R \cos \theta} \leq y^c$. With $|s| = R$ and $|ds| = R d\theta$,

$$|I(y, T)| = \frac{1}{2\pi} \left| \int_{-\theta_0}^{\theta_0} \frac{y^{R e^{i\theta}}}{R e^{i\theta}} R e^{i\theta} i d\theta \right| \leq \frac{1}{2\pi} \int_{-\theta_0}^{\theta_0} y^{R \cos \theta} d\theta \leq \frac{1}{2\pi} \cdot 2\theta_0 \cdot y^c \leq \frac{y^c}{2}$$

using $2\theta_0 \leq \pi$. Since $\delta(y) = 0$ here, $|I(y, T) - \delta(y)| \leq y^c/2 \leq y^c$.

Suppose now $y > 1$. Let Γ' be the arc of $|s| = R$ from $c + iT$ to $c - iT$ passing through $s = -R$ on the left, that is, $s = Re^{i\theta}$ with θ increasing from θ_0 to $2\pi - \theta_0$. The segment (upward) together with Γ' bounds a region containing $s = 0$ and is traversed counterclockwise, so the residue theorem gives $I(y, T) + \frac{1}{2\pi i} \int_{\Gamma'} y^s/s ds = 1$, hence $I(y, T) - 1 = -\frac{1}{2\pi i} \int_{\Gamma'} y^s/s ds$. On Γ' , $\operatorname{Re}(s) = R \cos \theta$ with $\theta \in [\theta_0, 2\pi - \theta_0]$, so $\cos \theta \leq \cos \theta_0 = c/R$, giving $R \cos \theta \leq c$; because $y > 1$, this forces $|y^s| = y^{R \cos \theta} \leq y^c$. The angular length of Γ' is $2\pi - 2\theta_0 \leq 2\pi$, so

$$|I(y, T) - 1| \leq \frac{1}{2\pi} \int_{\theta_0}^{2\pi - \theta_0} y^{R \cos \theta} d\theta \leq \frac{1}{2\pi} (2\pi - 2\theta_0) y^c \leq y^c$$

Since $\delta(y) = 1$ here, this is $|I(y, T) - \delta(y)| \leq y^c$.

Step 4: Combine and pass to the limit. For $y \neq 1$, Steps 2 and 3 give $|I(y, T) - \delta(y)| \leq y^c/(\pi T |\log y|)$ and $|I(y, T) - \delta(y)| \leq y^c$ respectively; the smaller of the two is the asserted bound $y^c \min(1, 1/(\pi T |\log y|))$. Letting $T \rightarrow \infty$ in this bound, with $y \neq 1$ fixed so that $1/(\pi T |\log y|) \rightarrow 0$, gives $I(y, T) \rightarrow \delta(y)$, which establishes $I(y) = 1$ for $y > 1$ and $I(y) = 0$ for $0 < y < 1$. Together with $I(1) = \frac{1}{2}$ from Step 1, all assertions are proved, completing the proof.

The lemma is the analytic mechanism behind everything in this section. The kernel y^s/s has a pole at $s = 0$ with residue 1, and integrating up a vertical line either captures that residue (when the contour can be closed to the left, which the decay of y^s permits exactly when $y > 1$) or does not (when it must be closed to the right, which is forced when $y < 1$). The jump of $I(y)$ from 0 to 1 as y crosses 1 is therefore the residue of y^s/s at the origin, switched on or off by the direction in which the integrand decays. The truncation estimate is the quantitative form: the error in replacing the limit by a finite integral is controlled by $1/(T |\log y|)$, so the integral resolves the indicator sharply except for ratios y extremely close to 1, where the second bound y^c takes over. This dichotomy, sharp resolution away from the diagonal $y = 1$ and a crude bound near it, is exactly what governs the error in the truncated form of Perron's formula below.

Applying the lemma term by term to a Dirichlet series converts its coefficient sum into a contour integral. The ratio y becomes x/n , so $I(x/n) = 1$ when $n < x$, $I(x/n) = 0$ when $n > x$, and the indicator $\sum_{n \leq x}$ emerges from the integral.

Theorem 2.5.2: Perron's Formula

Let $F(s) = \sum_{n \geq 1} a_n n^{-s}$ be a Dirichlet series with abscissa of absolute convergence σ_a (definition 2.1.3). Let $c > \max(0, \sigma_a)$ and $x > 0$. If x is not an integer, then

$$\sum_{n \leq x} a_n = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s) \frac{x^s}{s} ds$$

the integral being taken up the line $\operatorname{Re}(s) = c$ and converging as the symmetric limit $\lim_{T \rightarrow \infty} \int_{c-iT}^{c+iT}$. If x is an integer, the same identity holds with the left side replaced by the half-weighted sum $\sum_{n < x} a_n + \frac{1}{2}a_x$.

Proof:

Write $\sigma = \operatorname{Re}(s) = c$ throughout, and recall $I(y, T) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} y^s/s \, ds$ from lemma 2.5.1.

Step 1: Interchange sum and integral on the truncated contour. Fix $T > 0$. On the segment $[c - iT, c + iT]$ the series for F may be inserted and the order of summation and integration exchanged:

$$\frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s) \frac{x^s}{s} \, ds = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} \sum_{n=1}^{\infty} a_n \left(\frac{x}{n}\right)^s \frac{ds}{s} = \sum_{n=1}^{\infty} a_n \cdot I\left(\frac{x}{n}, T\right)$$

where $F(s)x^s = \sum_n a_n n^{-s} x^s = \sum_n a_n (x/n)^s$. The interchange is justified by absolute convergence over the product of counting measure on $\mathbb{Z}_{\geq 1}$ and Lebesgue measure on the finite segment: the modulus of the (n, s) -integrand is $|a_n|(x/n)^\sigma/|s|$, and

$$\sum_{n=1}^{\infty} \int_{c-iT}^{c+iT} \frac{|a_n|(x/n)^\sigma}{|s|} |ds| = \left(x^\sigma \sum_{n=1}^{\infty} \frac{|a_n|}{n^\sigma} \right) \left(\int_{-T}^T \frac{dt}{\sqrt{c^2 + t^2}} \right)$$

is finite: the first factor converges because $\sigma = c > \sigma_a$ (definition 2.1.3), and the second is the finite integral of a bounded function over a bounded interval. Fubini's theorem [6] therefore permits the interchange.

Step 2: Split off the finitely many terms with $n \leq x$. The sum $\sum_n a_n I(x/n, T)$ is split according to whether $n \leq x$ or $n > x$. Since x is fixed and finite, only finitely many indices satisfy $n \leq x$, so

$$\sum_{n=1}^{\infty} a_n I\left(\frac{x}{n}, T\right) = \sum_{n \leq x} a_n I\left(\frac{x}{n}, T\right) + \sum_{n > x} a_n I\left(\frac{x}{n}, T\right)$$

In the first sum, a finite sum, each term converges as $T \rightarrow \infty$ by lemma 2.5.1: for $n < x$ one has $x/n > 1$ and $I(x/n, T) \rightarrow 1$; if x is an integer the single term $n = x$ has $x/n = 1$ and $I(1, T) \rightarrow \frac{1}{2}$. Therefore

$$\lim_{T \rightarrow \infty} \sum_{n \leq x} a_n I\left(\frac{x}{n}, T\right) = \begin{cases} \sum_{n \leq x} a_n & x \notin \mathbb{Z} \\ \sum_{n < x} a_n + \frac{1}{2} a_x & x \in \mathbb{Z} \end{cases}$$

the cases differing only in the treatment of the boundary index $n = x$.

Step 3: The tail vanishes in the limit. For the tail $\sum_{n > x} a_n I(x/n, T)$, each index has $x/n < 1$, so $\delta(x/n) = 0$ and, by the trivial bound in lemma 2.5.1, $|I(x/n, T)| \leq (x/n)^c$ for every T . This dominating bound is summable, since

$$\sum_{n > x} |a_n| \left(\frac{x}{n}\right)^c = x^c \sum_{n > x} \frac{|a_n|}{n^c} < \infty$$

again because $c > \sigma_a$. For each fixed $n > x$, lemma 2.5.1 gives $I(x/n, T) \rightarrow \delta(x/n) = 0$ as $T \rightarrow \infty$, because $x/n < 1$ is fixed and bounded away from 1, so $1/(\pi T |\log(x/n)|) \rightarrow 0$. The

dominated convergence theorem for series [6] (counting measure on $\{n > x\}$, with dominating summable bound $|a_n|(x/n)^c$) then yields

$$\lim_{T \rightarrow \infty} \sum_{n > x} a_n I\left(\frac{x}{n}, T\right) = \sum_{n > x} a_n \cdot 0 = 0$$

Step 4: Assemble. Combining Steps 1 through 3, the symmetric limit of the truncated contour integral exists and equals

$$\lim_{T \rightarrow \infty} \frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s) \frac{x^s}{s} ds = \begin{cases} \sum_{n \leq x} a_n & x \notin \mathbb{Z} \\ \sum_{n < x} a_n + \frac{1}{2}a_x & x \in \mathbb{Z} \end{cases}$$

which is the assertion of the theorem, completing the proof.

Perron's formula is the precise statement that the analytic behavior of F controls the asymptotics of its coefficient sum. The left side is a step function of x , jumping by a_n at each integer n ; the right side is a contour integral whose integrand $F(s)x^s/s$ inherits all the singularities of F , together with the pole of the kernel at $s = 0$. The contour runs up a vertical line in the region of absolute convergence, where F is analytic and the integral has no singularities to contend with. The information is extracted by moving the contour to the left, into the region where F has poles: each pole crossed contributes its residue, and the integral over the shifted contour is smaller because the factor $x^s = x^\sigma e^{it \log x}$ has smaller modulus once σ has decreased. The residue at a pole $s = \rho$ of F contributes a term proportional to x^ρ/ρ , so a pole at $s = 1$ gives a main term of size x , a pole at $s = 0$ gives a constant, and poles further left give lower-order terms. The asymptotic expansion of $\sum_{n \leq x} a_n$ is, in this way, read off from the poles of F and their residues; the rightmost pole supplies the dominant term, and the remaining poles and the bound on the shifted contour supply the error. This is the mechanism that the analytic proof of the Prime Number Theorem makes quantitative in chapter 5.

The formula as stated is not yet usable for that program, because the contour runs to infinity in both directions and the integral over the full line is only conditionally convergent. Quantitative work requires truncating the contour to a finite height T and controlling the resulting error explicitly. The truncated form is the workhorse of the subject.

Theorem 2.5.3: The Truncated Perron Formula

Let $F(s) = \sum_{n \geq 1} a_n n^{-s}$ have abscissa of absolute convergence σ_a , let $c > \max(0, \sigma_a)$, and let $x \geq 1$ and $T \geq 1$. Then

$$\sum_{n \leq x} a_n = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s) \frac{x^s}{s} ds + R$$

where, if x is not an integer, the error term satisfies

$$|R| \leq \sum_{n=1}^{\infty} |a_n| \left(\frac{x}{n}\right)^c \min\left(1, \frac{1}{T|\log(x/n)|}\right)$$

and if x is an integer the same bound holds for the half-weighted sum $\sum_{n < x} a_n + \frac{1}{2}a_x$ in place of $\sum_{n \leq x} a_n$, augmented by the single near-diagonal term

$$|R| \leq \sum_{\substack{n=1 \\ n \neq x}}^{\infty} |a_n| \left(\frac{x}{n}\right)^c \min\left(1, \frac{1}{T|\log(x/n)|}\right) + |a_x| \frac{c}{\pi T}$$

Proof:

The starting point is the exact finite- T identity established in Step 1 of the proof of theorem 2.5.2, valid for every $T \geq 1$ with no passage to a limit:

$$\frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s) \frac{x^s}{s} ds = \sum_{n=1}^{\infty} a_n I\left(\frac{x}{n}, T\right)$$

the interchange of sum and integral being justified exactly as there (absolute convergence over the product of counting measure and the finite segment, using $c > \sigma_a$). The error R is the difference between the coefficient sum and this integral.

Case $x \notin \mathbb{Z}$. Here $x/n \neq 1$ for every n , so $\delta(x/n) = 1$ when $n < x$ and $\delta(x/n) = 0$ when $n > x$; equivalently $\delta(x/n) = \mathbb{1}[n \leq x]$, and therefore $\sum_{n \leq x} a_n = \sum_n a_n \delta(x/n)$. Subtracting the identity above,

$$R = \sum_{n \leq x} a_n - \frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s) \frac{x^s}{s} ds = \sum_{n=1}^{\infty} a_n \left(\delta\left(\frac{x}{n}\right) - I\left(\frac{x}{n}, T\right) \right)$$

The rearrangement is legitimate because both series converge absolutely: $\sum_n |a_n| |\delta(x/n)|$ is the finite sum over $n \leq x$, and $\sum_n |a_n| |I(x/n, T)|$ is dominated by $\sum_n |a_n| (x/n)^c = x^c \sum_n |a_n| n^{-c} < \infty$ via the trivial bound of lemma 2.5.1. Taking absolute values and applying the truncation estimate of lemma 2.5.1 to each term, with $|\delta(x/n) - I(x/n, T)| \leq (x/n)^c \min(1, 1/(\pi T |\log(x/n)|))$,

$$|R| \leq \sum_{n=1}^{\infty} |a_n| \left(\frac{x}{n}\right)^c \min\left(1, \frac{1}{\pi T |\log(x/n)|}\right) \leq \sum_{n=1}^{\infty} |a_n| \left(\frac{x}{n}\right)^c \min\left(1, \frac{1}{T |\log(x/n)|}\right)$$

the last inequality discarding the factor $1/\pi \leq 1$ inside the second argument of the minimum to give the form stated. This proves the non-integer case.

Case $x \in \mathbb{Z}$. Now exactly one index has $x/n = 1$, namely $n = x$, and for it δ takes the boundary value $\frac{1}{2}$ consistent with the half-weighted sum. For every other index $\delta(x/n) = \mathbb{1}[n < x]$, so the half-weighted sum is $\sum_{n < x} a_n + \frac{1}{2}a_x = \sum_n a_n \delta(x/n)$ with the convention $\delta(1) = \frac{1}{2}$. As before,

$$R = \sum_{n=1}^{\infty} a_n \left(\delta\left(\frac{x}{n}\right) - I\left(\frac{x}{n}, T\right) \right)$$

The single term $n = x$ contributes $a_x(\frac{1}{2} - I(1, T))$, bounded in modulus by $|a_x| c/(\pi T)$ by the boundary estimate of lemma 2.5.1. Every other term has $x/n \neq 1$ and is bounded as in the non-integer case. Hence

$$|R| \leq \sum_{\substack{n=1 \\ n \neq x}}^{\infty} |a_n| \left(\frac{x}{n}\right)^c \min\left(1, \frac{1}{T|\log(x/n)|}\right) + |a_x| \frac{c}{\pi T}$$

as required.

The structure of the error bound is what makes the truncated formula effective. The minimum has two regimes. For n far from x , the ratio x/n is bounded away from 1, $|\log(x/n)|$ is of order 1 or larger, and the term contributes at most $|a_n|(x/n)^c/T$, which is small because of the factor $1/T$. For n close to x , where $|\log(x/n)|$ is small and the first regime would give a useless bound, the minimum switches to the constant 1, and the term contributes at most $|a_n|(x/n)^c$; there are few such n , and the convergence factor $(x/n)^c$ keeps the total controlled. In applications the parameters c and T are chosen as functions of x to balance the contour integral against this error, and the explicit dependence on T in the bound is precisely what allows the optimization. The near-diagonal term in the integer case, of size $|a_x|/T$, is the price of evaluating the formula at a jump of the step function, and it is the reason the cleanest statements are made for non-integer x or after the half-weighting convention is adopted.

The application that motivates the entire development is the Chebyshev function ψ . Taking $a_n = \Lambda(n)$, the von Mangoldt function (definition 1.1.8), the generating Dirichlet series is the logarithmic derivative of ζ computed in theorem 2.2.6, and Perron's formula presents $\psi(x)$ as a contour integral of it.

Example 2.5.4: Perron's Formula for $\psi(x)$

Let $a_n = \Lambda(n)$, so that the summatory function is the Chebyshev function $\psi(x) = \sum_{n \leq x} \Lambda(n)$. By theorem 2.2.6, the generating Dirichlet series is

$$F(s) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} = -\frac{\zeta'(s)}{\zeta(s)}$$

valid on $\operatorname{Re}(s) > 1$, where this is also the abscissa of absolute convergence: $0 \leq \Lambda(n) \leq \log n$ gives $\sum_n \Lambda(n)n^{-\sigma} \leq \sum_n (\log n)n^{-\sigma} < \infty$ for every $\sigma > 1$, while at $\sigma = 1$ the series $\sum_n \Lambda(n)/n$ diverges, so $\sigma_a = 1$. Choosing any $c > 1$ and any non-integer $x > 0$, theorem 2.5.2 yields

$$\psi(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \left(-\frac{\zeta'(s)}{\zeta(s)} \right) \frac{x^s}{s} ds$$

and the truncated form of theorem 2.5.3 gives the same integral over $[c - iT, c + iT]$ together with an explicit error governed by $\sum_n \Lambda(n)(x/n)^c \min(1, 1/(T|\log(x/n)|))$. This integral is the analytic

starting point of the proof of the Prime Number Theorem. The integrand $-\zeta'(s)/\zeta(s) \cdot x^s/s$ carries a pole at $s = 1$ coming from the simple pole of $\zeta(s)$ there: since $-\zeta'(s)/\zeta(s) \sim 1/(s-1)$ near $s = 1$, the residue of the integrand at $s = 1$ is $x^1/1 = x$, which is the leading term $\psi(x) \sim x$ once the contour is shifted past this pole. The kernel contributes a further pole at $s = 0$, and the nontrivial zeros of $\zeta(s)$ in the critical strip contribute poles of $-\zeta'(s)/\zeta(s)$ whose residues produce the oscillatory terms $-x^\rho/\rho$ in the explicit formula. Evaluating the shifted integral, which requires the analytic continuation of $\zeta(s)$ past $\operatorname{Re}(s) = 1$ and a zero-free region near that line, is carried out in chapter 5; the present example records only the integral representation that initiates it.

Elementary Results on Prime Distribution

chapter 2 ended with the identity that lies at the center of the subject: for $\operatorname{Re}(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

(theorem 2.2.6). The right side is the Dirichlet series of the von Mangoldt function Λ , whose summatory function counts prime powers weighted by $\log p$; the left side is an analytic object whose poles and zeros are accessible to complex analysis. The full exploitation of this identity, locating the pole of $\zeta(s)$ at $s = 1$ and excluding zeros on the line $\operatorname{Re}(s) = 1$, yields the Prime Number Theorem and occupies chapter 5. That route requires contour integration and the analytic continuation of $\zeta(s)$.

This chapter takes the elementary road. Before deploying complex analysis, one can already pin down a great deal about the distribution of primes using nothing more than the unique factorization of integers, the asymptotics of the harmonic sum, and Abel's summation formula from chapter 1. The arguments of Chebyshev fix the correct *order of magnitude* of the prime counting function $\pi(x)$, showing that $\pi(x)$ is bounded above and below by constant multiples of $x/\log x$ without determining the constant. The theorems of Mertens evaluate the key sums over primes, $\sum_{p \leq x} (\log p)/p$ and $\sum_{p \leq x} 1/p$ and $\prod_{p \leq x} (1 - 1/p)$, exactly to leading order. Bertrand's postulate, that a prime always lies between n and $2n$, follows from a careful analysis of the central binomial coefficient. None of these results requires a single contour integral; each rests on counting prime powers inside factorials and binomial coefficients and feeding the count through partial summation. The prerequisites are ?? and the elementary parts of chapter 1, in particular the von Mangoldt function (definition 1.1.8), the summatory-function language (definition 1.3.1), and Abel summation (theorem 1.4.1); the analytic machinery of chapter 2 is not used until the final analytic proof in chapter 5.

3.1 Chebyshev's Functions

The Prime Number Theorem is most often stated as $\pi(x) \sim x/\log x$, a statement about the counting function π . The function π is a step function with a unit jump at each prime, and the irregular spacing of the primes makes its growth difficult to analyze directly. Chebyshev's insight was that two weighted counting functions, attaching the weight $\log p$ to each prime rather than the weight 1, are far more tractable: they are the summatory functions of arithmetic functions already studied in chapter 1, they are tied directly to the factorization of factorials, and their asymptotics are equivalent to those of π . This section introduces π , ϑ , and ψ , records their numerical behavior, relates ψ to ϑ , and proves that the three natural asymptotic statements about them stand or fall together. The equivalence is what licenses the analytic theory of chapter 5 to target the most convenient of the three, ψ , and transfer the conclusion back to π .

Definition 3.1.1: The Prime Counting Function

The *prime counting function* $\pi: [1, \infty) \rightarrow \mathbb{Z}_{\geq 0}$ is

$$\pi(x) = \#\{p \leq x \mid p \text{ prime}\}$$

the number of primes not exceeding x .

The function π has appeared informally since ?? , where Euclid's theorem was read as the statement $\pi(x) \rightarrow \infty$, and again in example 1.4.7, where Abel summation converted π into the sum $\sum_{p \leq x} 1/p$. It is a nondecreasing step function, constant on each interval between consecutive primes and jumping by 1 at each prime. Its central feature is the slowness of its growth: the Prime Number Theorem will show $\pi(x) \sim x/\log x$, so that the proportion $\pi(x)/x$ of integers up to x that are prime decays like $1/\log x$, tending to zero but only logarithmically.

The two functions that organize the elementary theory weight primes by $\log p$.

Definition 3.1.2: Chebyshev's Theta and Psi Functions

Chebyshev's theta function and *Chebyshev's psi function* are

$$\vartheta(x) = \sum_{p \leq x} \log p, \quad \psi(x) = \sum_{n \leq x} \Lambda(n) = \sum_{p^k \leq x} \log p$$

where in the sum for ϑ the index p runs over primes, in the first sum for ψ the index n runs over positive integers with Λ the von Mangoldt function (definition 1.1.8), and in the second sum the indices p, k run over primes p and exponents $k \geq 1$ with $p^k \leq x$.

The two expressions for ψ agree because $\Lambda(n)$ is $\log p$ when $n = p^k$ is a prime power and 0 otherwise, so summing $\Lambda(n)$ over $n \leq x$ is the same as summing $\log p$ over all prime powers $p^k \leq x$. Thus ψ is precisely the summatory function (definition 1.3.1) of the von Mangoldt function, and this is the link to the analytic side: ψ is the partial-sum counterpart of the Dirichlet series $\sum \Lambda(n)n^{-s} = -\zeta'(s)/\zeta(s)$ of theorem 2.2.6. The function ϑ discards the higher prime powers, summing $\log p$ only over primes $p \leq x$ rather than over all prime powers. The difference between the two is therefore the contribution of the proper prime powers p^k with $k \geq 2$, a contribution that proposition 3.1.4 will show is small.

Both functions are weighted forms of π . Where $\pi(x)$ adds 1 for each prime, $\vartheta(x)$ adds $\log p$, so ϑ is

heavier toward the large primes near x ; the weighting is exactly what makes $\vartheta(x)$ grow like x rather than like $x/\log x$. The heuristic is that the primes up to x have “average” logarithm of size $\log x$, so $\vartheta(x) \approx \pi(x) \log x \approx x$, while $\pi(x) \approx \vartheta(x)/\log x \approx x/\log x$. Making this heuristic precise is the content of theorem 3.1.5.

A table of values makes the comparison concrete and exhibits the tracking of $\vartheta(x)$ and $\psi(x)$ with x .

Example 3.1.3: Numerical Values of ϑ , ψ , and π

Direct computation, summing over the primes and prime powers up to each cutoff, gives the following values, rounded to four decimals:

x	$\pi(x)$	$\vartheta(x)$	$\psi(x)$	$\vartheta(x)/x$	$\psi(x)/x$
10	4	5.3471	7.8320	0.5347	0.7832
100	25	83.7284	94.0453	0.8373	0.9405
1000	168	956.2453	996.6809	0.9562	0.9967
10000	1229	9895.9914	10013.3967	0.9896	1.0013

The ratios $\vartheta(x)/x$ and $\psi(x)/x$ both approach 1 as x grows, the convergence of $\psi(x)/x$ being visibly faster, consistent with $\psi(x) - \vartheta(x)$ being a lower-order quantity. At $x = 10^4$ the value $\psi(x)/x = 1.0013$ already sits within two parts in a thousand of its conjectured limit. By contrast $\pi(x)$ is far from x ; it is $x/\log x$ that $\pi(x)$ tracks, and at $x = 10^4$ one has $x/\log x \approx 1085.7$ against $\pi(x) = 1229$, the ratio $\pi(x) \log x/x \approx 1.13$ still drifting slowly toward 1. The contribution of proper prime powers to ψ is small but nonzero: at $x = 100$ the gap $\psi(100) - \vartheta(100) = 10.3169$ comes from the squares 4, 9, 25, 49, the cubes 8, 27, and the higher prime powers 16, 32, 64, 81 below 100.

The table suggests that ψ and ϑ differ by a quantity of lower order. The next result makes this exact, expressing ψ as a sum of ϑ over the roots $x^{1/m}$ and bounding the difference. The estimate is needed to transfer asymptotics between ψ and ϑ , and it must be obtained without assuming any bound on ϑ itself, since the bound $\vartheta(x) = O(x)$ is proved only in the next section; using it here would be circular.

Proposition 3.1.4: Comparison of ψ and ϑ

For $x \geq 1$,

$$\psi(x) = \sum_{m \geq 1} \vartheta(x^{1/m})$$

where the sum is finite, the terms vanishing once $m > \log_2 x$. Moreover

$$0 \leq \psi(x) - \vartheta(x) = O(\sqrt{x} \log x)$$

Proof:

Step 1: The identity. Group the prime powers $p^k \leq x$ appearing in $\psi(x) = \sum_{p^k \leq x} \log p$ according to the exponent $k = m$. For a fixed $m \geq 1$, the primes p contributing are exactly those with $p^m \leq x$, that is, $p \leq x^{1/m}$, and each such prime contributes $\log p$. Hence the contribution of exponent m is $\sum_{p \leq x^{1/m}} \log p = \vartheta(x^{1/m})$, and summing over m gives

$$\psi(x) = \sum_{m \geq 1} \sum_{p \leq x^{1/m}} \log p = \sum_{m \geq 1} \vartheta(x^{1/m})$$

A term $\vartheta(x^{1/m})$ is nonzero precisely when there is a prime $p \leq x^{1/m}$, which requires $x^{1/m} \geq 2$, that is, $2^m \leq x$, equivalently $m \leq \log_2 x$. For $m > \log_2 x$ the argument $x^{1/m}$ is less than 2 and $\vartheta(x^{1/m}) = 0$, so the sum is finite with at most $\lfloor \log_2 x \rfloor$ nonzero terms.

Step 2: Nonnegativity and isolating the difference. The $m = 1$ term is $\vartheta(x)$, so

$$\psi(x) - \vartheta(x) = \sum_{m \geq 2} \vartheta(x^{1/m})$$

Every term $\vartheta(x^{1/m})$ is a sum of values $\log p \geq \log 2 > 0$, hence nonnegative, giving $\psi(x) - \vartheta(x) \geq 0$.

Step 3: The bound, via a trivial estimate for ϑ . The bound $\vartheta(y) = O(y)$ is not yet available, so a cruder estimate is used: each of the $\pi(y)$ primes $p \leq y$ satisfies $\log p \leq \log y$, whence

$$\vartheta(y) = \sum_{p \leq y} \log p \leq \pi(y) \log y \leq y \log y$$

the last step using $\pi(y) \leq y$ (there are at most y positive integers up to y , hence at most y primes). Apply this with $y = x^{1/m}$, so $\vartheta(x^{1/m}) \leq x^{1/m} \log(x^{1/m}) = \frac{1}{m} x^{1/m} \log x$. For $m \geq 2$ one has $x^{1/m} \leq x^{1/2} = \sqrt{x}$, and discarding the factor $1/m \leq 1$ gives the per-term bound $\vartheta(x^{1/m}) \leq \sqrt{x} \log x$. Summing over the at most $\log_2 x$ nonzero terms with $m \geq 2$,

$$0 \leq \psi(x) - \vartheta(x) = \sum_{2 \leq m \leq \log_2 x} \vartheta(x^{1/m}) \leq (\log_2 x) \cdot \sqrt{x} \log x = \frac{\sqrt{x} (\log x)^2}{\log 2}$$

This already gives $\psi(x) - \vartheta(x) = O(\sqrt{x} (\log x)^2)$. The sharper bound $O(\sqrt{x} \log x)$ follows by treating $m = 2$ separately: the single term $\vartheta(\sqrt{x}) \leq \sqrt{x} \log \sqrt{x} = \frac{1}{2} \sqrt{x} \log x$, while for $m \geq 3$ one has $x^{1/m} \leq x^{1/3}$, so

$$\sum_{3 \leq m \leq \log_2 x} \vartheta(x^{1/m}) \leq (\log_2 x) \cdot x^{1/3} \log x = O(x^{1/3} (\log x)^2)$$

and since $x^{1/3} (\log x)^2 = O(\sqrt{x} \log x)$ (because $x^{1/3-1/2} \log x = x^{-1/6} \log x \rightarrow 0$), the $m \geq 3$ tail is absorbed into the $m = 2$ term. Combining the $m = 2$ contribution with the tail gives $\psi(x) - \vartheta(x) = O(\sqrt{x} \log x)$, as we sought to show.

The proposition has two consequences worth separating. First, ψ and ϑ are interchangeable to leading order: since $\sqrt{x} \log x = o(x)$, the relation $0 \leq \psi(x) - \vartheta(x) = O(\sqrt{x} \log x)$ means $\psi(x) = \vartheta(x) + o(x)$, so $\psi(x)/x$ and $\vartheta(x)/x$ have the same limiting behavior and any asymptotic of the form $\psi(x) \sim cx$ holds if and only if $\vartheta(x) \sim cx$ with the same constant. Second, the proper prime powers are genuinely negligible: the entire contribution of squares, cubes, and higher powers of primes to ψ is concentrated in the term $\vartheta(\sqrt{x})$ of size $O(\sqrt{x} \log x)$, dominated overwhelmingly by the squares, since the cubes and beyond contribute only $O(x^{1/3} (\log x)^2)$. This is why $\psi(x)/x$ in example 3.1.3 converges faster than $\vartheta(x)/x$: the higher-power correction it carries is positive and shrinking, nudging ψ above ϑ by an amount that vanishes relative to x . The bound also explains why the analytic theory works with ψ rather than ϑ at no cost in the final statement about primes.

With ψ and ϑ tied together, the remaining task is to connect them to π . The three asymptotic statements, one for each function, turn out to be equivalent, and the bridge between the unweighted π and the weighted ϑ is supplied by Abel summation.

Theorem 3.1.5: Equivalence of the Prime Number Theorem

As $x \rightarrow \infty$, the following three statements are equivalent:

1. $\pi(x) \sim \frac{x}{\log x}$;
2. $\vartheta(x) \sim x$;
3. $\psi(x) \sim x$.

Proof:

The proof establishes the equivalence (ii) $\Leftrightarrow$ (iii) from the comparison of ψ and ϑ , then the equivalence (i) $\Leftrightarrow$ (ii) by partial summation in each direction.

Step 1: (ii) $\Leftrightarrow$ (iii). By proposition 3.1.4,

$$\frac{\psi(x)}{x} = \frac{\vartheta(x)}{x} + \frac{\psi(x) - \vartheta(x)}{x}, \quad 0 \leq \frac{\psi(x) - \vartheta(x)}{x} \leq \frac{C\sqrt{x} \log x}{x} = \frac{C \log x}{\sqrt{x}}$$

for a constant C and all large x . The right-hand bound tends to 0, so $\psi(x)/x$ and $\vartheta(x)/x$ have the same limit if either exists. Hence $\vartheta(x) \sim x$ if and only if $\psi(x) \sim x$.

Step 2: (ii) $\Rightarrow$ (i), *expressing π through ϑ* . Apply Abel summation (theorem 1.4.1) to recover π from ϑ . Take the coefficients $a_n = \log n$ when n is prime and $a_n = 0$ otherwise, so that the summatory function is $A(t) = \sum_{n \leq t} a_n = \vartheta(t)$, and take the test function $f(t) = 1/\log t$, which is continuously differentiable on $[2, \infty)$ with $f'(t) = -1/(t(\log t)^2)$. Then $a_n f(n) = 1$ for each prime n and 0 otherwise, so $\sum_{n \leq x} a_n f(n) = \pi(x) - \pi(2) + 1$; carrying out the summation from the base point 2, theorem 1.4.1 gives

$$\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(t)}{t(\log t)^2} dt + c_0$$

for a bounded constant c_0 absorbing the contribution below 2. Assume now $\vartheta(t) \sim t$, say $\vartheta(t) = t + r(t)$ with $r(t) = o(t)$. The leading term is $\vartheta(x)/\log x \sim x/\log x$. For the integral, split $\vartheta(t) = t + r(t)$:

$$\int_2^x \frac{\vartheta(t)}{t(\log t)^2} dt = \int_2^x \frac{dt}{(\log t)^2} + \int_2^x \frac{r(t)}{t(\log t)^2} dt$$

The first integral satisfies $\int_2^x (\log t)^{-2} dt = O(x/(\log x)^2) = o(x/\log x)$, by the standard estimate that splits the range at $\sqrt{x}$: on $[2, \sqrt{x}]$ the integrand is at most $(\log 2)^{-2}$, contributing $O(\sqrt{x})$, while on $[\sqrt{x}, x]$ one has $\log t \geq \frac{1}{2} \log x$, contributing at most $4x/(\log x)^2$. For the second integral, given $\varepsilon > 0$ choose T with $|r(t)| \leq \varepsilon t$ for $t \geq T$; then $\int_T^x |r(t)|/(t(\log t)^2) dt \leq \varepsilon \int_T^x (\log t)^{-2} dt = \varepsilon \cdot o(x/\log x)$, while $\int_2^T$ is a constant, so the second integral is $o(x/\log x)$ as well. Combining,

$\pi(x) = x/\log x + o(x/\log x)$, which is statement (i).

Step 3: (i) $\Rightarrow$ (ii), *expressing ϑ through π .* Run Abel summation in the reverse direction. Take $a_n = 1$ when n is prime and 0 otherwise, so $A(t) = \pi(t)$, and take $f(t) = \log t$, with $f'(t) = 1/t$. Then $a_n f(n) = \log n$ for each prime n , so $\sum_{n \leq x} a_n f(n) = \vartheta(x)$, and theorem 1.4.1 gives

$$\vartheta(x) = \pi(x) \log x - \int_2^x \frac{\pi(t)}{t} dt$$

Assume now $\pi(t) \sim t/\log t$, say $\pi(t) = t/\log t + e(t)$ with $e(t) = o(t/\log t)$. The leading term is $\pi(x) \log x \sim x$. For the integral, substitute $\pi(t) = t/\log t + e(t)$:

$$\int_2^x \frac{\pi(t)}{t} dt = \int_2^x \frac{dt}{\log t} + \int_2^x \frac{e(t)}{t} dt$$

The first integral is $\text{Li}(x) + O(1)$, and $\text{Li}(x) = O(x/\log x) = o(x)$ by the same split-at- $\sqrt{x}$ estimate applied to $\int_2^x (\log t)^{-1} dt$. For the second integral, given $\varepsilon > 0$ choose T with $|e(t)| \leq \varepsilon t/\log t$ for $t \geq T$; then $\int_T^x |e(t)|/t dt \leq \varepsilon \int_T^x (\log t)^{-1} dt = \varepsilon \cdot o(x)$, and $\int_2^T$ is bounded, so the second integral is $o(x)$. Hence $\vartheta(x) = \pi(x) \log x - o(x) = x + o(x)$, which is statement (ii).

Steps 2 and 3 together give (i) $\Leftrightarrow$ (ii), and Step 1 gives (ii) $\Leftrightarrow$ (iii), so all three statements are equivalent, completing the proof.

The equivalence reorganizes the entire project. Three asymptotic statements that look different, one counting primes, one summing $\log p$ over primes, one summing $\Lambda(n)$ over all integers, are a single statement viewed through three lenses, and the constant in each is forced to be 1 once it is 1 in any one of them. The freedom this buys is decisive for the analytic theory. Of the three functions, ψ is the most natural target for complex analysis: it is the summatory function of Λ , and Λ is exactly the coefficient sequence of $-\zeta'(s)/\zeta(s)$ by theorem 2.2.6, so the asymptotic $\psi(x) \sim x$ is read directly off the analytic behavior of the logarithmic derivative of ζ , with the leading term x produced by the simple pole of $-\zeta'(s)/\zeta(s)$ at $s = 1$ with residue 1. The function π , by contrast, has no such clean Dirichlet series. The strategy of chapter 5 is therefore to prove $\psi(x) \sim x$ analytically and invoke the present theorem to transfer the conclusion first to ϑ and then to the statement $\pi(x) \sim x/\log x$ that one actually wants. The elementary results of the remaining sections of this chapter operate on the same principle from the other side: Chebyshev bounds ψ and ϑ between constant multiples of x , and through this theorem those bounds become bounds on $\pi(x)$ of the correct order $x/\log x$.

3.2 Chebyshev's Estimates

The equivalence of theorem 3.1.5 reduces the Prime Number Theorem to the single asymptotic $\psi(x) \sim x$, but it does not produce that asymptotic; the limit $\psi(x)/x \rightarrow 1$ lies beyond elementary reach and is the work of the analytic theory in chapter 5. What can be extracted by elementary means alone is one step weaker and was obtained by Chebyshev in 1852, half a century before the analytic proof: the *order of magnitude* of the three counting functions is pinned down exactly. There are constants $0 < c_1 < c_2$ with $c_1 x \leq \vartheta(x) \leq c_2 x$, and likewise for ψ , so that $\vartheta(x)$ and $\psi(x)$ each grow linearly in x and $\pi(x)$ grows like $x/\log x$, with explicit numerical constants bracketing the

eventual limit.

The entire argument rests on a single arithmetic object, the central binomial coefficient $\binom{2n}{n}$. Its size is controlled on both sides by elementary bounds, $4^n/(2n+1) \leq \binom{2n}{n} \leq 4^n$, while its prime factorization is controlled by Legendre's formula (proposition 0.2.11): every prime power dividing $\binom{2n}{n}$ is at most $2n$, and every prime in the range $(n, 2n]$ divides it exactly once. Comparing the two descriptions of the same integer forces ψ and ϑ into a linear corridor. The upper bound comes first, in the form of a bound on the product of all primes up to x .

Lemma 3.2.1: Prime Content of the Central Binomial Coefficient

Let $n \geq 1$ and let p be a prime. Then

$$v_p\left(\binom{2n}{n}\right) = \sum_{k \geq 1} \left(\left\lfloor \frac{2n}{p^k} \right\rfloor - 2 \left\lfloor \frac{n}{p^k} \right\rfloor \right)$$

Each summand is 0 or 1, the summand vanishes for $p^k > 2n$, and consequently

$$p^{v_p\left(\binom{2n}{n}\right)} \leq 2n$$

for every prime p . In particular no prime power exceeding $2n$ divides $\binom{2n}{n}$.

Proof:

Step 1: The valuation formula. Since $\binom{2n}{n} = (2n)!/(n!n!)$, the additivity of the p -adic valuation gives $v_p\left(\binom{2n}{n}\right) = v_p((2n)!) - 2v_p(n!)$. Applying Legendre's formula (proposition 0.2.11) to each factorial,

$$v_p\left(\binom{2n}{n}\right) = \sum_{k \geq 1} \left\lfloor \frac{2n}{p^k} \right\rfloor - 2 \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor = \sum_{k \geq 1} \left(\left\lfloor \frac{2n}{p^k} \right\rfloor - 2 \left\lfloor \frac{n}{p^k} \right\rfloor \right)$$

both Legendre sums being finite, so their term-by-term difference is legitimate.

Step 2: Each summand is 0 or 1. For any real y write $y = \lfloor y \rfloor + \{y\}$ with $0 \leq \{y\} < 1$. Then

$$\lfloor 2y \rfloor - 2\lfloor y \rfloor = (\lfloor 2y \rfloor + \lfloor 2\{y\} \rfloor) - 2\lfloor y \rfloor = \lfloor 2\{y\} \rfloor$$

using $\lfloor 2y \rfloor = \lfloor 2\lfloor y \rfloor + 2\{y\} \rfloor = 2\lfloor y \rfloor + \lfloor 2\{y\} \rfloor$ since $2\lfloor y \rfloor$ is an integer. As $0 \leq 2\{y\} < 2$, the value $\lfloor 2\{y\} \rfloor$ is 0 when $\{y\} < 1/2$ and 1 when $\{y\} \geq 1/2$. Taking $y = n/p^k$ shows each summand $\lfloor 2n/p^k \rfloor - 2\lfloor n/p^k \rfloor$ equals 0 or 1.

Step 3: The bound on prime powers. A summand can be nonzero only while its terms are nonzero, and $\lfloor 2n/p^k \rfloor = 0$ once $p^k > 2n$. Hence the nonzero summands occur only for those k with $p^k \leq 2n$, that is, $k \leq \log_p(2n)$, and there are at most $\lfloor \log_p(2n) \rfloor$ of them. Since each contributes at most 1,

$$v_p\left(\binom{2n}{n}\right) \leq \#\{k \geq 1 \mid p^k \leq 2n\} = \lfloor \log_p(2n) \rfloor \leq \log_p(2n)$$

Exponentiating base p , the inequality $v_p\left(\binom{2n}{n}\right) \leq \log_p(2n)$ becomes $p^{v_p\left(\binom{2n}{n}\right)} \leq p^{\log_p(2n)} = 2n$, as we sought to show.

The lemma records two facts that will be used in opposite directions. The bound $p^{v_p(\binom{2n}{n})} \leq 2n$ says the central binomial coefficient is, prime by prime, “flat”: no single prime contributes a factor larger than $2n$, no matter how large n is. This is what limits the contribution of small primes when $\binom{2n}{n}$ is decomposed, and it drives the lower bound on ψ . The valuation formula itself, read in the range $n < p \leq 2n$, says something complementary: for such a prime the single term $k = 1$ gives $\lfloor 2n/p \rfloor - 2\lfloor n/p \rfloor = 1 - 0 = 1$ and all higher terms vanish (because $p^2 > n^2 \geq 2n$ for $n \geq 2$), so p divides $\binom{2n}{n}$ exactly once. The primes in $(n, 2n]$ therefore appear in $\binom{2n}{n}$ to the first power, and their product divides it. This second reading is the mechanism behind the primorial bound, to which the discussion now turns.

The upper estimate $\vartheta(x) = O(x)$ is equivalent to a bound on the product of all primes up to x , the *primorial*. Bounding this product is the heart of the upper Chebyshev estimate, and the argument is an induction in which the inductive step is supplied by the elementary inequality $\binom{2m+1}{m} \leq 4^m$ together with the divisibility just observed.

Proposition 3.2.2: The Primorial Bound

For every real $x \geq 1$,

$$\prod_{p \leq x} p \leq 4^x$$

the product running over primes $p \leq x$. Equivalently, $\vartheta(x) \leq (\log 4)x$ for all $x \geq 1$.

Proof:

The two formulations are equivalent: taking logarithms turns $\prod_{p \leq x} p \leq 4^x$ into $\sum_{p \leq x} \log p \leq x \log 4$, and the left side is exactly $\vartheta(x)$ by definition 3.1.2. It therefore suffices to prove the product bound. Write $\Pi(x) = \prod_{p \leq x} p$.

Step 1: Reduction to integers. The product $\Pi(x)$ depends only on $\lfloor x \rfloor$, since no prime lies strictly between $\lfloor x \rfloor$ and x . The target 4^x is nondecreasing in x , so if $\Pi(n) \leq 4^n$ holds for every integer $n \geq 1$ then for general $x \geq 1$ with $n = \lfloor x \rfloor$,

$$\Pi(x) = \Pi(n) \leq 4^n \leq 4^x$$

The claim is thus reduced to the inequality $\Pi(n) \leq 4^n$ for integers $n \geq 1$, which is proved by strong induction on n .

Step 2: Base cases. For $n = 1$ the product is empty, $\Pi(1) = 1 \leq 4$. For $n = 2$, $\Pi(2) = 2 \leq 16$. These anchor the induction; the inductive step below treats $n \geq 3$ and refers back to strictly smaller indices.

Step 3: Even n . Suppose $n = 2m$ with $m \geq 2$, so $n \geq 4$. Since $2m$ is composite for $m \geq 2$, it is not prime, and no prime satisfies $2m - 1 < p \leq 2m$. Hence $\Pi(2m) = \Pi(2m - 1)$. By the induction hypothesis applied to $2m - 1 < 2m$,

$$\Pi(2m) = \Pi(2m - 1) \leq 4^{2m-1} \leq 4^{2m}$$

so the bound for $n = 2m$ follows from the bound for the smaller odd index $2m - 1$.

Step 4: Odd n . Suppose $n = 2m + 1$ with $m \geq 1$, so $n \geq 3$. Split the primes $p \leq 2m + 1$ at $m + 1$:

$$\Pi(2m + 1) = \left(\prod_{p \leq m+1} p \right) \left(\prod_{m+1 < p \leq 2m+1} p \right) = \Pi(m + 1) \cdot \prod_{m+1 < p \leq 2m+1} p$$

The first factor is bounded by the induction hypothesis at $m + 1 \leq 2m < 2m + 1$:

$$\Pi(m + 1) \leq 4^{m+1}$$

For the second factor, every prime p with $m + 1 < p \leq 2m + 1$ divides the integer $\binom{2m+1}{m} = (2m + 1)! / (m! (m + 1)!)$. Indeed such a p divides the numerator $(2m + 1)!$, since $p \leq 2m + 1$, but divides neither $m!$ nor $(m + 1)!$, since $p > m + 1$ exceeds every factor in those products; by the additivity of v_p , $v_p \binom{2m+1}{m} = v_p((2m + 1)!) \geq 1$. These primes are distinct, so their product divides $\binom{2m+1}{m}$, whence

$$\prod_{m+1 < p \leq 2m+1} p \leq \binom{2m+1}{m}$$

It remains to bound the binomial coefficient. The two coefficients $\binom{2m+1}{m}$ and $\binom{2m+1}{m+1}$ are equal and both appear in the binomial expansion $\sum_{j=0}^{2m+1} \binom{2m+1}{j} = 2^{2m+1}$, so

$$2 \binom{2m+1}{m} = \binom{2m+1}{m} + \binom{2m+1}{m+1} \leq 2^{2m+1}$$

giving $\binom{2m+1}{m} \leq 2^{2m} = 4^m$. Combining the three bounds,

$$\Pi(2m + 1) = \Pi(m + 1) \cdot \prod_{m+1 < p \leq 2m+1} p \leq 4^{m+1} \cdot 4^m = 4^{2m+1}$$

which is the bound for $n = 2m + 1$. The even and odd steps together complete the induction, completing the proof.

The primorial bound is the upper half of the Chebyshev estimates already in finished form: $\vartheta(x) \leq (\log 4)x$ is a linear bound with the explicit constant $\log 4 \approx 1.386$. The inductive engine is the same mechanism that reappears, sharpened, in the proof of Bertrand's postulate (theorem 3.4.3) later in this chapter. The primes in the window $(m + 1, 2m + 1]$ are forced into the central binomial coefficient, whose size is capped by 4^m ; the product of those primes therefore cannot grow faster than 4^m , and feeding this back through the recursion bounds the product of all primes. The lower estimate runs the comparison in reverse: rather than asking how few primes can fit inside $\binom{2n}{n}$, it asks how the known size $\binom{2n}{n} \geq 4^n / (2n + 1)$ must be accounted for by primes, each capped at $2n$ by lemma 3.2.1.

Theorem 3.2.3: Chebyshev's Estimates

There exist absolute constants $0 < c_1 < c_2$ such that for all $x \geq 2$,

$$c_1 x \leq \vartheta(x) \leq \psi(x) \leq c_2 x$$

Explicitly:

1. (*Upper bound.*) $\vartheta(x) \leq (\log 4)x$ for all $x \geq 1$, and $\psi(x) \leq (\log 4)x + O(\sqrt{x} \log x)$, so $\psi(x) \leq c_2 x$ holds for any fixed $c_2 > \log 4$ and all large x .
2. (*Lower bound.*) $\psi(x) \geq (\log 2)x - O(\log x)$ and $\vartheta(x) \geq (\log 2)x - O(\sqrt{x} \log x)$, so each of $\psi(x) \geq c_1 x$ and $\vartheta(x) \geq c_1 x$ holds for any fixed $c_1 < \log 2$ and all large x .

In particular $\vartheta(x)$ and $\psi(x)$ are each of exact order x , with $\log 2 \leq \liminf \psi(x)/x \leq \limsup \psi(x)/x \leq \log 4$ and the same for ϑ .

Proof:

The relation $\psi(x) \geq \vartheta(x)$ is available throughout: proposition 3.1.4 gives $\psi(x) - \vartheta(x) = \sum_{m \geq 2} \vartheta(x^{1/m}) \geq 0$, together with the estimate $0 \leq \psi(x) - \vartheta(x) = O(\sqrt{x} \log x)$, which transfers any bound between ψ and ϑ up to a $o(x)$ error. The two sub-statements are proved in turn.

1. *Upper bound.* proposition 3.2.2 gives $\vartheta(x) \leq (\log 4)x$ for all $x \geq 1$ directly. For ψ , the comparison $\psi(x) = \vartheta(x) + (\psi(x) - \vartheta(x))$ with $0 \leq \psi(x) - \vartheta(x) = O(\sqrt{x} \log x)$ yields

$$\psi(x) \leq (\log 4)x + O(\sqrt{x} \log x)$$

Since $\sqrt{x} \log x = o(x)$, for any fixed $c_2 > \log 4$ the error is eventually dominated by $(c_2 - \log 4)x$, so $\psi(x) \leq c_2 x$ for all large x , and after enlarging c_2 once more the bound $\psi(x) \leq c_2 x$ holds for all $x \geq 2$. This is the right-hand inequality with c_2 an absolute constant.

2. *Lower bound.* The argument compares two descriptions of the central binomial coefficient $N = \binom{2n}{n}$ for an integer $n \geq 1$, then passes from even integers to all x .

Size of N . The central coefficient is the largest of the $2n + 1$ terms $\binom{2n}{j}$ that sum to $(1 + 1)^{2n} = 4^n$, so

$$N = \binom{2n}{n} \geq \frac{1}{2n + 1} \sum_{j=0}^{2n} \binom{2n}{j} = \frac{4^n}{2n + 1}$$

Taking logarithms, $\log N \geq n \log 4 - \log(2n + 1) = 2n \log 2 - \log(2n + 1)$.

Primes in N . By unique factorization $N = \prod_{p \leq 2n} p^{v_p(N)}$, the product over primes $p \leq 2n$ because lemma 3.2.1 forbids any prime power exceeding $2n$ from dividing N . For each such prime set $K_p = v_p(N)$; the same lemma gives $p^k \leq 2n$ for every k with $1 \leq k \leq K_p$. Hence

$$\log N = \sum_{p \leq 2n} K_p \log p = \sum_{p \leq 2n} \sum_{k=1}^{K_p} \log p \leq \sum_{\substack{p, k \geq 1 \\ p^k \leq 2n}} \log p = \psi(2n)$$

where the inequality holds because each pair (p, k) with $1 \leq k \leq K_p$ satisfies $p^k \leq 2n$ and so appears among the prime powers counted by $\psi(2n)$, while the right-hand sum may contain additional nonnegative terms. Combining the two displays,

$$\psi(2n) \geq \log N \geq 2n \log 2 - \log(2n + 1)$$

for every integer $n \geq 1$.

Passage to all x . Given a real $x \geq 2$, set $n = \lfloor x/2 \rfloor \geq 1$, so $2n \leq x$ and $2n > x - 2$. As ψ is nondecreasing, $\psi(x) \geq \psi(2n) \geq 2n \log 2 - \log(2n + 1)$, and using $2n > x - 2$ together with $2n + 1 \leq x + 1 \leq 2x$,

$$\psi(x) \geq (x - 2) \log 2 - \log(2x) = (\log 2)x - (2 \log 2 + \log(2x))$$

The subtracted quantity is $O(\log x)$, giving $\psi(x) \geq (\log 2)x - O(\log x)$. For ϑ , the estimate $0 \leq \psi(x) - \vartheta(x) = O(\sqrt{x} \log x)$ of proposition 3.1.4 yields $\vartheta(x) = \psi(x) - O(\sqrt{x} \log x) \geq (\log 2)x - O(\sqrt{x} \log x)$. Since both error terms are $o(x)$, for any fixed $c_1 < \log 2$ each of $\psi(x) \geq c_1 x$ and $\vartheta(x) \geq c_1 x$ holds for all large x .

Sub-statements (1) and (2) together furnish constants $0 < c_1 < c_2$ with $c_1 x \leq \vartheta(x) \leq \psi(x) \leq c_2 x$ for all $x \geq 2$, so both functions are of exact order x , as we sought to show.

The estimates fix the order of growth without fixing the constant. The corridor they cut out, $(\log 2)x \leq \psi(x) \leq (\log 4)x$ to leading order, is wide: the lower constant $\log 2 \approx 0.693$ and the upper constant $\log 4 \approx 1.386$ differ by a factor of exactly two, and the truth $\psi(x) \sim x$ of chapter 5 sits strictly between them. Chebyshev's method, resting only on the size and factorization of one binomial coefficient, cannot close this gap; the two constants are artifacts of comparing $\binom{2n}{n}$ against 4^n on one side and against the cap $2n$ on prime powers on the other, and no elementary sharpening of these comparisons reaches the constant 1. That value is analytic in origin, produced by the simple pole of $-\zeta'(s)/\zeta(s)$ at $s = 1$ with residue 1. What the estimates do settle is decisive for everything that follows: ψ , ϑ , and (through the next corollary) π have no anomalous growth, no logarithmic or power-law deviation from linear order, and any sum over primes built from these functions by partial summation inherits a bound of the correct shape.

The order of π now follows by converting the bounds on ϑ into bounds on π , exactly as in the proof of theorem 3.1.5 but at the level of inequalities rather than asymptotic equivalences. This is the bound foreshadowed in ?? , where $\pi(x) = O(x/\log x)$ was promised as a consequence of Chebyshev's work.

Corollary 3.2.4: The Order of the Prime Counting Function

The prime counting function is of exact order $x/\log x$: there are absolute constants $0 < a_1 < a_2$ with

$$a_1 \frac{x}{\log x} \leq \pi(x) \leq a_2 \frac{x}{\log x}$$

for all $x \geq 2$. More precisely, the constants may be taken arbitrarily close to $\log 2$ and $\log 4$:

$$(\log 2 + o(1)) \frac{x}{\log x} \leq \pi(x) \leq (\log 4 + o(1)) \frac{x}{\log x}$$

as $x \rightarrow \infty$. In particular $\pi(x) = O(x/\log x)$.

Proof:

The two inequalities come from the two bounds of theorem 3.2.3, transferred through the relation between π and ϑ established in theorem 3.1.5.

Step 1: The upper bound on π . Comparing π and ϑ directly, the primes in the range $\sqrt{x} < p \leq x$ each contribute a logarithm at least $\log \sqrt{x} = \frac{1}{2} \log x$, so

$$\vartheta(x) = \sum_{p \leq x} \log p \geq \sum_{\sqrt{x} < p \leq x} \log p \geq (\pi(x) - \pi(\sqrt{x})) \cdot \frac{1}{2} \log x$$

Rearranging and using $\pi(\sqrt{x}) \leq \sqrt{x}$,

$$\pi(x) \leq \frac{2\vartheta(x)}{\log x} + \pi(\sqrt{x}) \leq \frac{2\vartheta(x)}{\log x} + \sqrt{x}$$

By the upper bound $\vartheta(x) \leq (\log 4)x$ of theorem 3.2.3, the first term is at most $2(\log 4)x/\log x$, and the second satisfies $\sqrt{x} = o(x/\log x)$. Hence $\pi(x) \leq 2(\log 4)x/\log x + o(x/\log x)$, giving $\pi(x) = O(x/\log x)$ with an absolute constant a_2 . The sharper constant $\log 4 + o(1)$ is obtained below by partial summation.

Step 2: The lower bound on π . In the other direction, weighting each prime by at most $\log x$ gives

$$\vartheta(x) = \sum_{p \leq x} \log p \leq \sum_{p \leq x} \log x = \pi(x) \log x$$

whence $\pi(x) \geq \vartheta(x)/\log x$. By the lower bound $\vartheta(x) \geq (\log 2)x - O(\sqrt{x} \log x)$ of theorem 3.2.3,

$$\pi(x) \geq \frac{\vartheta(x)}{\log x} \geq \frac{(\log 2)x}{\log x} - O(\sqrt{x}) = (\log 2 + o(1)) \frac{x}{\log x}$$

since $O(\sqrt{x}) = o(x/\log x)$. This gives the lower inequality with constant a_1 arbitrarily close to $\log 2$.

Step 3: Sharpening the upper constant by partial summation. The crude factor 2 in Step 1 is removed by recovering π from ϑ through Abel summation (theorem 1.4.1), the same device used

in theorem 3.1.5. With $a_n = \log n$ for n prime and $a_n = 0$ otherwise, so that $A(t) = \vartheta(t)$, and with $f(t) = 1/\log t$ on $[2, \infty)$, where $f'(t) = -1/(t(\log t)^2)$, theorem 1.4.1 gives

$$\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(t)}{t(\log t)^2} dt + c_0$$

with c_0 a bounded constant. The bound $\vartheta(t) \leq (\log 4)t$ from theorem 3.2.3 controls both pieces: the leading term is at most $(\log 4)x/\log x$, and the integral is at most

$$(\log 4) \int_2^x \frac{dt}{(\log t)^2} = (\log 4) \cdot O\left(\frac{x}{(\log x)^2}\right) = o\left(\frac{x}{\log x}\right)$$

by the split-at- $\sqrt{x}$ estimate for $\int_2^x (\log t)^{-2} dt$ recorded in the proof of theorem 3.1.5. Therefore $\pi(x) \leq (\log 4)x/\log x + o(x/\log x) = (\log 4 + o(1))x/\log x$, which is the sharpened upper bound. Steps 2 and 3 together give the two-sided estimate with constants approaching $\log 2$ and $\log 4$, as we sought to show.

The corollary is the elementary substitute for the Prime Number Theorem, and it suffices for many purposes that do not require the exact constant. The proportion $\pi(x)/x$ of integers up to x that are prime lies between $(\log 2)/\log x$ and $(\log 4)/\log x$ for large x , so it decays like $1/\log x$, confirming the logarithmic thinning of the primes announced when π was first defined in definition 3.1.1. The bound $\pi(x) = O(x/\log x)$ is precisely the statement deferred in ??, now established without any appeal to complex analysis. Its converse direction, $\pi(x) \gg x/\log x$, is equally elementary and is the quantitative form of Euclid's theorem: not only are there infinitely many primes, but they are at least a constant multiple of $x/\log x$ in number up to x . The linear bound $\vartheta(x) = O(x)$ is also the input that Mertens' theorems require, converting by partial summation into exact leading-order evaluations of $\sum_{p \leq x} (\log p)/p$ and $\sum_{p \leq x} 1/p$.

The width of the Chebyshev corridor is best appreciated numerically, against the limit that the Prime Number Theorem will eventually supply.

Example 3.2.5: Chebyshev's Constants versus the Truth

The two constants delivered by theorem 3.2.3 are

$$\log 2 = 0.69314\dots, \quad \log 4 = 2 \log 2 = 1.38629\dots$$

bracketing the eventual limit $\psi(x)/x \rightarrow 1$ of chapter 5 from below and above. The bracket for ψ is direct: the theorem gives $(\log 2 + o(1))x \leq \psi(x) \leq (\log 4)x$, and the table of example 3.1.3 shows $\psi(x)/x$ already at 0.9405 for $x = 100$ and 1.0013 for $x = 10^4$, comfortably inside $[0.693, 1.386]$ and converging to 1 rather than to either Chebyshev constant.

For π , the corresponding bracket from corollary 3.2.4 is $(\log 2 + o(1))x/\log x \leq \pi(x) \leq (\log 4 + o(1))x/\log x$. Computing the ratio $\pi(x) \log x/x$, whose limit the Prime Number Theorem asserts

to be 1:

x	$\pi(x)$	$\frac{x}{\log x}$	$\frac{\pi(x) \log x}{x}$
10^2	25	21.71	1.1513
10^3	168	144.76	1.1605
10^4	1229	1085.74	1.1320
10^5	9592	8685.89	1.1043
10^6	78498	72382.41	1.0845

Every entry in the last column lies strictly between $\log 2 = 0.693$ and $\log 4 = 1.386$, as Chebyshev's bounds require, and the column drifts slowly toward 1, the drift being the slow $o(1)$ correction that only the analytic theory resolves. The gap between Chebyshev's two constants is a factor of 2, whereas the data already pin the true ratio to within roughly ten percent of 1 by $x = 10^6$; the elementary method captures the order of magnitude but not this finer convergence.

3.3 Mertens' Theorems

Chebyshev's estimates of section 3.2 fix the order of magnitude of ψ , ϑ , and π but not the leading constant. The three theorems of Mertens, proved in 1874 by the same elementary toolkit, achieve something the order-of-magnitude bounds cannot: they evaluate three sums over primes *exactly to leading order*, with an explicit error term and an explicit constant. The sums in question are

$$\sum_{p \leq x} \frac{\log p}{p}, \quad \sum_{p \leq x} \frac{1}{p}, \quad \prod_{p \leq x} \left(1 - \frac{1}{p}\right)$$

and Mertens' theorems assert, respectively, that the first is $\log x + O(1)$, the second is $\log \log x + M + O(1/\log x)$ for a specific constant M , and the third is asymptotic to $e^{-\gamma}/\log x$. Each statement is sharper than a bound of the form " $\asymp \log x$ " or " $\asymp 1/\log x$ ": the first pins the coefficient of $\log x$ to be exactly 1, the second produces the second-order constant M , and the third produces the constant $e^{-\gamma}$ with the Euler–Mascheroni γ in the exponent.

What makes this reach possible without analytic continuation is that Chebyshev's bound $\psi(x) = O(x)$ already controls the error when the factorization of $[x]!$ is fed through Abel summation. The route is largely uniform: an exact identity for $\log([x]!)$ combined with Stirling's formula gives Mertens' first theorem; partial summation against the weight $1/\log t$ converts the first theorem into the second; comparison of $\sum 1/p$ with $-\sum \log(1 - 1/p)$ reduces the constant M to the Euler–Mascheroni γ ; and exponentiating the second theorem collapses that closed form to $e^{-\gamma}$, yielding the third. Only one step reaches past the Chebyshev-level toolkit, the identification of the additive constant as exactly γ rather than an unspecified number: it draws on the Euler product and the behavior of ζ near its pole, both already in hand from chapter 2 and used only in the convergent range $\text{Re } s > 1$, and it turns on a single Gamma-function identity isolated and cited at the point of use. The constant M produced here is the same $B = \gamma + \sum_p (\log(1 - 1/p) + 1/p) = 0.2614 \dots$ that appeared in the average order of ω (proposition 1.3.9); this section establishes that identification.

The starting point is an exact arithmetic identity for the logarithm of a factorial, the same factorization idea used for the central binomial coefficient in section 3.2, now applied to $[x]!$ itself. It expresses $\log([x]!)$ as a weighted sum of the von Mangoldt function (definition 1.1.8), and pairing it with Stirling's formula (theorem 1.5.7) gives the asymptotic that drives the first theorem.

Lemma 3.3.1: The Prime-Power Decomposition of $\log n!$

For every real $x \geq 1$,

$$\log([x]!) = \sum_{n \leq x} \Lambda(n) \left\lfloor \frac{x}{n} \right\rfloor$$

where Λ is the von Mangoldt function and the sum runs over positive integers $n \leq x$. Consequently

$$\log([x]!) = x \log x - x + O(\log x)$$

as $x \rightarrow \infty$.

Proof:

Write $N = [x]$, so that $[x]! = N!$ and $[x/n] = [N/n]$ for every positive integer n (both count the multiples of n in $\{1, \dots, N\}$, and no multiple of n lies strictly between N and x). The two displayed claims are proved in turn.

Step 1: The identity. By unique factorization, $N! = \prod_{p \leq N} p^{v_p(N!)}$, the product over primes $p \leq N$. Taking logarithms and applying Legendre's formula (proposition 0.2.11) to each exponent,

$$\log(N!) = \sum_{p \leq N} v_p(N!) \log p = \sum_{p \leq N} \left(\sum_{k \geq 1} \left\lfloor \frac{N}{p^k} \right\rfloor \right) \log p = \sum_{p \leq N} \sum_{k \geq 1} \left\lfloor \frac{N}{p^k} \right\rfloor \log p$$

Each Legendre sum is finite (the terms vanish once $p^k > N$), so the double sum has finitely many nonzero terms and may be reindexed. Group the terms by the value $n = p^k$ of the prime power. For each prime power $n = p^k \leq N$ the corresponding summand is $[N/n] \log p$, and $\log p = \Lambda(n)$ by the definition of the von Mangoldt function (definition 1.1.8); for every $n \leq N$ that is *not* a prime power, $\Lambda(n) = 0$ contributes nothing. Hence

$$\log(N!) = \sum_{\substack{n \leq N \\ n=p^k}} \left\lfloor \frac{N}{n} \right\rfloor \log p = \sum_{n \leq N} \Lambda(n) \left\lfloor \frac{N}{n} \right\rfloor = \sum_{n \leq x} \Lambda(n) \left\lfloor \frac{x}{n} \right\rfloor$$

the last equality by $[N/n] = [x/n]$ and $N = [x]$. This is the identity.

Step 2: The asymptotic via Stirling. Stirling's formula (theorem 1.5.7) gives $\log(N!) = N \log N - N + \frac{1}{2} \log(2\pi N) + O(1/N)$, so in particular $\log(N!) = N \log N - N + O(\log N)$. It remains to replace $N = [x]$ by x at a cost of $O(\log x)$. Write $x = N + \theta$ with $0 \leq \theta < 1$. Then

$$x \log x - x = (N + \theta) \log(N + \theta) - (N + \theta)$$

The function $t \mapsto t \log t - t$ has derivative $\log t$, so by the mean value theorem $x \log x - x - (N \log N - N) = \theta \log \xi$ for some $\xi \in [N, x]$, whence

$$|(x \log x - x) - (N \log N - N)| = \theta \log \xi \leq \log(N + 1) = O(\log x)$$

Combining with $\log(N!) = N \log N - N + O(\log N)$ and $\log N = O(\log x)$,

$$\log([x]!) = N \log N - N + O(\log N) = x \log x - x + O(\log x)$$

as we sought to show.

The identity is the elementary counterpart of the analytic identity $-\zeta'(s)/\zeta(s) = \sum_n \Lambda(n)n^{-s}$ from theorem 2.2.6: both extract the von Mangoldt function from a global object, there the logarithmic derivative of ζ , here the logarithm of a factorial. The factor $\lfloor x/n \rfloor$ counts how many of the integers $1, \dots, \lfloor x \rfloor$ are divisible by n , so the identity says that the total logarithm $\log(\lfloor x \rfloor!) = \sum_{m \leq x} \log m$ is reorganized by collecting, for each prime power $n = p^k$, the contribution $\log p$ once for every multiple of n up to x . Stirling's formula then evaluates the left side to within $O(\log x)$, and the resulting estimate $\sum_{n \leq x} \Lambda(n) \lfloor x/n \rfloor = x \log x - x + O(\log x)$ is the bridge to the first theorem: the floor $\lfloor x/n \rfloor$ is within 1 of x/n , and replacing it by x/n produces exactly the sum $\sum \Lambda(n)/n$ whose evaluation is sought.

Theorem 3.3.2: Mertens' First Theorem

As $x \rightarrow \infty$,

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} = \log x + O(1)$$

and equivalently, summing over primes only,

$$\sum_{p \leq x} \frac{\log p}{p} = \log x + O(1)$$

Proof:

The two statements differ by the contribution of the proper prime powers, treated at the end. The main work is the first estimate.

Step 1: Removing the floor. Start from the identity of lemma 3.3.1, $\log(\lfloor x \rfloor!) = \sum_{n \leq x} \Lambda(n) \lfloor x/n \rfloor$. Writing $\lfloor x/n \rfloor = x/n - \{x/n\}$ with $0 \leq \{x/n\} < 1$ separates the main term from an error:

$$\sum_{n \leq x} \Lambda(n) \left\lfloor \frac{x}{n} \right\rfloor = x \sum_{n \leq x} \frac{\Lambda(n)}{n} - \sum_{n \leq x} \Lambda(n) \left\{ \frac{x}{n} \right\}$$

The error sum is controlled by Chebyshev's bound. Since $0 \leq \{x/n\} < 1$ and $\Lambda(n) \geq 0$,

$$0 \leq \sum_{n \leq x} \Lambda(n) \left\{ \frac{x}{n} \right\} \leq \sum_{n \leq x} \Lambda(n) = \psi(x)$$

by the definition of ψ (definition 3.1.2), and $\psi(x) = O(x)$ by Chebyshev's estimates (theorem 3.2.3). Hence the error sum is $O(x)$.

Step 2: Solving for the sum. Combining Step 1 with the asymptotic $\log(\lfloor x \rfloor!) = x \log x - x + O(\log x)$ of lemma 3.3.1,

$$x \sum_{n \leq x} \frac{\Lambda(n)}{n} = \log(\lfloor x \rfloor!) + \sum_{n \leq x} \Lambda(n) \left\{ \frac{x}{n} \right\} = (x \log x - x + O(\log x)) + O(x) = x \log x + O(x)$$

the term $-x$ and the error $O(\log x)$ both being absorbed into $O(x)$. Dividing by x ,

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} = \log x + O(1)$$

which is the first statement.

Step 3: Passage to primes. The sum $\sum_{n \leq x} \Lambda(n)/n$ runs over all prime powers; isolating the primes ($k = 1$) leaves the proper prime powers ($k \geq 2$):

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} = \sum_{p \leq x} \frac{\log p}{p} + \sum_{p^k \leq x, k \geq 2} \frac{\log p}{p^k}$$

The second sum is bounded by the full convergent series over all $k \geq 2$ and all primes:

$$0 \leq \sum_{\substack{p^k \leq x \\ k \geq 2}} \frac{\log p}{p^k} \leq \sum_p \log p \sum_{k \geq 2} \frac{1}{p^k} = \sum_p \log p \cdot \frac{1/p^2}{1 - 1/p} = \sum_p \frac{\log p}{p(p-1)}$$

using the geometric series $\sum_{k \geq 2} p^{-k} = p^{-2}/(1 - p^{-1}) = 1/(p(p-1))$. The final series converges: its terms satisfy $\log p/(p(p-1)) \leq \log p/(p-1)^2 \leq C/n^{3/2}$ for a constant C and all $n = p$, since $\log p = O(p^{1/2})$, and $\sum_n n^{-3/2} < \infty$. Therefore the proper-prime-power contribution is a bounded quantity, $O(1)$, and subtracting it from $\sum_{n \leq x} \Lambda(n)/n = \log x + O(1)$ leaves

$$\sum_{p \leq x} \frac{\log p}{p} = \log x + O(1)$$

completing the proof.

The first theorem already separates the elementary theory from the order-of-magnitude estimates of the previous section: where Chebyshev gives $\sum_{p \leq x} (\log p)/p \asymp \log x$ with unspecified constant, Mertens fixes the coefficient of $\log x$ to be exactly 1. The reason the constant is accessible while $\psi(x) \sim x$ is not lies in the weighting. Dividing $\Lambda(n)$ by n converts the identity $\sum \Lambda(n)[x/n] = x \log x - x + O(\log x)$, whose main term is of size $x \log x$, into a statement whose main term is of size $\log x$; the Chebyshev error $\psi(x) = O(x)$, fatal at the scale of $x \log x$, becomes a harmless $O(1)$ after division by x . The first theorem is thus an averaged statement, robust to the $O(x)$ uncertainty in ψ that the elementary method cannot remove. It is the input from which the remaining two theorems follow by partial summation and exponentiation, with no further arithmetic.

The passage from $\sum (\log p)/p$ to $\sum 1/p$ is a textbook application of Abel summation: the summand $1/p$ is $(\log p)/p$ divided by $\log p$, so summing against the weight $1/\log t$ against the known partial sums of $(\log p)/p$ recovers $\sum_{p \leq x} 1/p$. This is the device foreshadowed at corollary 1.4.2 in chapter 1.

Theorem 3.3.3: Mertens' Second Theorem

There is a constant M such that, as $x \rightarrow \infty$,

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + M + O\left(\frac{1}{\log x}\right)$$

The constant M is the *Mertens constant*, identified in closed form in definition 3.3.4.

Proof:

Step 1: Setting up Abel summation. Define the coefficients $a_n = (\log n)/n$ when n is prime and $a_n = 0$ otherwise, so that the summatory function is

$$A(t) = \sum_{n \leq t} a_n = \sum_{p \leq t} \frac{\log p}{p}$$

By theorem 3.3.2, $A(t) = \log t + R(t)$ with $R(t) = O(1)$; fix a constant C_0 with $|R(t)| \leq C_0$ for all $t \geq 2$. Take the test function $f(t) = 1/\log t$ on $[2, \infty)$, which is continuously differentiable with $f'(t) = -1/(t(\log t)^2)$. For a prime $n = p$, the product $a_n f(n) = \frac{\log p}{p} \cdot \frac{1}{\log p} = \frac{1}{p}$, so that $\sum_{n \leq x} a_n f(n) = \sum_{p \leq x} 1/p$.

Step 2: Applying the formula. Abel summation (theorem 1.4.1), carried out from the base point 2 rather than 1 (no prime lies below 2, so $A(t) = 0$ on $[1, 2)$ and the integral over $[1, 2)$ vanishes), gives

$$\sum_{p \leq x} \frac{1}{p} = \frac{A(x)}{\log x} + \int_2^x \frac{A(t)}{t(\log t)^2} dt$$

Substitute $A(t) = \log t + R(t)$ into both pieces. The boundary term is

$$\frac{A(x)}{\log x} = \frac{\log x + R(x)}{\log x} = 1 + O\left(\frac{1}{\log x}\right)$$

since $R(x) = O(1)$. The integral splits as

$$\int_2^x \frac{\log t + R(t)}{t(\log t)^2} dt = \int_2^x \frac{dt}{t \log t} + \int_2^x \frac{R(t)}{t(\log t)^2} dt$$

Step 3: The two integrals. The first integral is elementary: with the substitution $u = \log t$, $du = dt/t$,

$$\int_2^x \frac{dt}{t \log t} = \int_{\log 2}^{\log x} \frac{du}{u} = \log \log x - \log \log 2$$

For the second integral, the integrand is absolutely bounded by $C_0/(t(\log t)^2)$, and

$$\int_2^\infty \frac{dt}{t(\log t)^2} = \int_{\log 2}^\infty \frac{du}{u^2} = \frac{1}{\log 2} < \infty$$

so $\int_2^\infty R(t)/(t(\log t)^2) dt$ converges absolutely to some real number, call it I . The tail past x is estimated the same way:

$$\left| \int_x^\infty \frac{R(t)}{t(\log t)^2} dt \right| \leq C_0 \int_x^\infty \frac{dt}{t(\log t)^2} = \frac{C_0}{\log x} = O\left(\frac{1}{\log x}\right)$$

whence $\int_2^x R(t)/(t(\log t)^2) dt = I + O(1/\log x)$.

Step 4: Assembling the constant. Collecting the boundary term and the two integrals,

$$\sum_{p \leq x} \frac{1}{p} = \left(1 + O\left(\frac{1}{\log x}\right)\right) + (\log \log x - \log \log 2) + \left(I + O\left(\frac{1}{\log x}\right)\right)$$

Setting $M = 1 - \log \log 2 + I$, the constant terms combine and the two error terms merge into a single $O(1/\log x)$, giving

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + M + O\left(\frac{1}{\log x}\right)$$

as we sought to show.

The second theorem is the quantitative form of the divergence of $\sum 1/p$, Euler's theorem that the primes are "dense enough" for their reciprocals to sum to infinity. The growth rate is now exact: the partial sums increase like $\log \log x$, doubly logarithmically, so the divergence is extraordinarily slow. To reach $\sum_{p \leq x} 1/p = 4$ requires x on the order of $e^{e^4} \approx 5 \times 10^{23}$. The presence of the second-order constant M is what distinguishes Mertens' statement from the bare divergence: $\sum_{p \leq x} 1/p - \log \log x$ converges to a definite limit M , at the rate $O(1/\log x)$. The proof produced $M = 1 - \log \log 2 + I$ as an integral expression depending on the error term $R(t)$ in the first theorem; this form is correct but opaque. The next definition gives the closed form, the same constant B that already appeared in proposition 1.3.9 as the second-order term in the average of ω , and identifying the two is the substantive content of what follows.

The constant M in the second theorem was produced as $1 - \log \log 2 + I$, with I an integral of the first-theorem error. Establishing its closed form proceeds in two stages. Comparing the divergent sum $\sum 1/p$ with the divergent sum $-\sum \log(1 - 1/p)$, whose difference is the convergent prime correction S , isolates the part of M contributed by the higher prime powers. The Euler–Mascheroni γ then enters separately, through the behavior of the zeta function at its pole, $\zeta(1 + \rho) = 1/\rho + \gamma + o(1)$, where the γ is the same harmonic-sum constant already computed in example 1.4.4; matching the smoothly damped prime sum that ζ governs against the sharp partial sums of $\sum 1/p$ is what pins the additive constant to γ .

Definition 3.3.4: The Mertens Constant

The *Mertens constant* is

$$M = \gamma + \sum_p \left(\log \left(1 - \frac{1}{p} \right) + \frac{1}{p} \right)$$

where γ is the Euler–Mascheroni constant (example 1.4.4) and the sum runs over all primes. This is the same constant as the B of proposition 1.3.9, $M = B = 0.2614\dots$, and it equals the constant M appearing in Mertens' second theorem (theorem 3.3.3).

The series defining M converges absolutely: for $|t| \leq 1/2$ the Taylor expansion $\log(1 - t) = -t - t^2/2 - t^3/3 - \dots$ gives $\log(1 - t) + t = -\sum_{j \geq 2} t^j/j$, so $|\log(1 - 1/p) + 1/p| \leq \sum_{j \geq 2} p^{-j}/j \leq \frac{1}{2} \sum_{j \geq 2} p^{-j} = \frac{1}{2p(p-1)} \leq 1/p^2$ for every prime $p \geq 2$, and $\sum_p p^{-2}$ converges. The claim requiring proof is that this closed-form constant is the same number M delivered by the proof of theorem 3.3.3.

Proof:

Let $M_2 = \lim_{x \rightarrow \infty} (\sum_{p \leq x} 1/p - \log \log x)$ denote the constant produced in theorem 3.3.3 (the limit exists because the error there is $O(1/\log x) \rightarrow 0$), and let $M = \gamma + \sum_p (\log(1 - 1/p) + 1/p)$ denote the closed form above. The two are shown equal.

Step 1: Logarithm of the truncated Euler product. For a fixed $x \geq 2$, take logarithms of the finite product over primes $p \leq x$ and expand each factor by the Taylor series for $\log(1 - 1/p)$:

$$\sum_{p \leq x} \log \left(1 - \frac{1}{p} \right) = - \sum_{p \leq x} \frac{1}{p} - \sum_{p \leq x} \sum_{j \geq 2} \frac{1}{j p^j} = - \sum_{p \leq x} \frac{1}{p} - S(x)$$

where $S(x) = \sum_{p \leq x} \sum_{j \geq 2} \frac{1}{j p^j} = - \sum_{p \leq x} (\log(1 - 1/p) + 1/p)$. As established above, the double series $\sum_p \sum_{j \geq 2} (j p^j)^{-1}$ converges absolutely, so $S(x) \rightarrow S := - \sum_p (\log(1 - 1/p) + 1/p)$ as $x \rightarrow \infty$, with tail

$$0 \leq S - S(x) = \sum_{p > x} \sum_{j \geq 2} \frac{1}{j p^j} \leq \sum_{p > x} \frac{1}{p^2} \leq \sum_{n > x} \frac{1}{n^2} = O\left(\frac{1}{x}\right)$$

Rearranging the displayed identity,

$$\sum_{p \leq x} \frac{1}{p} = - \sum_{p \leq x} \log \left(1 - \frac{1}{p} \right) - S(x)$$

Step 2: The regularized prime sum and the entry of γ . Write $P(x) = \prod_{p \leq x} (1 - 1/p)^{-1}$, so that $\log P(x) = - \sum_{p \leq x} \log(1 - 1/p)$ and the identity of Step 1 becomes

$$\sum_{p \leq x} \frac{1}{p} = \log P(x) - S(x) \tag{3.1}$$

The number M_2 to be evaluated is $\lim_{x \rightarrow \infty} (\sum_{p \leq x} 1/p - \log \log x)$, and (3.1) together with $S(x) \rightarrow S$ rewrites it as $M_2 = \lim_{x \rightarrow \infty} (\log P(x) - \log \log x) - S$. The whole task is therefore to show that the additive constant in $\log P(x) = \log \log x + (\text{const}) + o(1)$ is exactly the Euler–Mascheroni γ . This constant is not produced by any termwise manipulation of the product; it is the precise discrepancy between two regularizations of the divergent sum $\sum_p 1/p$, the sharp cutoff $\sum_{p \leq x} 1/p$ and the analytic cutoff $F(\rho) := \sum_p p^{-1-\rho}$ as $\rho \rightarrow 0^+$. The strategy is to evaluate the analytic cutoff in closed form, where γ appears in-chapter, and then compare the two cutoffs.

The analytic regularization is evaluated first. Taking logarithms of the Euler product $\zeta(1 + \rho) = \prod_p (1 - p^{-1-\rho})^{-1}$ (valid for $\rho > 0$, where both sides converge absolutely) and expanding each factor exactly as in Step 1,

$$\log \zeta(1 + \rho) = \sum_p \sum_{j \geq 1} \frac{1}{j p^{j(1+\rho)}} = F(\rho) + S(\rho), \quad S(\rho) := \sum_p \sum_{j \geq 2} \frac{1}{j p^{j(1+\rho)}}$$

Since $0 \leq S(\rho) \leq S(0) = S$ for $\rho \geq 0$ and each term increases to its $\rho = 0$ value as $\rho \rightarrow 0^+$, monotone convergence gives $S(\rho) \rightarrow S$. The constant γ now enters through $\zeta(1 + \rho)$ as $\rho \rightarrow 0^+$,

evaluated entirely within the convergent range $\text{Re } s > 1$, with no appeal to continuation past the line. By the closed form of proposition 1.4.6, valid for $\text{Re } s > 1$, $\zeta(s) = \frac{1}{s-1} + 1 - s \int_1^\infty \{t\} t^{-s-1} dt$, and at $s = 1 + \rho$ this reads

$$\zeta(1 + \rho) = \frac{1}{\rho} + 1 - (1 + \rho) \int_1^\infty \frac{\{t\}}{t^{2+\rho}} dt = \frac{1}{\rho} + \gamma + o(1) \quad (\rho \rightarrow 0^+)$$

the limit by dominated convergence ($\{t\} t^{-2-\rho} \leq t^{-2}$ for $\rho \geq 0$) and the value $\int_1^\infty \{t\} t^{-2} dt = 1 - \gamma$, which is exactly the evaluation of the Euler–Mascheroni constant in example 1.4.4. Hence $\log \zeta(1 + \rho) = \log(1/\rho) + \log(1 + \gamma\rho + o(\rho)) = \log(1/\rho) + o(1)$, and subtracting $S(\rho) \rightarrow S$,

$$F(\rho) = \sum_p \frac{1}{p^{1+\rho}} = \log \frac{1}{\rho} - S + o(1) \quad (\rho \rightarrow 0^+) \quad (3.2)$$

This is the analytic regularization in closed form, with γ already discharged into the relation $M_2 = \gamma - S$ once the two regularizations are matched.

Step 3: Matching the two regularizations. It remains to compare the sharp cutoff with the analytic one at the calibration $\rho = 1/\log x$. The comparison is

$$\sum_{p \leq x} \frac{1}{p} - F\left(\frac{1}{\log x}\right) = \gamma + o(1) \quad (x \rightarrow \infty) \quad (3.3)$$

This single comparison is the one analytic input not delivered by the tools of this chapter, and it is isolated here as such. Its content is that replacing the sharp truncation $1_{p \leq x}$ by the smooth damping $p^{-1/\log x}$ shifts the (divergent) prime sum by exactly γ in the limit. The shift is computed by Mertens' own method: the partial sums of theorem 3.3.2 are fed through partial summation against the two cutoff weights, the difference is expressed as an integral over a smooth integrand by the Euler–Maclaurin formula (theorem 1.5.6), and the limit $\rho \rightarrow 0^+$ is taken. The constant that survives is the value of the integral $\int_0^\infty \left(\frac{1}{e^t-1} - \frac{1}{te^t}\right) dt = \gamma$, equivalently $\Gamma'(1) = -\gamma$, an identity for the Gamma function rather than for ζ . The full computation, with explicit error bounds and the evaluation of this integral, is carried out in [8, §I.1.6] and in [2, Theorem 429]; the historically faithful version of Mertens' 1874 argument, with the same integral producing γ , is reproduced in detail in [10]. The mechanical tools that (3.3) rests on, the Euler product, partial summation, and Euler–Maclaurin, are all in hand; only the Gamma-function constant lies outside the chapter, and it is a fact about the Gamma function, not a disguised appeal to the analytic theory of ζ developed in chapter 2.

Granting (3.3), substitute (3.2) at $\rho = 1/\log x$, where $\log(1/\rho) = \log \log x$, to obtain

$$\sum_{p \leq x} \frac{1}{p} = F\left(\frac{1}{\log x}\right) + \gamma + o(1) = (\log \log x - S + o(1)) + \gamma + o(1) = \log \log x + (\gamma - S) + o(1)$$

Hence $\lim_{x \rightarrow \infty} (\sum_{p \leq x} 1/p - \log \log x) = \gamma - S$. By theorem 3.3.3 the same limit equals M_2 , and limits are unique, so

$$M_2 = \gamma - S = \gamma + \sum_p \left(\log \left(1 - \frac{1}{p} \right) + \frac{1}{p} \right) = M$$

the middle equality by $S = -\sum_p (\log(1 - 1/p) + 1/p)$ from Step 1. This is the closed form claimed for the Mertens constant. The equality $M = B$ with the constant of proposition 1.3.9 is immediate, since B was defined there by the identical expression $\gamma + \sum_p (\log(1 - 1/p) + 1/p)$, completing the proof.

The closed form makes the Mertens constant computable and ties together three appearances of the same number: the second-order term in $\sum_{p \leq x} 1/p$, the second-order term in the average of ω from proposition 1.3.9, and (through theorem 3.3.5 below) the constant governing the product $\prod(1 - 1/p)$. The expression $M = \gamma + \sum_p (\log(1 - 1/p) + 1/p)$ exhibits M as the Euler–Mascheroni constant corrected by a rapidly convergent prime sum: each correction term $\log(1 - 1/p) + 1/p$ is $-1/(2p^2) + O(p^{-3})$, so the sum is dominated by its first few terms and $M = 0.2614972128\dots$ is computed to high precision from a short truncation. The dependence on γ is not incidental. The constant γ entered through the value $\zeta(1 + \rho) = 1/\rho + \gamma + o(1)$ of the zeta function near its pole, which is itself the harmonic-sum constant of example 1.4.4 read off from the analytic continuation in proposition 1.4.6; the prime sum $\sum 1/p$ inherits γ as the precise gap between its sharp partial sums and the smoothly damped sum $\sum_p p^{-1-\rho}$, the two regularizations of the same divergent series whose difference (3.3) pins to γ . The convergent correction $-S$ is a separate, smaller adjustment: it measures only the gap between $\sum 1/p$ and $\log \prod(1 - 1/p)^{-1}$ caused by the higher prime powers, not the appearance of γ itself.

The product form is the most frequently cited of the three theorems, appearing throughout sieve theory and the theory of arithmetic functions as the density of integers free of small prime factors. Taking the logarithm of the product converts it into the sum $\sum_{p \leq x} \log(1 - 1/p)$, which the second theorem and the closed form of M evaluate; the constant $e^{-\gamma}$ emerges because the convergent prime correction in M cancels precisely against the convergent correction in $\sum \log(1 - 1/p)$, leaving only $-\gamma$.

Theorem 3.3.5: Mertens' Third Theorem

As $x \rightarrow \infty$,

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right) \sim \frac{e^{-\gamma}}{\log x}$$

where γ is the Euler–Mascheroni constant. Equivalently,

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right)^{-1} = e^{\gamma} \log x (1 + o(1))$$

Proof:

Step 1: Taking logarithms. Let $L(x) = \log \prod_{p \leq x} (1 - 1/p) = \sum_{p \leq x} \log(1 - 1/p)$. The claim $\prod_{p \leq x} (1 - 1/p) \sim e^{-\gamma} / \log x$ is equivalent, after taking logarithms, to

$$L(x) = -\gamma - \log \log x + o(1)$$

since $\log(e^{-\gamma} / \log x) = -\gamma - \log \log x$ and the asymptotic $\sim$ between positive quantities is equivalent to the difference of their logarithms tending to 0.

Step 2: Splitting off the convergent correction. For each prime p write $\log(1 - 1/p) = -1/p +$

$(\log(1 - 1/p) + 1/p)$, and sum over $p \leq x$:

$$L(x) = - \sum_{p \leq x} \frac{1}{p} + \sum_{p \leq x} \left(\log \left(1 - \frac{1}{p} \right) + \frac{1}{p} \right)$$

The second sum converges as $x \rightarrow \infty$: by the bound established after definition 3.3.4, $|\log(1 - 1/p) + 1/p| \leq 1/p^2$, so the series $\sum_p (\log(1 - 1/p) + 1/p)$ converges absolutely, with value $M - \gamma$ by the definition of the Mertens constant (definition 3.3.4). Writing the partial sum as the full sum minus a tail,

$$\sum_{p \leq x} \left(\log \left(1 - \frac{1}{p} \right) + \frac{1}{p} \right) = (M - \gamma) - \sum_{p > x} \left(\log \left(1 - \frac{1}{p} \right) + \frac{1}{p} \right) = (M - \gamma) + o(1)$$

since the tail of a convergent series tends to 0 (explicitly, $|\sum_{p > x} (\dots)| \leq \sum_{n > x} n^{-2} = O(1/x)$).

Step 3: Applying the second theorem. By Mertens' second theorem (theorem 3.3.3) with the closed form M of definition 3.3.4,

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + M + O\left(\frac{1}{\log x}\right)$$

Substituting both evaluations into the expression for $L(x)$ from Step 2,

$$L(x) = - \left(\log \log x + M + O\left(\frac{1}{\log x}\right) \right) + ((M - \gamma) + o(1))$$

The constant M cancels: $-M + (M - \gamma) = -\gamma$. The error terms $O(1/\log x)$ and $o(1)$ combine into $o(1)$. Hence

$$L(x) = -\log \log x - \gamma + o(1)$$

which is exactly the equivalent form from Step 1.

Step 4: Exponentiating. Exponentiating $L(x) = -\gamma - \log \log x + o(1)$,

$$\prod_{p \leq x} \left(1 - \frac{1}{p} \right) = e^{L(x)} = e^{-\gamma - \log \log x + o(1)} = \frac{e^{-\gamma}}{\log x} \cdot e^{o(1)} = \frac{e^{-\gamma}}{\log x} (1 + o(1))$$

using $e^{o(1)} = 1 + o(1)$, which is the first stated form. Taking reciprocals gives $\prod_{p \leq x} (1 - 1/p)^{-1} = e^{\gamma} \log x (1 + o(1))$, the second form, completing the proof.

The cancellation in Step 3 is the mechanism that produces $e^{-\gamma}$, and it shows precisely why the Mertens constant had to be defined with the γ term: the constant M in $\sum 1/p$ and the convergent sum $\sum (\log(1 - 1/p) + 1/p) = M - \gamma$ in $\log \prod (1 - 1/p)$ share the prime-sum part, which cancels, while the γ that distinguishes them survives as the exponent. The third theorem quantifies a competition between two effects in the product $\prod_{p \leq x} (1 - 1/p)$: each factor $1 - 1/p$ is slightly less than 1, so the product decreases, but the factors approach 1 as p grows, so the decrease slows. The outcome $\sim e^{-\gamma}/\log x$ says the product tends to 0, but only as slowly as $1/\log x$, with the precise rate fixed by $e^{-\gamma} \approx 0.5615$. This product is the proportion of integers surviving the sieve that removes every

multiple of every prime up to x ; its value $e^{-\gamma}/\log x$, rather than the naive $1/\log x$ one might guess from the heuristic that “a $1/\log x$ fraction of integers near x are prime,” is the source of the constant $2e^{-\gamma}$ in the Mertens correction to sieve estimates and the reason the twin-prime constant and similar products carry factors of $e^{-\gamma}$.

A numerical check confirms that the two nontrivial constants, M in the second theorem and $e^{-\gamma}$ in the third, are the ones the data demand, and exhibits the slow convergence that the error terms predict.

Example 3.3.6: Numerical Check of Mertens' Theorems

The constants entering the second and third theorems are

$$M = 0.26150\dots, \quad \gamma = 0.57722\dots, \quad e^{-\gamma} = 0.56146\dots$$

For the second theorem, the table compares the partial sum $\sum_{p \leq x} 1/p$ against its predicted value $\log \log x + M$, recording the discrepancy:

x	$\sum_{p \leq x} 1/p$	$\log \log x + M$	difference
10^2	1.80282	1.78868	+0.01414
10^4	2.48306	2.48182	+0.00124
10^6	2.88733	2.88729	+0.00004

The difference $\sum_{p \leq x} 1/p - (\log \log x + M)$ shrinks steadily, from about 1.4×10^{-2} at $x = 10^2$ to 4×10^{-5} at $x = 10^6$, consistent with the $O(1/\log x)$ error term of theorem 3.3.3: the ratio of successive differences tracks the ratio of successive values of $1/\log x$, the predicted decay. For the third theorem, the table compares $\prod_{p \leq x} (1 - 1/p) \cdot \log x$ against the predicted limit $e^{-\gamma} = 0.56146$:

x	$\prod_{p \leq x} (1 - 1/p)$	$(\prod_{p \leq x} (1 - 1/p)) \log x$	$e^{-\gamma}$
10^2	0.12032	0.55408	0.56146
10^4	0.06088	0.56077	0.56146
10^6	0.04064	0.56144	0.56146

The product $(\prod_{p \leq x} (1 - 1/p)) \log x$ approaches $e^{-\gamma}$ from below, sitting within two parts in 10^5 of $e^{-\gamma}$ at $x = 10^6$. Both tables confirm the predicted constants, M in the second theorem and $e^{-\gamma}$ in the third, and both exhibit the slow convergence characteristic of error terms that decay only as a power of $1/\log x$ rather than a power of $1/x$.

3.4 Bertrand's Postulate

Chebyshev's estimates of section 3.2 pin the global count of primes up to x to the correct order $x/\log x$, but a global count says nothing about where the primes sit. A statement like $\pi(2n) - \pi(n) \approx n/\log n$ would assert, on average, many primes between n and $2n$, yet an average leaves open the possibility of long prime-free stretches compensated by dense clusters elsewhere. Bertrand's postulate is the first genuine *localization* result: for every integer $n \geq 1$ there is at least one prime in the interval $(n, 2n]$, with no exceptions and no averaging. Conjectured by Bertrand in 1845 on numerical evidence up to several million and proved by Chebyshev in 1852, it guarantees a prime in an interval whose length is comparable to its left endpoint, a far shorter window than the global count alone delivers.

The proof is a refinement of the binomial method already used for Chebyshev's bounds. The central binomial coefficient $\binom{2n}{n}$ is large, of size roughly 4^n , and section 3.2 controlled its prime factorization through Legendre's formula. Sharpening that control shows that if no prime fell in $(n, 2n]$, then $\binom{2n}{n}$ would be forced to assemble its entire size out of small primes alone, each contributing only a bounded factor; the product of those small contributions is too small to account for 4^n once n is large. The contradiction proves the postulate for all large n , and the finitely many remaining cases are settled by exhibiting an explicit chain of primes. The argument here is the one found by Erdős in 1932, which replaced Chebyshev's analytic estimates of the Gamma function by the elementary bounds of lemma 3.4.1.

Lemma 3.4.1: Bounds on the Central Binomial Coefficient

For every integer $n \geq 1$,

$$\frac{4^n}{2n+1} \leq \binom{2n}{n} \leq 4^n$$

Proof:

Both bounds compare $\binom{2n}{n}$ with the full binomial sum $\sum_{j=0}^{2n} \binom{2n}{j} = (1+1)^{2n} = 4^n$.

Step 1: The upper bound. The coefficient $\binom{2n}{n}$ is one of the $2n+1$ nonnegative terms $\binom{2n}{0}, \binom{2n}{1}, \dots, \binom{2n}{2n}$ whose sum is 4^n . A single term of a sum of nonnegative reals cannot exceed the whole sum, so

$$\binom{2n}{n} \leq \sum_{j=0}^{2n} \binom{2n}{j} = 4^n$$

which is the upper bound.

Step 2: The lower bound. The same $2n+1$ terms have $\binom{2n}{n}$ as their largest. Indeed the ratio of consecutive terms is

$$\frac{\binom{2n}{j+1}}{\binom{2n}{j}} = \frac{2n-j}{j+1}$$

which exceeds 1 exactly when $2n-j > j+1$, that is, $j < (2n-1)/2 = n - \frac{1}{2}$, and is less than 1 when $j > n - \frac{1}{2}$. Thus the terms strictly increase up to $j = n$ and strictly decrease after it, so $\binom{2n}{n}$ is the unique maximum. Being the largest of $2n+1$ terms, it is at least their average:

$$\binom{2n}{n} \geq \frac{1}{2n+1} \sum_{j=0}^{2n} \binom{2n}{j} = \frac{4^n}{2n+1}$$

which is the lower bound, completing the proof.

The two bounds bracket $\binom{2n}{n}$ between $4^n/(2n+1)$ and 4^n , a window narrow on the logarithmic scale: $\log \binom{2n}{n} = n \log 4 + O(\log n)$, so to leading order $\binom{2n}{n}$ behaves like $4^n = 2^{2n}$. The lower bound is the operative one for Bertrand's postulate, since the argument proceeds by showing that a hypothetical absence of primes in $(n, 2n]$ makes $\binom{2n}{n}$ too small, and "too small" is measured against $4^n/(2n+1)$. The factor $2n+1$ in the denominator is wasteful (Stirling's formula theorem 1.5.7 gives the sharper

$\binom{2n}{n} \sim 4^n / \sqrt{\pi n}$, but the crude bound suffices and keeps the proof elementary. The next lemma supplies the matching upper control on $\binom{2n}{n}$, not through its size but through its prime factorization.

The mechanism of Bertrand's postulate is a restriction on *which* primes can divide $\binom{2n}{n}$, and to what power. Three facts are needed. The first removes a whole band of primes just below n from the factorization; the second caps the exponent of every prime past $\sqrt{2n}$ at one; the third, already proved in section 3.2, caps every prime power at $2n$. Together they bound $\binom{2n}{n}$ from above by a product whose factors are small primes raised to controlled powers.

Lemma 3.4.2: Primes in the Central Binomial Coefficient

Let $n \geq 3$ be an integer and let p be a prime.

- (i) If $\frac{2n}{3} < p \leq n$, then $p \nmid \binom{2n}{n}$.
- (ii) If $\sqrt{2n} < p \leq 2n$, then $v_p\left(\binom{2n}{n}\right) \leq 1$, so p divides $\binom{2n}{n}$ to at most the first power.
- (iii) For every prime p , the prime power $p^{v_p\left(\binom{2n}{n}\right)}$ satisfies $p^{v_p\left(\binom{2n}{n}\right)} \leq 2n$.

Proof:

Throughout, the valuation formula of lemma 3.2.1 is used in the form

$$v_p\left(\binom{2n}{n}\right) = \sum_{k \geq 1} \left(\left\lfloor \frac{2n}{p^k} \right\rfloor - 2 \left\lfloor \frac{n}{p^k} \right\rfloor \right)$$

in which every summand is 0 or 1 and the summand vanishes once $p^k > 2n$.

- (i) *The band $(2n/3, n]$.* Assume $\frac{2n}{3} < p \leq n$. The hypotheses on p and $n \geq 3$ force $p \geq 3$: from $p > 2n/3 \geq 2$ one gets $p \geq 3$. The aim is to show every summand in the valuation formula vanishes. Consider the terms by exponent k .

For $k = 1$: from $p \leq n$ one has $n/p \geq 1$, and from $p > 2n/3$ one has $n/p < 3/2$, so $1 \leq n/p < 3/2$ and $\lfloor n/p \rfloor = 1$. Likewise $2 \leq 2n/p < 3$, so $\lfloor 2n/p \rfloor = 2$. The $k = 1$ summand is $\lfloor 2n/p \rfloor - 2\lfloor n/p \rfloor = 2 - 2 \cdot 1 = 0$.

For $k \geq 2$: since $p > 2n/3$ and $p \geq 3$,

$$p^2 > \frac{2n}{3} \cdot p \geq \frac{2n}{3} \cdot 3 = 2n$$

so $p^k \geq p^2 > 2n$ for all $k \geq 2$, and every such summand vanishes by lemma 3.2.1.

All summands being zero, $v_p\left(\binom{2n}{n}\right) = 0$, that is, $p \nmid \binom{2n}{n}$.

- (ii) *Primes past $\sqrt{2n}$.* Assume $\sqrt{2n} < p \leq 2n$. Then $p^2 > 2n$, so for every $k \geq 2$ the term $\lfloor 2n/p^k \rfloor = 0$ and the corresponding summand vanishes. Only the $k = 1$ summand can survive, and it is 0 or 1 by lemma 3.2.1. Hence $v_p\left(\binom{2n}{n}\right) \leq 1$.
- (iii) *The cap on prime powers.* This is exactly the bound $p^{v_p\left(\binom{2n}{n}\right)} \leq 2n$ proved in lemma 3.2.1, valid for every prime p and every $n \geq 1$, recorded here for use alongside (i) and (ii).

The three statements together control the factorization as required.

The three statements act on disjoint parts of the prime range. Statement (iii) is universal but coarse: it bounds each prime's contribution by $2n$ regardless of the prime. Statement (ii) improves this for primes past $\sqrt{2n}$, where the exponent drops to at most one, so the contribution is just p rather than a power. Statement (i) is the new ingredient, absent from the Chebyshev bounds of section 3.2: it asserts that the primes in the band $(2n/3, n]$ are entirely missing from $\binom{2n}{n}$. The reason is arithmetic cancellation. Such a prime p appears once in the numerator $(2n)!$ as the factor p and once more as $2p$ (since $2p \leq 2n$ when $p \leq n$), contributing two to $v_p((2n)!)$; in the denominator $n!n!$ it appears once in each factorial as p (since $p \leq n$), contributing $2 \cdot 1 = 2$ to $2v_p(n!)$. The two cancel exactly. The constraint $p > 2n/3$ ensures that $3p > 2n$, so $3p$ is not among the factors of $(2n)!$ and no third copy disturbs the balance. This band is precisely the range just below n where one might have hoped to find the divisors of $\binom{2n}{n}$; their absence is what forces the coefficient to draw on smaller primes and ultimately produces the contradiction.

With the size of $\binom{2n}{n}$ bounded below and its factorization bounded above, the postulate follows by comparing the two. The comparison yields a contradiction for all sufficiently large n ; the threshold is explicit, and the finitely many smaller values are checked by an explicit list of primes.

Theorem 3.4.3: Bertrand's Postulate

For every integer $n \geq 1$ there exists a prime p with $n < p \leq 2n$.

Proof:

The argument has two parts: a proof for all $n \geq 512$ by contradiction, and a direct verification of $1 \leq n \leq 511$ by an explicit chain of primes.

Step 1: Assume no prime lies in $(n, 2n]$ for some $n \geq 512$. Under this assumption the primes dividing $\binom{2n}{n}$ are confined to small values. By lemma 3.2.1 every prime dividing $\binom{2n}{n}$ is at most $2n$. The assumption removes the primes in $(n, 2n]$, and part (i) of lemma 3.4.2 removes the primes in $(2n/3, n]$ (here $n \geq 512 \geq 3$). What remains is

$$\binom{2n}{n} = \prod_{p \leq 2n/3} p^{v_p(\binom{2n}{n})}$$

the product over primes $p \leq 2n/3$ only.

Step 2: Bound the product from above. Split the surviving primes at $\sqrt{2n}$.

For primes $p \leq \sqrt{2n}$, part (iii) of lemma 3.4.2 gives $p^{v_p(\binom{2n}{n})} \leq 2n$. The number of primes $p \leq \sqrt{2n}$ is at most $\sqrt{2n}$ (there are at most $\sqrt{2n}$ positive integers up to $\sqrt{2n}$), so their combined contribution is

$$\prod_{p \leq \sqrt{2n}} p^{v_p(\binom{2n}{n})} \leq (2n)^{\sqrt{2n}}$$

For primes $\sqrt{2n} < p \leq 2n/3$, part (ii) of lemma 3.4.2 gives $v_p(\binom{2n}{n}) \leq 1$, so each contributes at most p itself, and their product is at most the product of all primes up to $2n/3$. The primorial bound proposition 3.2.2 controls this:

$$\prod_{\sqrt{2n} < p \leq 2n/3} p^{v_p(\binom{2n}{n})} \leq \prod_{p \leq 2n/3} p \leq 4^{2n/3}$$

Multiplying the two ranges,

$$\binom{2n}{n} \leq (2n)^{\sqrt{2n}} \cdot 4^{2n/3}$$

is the upper bound forced by the assumption.

Step 3: Contradiction with the lower bound. The lower bound of lemma 3.4.1 gives $\binom{2n}{n} \geq 4^n/(2n+1)$. Combined with Step 2,

$$\frac{4^n}{2n+1} \leq (2n)^{\sqrt{2n}} \cdot 4^{2n/3}$$

Dividing by $4^{2n/3}$ and noting $4^n/4^{2n/3} = 4^{n/3}$,

$$4^{n/3} \leq (2n+1)(2n)^{\sqrt{2n}}$$

Take logarithms. With $2n+1 \leq (2n)^2$ for $n \geq 1$ (since $2n+1 \leq 4n^2$ once $n \geq 1$), the right side satisfies $\log((2n+1)(2n)^{\sqrt{2n}}) \leq (\sqrt{2n}+2)\log(2n)$, so

$$\frac{n}{3} \log 4 \leq (\sqrt{2n}+2)\log(2n)$$

The left side grows linearly in n while the right side grows like $\sqrt{n} \log n$, so the inequality fails for all large n ; it remains to locate the threshold. Substitute $m = \sqrt{2n}$, so $n = m^2/2$ and the inequality reads

$$\frac{m^2}{6} \log 4 \leq (m+2) \cdot 2 \log m$$

that is, $m \log 4 \leq 12(1+2/m) \log m$. For $m \geq 32$ (which corresponds to $n \geq 512$) the factor $1+2/m \leq 1+1/16 = 17/16$, so the right side is at most $12 \cdot \frac{17}{16} \log m = \frac{51}{4} \log m$, and the inequality would force

$$m \log 4 \leq \frac{51}{4} \log m, \quad \text{i.e.} \quad \frac{m}{\log m} \leq \frac{51}{4 \log 4} = \frac{51}{4 \cdot 1.3862 \dots} = 9.197 \dots$$

But $m/\log m$ is increasing for $m \geq 3$, and at $m = 32$ it already equals $32/\log 32 = 32/3.4657 \dots = 9.233 \dots > 9.197 \dots$. Hence the inequality fails for $m = 32$, and being increasing, $m/\log m$ exceeds the bound for all $m \geq 32$. This contradicts the displayed inequality for every $n \geq 512$. Therefore the assumption is untenable: a prime exists in $(n, 2n]$ for every $n \geq 512$.

Step 4: The range $1 \leq n \leq 511$. Consider the increasing chain of primes

$$2, 3, 5, 7, 13, 23, 43, 83, 163, 317, 631$$

Each prime in the list is less than twice its predecessor: $3 < 4$, $5 < 6$, $7 < 10$, $13 < 14$, $23 < 26$, $43 < 46$, $83 < 86$, $163 < 166$, $317 < 326$, $631 < 634$. Write the chain as $q_0 = 2 < q_1 = 3 < \dots < q_{10} = 631$, so that $q_i < 2q_{i-1}$ for each $i \geq 1$. Given an integer n with $1 \leq n \leq 511$, choose the smallest chain prime q_i with $q_i > n$; such an index exists because $q_{10} = 631 > 511 \geq n$. Minimality gives $q_{i-1} \leq n$ when $i \geq 1$, and when $n = 1$ the prime $q_0 = 2$ already satisfies $1 < 2 \leq 2 = 2n$. For $i \geq 1$,

$$q_i < 2q_{i-1} \leq 2n$$

so $n < q_i \leq 2n$, exhibiting a prime in $(n, 2n]$. This covers every n in $[1, 511]$.

Steps 1 through 3 establish the postulate for $n \geq 512$ and Step 4 for $1 \leq n \leq 511$, so a prime lies in $(n, 2n]$ for every integer $n \geq 1$, as we sought to show.

Bertrand's postulate sharpens the picture of the primes from a global count to a guarantee on every dyadic scale. Where Chebyshev's bounds (theorem 3.2.3) assert that the primes up to x number a constant multiple of $x/\log x$, leaving open how they bunch, the postulate forbids any gap from doubling: between n and $2n$ there is always a prime, so the ratio of consecutive primes is bounded. The two results draw on the same object, the central binomial coefficient, and differ only in how finely its factorization is dissected. Chebyshev compared the size $4^n/(2n+1) \leq \binom{2n}{n} \leq 4^n$ against the universal cap $p^{v_p} \leq 2n$ on prime powers; Bertrand adds the observation that the primes in the band $(2n/3, n]$ contribute nothing, which concentrates the entire size of $\binom{2n}{n}$ onto primes below $2n/3$ and onto the at-most-once primes between $\sqrt{2n}$ and $2n/3$, controlled by the primorial bound $\prod_{p \leq 2n/3} p \leq 4^{2n/3}$ of proposition 3.2.2. The exponent $2n/3$ against the size 4^n leaves a deficit of $4^{n/3}$ that only primes in $(n, 2n]$ could supply, and that is exactly the interval whose nonemptiness the theorem asserts. The threshold $n \geq 512$ is not optimal; the inequality of Step 3 already fails for $n \geq 468$, and tightening the prime-power bookkeeping lowers it further, but a clean power of two suffices and the explicit chain disposes of the remainder. In the language of Chebyshev's functions (definition 3.1.2), the postulate is the statement that $\vartheta(2n) - \vartheta(n) > 0$ for all $n \geq 1$, a positivity that the order-of-magnitude estimate $\vartheta(x) \asymp x$ suggests but does not by itself prove.

The postulate yields immediate bounds on the growth of the prime sequence, both on the gap between consecutive primes and on the size of the n -th prime.

Corollary 3.4.4: Consequences of Bertrand's Postulate

Let $p_1 = 2 < p_2 = 3 < p_3 = 5 < \dots$ enumerate the primes in increasing order. Then for every $n \geq 1$:

- (i) the consecutive primes satisfy $p_{n+1} < 2p_n$;
- (ii) the n -th prime satisfies $p_n \leq 2^n$, with equality only at $n = 1$.

Proof:

The two statements are proved in turn, the second by induction using the first.

- (i) Apply theorem 3.4.3 with n replaced by p_n : there is a prime q with $p_n < q \leq 2p_n$. Since $q > p_n$, the prime q is one of $p_{n+1}, p_{n+2}, \dots$, so $p_{n+1} \leq q$ as p_{n+1} is the least prime exceeding p_n . The strict inequality $q \leq 2p_n$ combined with $2p_n$ being even and hence not prime (as $p_n \geq 2$ gives $2p_n \geq 4 > 2$) yields $q < 2p_n$, whence $p_{n+1} \leq q < 2p_n$.
- (ii) Induct on n . The base case $n = 1$ is the equality $p_1 = 2 = 2^1$, which gives $p_1 \leq 2^1$. Assume $p_n \leq 2^n$ for some $n \geq 1$. By part (i), $p_{n+1} < 2p_n \leq 2 \cdot 2^n = 2^{n+1}$, so $p_{n+1} \leq 2^{n+1}$, which is the claim for $n + 1$. By induction $p_n \leq 2^n$ for all $n \geq 1$. Equality occurs only at $n = 1$: for $n \geq 2$ the inductive step delivers the strict $p_{n+1} < 2^{n+1}$, so $p_n < 2^n$ for every $n \geq 2$.

Both bounds follow, completing the proof.

The bound $p_{n+1} < 2p_n$ records that the primes never thin out faster than geometrically: each prime is followed by another before the value doubles. The bound $p_n \leq 2^n$ (strict for $n \geq 2$) is a crude but unconditional handle on the size of the n -th prime, obtained with no analytic input. It is far from the truth, since the Prime Number Theorem (chapter 5, foreshadowed) gives $p_n \sim n \log n$, a function of much slower growth than 2^n ; the gap measures how much is lost by iterating a doubling bound

rather than tracking the actual density of primes. The value of $p_n \leq 2^n$ is its independence from any analytic input: it follows from Bertrand's postulate alone, which follows in turn from the size and factorization of a single binomial coefficient.

The Riemann Zeta Function

The two preceding chapters approached the zeta function from opposite directions. chapter 2 built $\zeta(s) = \sum_{n \geq 1} n^{-s}$ as a Dirichlet series, established its Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ as the analytic form of unique factorization (corollary 2.3.2), and produced the integral representation $\Gamma(s)\zeta(s) = \int_0^\infty t^{s-1}/(e^t - 1) dt$ (theorem 2.4.5), all on the half-plane $\operatorname{Re}(s) > 1$ where the defining series converges absolutely. chapter 3 took the elementary road, fixing the order of magnitude of $\pi(x)$ and evaluating the sums $\sum_{p \leq x} (\log p)/p$ and $\sum_{p \leq x} 1/p$ to leading order by counting prime powers, without any appeal to complex analysis. What both chapters left untouched is the behavior of $\zeta(s)$ as a function of a *complex* variable outside the half-plane of convergence. The series $\sum n^{-s}$ diverges at $s = 1$ and gives no value there, yet $\zeta(s)$ extends to a meromorphic function on a larger region, and it is the analytic structure of that extension, not the series, that controls the primes.

This chapter studies $\zeta(s)$ as a holomorphic function. The central facts are three. First, $\zeta(s)$ continues meromorphically to the half-plane $\operatorname{Re}(s) > 0$, holomorphic everywhere except for a single simple pole at $s = 1$ with residue 1; the continuation is already implicit in the partial-sum formula of chapter 1 and requires no new machinery. Second, the logarithmic derivative $-\zeta'(s)/\zeta(s) = \sum_n \Lambda(n)n^{-s}$ inherits a simple pole at $s = 1$ with residue 1, and this pole is the analytic source of the leading term $\psi(x) \sim x$ of the Prime Number Theorem. Third, $\zeta(s)$ does not vanish on the line $\operatorname{Re}(s) = 1$, a fact that lies just outside the reach of the elementary Euler-product argument of chapter 2 and demands a new idea. These three results are the analytic foundation on which the proof of the Prime Number Theorem rests in chapter 5; the deeper continuation to all of $\mathbb{C}$, together with the functional equation relating $\zeta(s)$ to $\zeta(1-s)$, is taken up in ??.

4.1 Definition and the Euler Product

The starting point is the zeta function on its native half-plane $\operatorname{Re}(s) > 1$, where the series, the product, and the integral of chapter 2 all converge and agree. This section consolidates what is

already known there, recording the elementary analytic properties of $\zeta(s)$ as a holomorphic function of s , restating the Euler product and the non-vanishing it forces, and computing a few special values. The Euler product and the integral representation are not re-derived; they are imported from chapter 2 and cited. The one formula carried forward in full is the partial-sum identity of chapter 1, which already contains the continuation to $\operatorname{Re}(s) > 0$ and which the next section turns into the continuation theorem.

The defining series and the meaning of $\zeta(s)$ on $\operatorname{Re}(s) > 1$ were fixed in chapter 2; recall that $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ is the Dirichlet series of the constant function **1** (definition 2.1.1), with abscissa of absolute convergence $\sigma_a = 1$. The first task is to assemble the elementary consequences of this description into a single statement about ζ as a function.

Proposition 4.1.1: Analytic Properties of ζ on $\operatorname{Re}(s) > 1$

The Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ has the following properties on the half-plane $\operatorname{Re}(s) > 1$.

1. It is holomorphic, and its derivative is obtained by term-by-term differentiation:

$$\zeta'(s) = - \sum_{n=1}^{\infty} \frac{\log n}{n^s}$$

2. For every s with $\operatorname{Re}(s) = \sigma > 1$,

$$|\zeta(s) - 1| \leq \sum_{n=2}^{\infty} \frac{1}{n^{\sigma}}$$

and the right-hand side tends to 0 as $\sigma \rightarrow +\infty$; hence $\zeta(s) \rightarrow 1$ as $\operatorname{Re}(s) \rightarrow +\infty$.

3. On the real axis, $\zeta(\sigma)$ is real and strictly decreasing for $\sigma > 1$, with $\zeta(\sigma) > 1$ throughout, $\zeta(\sigma) \rightarrow 1$ as $\sigma \rightarrow +\infty$, and $\zeta(\sigma) \rightarrow +\infty$ as $\sigma \rightarrow 1^+$.

Proof:

The three parts are proved in turn.

1. The function ζ is the Dirichlet series $D(\mathbf{1}, s)$, whose abscissa of convergence is $\sigma_c = 1$ (example 2.1.9). By theorem 2.1.7, a Dirichlet series is holomorphic on the half-plane to the right of its abscissa of convergence, and there it may be differentiated term by term, with $D'(f, s) = \sum_n f(n)(-\log n)n^{-s}$. Taking $f = \mathbf{1}$ gives holomorphy on $\operatorname{Re}(s) > 1$ and

$$\zeta'(s) = \sum_{n=1}^{\infty} \frac{-\log n}{n^s} = - \sum_{n=1}^{\infty} \frac{\log n}{n^s}$$

the $n = 1$ term vanishing since $\log 1 = 0$.

2. Separating the $n = 1$ term, $\zeta(s) - 1 = \sum_{n \geq 2} n^{-s}$. Since $|n^{-s}| = n^{-\sigma}$ for $\operatorname{Re}(s) = \sigma$, the triangle inequality gives

$$|\zeta(s) - 1| = \left| \sum_{n=2}^{\infty} \frac{1}{n^s} \right| \leq \sum_{n=2}^{\infty} \frac{1}{n^{\sigma}}$$

which depends only on σ . The tail $\sum_{n \geq 2} n^{-\sigma}$ is bounded by the integral comparison $\sum_{n \geq 2} n^{-\sigma} \leq \int_1^\infty t^{-\sigma} dt = 1/(\sigma - 1)$, valid for $\sigma > 1$, and $1/(\sigma - 1) \rightarrow 0$ as $\sigma \rightarrow +\infty$. Hence $|\zeta(s) - 1| \rightarrow 0$, that is, $\zeta(s) \rightarrow 1$, uniformly in $\text{Im}(s)$ as $\text{Re}(s) \rightarrow +\infty$.

3. For real $s = \sigma > 1$, every term $n^{-\sigma}$ is a positive real number, so $\zeta(\sigma) = \sum_n n^{-\sigma}$ is real and exceeds its first term 1, giving $\zeta(\sigma) > 1$. For $1 < \sigma_1 < \sigma_2$ and each $n \geq 2$, the function $\sigma \mapsto n^{-\sigma} = e^{-\sigma \log n}$ is strictly decreasing because $\log n > 0$, so $n^{-\sigma_2} < n^{-\sigma_1}$; summing over $n \geq 2$ (the $n = 1$ term being constant) yields $\zeta(\sigma_2) < \zeta(\sigma_1)$, so ζ is strictly decreasing on $(1, \infty)$. The limit $\zeta(\sigma) \rightarrow 1$ as $\sigma \rightarrow +\infty$ is the real-axis case of part (2). For the limit at 1^+ , fix any N ; for $1 < \sigma$ the partial sum gives $\zeta(\sigma) \geq \sum_{n=1}^N n^{-\sigma}$, and letting $\sigma \rightarrow 1^+$ this finite sum tends to $\sum_{n=1}^N 1/n$, so $\liminf_{\sigma \rightarrow 1^+} \zeta(\sigma) \geq \sum_{n=1}^N 1/n$. Since N is arbitrary and the harmonic series $\sum_n 1/n$ diverges (example 1.4.4), the right-hand side is unbounded, forcing $\zeta(\sigma) \rightarrow +\infty$ as $\sigma \rightarrow 1^+$, as we sought to show.

These properties already record the qualitative shape of ζ along the real axis: a function descending strictly from $+\infty$ at the right edge of the pole to the horizontal asymptote $\zeta = 1$ at infinity. The blow-up at $\sigma = 1^+$ is the real-axis trace of the simple pole that the next section locates precisely, and it is inseparable from the divergence of the harmonic series, the most elementary statement about the density of the integers. Part (2) is the analytic counterpart of the fact that, far to the right, the $n = 1$ term dominates the series and every higher term is negligible; this is the same decay-at-different-rates phenomenon that made the coefficients of a Dirichlet series recoverable one at a time in corollary 2.1.8, and it will reappear as the normalization $\log \zeta(s) \rightarrow 0$ that pins the branch of the logarithm used in the non-vanishing argument of section 4.4. The holomorphy in part (1) is what entitles one to speak of the zeros and poles of ζ at all, and term-by-term differentiation makes $\zeta'(s)$ a Dirichlet series in its own right, the series of $-\log$, whose ratio with ζ produces the von Mangoldt series that carries the arithmetic.

The product form of ζ , established in chapter 2, is the single feature that ties the function to the primes; it is recalled here together with the non-vanishing it forces on the half-plane of absolute convergence.

Corollary 4.1.2: The Euler Product and Non-Vanishing on $\text{Re}(s) > 1$

For $\text{Re}(s) > 1$, the zeta function factors as an absolutely convergent product over the primes,

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}$$

and $\zeta(s) \neq 0$ on this half-plane.

Proof:

The factorization is corollary 2.3.2, the Euler product for ζ , obtained by applying the Euler product for multiplicative functions to the totally multiplicative function 1. The non-vanishing is proposition 2.3.4: an absolutely convergent infinite product whose factors are all nonzero has a nonzero value, and the local factors $(1 - p^{-s})^{-1}$ are reciprocals, hence never zero. Therefore $\zeta(s) = \prod_p (1 - p^{-s})^{-1} \neq 0$ for $\text{Re}(s) > 1$, as required.

The Euler product is the reason ζ encodes the primes at all: it expresses the global function as an

assembly of one local factor per prime, so that every analytic statement about ζ on $\operatorname{Re}(s) > 1$ is simultaneously a statement about the primes, and conversely. The non-vanishing recalled here is the easy half of the story. It holds throughout the region of absolute convergence precisely because the obstruction to a zero is local, a single vanishing factor, and no factor vanishes; this is the structural argument of proposition 2.3.4, sharper than the earlier proof through the Möbius series in example 2.2.5. The non-vanishing that matters for the distribution of primes, on the boundary line $\operatorname{Re}(s) = 1$, lies outside this region and is inaccessible to the product argument; it is the subject of section 4.4.

A few numerical values anchor the qualitative picture and connect ζ to constants from elsewhere in mathematics.

Example 4.1.3: Special Values of ζ

The two best-known special values are the even-argument evaluations

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \quad \zeta(4) = \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

The first is the Basel problem, solved by Euler; both are proved in [13] (the standard derivations use the Fourier expansion of a periodic function or the product formula for $\sin$, neither of which is developed here). Numerically $\zeta(2) \approx 1.6449$ and $\zeta(4) \approx 1.0823$. The monotone decrease of proposition 4.1.1(3) toward the asymptote $\zeta = 1$ is visible in a short table of real values:

s	1.5	2	3	4	6	10
$\zeta(s)$	2.6124	1.6449	1.2021	1.0823	1.0173	1.0010

Each entry exceeds the next, and the values approach 1 as s grows, the contribution beyond the $n = 1$ term shrinking like 2^{-s} . The value $\zeta(3) = 1.2021\dots$, the constant appearing at $s = 3$, is *Apéry's constant*; Apéry proved it irrational in 1979. The even values $\zeta(2k)$ are all rational multiples of π^{2k} , a pattern that the functional equation of ?? explains in one stroke, whereas the odd values $\zeta(3), \zeta(5), \dots$ have no such closed form and remain arithmetically obscure.

The contrast between the even and odd values is the first sign that ζ carries arithmetic information no elementary manipulation of the series reveals. The even values fall out of the functional equation because that equation relates $\zeta(2k)$ to $\zeta(1 - 2k)$, a value the equation forces to be rational; the odd values are paired by the same equation with $\zeta(-2k)$, which the functional equation makes vanish, so no comparable evaluation results. This asymmetry is what makes $\zeta(3)$ and its odd siblings difficult, and it foreshadows the role of the functional equation as the organizing principle of the next two chapters.

The single tool from chapter 1 that the continuation rests on is the partial-sum formula for $\sum_{n \leq x} n^{-s}$. For $\operatorname{Re}(s) > 0$ with $s \neq 1$, proposition 1.4.6 gives

$$\sum_{n \leq x} \frac{1}{n^s} = \frac{x^{1-s}}{1-s} + \zeta(s) + O(x^{-\operatorname{Re}(s)})$$

where the constant $\zeta(s)$ appearing here is, for $\operatorname{Re}(s) > 1$, the value of the defining series, and is given on the wider half-plane $\operatorname{Re}(s) > 0$ by the convergent integral

$$\zeta(s) = \frac{s}{s-1} - s \int_1^{\infty} \frac{\{t\}}{t^{s+1}} dt$$

with $\{t\}$ the fractional part of t . The integral converges absolutely for $\operatorname{Re}(s) > 0$ because $|\{t\}t^{-s-1}| \leq t^{-\operatorname{Re}(s)-1}$ and $\int_1^\infty t^{-\operatorname{Re}(s)-1} dt < \infty$ there. The right-hand side of this formula is defined and finite for every s with $\operatorname{Re}(s) > 0$ except $s = 1$, where the term $s/(s-1)$ has a simple pole; it therefore furnishes a value for $\zeta(s)$ on all of $\operatorname{Re}(s) > 0$, agreeing with the series where the series converges. This formula is the engine of the continuation, and the next section establishes that the function it defines is holomorphic on $\operatorname{Re}(s) > 0 \setminus \{1\}$ with a simple pole of residue 1 at $s = 1$, thereby extending ζ past the line $\operatorname{Re}(s) = 1$ to which the series confined it.

4.2 Meromorphic Continuation to $\operatorname{Re}(s) > 0$

The Dirichlet series $\sum_{n \geq 1} n^{-s}$ of definition 2.1.1 converges only on the half-plane $\operatorname{Re}(s) > 1$, and there it defines $\zeta(s)$ as a holomorphic function (proposition 4.1.1). Outside that half-plane the series is silent: it diverges, and the symbol $\zeta(s)$ has, as yet, no meaning. The whole analytic theory of the primes depends on giving it one. The Abel-summation analysis of proposition 1.4.6 already does so. Comparing the partial sums $\sum_{n \leq x} n^{-s}$ against the integral $\int_1^x t^{-s} dt$ produced the closed form

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \frac{\{t\}}{t^{s+1}} dt$$

in which the right-hand side makes sense on the larger half-plane $\operatorname{Re}(s) > 0$, because the integral converges there. The series-defined ζ on $\operatorname{Re}(s) > 1$ and this integral expression agree where both are defined, so the latter extends the former.

This section records that continuation as a theorem and reads off its local structure. The two elementary pieces on the right behave very differently near the line $\operatorname{Re}(s) = 1$: the rational term $s/(s-1)$ carries a single pole at $s = 1$, while the integral term is holomorphic across the whole half-plane and contributes nothing singular. Isolating the pole and computing the first two coefficients of the Laurent expansion of ζ at $s = 1$ is the work that remains, and it is the structural input that the proof of the Prime Number Theorem will draw on.

Theorem 4.2.1: Meromorphic Continuation of ζ to $\operatorname{Re}(s) > 0$

The function $\zeta(s)$, defined for $\operatorname{Re}(s) > 1$ by the series $\sum_{n=1}^\infty n^{-s}$, extends to a meromorphic function on the half-plane $\operatorname{Re}(s) > 0$ whose only singularity there is a simple pole at $s = 1$ with residue 1. Explicitly, for all s with $\operatorname{Re}(s) > 0$ and $s \neq 1$,

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \frac{\{t\}}{t^{s+1}} dt \quad (4.1)$$

where $\{t\} = t - [t]$ denotes the fractional part.

Proof:

Write $I(s) = \int_1^\infty \{t\} t^{-s-1} dt$ for the integral on the right of (4.1), so that the asserted identity reads $\zeta(s) = s/(s-1) - sI(s)$. The argument has three stages: the integral defines a holomorphic function on $\operatorname{Re}(s) > 0$; the right-hand side of (4.1) is therefore meromorphic on $\operatorname{Re}(s) > 0$ with a single simple pole at $s = 1$; and it agrees with the series on $\operatorname{Re}(s) > 1$, so it is the continuation. The residue is computed at the end.

Step 1: The integral $I(s)$ is holomorphic on $\operatorname{Re}(s) > 0$. The integrand $g(s, t) = \{t\} t^{-s-1}$ is, for each fixed $t \geq 1$, an entire function of s , and for each fixed s it is continuous in t . Since $0 \leq \{t\} < 1$, it is dominated by

$$|g(s, t)| = \{t\} t^{-\operatorname{Re}(s)-1} \leq t^{-\operatorname{Re}(s)-1}$$

Fix a compact set $K \subset \{\operatorname{Re}(s) > 0\}$ and let $\sigma_0 = \min_{s \in K} \operatorname{Re}(s) > 0$. Then $|g(s, t)| \leq t^{-\sigma_0-1}$ for all $s \in K$ and all $t \geq 1$, and $\int_1^\infty t^{-\sigma_0-1} dt = 1/\sigma_0 < \infty$. The dominating function $t^{-\sigma_0-1}$ is independent of s and integrable on $[1, \infty)$, so the integral defining $I(s)$ converges absolutely and uniformly for $s \in K$. A locally uniform limit of holomorphic functions is holomorphic, and a parameter integral with holomorphic integrand and a locally uniform integrable bound is holomorphic in the parameter, with differentiation permissible under the integral sign; this is the standard theorem on holomorphy of integrals depending on a parameter ([13]). Applying it on each such K , and noting that the half-plane $\operatorname{Re}(s) > 0$ is exhausted by such compact sets, shows that I is holomorphic on $\operatorname{Re}(s) > 0$.

Step 2: The right-hand side is meromorphic with a simple pole at $s = 1$. On $\operatorname{Re}(s) > 0$ the term $sI(s)$ is holomorphic, being the product of the entire function s with the holomorphic function $I(s)$ of Step 1. The term $s/(s-1)$ is a rational function whose only singularity is a simple pole at $s = 1$: writing $s/(s-1) = 1 + 1/(s-1)$ exhibits the principal part $1/(s-1)$ explicitly. The sum

$$R(s) := \frac{s}{s-1} - sI(s)$$

is therefore meromorphic on $\operatorname{Re}(s) > 0$, holomorphic away from $s = 1$, and has at $s = 1$ a simple pole, since the holomorphic term $sI(s)$ cannot cancel the simple pole of $s/(s-1)$.

Step 3: R agrees with ζ on $\operatorname{Re}(s) > 1$ and is the continuation. By proposition 1.4.6, for every s with $\operatorname{Re}(s) > 1$ the constant term in the asymptotic expansion of $\sum_{n \leq x} n^{-s}$ is exactly $R(s)$, and that constant equals $\lim_{x \rightarrow \infty} \sum_{n \leq x} n^{-s} = \sum_{n=1}^\infty n^{-s} = \zeta(s)$. Hence $R(s) = \zeta(s)$ for all s with $\operatorname{Re}(s) > 1$. The function ζ is holomorphic on the connected open set $\operatorname{Re}(s) > 1$, and R is meromorphic on the connected open set $\operatorname{Re}(s) > 0$ with its only pole at $s = 1$; the two agree on the nonempty open subset $\operatorname{Re}(s) > 1$, which has accumulation points in $\operatorname{Re}(s) > 0$. By the identity theorem ([13]), a meromorphic function on a connected open set is determined by its values on any subset with an accumulation point, so R is the unique meromorphic extension of ζ to $\operatorname{Re}(s) > 0$. This establishes (4.1) and shows that the only singularity of the continued ζ on $\operatorname{Re}(s) > 0$ is the simple pole of R at $s = 1$.

Step 4: The residue at $s = 1$ is 1. For a simple pole the residue is the limit of $(s-1)$ times the function. Using (4.1) and the holomorphy of $sI(s)$ at $s = 1$,

$$\lim_{s \rightarrow 1} (s-1)\zeta(s) = \lim_{s \rightarrow 1} \left[(s-1) \cdot \frac{s}{s-1} - (s-1)sI(s) \right] = \lim_{s \rightarrow 1} [s - (s-1)sI(s)] = 1$$

the second term vanishing in the limit because $sI(s)$ is finite at $s = 1$. Thus the pole is simple with residue 1, completing the proof.

The continuation is what lets the analytic machinery reach the line $\operatorname{Re}(s) = 1$ and the point $s = 1$, where the series itself gives nothing. Every later argument that extracts arithmetic from $\zeta(s)$ near

this line, the non-vanishing on $\operatorname{Re}(s) = 1$, the behavior of the logarithmic derivative, and ultimately the asymptotic $\psi(x) \sim x$, presupposes that ζ is defined and analytic there. The single simple pole at $s = 1$ is the one feature of this extended function that survives into the Prime Number Theorem: when Perron's formula expresses the Chebyshev sum $\psi(x)$ as a contour integral of $-\zeta'(s)/\zeta(s) \cdot x^s/s$, the pole of ζ at $s = 1$ becomes a pole of the integrand with residue x , and that residue is the main term $\psi(x) \sim x$ (chapter 5). That the residue of ζ at the pole is exactly 1, rather than some other constant, is what makes the main term x rather than a multiple of it. The pole is, in this precise sense, the analytic source of the leading order of the distribution of primes.

The continuation obtained here reaches only the half-plane $\operatorname{Re}(s) > 0$, governed by the convergence of the integral in (4.1), which fails once $\operatorname{Re}(s) \leq 0$. Extending ζ to all of $\mathbb{C}$ requires a different mechanism, the functional equation relating $\zeta(s)$ to $\zeta(1-s)$, developed in ???. An alternative route to the same half-plane runs through the Dirichlet eta function: the integral representation of example 2.4.6 expresses $(1 - 2^{1-s})\Gamma(s)\zeta(s)$ by an integral that already converges on $\operatorname{Re}(s) > 0$, and dividing by the elementary factor $(1 - 2^{1-s})\Gamma(s)$ continues ζ across the line $\operatorname{Re}(s) = 1$ as well. The Abel-summation route taken here is the more elementary of the two, requiring no special functions, and it has the advantage of displaying the pole and its residue directly in the formula.

The pole at $s = 1$ is significant enough to isolate as a corollary in the form most often invoked, the value of the limit defining the residue.

Corollary 4.2.2: The Simple Pole at $s = 1$

The continued function $\zeta(s)$ has a simple pole at $s = 1$, and

$$\lim_{s \rightarrow 1} (s - 1) \zeta(s) = 1$$

Proof:

This is the statement of the pole and residue in theorem 4.2.1: the residue of a simple pole at $s = 1$ is by definition $\lim_{s \rightarrow 1} (s - 1)\zeta(s)$, and theorem 4.2.1 computes this limit to be 1, as required.

Knowing that the pole is simple with residue 1 fixes the leading term $1/(s - 1)$ of the Laurent expansion of ζ at $s = 1$. The constant term is the next datum, and it is not arbitrary: it is the Euler–Mascheroni constant γ , the same number that measures the gap between the harmonic sum $\sum_{n \leq x} 1/n$ and $\log x$ (example 1.4.4). The coincidence is not accidental. Both quantities arise from comparing a sum of n^{-s} against its integral, the harmonic sum at $s = 1$ and the zeta values as $s \rightarrow 1$, and the constant γ is the common discrepancy. The formula (4.1) makes the connection precise, because the integral $I(s)$ that appears in it specializes at $s = 1$ to the very integral whose value computes γ .

Proposition 4.2.3: The Laurent Expansion at $s = 1$

Near $s = 1$ the continued zeta function has the Laurent expansion

$$\zeta(s) = \frac{1}{s - 1} + \gamma + O(s - 1)$$

where γ is the Euler–Mascheroni constant.

Proof:

Start from the continuation formula (4.1) of theorem 4.2.1, valid for $\operatorname{Re}(s) > 0$ with $s \neq 1$, and split off the principal part of the rational term by writing $s/(s-1) = 1 + 1/(s-1)$. With $I(s) = \int_1^\infty \{t\} t^{-s-1} dt$ as before, this gives

$$\zeta(s) = \frac{1}{s-1} + 1 - sI(s) \quad (4.2)$$

The leading term $1/(s-1)$ is the principal part already identified in corollary 4.2.2; the remaining part $1 - sI(s)$ is holomorphic at $s = 1$ by Step 1 of the proof of theorem 4.2.1, so its value at $s = 1$ is the constant term of the Laurent expansion and its Taylor expansion supplies the higher-order terms. Evaluating at $s = 1$ requires the single number $I(1)$.

The function I is continuous at $s = 1$, again by the holomorphy of Step 1, so $I(1) = \int_1^\infty \{t\} t^{-2} dt$, the integral obtained by setting $s = 1$ in the integrand. This is exactly the integral whose value was computed in example 1.4.4 in the course of evaluating the Euler–Mascheroni constant, namely

$$\int_1^\infty \frac{\{t\}}{t^2} dt = 1 - \gamma$$

Substituting $s = 1$ and $I(1) = 1 - \gamma$ into the holomorphic part $1 - sI(s)$ of (4.2) gives the constant term

$$1 - 1 \cdot I(1) = 1 - (1 - \gamma) = \gamma$$

Since $1 - sI(s)$ is holomorphic at $s = 1$, it equals $\gamma + O(s-1)$ near $s = 1$, and combining with the principal part yields $\zeta(s) = 1/(s-1) + \gamma + O(s-1)$, as we sought to show.

The constant term γ ties the analytic behavior of ζ at its pole to the elementary asymptotics of the harmonic series, and this is the form in which the pole enters several refined estimates: the closed-form evaluation of the Mertens constant in theorem 3.3.3, for instance, isolates γ precisely as the value $\zeta(1+\rho) = 1/\rho + \gamma + o(1)$ read off from this expansion as $\rho \rightarrow 0^+$. The higher coefficients of the Laurent expansion are the *Stieltjes constants* γ_k , defined by $\zeta(s) = 1/(s-1) + \sum_{k \geq 0} \frac{(-1)^k}{k!} \gamma_k (s-1)^k$ with $\gamma_0 = \gamma$; they admit closed forms as regularized limits of $\sum_{n \leq x} (\log n)^k / n$ but do not enter the leading-order theory of the primes.

4.3 First Arithmetic Consequences

The continuation theorem of the previous section gives $\zeta(s)$ a single simple pole at $s = 1$ with residue 1 (theorem 4.2.1 and corollary 4.2.2), and the Laurent expansion $\zeta(s) = 1/(s-1) + \gamma + O(s-1)$ records its leading behavior there (proposition 4.2.3). This is a statement about ζ as an analytic function; on its own it says nothing about primes. The bridge to arithmetic is the Euler product, which on $\operatorname{Re}(s) > 1$ ties ζ to the primes one factor at a time, and the two operations that read the primes off the product, the logarithmic derivative $-\zeta'/\zeta$ and the logarithm $\log \zeta$. Applying these operations to a function with a known pole transfers that pole into analytic statements about prime-power sums.

This section carries out three such transfers. The logarithmic derivative $-\zeta'/\zeta$ acquires its own simple pole at $s = 1$, again of residue 1, and its Dirichlet series is the von Mangoldt series $\sum_n \Lambda(n) n^{-s}$

(theorem 2.2.6); this is the analytic object Perron's formula integrates to produce $\psi(x)$. Stripping the higher prime powers from $-\zeta'/\zeta$ isolates the prime series $\Phi(s) = \sum_p (\log p)p^{-s}$, which inherits the same pole. And the logarithm $\log \zeta$, expanded over prime powers, yields an analytic reproof that $\sum_p 1/p$ diverges, with the precise rate $\sum_p p^{-s} = \log(1/(s-1)) + O(1)$ as $s \rightarrow 1^+$. Each of these is a meromorphic fact about a half-plane; converting any of them into an asymptotic for $\psi(x)$ requires control of these functions on the boundary line $\operatorname{Re}(s) = 1$, supplied in the next section and exploited in chapter 5.

The first transfer is the most direct. The logarithmic derivative of a meromorphic function turns a pole or zero into a simple pole whose residue counts its order, and ζ has exactly one pole near $s = 1$ and (by the continuation) no zero there. The order of that pole is 1, so $-\zeta'/\zeta$ has a simple pole of residue 1.

Proposition 4.3.1: The Pole of the Logarithmic Derivative at $s = 1$

On the half-plane $\operatorname{Re}(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

where Λ is the von Mangoldt function (definition 1.1.8). Moreover $-\zeta'/\zeta$ continues to a meromorphic function on a neighborhood of $s = 1$, holomorphic there apart from a simple pole at $s = 1$, and that pole has residue 1.

Proof:

The Dirichlet series identity on $\operatorname{Re}(s) > 1$ is theorem 2.2.6: the von Mangoldt series $\sum_n \Lambda(n)n^{-s}$ converges absolutely on $\operatorname{Re}(s) > 1$ and equals $-\zeta'(s)/\zeta(s)$ there. It remains to analyze the behavior near $s = 1$.

By the continuation theorem (theorem 4.2.1) and the Laurent expansion of proposition 4.2.3, on a punctured neighborhood of $s = 1$ the zeta function has the form

$$\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1)$$

Factor out the principal part: writing

$$\zeta(s) = \frac{1}{s-1} (1 + \gamma(s-1) + O((s-1)^2)) = \frac{1}{s-1} g(s)$$

defines $g(s) = 1 + \gamma(s-1) + O((s-1)^2)$, which is holomorphic at $s = 1$ with $g(1) = 1 \neq 0$. Since g is holomorphic and nonzero at $s = 1$, it is nonzero on some neighborhood U of $s = 1$, and on U the function $\zeta(s) = g(s)/(s-1)$ is meromorphic with its only pole the simple pole at $s = 1$ contributed by the factor $1/(s-1)$; in particular ζ has no zero on U .

On $U \setminus \{1\}$ the logarithmic derivative is computed from the factorization. Taking logarithmic derivatives turns the product $\zeta = (s-1)^{-1}g$ into a sum,

$$\frac{\zeta'(s)}{\zeta(s)} = \frac{d}{ds} \log((s-1)^{-1}) + \frac{g'(s)}{g(s)} = -\frac{1}{s-1} + \frac{g'(s)}{g(s)}$$

the first term being the logarithmic derivative of $(s-1)^{-1}$. Because g is holomorphic and nonvanishing on U , the quotient g'/g is holomorphic on U ; denote it $h(s)$, so that h is holomorphic at $s=1$. Negating,

$$-\frac{\zeta'(s)}{\zeta(s)} = \frac{1}{s-1} - h(s)$$

on $U \setminus \{1\}$. The right-hand side is meromorphic on U , holomorphic away from $s=1$, and has at $s=1$ a simple pole whose principal part is $1/(s-1)$, since the holomorphic term $-h(s)$ contributes nothing to the principal part. The coefficient of $(s-1)^{-1}$ is 1, so the residue is

$$\operatorname{Res}_{s=1} \left(-\frac{\zeta'(s)}{\zeta(s)} \right) = 1$$

This identifies the continuation of $-\zeta'/\zeta$ near $s=1$ as a simple pole of residue 1, as we sought to show.

The pole of $-\zeta'/\zeta$ at $s=1$ is the single analytic fact on which the leading order of the Prime Number Theorem rests. Perron's formula applied to the coefficients $a_n = \Lambda(n)$ expresses the Chebyshev function as the contour integral

$$\psi(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \left(-\frac{\zeta'(s)}{\zeta(s)} \right) \frac{x^s}{s} ds$$

for any $c > 1$ (theorem 2.5.2 and example 2.5.4). Shifting the contour to the left past the line $\operatorname{Re}(s) = 1$ picks up the residue of the integrand at the pole $s=1$; since $-\zeta'/\zeta$ has residue 1 there, the integrand $-\zeta'(s)/\zeta(s) \cdot x^s/s$ has residue $x^1/1 = x$, and this residue is exactly the main term $\psi(x) \sim x$. The agreement of the two residue-1 computations, the pole of ζ and the pole of $-\zeta'/\zeta$, is what fixes the main term as x with coefficient 1 rather than some other multiple. What the present proposition does not supply is the legitimacy of the contour shift: moving the line of integration from $\operatorname{Re}(s) = c > 1$ to a line $\operatorname{Re}(s) < 1$ requires that $-\zeta'/\zeta$ be holomorphic on the line $\operatorname{Re}(s) = 1$ itself, away from $s=1$, which in turn requires ζ to have no zeros there. That non-vanishing is the missing input, established in section 4.4, and the quantitative contour estimate that turns it into $\psi(x) \sim x$ is the work of chapter 5.

The von Mangoldt series $\sum_n \Lambda(n)n^{-s}$ runs over all prime powers p^k , weighting each by $\log p$. The contribution of the genuine primes, the terms with $k=1$, is a series over primes alone, and isolating it produces an object closer to the prime-counting function than $-\zeta'/\zeta$ is.

Definition 4.3.2: The Prime Series $\Phi(s)$

The *prime series* is

$$\Phi(s) = \sum_p \frac{\log p}{p^s}$$

the sum running over all primes p , defined for $\operatorname{Re}(s) > 1$.

The series converges absolutely on $\operatorname{Re}(s) > 1$: its terms are a subseries of the von Mangoldt series $\sum_n \Lambda(n)n^{-s}$, which converges absolutely there (theorem 2.2.6), so Φ is holomorphic on $\operatorname{Re}(s) > 1$. The function Φ is the Dirichlet generating function of the coefficients $(\log p)$ supported on primes, and its summatory function is the Chebyshev theta function $\vartheta(x) = \sum_{p \leq x} \log p$ of definition 3.1.2, just as $-\zeta'/\zeta$ generates $\psi(x) = \sum_{n \leq x} \Lambda(n)$. The two Chebyshev functions differ only by the prime-power

terms $\psi(x) - \vartheta(x) = \sum_{k \geq 2} \sum_{p^k \leq x} \log p$, and theorem 3.1.5 shows this difference to be of lower order, so $\vartheta(x) \sim x$ and $\psi(x) \sim x$ stand or fall together. The same redundancy appears analytically: Φ and $-\zeta'/\zeta$ differ by a function holomorphic well past the line $\operatorname{Re}(s) = 1$, so they carry the same singular data near $s = 1$.

Example 4.3.3: The Prime Series at $s = 2$

At $s = 2$ the series $\Phi(2) = \sum_p (\log p)p^{-2}$ converges, and its first terms are

$$\frac{\log 2}{4} + \frac{\log 3}{9} + \frac{\log 5}{25} + \frac{\log 7}{49} + \frac{\log 11}{121} + \dots$$

Numerically the first five terms contribute $0.1733 + 0.1221 + 0.0644 + 0.0397 + 0.0198 = 0.4193$, and the full sum is $\Phi(2) = 0.4931\dots$. The companion logarithmic-derivative value is $-\zeta'(2)/\zeta(2) = 0.5700\dots$, the full von Mangoldt sum $\sum_n \Lambda(n)n^{-2}$; the difference $-\zeta'(2)/\zeta(2) - \Phi(2) = 0.0769\dots$ collects the prime-power terms $\sum_p \sum_{k \geq 2} (\log p)p^{-2k}$, the largest of which is $(\log 2)(4^{-2} + 4^{-3} + \dots) = (\log 2)/12 = 0.0578$. The prime-power correction is small even at $s = 2$ and shrinks as $\operatorname{Re}(s)$ grows, the first sign of the general bound that makes Φ and $-\zeta'/\zeta$ interchangeable near the line $\operatorname{Re}(s) = 1$.

The relation between Φ and $-\zeta'/\zeta$ is exact and explicit: their difference is the sum of the prime-power terms with exponent $k \geq 2$, and that sum converges on a half-plane reaching well to the left of $\operatorname{Re}(s) = 1$. The difference is therefore holomorphic where it matters, and the pole of $-\zeta'/\zeta$ at $s = 1$ passes to Φ unchanged.

Proposition 4.3.4: Φ and the Logarithmic Derivative

For $\operatorname{Re}(s) > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \Phi(s) + R(s), \quad R(s) = \sum_p \sum_{k=2}^{\infty} \frac{\log p}{p^{ks}}$$

The remainder $R(s)$ converges absolutely on $\operatorname{Re}(s) > 1/2$ and is holomorphic there. Consequently $\Phi(s) - 1/(s-1)$ extends holomorphically to a neighborhood of $s = 1$: the prime series Φ has a simple pole at $s = 1$ with residue 1.

Proof:

Step 1: The decomposition. By theorem 2.2.6 the logarithmic derivative is the von Mangoldt series, which by definition 1.1.8 sums $\log p$ over all prime powers p^k with $k \geq 1$:

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} = \sum_p \sum_{k=1}^{\infty} \frac{\log p}{p^{ks}}$$

on $\operatorname{Re}(s) > 1$, where the rearrangement of the absolutely convergent series into a double sum over (p, k) is the bijection between prime powers and pairs (p, k) . Splitting off the $k = 1$ terms, which form $\Phi(s)$ by definition 4.3.2, leaves

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_p \frac{\log p}{p^s} + \sum_p \sum_{k=2}^{\infty} \frac{\log p}{p^{ks}} = \Phi(s) + R(s)$$

which is the asserted identity.

Step 2: R is holomorphic on $\operatorname{Re}(s) > 1/2$. Fix a compact set $K \subset \{\operatorname{Re}(s) > 1/2\}$ and let $\sigma_0 = \min_{s \in K} \operatorname{Re}(s) > 1/2$. For $s \in K$ the modulus of the (p, k) -term is $(\log p)p^{-k\sigma_0}$ or smaller, and summing the geometric inner series over $k \geq 2$,

$$\sum_{k=2}^{\infty} \frac{\log p}{p^{k\sigma_0}} = \frac{\log p}{p^{2\sigma_0}} \cdot \frac{1}{1 - p^{-\sigma_0}} \leq \frac{\log p}{p^{2\sigma_0}} \cdot \frac{1}{1 - 2^{-\sigma_0}}$$

using $p \geq 2$ so that $1 - p^{-\sigma_0} \geq 1 - 2^{-\sigma_0} > 0$. The constant $(1 - 2^{-\sigma_0})^{-1}$ is finite and independent of p , so it suffices that $\sum_p (\log p)p^{-2\sigma_0}$ converges. Since $2\sigma_0 > 1$, choose $\varepsilon > 0$ with $2\sigma_0 - \varepsilon > 1$; then $\log p \leq C_\varepsilon p^\varepsilon$ for a constant C_ε and all p , whence

$$\sum_p \frac{\log p}{p^{2\sigma_0}} \leq C_\varepsilon \sum_p \frac{1}{p^{2\sigma_0 - \varepsilon}} \leq C_\varepsilon \sum_{n=1}^{\infty} \frac{1}{n^{2\sigma_0 - \varepsilon}} < \infty$$

the last series converging because its exponent exceeds 1. The bound is independent of $s \in K$, so the double series for R converges absolutely and uniformly on K . Each summand $(\log p)p^{-ks}$ is entire in s , and a locally uniform limit of holomorphic functions is holomorphic ([13]); exhausting $\{\operatorname{Re}(s) > 1/2\}$ by compact sets K shows R is holomorphic on $\operatorname{Re}(s) > 1/2$.

Step 3: The pole of Φ . On $\operatorname{Re}(s) > 1$ the identity of Step 1 rearranges to $\Phi(s) = -\zeta'(s)/\zeta(s) - R(s)$. By proposition 4.3.1 the function $-\zeta'/\zeta$ continues to a neighborhood U of $s = 1$ as a meromorphic function whose only singularity is a simple pole of residue 1, with $-\zeta'(s)/\zeta(s) = 1/(s-1) - h(s)$ for some h holomorphic on U . By Step 2, R is holomorphic on the neighborhood $U \cap \{\operatorname{Re}(s) > 1/2\}$ of $s = 1$. Therefore on this neighborhood

$$\Phi(s) - \frac{1}{s-1} = \left(-\frac{\zeta'(s)}{\zeta(s)} - \frac{1}{s-1} \right) - R(s) = -h(s) - R(s)$$

is a difference of two functions holomorphic at $s = 1$, hence holomorphic at $s = 1$. Thus $\Phi(s) = 1/(s-1) + (\text{holomorphic near } s = 1)$, so Φ has a simple pole at $s = 1$ whose principal part is $1/(s-1)$, that is, a simple pole of residue 1, completing the proof.

The prime series Φ is the analytic object that most directly governs the primes themselves, as opposed to the prime powers, and its pole at $s = 1$ is the analytic form of the statement that the primes have density $1/\log x$ in the sense made precise by the Prime Number Theorem. Perron's formula applied to Φ , with coefficients $a_n = \log p$ for $n = p$ prime and $a_n = 0$ otherwise, produces the contour integral for the Chebyshev theta function $\vartheta(x) = \sum_{p \leq x} \log p$, and the residue of $\Phi(s)x^s/s$ at the pole $s = 1$ is again x , giving the main term $\vartheta(x) \sim x$. The decomposition $-\zeta'/\zeta = \Phi + R$ with R holomorphic past the line $\operatorname{Re}(s) = 1$ is the analytic counterpart of the elementary estimate $\psi(x) - \vartheta(x) = O(\sqrt{x} \log x)$: in both pictures the prime powers of exponent $k \geq 2$ contribute a lower-order, harmless correction, and the substance lies entirely with the primes. The non-vanishing of ζ on $\operatorname{Re}(s) = 1$ enters here exactly as it does for $-\zeta'/\zeta$, since Φ differs from it only by the holomorphic R ; once that non-vanishing is in hand, Φ and $-\zeta'/\zeta$ may be used interchangeably in the contour argument of chapter 5.

The third transfer works with $\log \zeta$ rather than its derivative, and it reproves, by analytic means, the oldest quantitative fact about the primes. The expansion of $\log \zeta$ over prime powers (proposition 2.3.5) has the prime sum $\sum_p p^{-s}$ as its leading piece, and the pole of ζ forces $\log \zeta$ to blow up as $s \rightarrow 1^+$. Since the higher prime-power terms stay bounded, the blow-up must come from the prime sum, and

the Laurent expansion of ζ makes the rate explicit.

Proposition 4.3.5: Analytic Divergence of the Prime Sum

As $s \rightarrow 1^+$ along the real axis,

$$\sum_p \frac{1}{p^s} = \log \frac{1}{s-1} + O(1)$$

In particular $\sum_p 1/p$ diverges.

Proof:

Step 1: Separate the prime sum from the prime powers in $\log \zeta$. For real $s > 1$, proposition 2.3.5 gives the absolutely convergent expansion

$$\log \zeta(s) = \sum_p \sum_{k=1}^{\infty} \frac{1}{k p^{ks}} = \sum_p \frac{1}{p^s} + \sum_p \sum_{k=2}^{\infty} \frac{1}{k p^{ks}}$$

where the logarithm is the real logarithm of $\zeta(s) > 1$, and the second sum splits off the prime-power terms with $k \geq 2$. Write $E(s)$ for that second sum.

Step 2: The prime-power tail is bounded. For $s \geq 1$ the bound established in example 2.3.6 applies: dropping the factor $1/k \leq 1$ and summing the geometric inner series,

$$0 \leq E(s) = \sum_p \sum_{k=2}^{\infty} \frac{1}{k p^{ks}} \leq \sum_p \sum_{k=2}^{\infty} \frac{1}{p^{ks}} \leq \sum_p \sum_{k=2}^{\infty} \frac{1}{p^k} = \sum_p \frac{p^{-2}}{1 - p^{-1}}$$

the middle inequality using $p^{ks} \geq p^k$ for $s \geq 1$. The final sum is bounded by a convergent series independent of s : since $p^{-2}/(1 - p^{-1}) = 1/(p^2 - p) = 1/(p(p-1))$, comparison with all integers gives

$$\sum_p \frac{1}{p(p-1)} \leq \sum_{n=2}^{\infty} \frac{1}{n(n-1)} = \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1$$

the sum telescoping. Hence $0 \leq E(s) \leq 1$ for all $s \geq 1$, so $E(s) = O(1)$ as $s \rightarrow 1^+$.

Step 3: The size of $\log \zeta(s)$ near $s = 1$. By the Laurent expansion of proposition 4.2.3, near $s = 1$,

$$\zeta(s) = \frac{1}{s-1} + \gamma + O(s-1) = \frac{1}{s-1} (1 + \gamma(s-1) + O((s-1)^2))$$

For real s slightly greater than 1 both sides are positive, so taking the real logarithm,

$$\log \zeta(s) = \log \frac{1}{s-1} + \log(1 + \gamma(s-1) + O((s-1)^2))$$

As $s \rightarrow 1^+$ the argument of the second logarithm tends to 1, so that term tends to $\log 1 = 0$ and is in particular $O(1)$. Therefore

$$\log \zeta(s) = \log \frac{1}{s-1} + O(1) \quad (s \rightarrow 1^+)$$

Step 4: Combine. Subtracting the bounded tail of Step 2 from the expansion of Step 1 and inserting Step 3,

$$\sum_p \frac{1}{p^s} = \log \zeta(s) - E(s) = \left(\log \frac{1}{s-1} + O(1) \right) - O(1) = \log \frac{1}{s-1} + O(1)$$

as $s \rightarrow 1^+$. The right-hand side tends to $+\infty$ as $s \rightarrow 1^+$, since $\log(1/(s-1)) \rightarrow +\infty$. If the series $\sum_p 1/p$ converged, then for every $s > 1$ the comparison $\sum_p p^{-s} \leq \sum_p p^{-1}$ would bound $\sum_p p^{-s}$ by a finite constant uniformly in s , contradicting the divergence to $+\infty$ just established. Hence $\sum_p 1/p$ diverges, completing the proof.

The divergence of $\sum_p 1/p$ is Euler's theorem, and the rate $\sum_p p^{-s} \sim \log(1/(s-1))$ is its analytic sharpening: the prime sum grows like $\log(1/(s-1))$ as s approaches the pole, the same logarithmic rate by which $\log \zeta(s)$ itself blows up. This is the Dirichlet-series shadow of the Mertens estimate $\sum_{p \leq x} 1/p = \log \log x + M + O(1/\log x)$ of theorem 3.3.3. The two are linked by the heuristic $s-1 \leftrightarrow 1/\log x$ that runs throughout the subject: setting $s = 1 + 1/\log x$ turns $\log(1/(s-1))$ into $\log \log x$, and the analytic statement becomes the leading term of the Mertens statement. The agreement is not a coincidence but a reflection of the same arithmetic seen through two summation methods, the sharp cutoff $\sum_{p \leq x}$ and the smooth analytic weighting $\sum_p p^{-s}$; the constant M and the error $O(1/\log x)$ are the finer data that the smooth version compresses into its $O(1)$. The Mertens theorem, proved by elementary partial summation in chapter 3, is the stronger statement, with an explicit constant and error term that the present proof does not reach. What the analytic proof offers in exchange is method: it derives the divergence from the single fact that ζ has a pole at $s = 1$, and the same argument applied to a Dirichlet L -function $L(s, \chi)$, whose non-vanishing at $s = 1$ replaces the pole of ζ , yields the divergence of $\sum_{p \equiv a} 1/p$ over primes in an arithmetic progression, the analytic core of Dirichlet's theorem in ??.

All three results of this section are statements about half-planes and about behavior as $s \rightarrow 1^+$ along the real axis. The pole of $-\zeta'/\zeta$, the pole of Φ , and the rate of divergence of $\sum_p p^{-s}$ each isolate the singular structure of an analytic function at the point $s = 1$. None of them yet delivers an asymptotic for $\psi(x)$, $\vartheta(x)$, or $\pi(x)$, because the passage from the singularity to the counting function runs through Perron's formula (theorem 2.5.2 and example 2.5.4), and that passage demands a contour shift across the line $\operatorname{Re}(s) = 1$. The shift is legitimate only where the integrand is holomorphic, and the integrand $-\zeta'(s)/\zeta(s) \cdot x^s/s$ is holomorphic on the line $\operatorname{Re}(s) = 1$ precisely when ζ has no zeros there. Establishing that $\zeta(1+it) \neq 0$ for all real t , the one piece of analytic input that lies beyond the elementary Euler-product arguments, is the subject of the next section, after which chapter 5 converts the pole at $s = 1$ into the asymptotic $\psi(x) \sim x$.

4.4 Non-Vanishing on the Line $\operatorname{Re}(s) = 1$

The preceding sections fixed the analytic structure of $\zeta(s)$ on the half-plane $\operatorname{Re}(s) > 0$: a single simple pole at $s = 1$ with residue 1 (theorem 4.2.1), and the logarithmic derivative $-\zeta'(s)/\zeta(s) = \sum_n \Lambda(n)n^{-s}$ carrying the arithmetic. What remains is the one analytic fact that the elementary Euler-product argument of chapter 2 could not reach. The non-vanishing of $\zeta(s)$ throughout $\operatorname{Re}(s) > 1$ followed from the factorization $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$, in which no local factor vanishes (corollary 4.1.2); but

that argument lives inside the region of absolute convergence and says nothing on the boundary line $\operatorname{Re}(s) = 1$, where the product no longer converges. The decisive analytic fact behind the Prime Number Theorem is that $\zeta(s)$ has no zeros on this line either: $\zeta(1 + it) \neq 0$ for every real t .

This section proves it by the positivity argument of Mertens and de la Vallée Poussin, organized around the elementary trigonometric inequality $3 + 4 \cos \theta + \cos 2\theta \geq 0$. The mechanism is the logarithmic expansion $\log \zeta(s) = \sum_{p,k} 1/(k p^{ks})$ of proposition 2.3.5, whose real part on a vertical line is a sum of cosines $\cos(kt \log p)$ weighted by positive coefficients. Feeding the three real parts at heights 0, t , and $2t$ into the inequality produces a single positive combination, and that positivity is incompatible with a zero of ζ on the line: a zero at $1 + it_0$ would force the combination to negative infinity as s approaches the line, a contradiction. The argument is short once the inequality is in hand.

Lemma 4.4.1: The Inequality $3 + 4 \cos \theta + \cos 2\theta \geq 0$

For every real number θ ,

$$3 + 4 \cos \theta + \cos 2\theta = 2(1 + \cos \theta)^2 \geq 0$$

Proof:

The double-angle identity $\cos 2\theta = 2 \cos^2 \theta - 1$ gives

$$3 + 4 \cos \theta + \cos 2\theta = 3 + 4 \cos \theta + (2 \cos^2 \theta - 1) = 2 \cos^2 \theta + 4 \cos \theta + 2$$

The right-hand side is $2(\cos^2 \theta + 2 \cos \theta + 1) = 2(\cos \theta + 1)^2$. A real square is nonnegative and the factor 2 is positive, so $2(1 + \cos \theta)^2 \geq 0$ for every real θ , as we sought to show.

The choice of coefficients 3, 4, 1 is not arbitrary. The argument below combines $\log \zeta$ at three heights, weighting the height 0 contribution by 3, the height t contribution by 4, and the height $2t$ contribution by 1, and the combination is useful only if the resulting trigonometric sum is nonnegative for every angle. Any nonnegative combination $a_0 + a_1 \cos \theta + a_2 \cos 2\theta$ with $a_0 = a_2$ doubled appropriately would serve in principle, but (3, 4, 1) is the lightest integer choice that both stays nonnegative and gives the height 0 term enough weight to overpower a single zero on the line; the equality case $\theta = \pi$, where the expression vanishes, shows the coefficients cannot be lowered while preserving positivity. The same inequality, applied to Dirichlet L -functions, drives the non-vanishing $L(1, \chi) \neq 0$ behind primes in arithmetic progressions (??).

The positivity now transfers from angles to the zeta function. The link is the expansion of $\log \zeta$ along a vertical line, whose real part is precisely a weighted sum of cosines.

Theorem 4.4.2: Non-Vanishing of ζ on $\operatorname{Re}(s) = 1$

The Riemann zeta function does not vanish on the line $\operatorname{Re}(s) = 1$ away from the pole: $\zeta(1 + it) \neq 0$ for every real $t \neq 0$.

Proof:

Throughout, write $s = \sigma + it$ with $\sigma > 1$ real, so that s lies in the half-plane of absolute convergence where the Euler product and its logarithm are valid. The argument has two stages: a positivity estimate valid for all $\sigma > 1$, and a contradiction derived from a hypothetical zero on the line.

Step 1: The fundamental inequality $|\zeta(\sigma)|^3 |\zeta(\sigma + it)|^4 |\zeta(\sigma + 2it)| \geq 1$ *for* $\sigma > 1$. By proposition 2.3.5, for $\operatorname{Re}(s) > 1$ the logarithm of ζ is given by the absolutely convergent double series

$$\log \zeta(s) = \sum_p \sum_{k=1}^{\infty} \frac{1}{k p^{ks}}$$

For $s = \sigma + it$ the summand is $k^{-1} p^{-k\sigma} p^{-ikt} = k^{-1} p^{-k\sigma} e^{-ikt \log p}$, whose real part is $k^{-1} p^{-k\sigma} \cos(kt \log p)$ since $\operatorname{Re}(e^{-i\alpha}) = \cos \alpha$. Absolute convergence permits taking the real part term by term, so

$$\operatorname{Re} \log \zeta(\sigma + it) = \sum_p \sum_{k=1}^{\infty} \frac{\cos(kt \log p)}{k p^{k\sigma}} \quad (4.3)$$

Now form the combination of (4.3) at the three heights 0, t , and $2t$ with weights 3, 4, 1. Each term carries the common positive coefficient $c_{p,k} = (k p^{k\sigma})^{-1}$, and writing $\theta_{p,k} = kt \log p$ for the angle at height t , the angle at height 0 is 0 and the angle at height $2t$ is $2\theta_{p,k}$. Hence

$$3\operatorname{Re} \log \zeta(\sigma) + 4\operatorname{Re} \log \zeta(\sigma + it) + \operatorname{Re} \log \zeta(\sigma + 2it) = \sum_p \sum_{k=1}^{\infty} c_{p,k} (3 + 4 \cos \theta_{p,k} + \cos 2\theta_{p,k})$$

the rearrangement of the three absolutely convergent series into one being permissible. By lemma 4.4.1 each bracket is nonnegative, and each coefficient $c_{p,k}$ is positive, so every term of the combined series is nonnegative and therefore

$$3\operatorname{Re} \log \zeta(\sigma) + 4\operatorname{Re} \log \zeta(\sigma + it) + \operatorname{Re} \log \zeta(\sigma + 2it) \geq 0 \quad (4.4)$$

For any nonzero complex number w and any branch of its logarithm, $\operatorname{Re} \log w = \log |w|$. Each ζ value here is nonzero, lying in the half-plane $\operatorname{Re}(s) > 1$ (corollary 4.1.2), so (4.4) reads $3 \log |\zeta(\sigma)| + 4 \log |\zeta(\sigma + it)| + \log |\zeta(\sigma + 2it)| \geq 0$. Exponentiating, and using that $\exp$ is increasing,

$$|\zeta(\sigma)|^3 |\zeta(\sigma + it)|^4 |\zeta(\sigma + 2it)| \geq 1 \quad (\sigma > 1) \quad (4.5)$$

This inequality holds for every real t and every $\sigma > 1$, and it is the only consequence of positivity that the argument needs.

Step 2: A zero on the line contradicts (4.5). Suppose, for contradiction, that $\zeta(1 + it_0) = 0$ for some real $t_0 \neq 0$. Apply (4.5) with $t = t_0$ and let $\sigma \rightarrow 1^+$ along the real axis; the three factors are estimated separately. The estimates compare each factor with a power of $\sigma - 1$, and the exponents are arranged so that the product tends to 0, contradicting the lower bound 1.

The first factor blows up at the controlled rate of the pole. By corollary 4.2.2 the continued ζ has a simple pole at $s = 1$ with $\lim_{\sigma \rightarrow 1^+} (\sigma - 1)\zeta(\sigma) = 1$, so $(\sigma - 1)\zeta(\sigma) \rightarrow 1$ and in particular $(\sigma - 1)|\zeta(\sigma)| \rightarrow 1$. Thus there is a constant $A > 0$ and a threshold $\sigma_1 \in (1, 2]$ such that

$$|\zeta(\sigma)| \leq \frac{A}{\sigma - 1} \quad (1 < \sigma \leq \sigma_1) \quad (4.6)$$

indeed any $A > 1$ works for σ near 1. Cubing, $|\zeta(\sigma)|^3 \leq A^3(\sigma - 1)^{-3}$.

The second factor vanishes at the controlled rate of the assumed zero. The point $1 + it_0$ is not the pole, since $t_0 \neq 0$, so by theorem 4.2.1 the function ζ is holomorphic in a neighborhood of $1 + it_0$. By hypothesis it vanishes there, hence vanishes to some finite order $m \geq 1$: there is a function h holomorphic near $1 + it_0$ with $h(1 + it_0) \neq 0$ and $\zeta(s) = (s - (1 + it_0))^m h(s)$ on that neighborhood (the local factorization of a holomorphic function at a zero, [13]). Restricting to $s = \sigma + it_0$ with $\sigma > 1$, the displacement from the zero is $s - (1 + it_0) = \sigma - 1$, a positive real number, so

$$\zeta(\sigma + it_0) = (\sigma - 1)^m h(\sigma + it_0)$$

As $\sigma \rightarrow 1^+$ the factor $h(\sigma + it_0) \rightarrow h(1 + it_0)$, a finite value, so $|h(\sigma + it_0)|$ is bounded by some constant $B > 0$ for σ in an interval $(1, \sigma_2]$ with $\sigma_2 \in (1, \sigma_1]$. Since $m \geq 1$ and $0 < \sigma - 1 < 1$ on this interval (shrinking σ_2 below 2 if needed), $(\sigma - 1)^m \leq (\sigma - 1)$, and therefore

$$|\zeta(\sigma + it_0)| = (\sigma - 1)^m |h(\sigma + it_0)| \leq B(\sigma - 1) \quad (1 < \sigma \leq \sigma_2) \quad (4.7)$$

Raising to the fourth power, $|\zeta(\sigma + it_0)|^4 \leq B^4(\sigma - 1)^4$.

The third factor stays bounded. The point $1 + 2it_0$ is also distinct from the pole, since $2t_0 \neq 0$, so by theorem 4.2.1 the function ζ is holomorphic at $1 + 2it_0$, hence continuous there. Consequently $\zeta(\sigma + 2it_0) \rightarrow \zeta(1 + 2it_0)$ as $\sigma \rightarrow 1^+$, a finite limit, and there is a constant $C > 0$ and a threshold $\sigma_3 \in (1, \sigma_2]$ with

$$|\zeta(\sigma + 2it_0)| \leq C \quad (1 < \sigma \leq \sigma_3) \quad (4.8)$$

No claim about the order of ζ at $1 + 2it_0$ is needed; mere boundedness suffices, and that is guaranteed by continuity whether or not ζ happens to vanish there.

Combining the three estimates (4.6), (4.7), and (4.8), valid simultaneously for $1 < \sigma \leq \sigma_3$, gives

$$|\zeta(\sigma)|^3 |\zeta(\sigma + it_0)|^4 |\zeta(\sigma + 2it_0)| \leq \frac{A^3}{(\sigma - 1)^3} \cdot B^4(\sigma - 1)^4 \cdot C = A^3 B^4 C (\sigma - 1)$$

The right-hand side tends to 0 as $\sigma \rightarrow 1^+$. Yet by (4.5) the left-hand side is at least 1 for every $\sigma > 1$. Letting $\sigma \rightarrow 1^+$ forces $1 \leq \lim_{\sigma \rightarrow 1^+} A^3 B^4 C (\sigma - 1) = 0$, an impossibility. The assumption $\zeta(1 + it_0) = 0$ is therefore untenable, and $\zeta(1 + it_0) \neq 0$ for every real $t_0 \neq 0$, completing the proof.

The arithmetic of the exponents is the heart of the matter. The pole contributes a singularity of order one, so the cube $|\zeta(\sigma)|^3$ grows like $(\sigma - 1)^{-3}$; a zero of order $m \geq 1$ makes the fourth power $|\zeta(\sigma + it_0)|^4$ vanish at least like $(\sigma - 1)^4$; and the bounded third factor contributes nothing. The net exponent $-3 + 4 = +1$ is positive, so the product vanishes, and this is exactly the contradiction. Had the weights been chosen differently, the balance could fail: weights $(1, 2, 1)$, for instance, would give net exponent $-1 + 2 = +1$ as well, but the corresponding trigonometric polynomial $1 + 2\cos\theta + \cos 2\theta = 2\cos^2\theta + 2\cos\theta$ is *not* nonnegative for all θ , so Step 1 would fail. The pair of requirements, nonnegativity of the cosine polynomial and a strictly positive net exponent, is what singles out $(3, 4, 1)$, and the equality $3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2$ is the smallest integer solution meeting both. A single zero on the line cannot survive this squeeze.

This non-vanishing is the analytic linchpin of the Prime Number Theorem. The proof of $\psi(x) \sim x$ in chapter 5 begins from the Perron representation $\psi(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} (-\zeta'(s)/\zeta(s)) x^s/s ds$ of example 2.5.4, taken on a vertical line $\operatorname{Re}(s) = c$ with $c > 1$, and proceeds by pushing the contour

leftward onto the line $\operatorname{Re}(s) = 1$. Each zero of ζ on or near that line would be a pole of the integrand $-\zeta'(s)/\zeta(s)$, an obstruction to the contour shift contributing an extra term to $\psi(x)$ of size comparable to the main term. theorem 4.4.2 is precisely the statement that there is no such obstruction on the line itself: the only pole of the integrand encountered on $\operatorname{Re}(s) = 1$ is the one at $s = 1$ coming from the pole of ζ (proposition 4.3.1), whose residue x is the main term $\psi(x) \sim x$. The non-vanishing is what licenses the contour to rest on $\operatorname{Re}(s) = 1$ with no pole obstruction except $s = 1$, and it is in this sense that the asymptotic distribution of primes is encoded in a single line of the complex plane.

4.4.3 Historical Note: Hadamard and de la Vallée Poussin The Prime Number Theorem was conjectured by Gauss and Legendre near the close of the eighteenth century and resisted proof for nearly a hundred years. It was settled independently in 1896 by Jacques Hadamard and Charles-Jean de la Vallée Poussin, each building on Riemann's 1859 program of studying $\pi(x)$ through the complex-analytic properties of $\zeta(s)$. The technical core of both proofs is the non-vanishing of $\zeta(s)$ on the line $\operatorname{Re}(s) = 1$; the positivity argument through the inequality $3 + 4 \cos \theta + \cos 2\theta \geq 0$ presented here is the streamlined form later extracted by Mertens and refined by de la Vallée Poussin, and is the version that has become standard. Qualitative non-vanishing on the line suffices to prove $\psi(x) \sim x$, but not to control the error term. De la Vallée Poussin went further, converting the same positivity into a *quantitative* zero-free region of the form $\sigma > 1 - c/\log |t|$ to the left of the line, which yields an explicit error bound in the Prime Number Theorem; these zero-free regions, and the error terms they produce, are developed in chapter 5.

The non-vanishing is recorded once more in the form in which the proof of the Prime Number Theorem invokes it, as a statement about the logarithmic derivative.

Corollary 4.4.4: Regularity of the Logarithmic Derivative on $\operatorname{Re}(s) = 1$

The logarithmic derivative $-\zeta'(s)/\zeta(s)$ is holomorphic on the line $\operatorname{Re}(s) = 1$ except for a simple pole at $s = 1$. Equivalently, $\zeta(s)$ has no zeros on $\operatorname{Re}(s) = 1$.

Proof:

By theorem 4.2.1 the function ζ is holomorphic on a neighborhood of the line $\operatorname{Re}(s) = 1$ except at $s = 1$, where it has a simple pole. The logarithmic derivative $-\zeta'/\zeta$ of a meromorphic function is holomorphic at every point that is neither a zero nor a pole of ζ ; at a zero of order m it has a simple pole with residue $-m$, and at a pole of order n it has a simple pole with residue n (the standard local computation for $-\zeta'/\zeta$, [13]). On the line $\operatorname{Re}(s) = 1$ the only pole of ζ is the simple pole at $s = 1$, which contributes a simple pole of $-\zeta'/\zeta$ at $s = 1$ with residue 1 (proposition 4.3.1). It remains to rule out zeros of ζ on the line, the other possible source of poles of $-\zeta'/\zeta$. The point $s = 1$ is the pole, not a zero. For every other point $1 + it$ with $t \neq 0$, theorem 4.4.2 gives $\zeta(1 + it) \neq 0$. Hence ζ has no zeros on $\operatorname{Re}(s) = 1$, so $-\zeta'/\zeta$ has no poles there beyond the simple pole at $s = 1$, and is holomorphic on the rest of the line, as required.

Index

- Ω Notation, [xxx](#)
- Θ Notation, [xxx](#)
- p -adic Valuation, [ix](#)
- Abel's Summation Formula, [17](#)
- Abel's Test, [21](#)
- Abscissa of Absolute Convergence, [32](#)
- Abscissa of Convergence, [32](#)
- Additive Function, [2](#)
- Analyticity of Dirichlet Series, [35](#)
- Arithmetic Function, [1](#)
- Asymptotic Equivalence, [xxx](#)
- Asymptotic Notation, [xxix](#)
- Average Order, [12](#)
 - of $\omega(n)$, [16](#)
- Bernoulli Numbers, [22](#)
- Bernoulli Polynomials, [23](#)
- Bertrand's Postulate, [93](#)
- Big- O Notation, [xxix](#)
- Bézout's Identity, [iv](#)
- Carmichael Function, [xxii](#)
- Central Binomial Coefficient
 - prime content, [73](#)
 - prime factorization, [92](#)
 - size bounds, [91](#)
- Chebyshev Functions, [68](#)
 - comparison of ψ and ϑ , [69](#)
- Chebyshev's Estimates, [75](#)
- Chinese Remainder Theorem, [xiv](#)
- Congruence, [xii](#)
- Coprime, [iii](#)
- Dirichlet Convolution, [6](#)
- Dirichlet Hyperbola Method, [12](#)
- Dirichlet Series, [32](#)
 - Multiplication Theorem, [37](#)
- Discontinuous Integral, [58](#)
- Divisibility, [ii](#)
- Division Algorithm, [i](#)
- Divisor Function, [3](#)
- Divisor Sum Asymptotic, [13](#)
- Euclidean Algorithm, [iii](#)
- Euler Product, [43](#)
 - Logarithm, [48](#)
 - Riemann Zeta Function, [45, 99](#)
- Euler Totient Function, [xiii](#)
- Euler's Criterion, [xxiv](#)
- Euler's Theorem, [xvi](#)
- Euler-Maclaurin Summation Formula, [25](#)
- Exact Divisibility, [xi](#)
- Fermat's Little Theorem, [xvi](#)
- Fundamental Theorem of Arithmetic, [viii](#)
- Gauss's Lemma, [xxv](#)
- Greatest Common Divisor, [iii](#)
- Irreducible, [vii](#)
- Jacobi Symbol, [xxviii](#)
- Kummer's Theorem, [xi](#)

- Landau's Theorem, 33
- Laplace Transform, 122
 - auxiliary function g , 124
 - of the Chebyshev function, 123
- Laurent Expansion
 - of ζ at $s = 1$, 103
- Legendre Symbol, xxiv
- Legendre's Formula, xi
- Linear Diophantine Equation, vi
- Liouville Function, 3
- Logarithmic Derivative of ζ
 - pole at $s = 1$, 105
- Logarithmic Derivative of Zeta, 41
 - regularity on $\text{Re}(s)=1$, 114
- Möbius Function, 3
- Möbius Inversion Formula, 9
- Mellin Transform, 51
 - Inversion Formula, 53
- Mertens Constant, 85
- Mertens Function, 14
- Mertens Inequality ($3+4\cos+\cos 2$), 111
- Mertens' Theorems
 - First, 82
 - Second, 83
 - Third, 88
- Multiplicative Function, 2
- Multiplicative Order, xviii
- Newman's Tauberian Theorem, 126
- Non-Vanishing
 - via Euler Product, 47
- Number of Distinct Prime Factors, 3
- Number of Prime Factors with Multiplicity, 3
- Partial Summation, 17
- Periodic Bernoulli Functions, 25
- Perron's Formula, 61
 - Truncated Form, 63
- Prime, vii
- Prime Counting Function, 68
 - order of growth, 77
- Prime Number Theorem, 120, 136
 - consequences, 138
 - equivalent forms, 71
 - error term, 145
 - key integral, 131
 - monotonicity argument, 134
- Prime Reciprocal Sum
 - analytic divergence, 109
- Prime Series Φ , 106
 - pole at $s = 1$, 107
- Prime-Power Decomposition of $\log n!$, 80
- Primitive Root, xviii
 - Existence, xix
- Primorial Bound, 74
- Quadratic Reciprocity, xxvi
- Quadratic Residue, xxiii
- Residue Class Ring, xii
- Riemann Zeta Function, 98
 - growth near $\text{Re}(s)=1$, 139
 - Integral Representation, 56
 - meromorphic continuation, 101
 - non-vanishing on $\text{Re}(s)=1$, 111
- Ring Isomorphism
 - Arithmetic Functions and Dirichlet Series, 40
- Ring of Arithmetic Functions, 7
- Sum-of-Divisors Function, 3
- Summatory Function, 12
- Totally Multiplicative Function, 2
- Vinogradov Notation, xxix
- Von Mangoldt Function, 4
 - Dirichlet Series, 41
- Wilson's Theorem, xvi
- Zero-Free Region
 - de la Vallée Poussin, 142

Bibliography

- [1] Harold Davenport. *Multiplicative Number Theory*. 3rd ed. Graduate Texts in Mathematics 74. Revised and with a preface by Hugh L. Montgomery. New York: Springer-Verlag, 2000. ISBN: 978-0-387-95097-6.
- [2] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*. Ed. by D. R. Heath-Brown and J. H. Silverman. 6th ed. Oxford: Oxford University Press, 2008. ISBN: 978-0-19-921986-5.
- [3] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I: Classical Theory*. Cambridge Studies in Advanced Mathematics 97. Cambridge: Cambridge University Press, 2007. ISBN: 978-0-521-84903-6.
- [4] D. J. Newman. “Simple Analytic Proof of the Prime Number Theorem”. In: *The American Mathematical Monthly* 87.9 (1980), pp. 693–696. DOI: [10.2307/2321853](https://doi.org/10.2307/2321853).
- [5] Nathanael Chwojko-Srawley. *Everything You Need To Know About Harmonic Analysis*. 1st ed. Vol. 1. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_harmonic_analysis.pdf.
- [6] Nathanael Chwojko-Srawley. *Everything You Need To Know About Real Analysis*. 1st ed. Vol. 1. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_real_analysis.pdf.
- [7] Nathanael Chwojko-Srawley. *Everything You Need To Know About Class Field Theory*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_class_field_theory.pdf.
- [8] Gérald Tenenbaum. *Introduction to Analytic and Probabilistic Number Theory*. 3rd ed. Graduate Studies in Mathematics 163. Providence, RI: American Mathematical Society, 2015. ISBN: 978-0-8218-9854-3.
- [9] E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function*. Ed. by D. R. Heath-Brown. 2nd ed. Revised by D. R. Heath-Brown. Oxford: Clarendon Press, Oxford University Press, 1986. ISBN: 978-0-19-853369-6.
- [10] Mark B. Villarino. “Mertens’ Proof of Mertens’ Theorem”. In: (2005). arXiv:math/0504289. arXiv: [math/0504289](https://arxiv.org/abs/math/0504289) [[math.HO](#)].

- [11] Nathanael Chwojko-Srawley. *Everything You Need To Know About Undergraduate Algebra*. 1st ed. n.d. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.
- [12] Nathanael Chwojko-Srawley. *Everything You Need To Know About Local Field Theory*. URL: <https://nathanaelsrawley.com/pdf-viewer?src=.%2F.%2Fassets%2Fpdfs%2Fbooks%2FEYNTKA-local-field-theory.pdf&name=Local%20Field%20Theory&source=books>.
- [13] Nathanael Chwojko-Srawley. *Everything You Need To Know About Complex Analysis*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_complex_analysis.pdf.
- [14] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Number Theory*. Vol. 1. URL: <https://nathanaelsrawley.com/pdf-viewer?src=.%2F.%2Fassets%2Fpdfs%2Fbooks%2FEYNTKA-algebraic-number-theory.pdf&name=Algebraic%20Number%20Theory&source=books>.
- [15] D. Zagier. “Newman’s Short Proof of the Prime Number Theorem”. In: *The American Mathematical Monthly* 104.8 (1997), pp. 705–708. DOI: [10.2307/2975232](https://doi.org/10.2307/2975232).

The Prime Number Theorem

The four preceding chapters assembled, piece by piece, every ingredient the proof of the Prime Number Theorem requires. chapter 3 reduced the statement $\pi(x) \sim x/\log x$ to the assertion $\psi(x) \sim x$ about the Chebyshev function, and to the equivalent assertion $\vartheta(x) \sim x$ about its prime-only counterpart, through the equivalence of theorem 3.1.5; the same chapter pinned the order of magnitude of these functions, proving $\vartheta(x) \asymp x$ by Chebyshev's elementary count of prime powers (theorem 3.2.3). chapter 4 then supplied the analytic facts that the elementary theory could not reach: the zeta function continues meromorphically across the line $\operatorname{Re}(s) = 1$ with a single simple pole at $s = 1$ of residue 1 (theorem 4.2.1), its logarithmic derivative $-\zeta'(s)/\zeta(s)$ inherits that pole (proposition 4.3.1), and, the one fact lying just beyond the Euler-product arguments, $\zeta(s)$ has no zeros on the line $\operatorname{Re}(s) = 1$ (theorem 4.4.2). The goal of this chapter is to combine these into the asymptotic $\psi(x) \sim x$, equivalently $\vartheta(x) \sim x$ and $\pi(x) \sim x/\log x$, which is the Prime Number Theorem.

The route is the analytic proof of Hadamard and de la Vallée Poussin from 1896, in the streamlined form due to D. J. Newman in 1980 [4] and presented by Zagier [15], which trades the heavy contour estimates of the original arguments for a single Tauberian theorem. The strategy is to package the primes into the Dirichlet series $\Phi(s) = \sum_p (\log p)p^{-s}$, to convert the non-vanishing of ζ on $\operatorname{Re}(s) = 1$ into a holomorphic extension of Φ to the closed half-plane $\operatorname{Re}(s) \geq 1$ except for one pole, and to feed that extension into a Tauberian theorem that forces the integral $\int_1^\infty (\vartheta(x) - x)/x^2 dx$ to converge; the convergence of this integral, together with the monotonicity of ϑ , yields $\vartheta(x) \sim x$. This first section lays out the plan in full and states the central analytic claim that the next three sections establish. The Laplace-transform machinery and Newman's theorem occupy section 5.2 and section 5.3, the proof itself section 5.4; the final section, section 5.5, sharpens qualitative non-vanishing into a quantitative zero-free region and reads off an error term, the first quantitative refinement of the asymptotic.

5.1 The Analytic Strategy

The Prime Number Theorem is a single asymptotic statement, but its proof draws on two streams of input developed in separate chapters, and the first task is to see precisely how they join. The elementary side, from chapter 3, contributes two facts: that the three counting functions ψ , ϑ , and π rise or fall together, so that proving the asymptotic for any one of them proves it for all three; and that $\vartheta(x)$ is already known to be of the correct order x , bounded above and below by constant multiples of it. These reduce the entire theorem to the one limit $\vartheta(x)/x \rightarrow 1$. The analytic side, from chapter 4, contributes the behavior of $\zeta(s)$ and of the prime series $\Phi(s)$ near and on the line $\operatorname{Re}(s) = 1$. This section records the reduction, then states the plan by which the analytic inputs produce the missing limit.

The reduction itself is a restatement of results already proved, collected here so that the rest of the chapter may speak only of ϑ .

Proposition 5.1.1: Reduction of the Prime Number Theorem to $\vartheta(x) \sim x$

The following three asymptotic statements are equivalent:

$$\pi(x) \sim \frac{x}{\log x}, \quad \vartheta(x) \sim x, \quad \psi(x) \sim x$$

Moreover $\vartheta(x) = O(x)$, and the ratio $\vartheta(x)/x$ is bounded. Consequently the Prime Number Theorem in any of its three forms is equivalent to the single statement $\vartheta(x) \sim x$, and to prove it it suffices to prove that $\vartheta(x)/x \rightarrow 1$.

Proof:

The equivalence of the three asymptotics is theorem 3.1.5, where $\pi(x) \sim x/\log x$, $\vartheta(x) \sim x$, and $\psi(x) \sim x$ are shown to imply one another by Abel summation transferring between the prime-counting function π (definition 3.1.1) and the Chebyshev functions ϑ and ψ (definition 3.1.2). The bound $\vartheta(x) = O(x)$ is the upper half of Chebyshev's estimates (theorem 3.2.3), which produces constants $0 < c_1 < c_2$ with $c_1x \leq \vartheta(x) \leq \psi(x) \leq c_2x$ for all large x ; the right-hand inequality gives $\vartheta(x) \leq c_2x$, so $\vartheta(x) = O(x)$ and $\vartheta(x)/x \leq c_2$ is bounded above, while $\vartheta(x)/x \geq c_1$ bounds it below. Since the three statements are equivalent and $\vartheta(x) \sim x$ is by definition the assertion $\vartheta(x)/x \rightarrow 1$, establishing that single limit establishes all three, completing the proof.

The proposition is the hinge of the chapter. It says that the entire content of the Prime Number Theorem is concentrated in the behavior of one function, $\vartheta(x) = \sum_{p \leq x} \log p$, and that the work already done has narrowed the target to a single limit: $\vartheta(x)/x$ is known to lie between two positive constants, and what remains is to show that it converges, and that its limit is exactly 1. The Chebyshev bounds do half the job, confining the ratio to a bounded interval; the analytic theory must do the other half, collapsing that interval to the point 1. The boundedness recorded here is not incidental to the proof that follows. It is the hypothesis that makes the Tauberian step legitimate: a Tauberian theorem extracts an asymptotic for $\vartheta(x)$ from the analytic behavior of its generating function only under a one-sided growth condition, and $\vartheta(x) = O(x)$ is exactly that condition.

With the target fixed as $\vartheta(x) \sim x$, the plan of the proof can be stated in three steps. Each step is the subject of one of the next three sections; the purpose here is to show how they fit together and where each draws on the analytic structure of chapter 4.

5.1.2 Key Ideas: The Plan of the Proof The proof of $\vartheta(x) \sim x$ proceeds in three steps.

1. *The prime series extends to the line.* The prime series $\Phi(s) = \sum_p (\log p) p^{-s}$ of definition 4.3.2, holomorphic on $\operatorname{Re}(s) > 1$, has a simple pole at $s = 1$ of residue 1 and extends holomorphically to the closed half-plane $\operatorname{Re}(s) \geq 1$ with no singularity other than that pole. This is the analytic heart of the argument, and it is where the non-vanishing of ζ enters: by proposition 4.3.4 the difference $-\zeta'(s)/\zeta(s) - \Phi(s)$ is holomorphic on $\operatorname{Re}(s) > 1/2$, so Φ and $-\zeta'/\zeta$ have the same singularities up to that line, and by corollary 4.4.4 the function $-\zeta'/\zeta$ is holomorphic on all of $\operatorname{Re}(s) = 1$ except for the simple pole at $s = 1$, precisely because ζ has no zeros on the line (theorem 4.4.2).
2. *An integral converges.* The extension of Φ to the line forces the convergence of the integral

$$\int_1^\infty \frac{\vartheta(x) - x}{x^2} dx$$

The passage from step (1) to this convergence runs through a Laplace transform: Φ is recast as the Laplace transform of a function built from ϑ (the subject of section 5.2), and the holomorphic extension of that transform to the boundary line is exactly the hypothesis of the Tauberian theorem of Newman established in section 5.3, whose conclusion is the convergence of the integral. This step is carried out in section 5.4.

3. *Convergence forces the asymptotic.* The convergence of $\int_1^\infty (\vartheta(x) - x)/x^2 dx$, combined with the fact that ϑ is nondecreasing, forces $\vartheta(x) \sim x$. A monotone function whose normalized deviation from x has a convergent integral cannot stay bounded away from x : were $\vartheta(x) \geq \lambda x$ for some $\lambda > 1$ on a sequence of points, or $\vartheta(x) \leq \lambda x$ for some $\lambda < 1$, monotonicity would make the integrand bounded below by a positive constant over intervals of fixed logarithmic length, contradicting convergence. This final step, also in section 5.4, yields $\vartheta(x)/x \rightarrow 1$ and hence, by proposition 5.1.1, the Prime Number Theorem.

The three steps trace a path from the complex plane to the real line: a holomorphic extension across $\operatorname{Re}(s) = 1$ in step (1), a statement about a real integral in step (2), and a real-variable asymptotic in step (3). The transitions between them are the substance of the next three sections, and each transition is where a piece of analysis does work that no elementary argument could. What the roadmap makes visible, and what is worth dwelling on before the details, is why the non-vanishing of ζ on the line $\operatorname{Re}(s) = 1$ is the one input on which everything turns.

The decisive point is step (1), and within it the holomorphic extension of Φ to the closed half-plane. The series $\Phi(s) = \sum_p (\log p) p^{-s}$ converges only on $\operatorname{Re}(s) > 1$; by itself it gives no value on the boundary, and the convergence of the integral in step (2) is a statement about the behavior of ϑ that is sensitive to exactly that boundary. The bridge is the identity $-\zeta'(s)/\zeta(s) = \Phi(s) + R(s)$ of proposition 4.3.4, in which R is holomorphic well past the line, on $\operatorname{Re}(s) > 1/2$. This identity transfers the question about Φ on the line into a question about $-\zeta'/\zeta$ on the line, and the singularities of $-\zeta'/\zeta$ are dictated entirely by the zeros and the pole of ζ . The pole of ζ at $s = 1$ produces the one permitted singularity, the simple pole of Φ of residue 1, which becomes the main term in the asymptotic. Every zero of ζ on the line would produce an additional pole of $-\zeta'/\zeta$, hence of Φ , and would destroy the holomorphic extension that step (1) asserts. A single zero at a point $1 + it_0$ with $t_0 \neq 0$ would be a pole of Φ off the real axis, the Laplace transform of step (2) would fail to extend holomorphically to that point of the boundary, the hypothesis of Newman's theorem would not hold,

and the integral would have no reason to converge. The non-vanishing $\zeta(1+it) \neq 0$ for all real $t \neq 0$, proved in theorem 4.4.2, is precisely the statement that no such obstruction exists: the only singularity of Φ on the closed half-plane $\operatorname{Re}(s) \geq 1$ is the pole at $s = 1$, and the extension of step (1) goes through. This is the sense in which the distribution of primes is governed by a single vertical line in the complex plane, and why the analytic input that the elementary theory of chapter 3 could not supply, the absence of zeros on that line, is the input that completes the proof.

5.2 The Laplace Transform

The strategy of section 5.1 reduces the Prime Number Theorem to the single asymptotic $\vartheta(x) \sim x$, and proposition 5.1.1 traces the chain that delivers $\pi(x) \sim x/\log x$ once that asymptotic is in hand. What remains is to prove $\vartheta(x) \sim x$ from the analytic data assembled in chapter 4: the prime series $\Phi(s) = \sum_p (\log p) p^{-s}$ (definition 4.3.2), its simple pole at $s = 1$ with residue 1 (proposition 4.3.4), and the non-vanishing of ζ on the line $\operatorname{Re}(s) = 1$ (corollary 4.4.4). The bridge between the Dirichlet-series object Φ and the counting function ϑ is an integral transform, the *Laplace transform*, and the route runs through the substitution $x = e^t$ that turns the counting function $\vartheta(x)$ on $[1, \infty)$ into a function of t on $[0, \infty)$.

This section records the Laplace transform and computes the two integrals the proof in section 5.3 and section 5.4 requires. The first expresses the transform of $\vartheta(e^t)$ in terms of $\Phi(s)/s$; the second packages this into an auxiliary function $g(s)$ whose holomorphic extension to the closed half-plane $\operatorname{Re}(s) \geq 0$ is the one analytic input Newman's tauberian theorem will consume.

Definition 5.2.1: The Laplace Transform

Let $f: [0, \infty) \rightarrow \mathbb{C}$ be locally integrable. The *Laplace transform* of f is the function

$$(\mathcal{L}f)(s) = \int_0^\infty f(t) e^{-st} dt$$

defined for those $s \in \mathbb{C}$ for which the integral converges.

The Laplace transform is the one-sided companion of the Mellin transform of section 2.4, and the two are linked by the same change of variables that linked the Mellin transform to a two-sided Laplace transform in proposition 2.4.3. Under the substitution $t = e^{-x}$, the Mellin transform $(\mathcal{M}F)(s) = \int_0^\infty F(t) t^{s-1} dt$ becomes the two-sided Laplace integral $\int_{-\infty}^\infty F(e^{-x}) e^{-sx} dx$; restricting a function to the half-line and integrating only over $t \geq 0$ gives the one-sided transform of definition 5.2.1. As with both of those transforms, the modulus $|e^{-st}| = e^{-\operatorname{Re}(s)t}$ depends only on $\operatorname{Re}(s)$, so the region of convergence is a half-plane $\operatorname{Re}(s) > \sigma_0$ invariant under vertical translation, the abscissa σ_0 depending on the growth of f .

Example 5.2.2: The Laplace Transform of a Constant

Take $f(t) = 1$ for all $t \geq 0$. The transform is

$$(\mathcal{L}1)(s) = \int_0^\infty e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_{t=0}^{t=\infty} = \frac{1}{s}$$

for $\operatorname{Re}(s) > 0$. The boundary evaluation at $t = \infty$ vanishes because $|e^{-st}| = e^{-\operatorname{Re}(s)t} \rightarrow 0$ exactly

when $\operatorname{Re}(s) > 0$, and the value at $t = 0$ is $-1/s$; the abscissa of convergence is $\sigma_0 = 0$. The transform $1/s$ is holomorphic on $\operatorname{Re}(s) > 0$ and continues meromorphically to $\mathbb{C}$ with a single simple pole at $s = 0$, the boundary point of the half-plane of convergence. This $1/s$ is the term that will subtract against the pole of $\Phi(s+1)/(s+1)$ at $s = 0$ in proposition 5.2.4.

The function whose transform the proof needs is $\vartheta(e^t)$, the Chebyshev theta function (definition 3.1.2) reparametrized by $t = \log x$. Its transform is, up to the factor $1/s$, the prime series Φ .

Proposition 5.2.3: The Laplace Transform of $\vartheta(e^t)$

For $\operatorname{Re}(s) > 1$,

$$\int_0^\infty \vartheta(e^t) e^{-st} dt = \int_1^\infty \vartheta(x) x^{-s-1} dx = \frac{\Phi(s)}{s}$$

where $\Phi(s) = \sum_p (\log p) p^{-s}$ is the prime series of definition 4.3.2.

Proof:

Throughout write $\sigma = \operatorname{Re}(s)$ and assume $\sigma > 1$.

Step 1: The change of variables $x = e^t$. On the integral $\int_0^\infty \vartheta(e^t) e^{-st} dt$ substitute $x = e^t$, so $t = \log x$, $dt = dx/x$, and $e^{-st} = x^{-s}$. As t runs over $[0, \infty)$ the variable x runs over $[1, \infty)$, and

$$\int_0^\infty \vartheta(e^t) e^{-st} dt = \int_1^\infty \vartheta(x) x^{-s} \frac{dx}{x} = \int_1^\infty \vartheta(x) x^{-s-1} dx$$

The substitution $x = e^t$ is a C^1 diffeomorphism of $[0, \infty)$ onto $[1, \infty)$, so it preserves absolute convergence; convergence of the resulting integral is verified in Step 3. This establishes the first equality.

Step 2: Abel summation against $d\vartheta$. The function ϑ is the summatory function of the coefficients $a_n = \log n$ for n prime and $a_n = 0$ otherwise, so $\vartheta(x) = \sum_{n \leq x} a_n$, and its Stieltjes measure $d\vartheta(x)$ places mass $\log p$ at each prime p . Apply Abel summation (theorem 1.4.1) in its Stieltjes form to the integral $\int_1^\infty \vartheta(x) x^{-s-1} dx$. Take the test function $f(x) = x^{-s}$, which is continuously differentiable on $[1, \infty)$ with $f'(x) = -s x^{-s-1}$, so that $x^{-s-1} = -f'(x)/s$. For a finite upper limit $X \geq 1$, integration by parts for the Riemann–Stieltjes integral gives

$$\int_1^X \vartheta(x) x^{-s-1} dx = -\frac{1}{s} \int_1^X \vartheta(x) f'(x) dx = -\frac{1}{s} \left(\left[\vartheta(x) f(x) \right]_1^X - \int_1^X f(x) d\vartheta(x) \right)$$

the boundary term and the Stieltjes integral arising from $\int u dv = [uv] - \int v du$ with $u = \vartheta$, $v = f$. Since $\vartheta(x) = 0$ for $x < 2$, the boundary contribution at $x = 1$ vanishes, and $\int_1^X f(x) d\vartheta(x) = \sum_{p \leq X} (\log p) p^{-s}$ because $d\vartheta$ assigns mass $\log p$ to each prime $p \leq X$ and the integrand f is evaluated at $x = p$. Therefore

$$\int_1^X \vartheta(x) x^{-s-1} dx = -\frac{1}{s} \vartheta(X) X^{-s} + \frac{1}{s} \sum_{p \leq X} \frac{\log p}{p^s}$$

Step 3: The boundary term vanishes and the limit is $\Phi(s)/s$. By Chebyshev's estimates (theorem 3.2.3), $\vartheta(x) = O(x)$, so there is a constant C with $\vartheta(X) \leq CX$ for all $X \geq 1$. Hence

$$|\vartheta(X) X^{-s}| = \vartheta(X) X^{-\sigma} \leq C X^{1-\sigma}$$

and since $\sigma > 1$ the exponent $1 - \sigma$ is negative, so $X^{1-\sigma} \rightarrow 0$ as $X \rightarrow \infty$; the boundary term tends to 0. The same bound shows the integral converges absolutely: $|\vartheta(x) x^{-s-1}| = \vartheta(x) x^{-\sigma-1} \leq Cx \cdot x^{-\sigma-1} = Cx^{-\sigma}$, and $\int_1^\infty x^{-\sigma} dx < \infty$ because $\sigma > 1$. Letting $X \rightarrow \infty$,

$$\int_1^\infty \vartheta(x) x^{-s-1} dx = \frac{1}{s} \sum_p \frac{\log p}{p^s} = \frac{\Phi(s)}{s}$$

the sum on the right being $\Phi(s)$ by definition 4.3.2, convergent on $\sigma > 1$. Combining with Step 1 gives all three equalities, as we sought to show.

The proposition is the analytic dictionary entry that the whole proof turns on: it converts the prime series Φ , an object built from the primes through a Dirichlet series, into the Laplace transform of the counting function $\vartheta(e^t)$, an object built from the same primes through their summatory behavior. The factor $1/s$ is the signature of summation: passing from the coefficients $\log p$ to their partial sums ϑ divides the generating function by s , exactly as the Laplace transform of a constant in example 5.2.2 produced $1/s$ from the summatory function of the unit mass at the origin. The pole of Φ at $s = 1$ (proposition 4.3.4) therefore becomes a pole of $\Phi(s)/s$ at $s = 1$, with the kernel factor $1/s$ contributing the value 1 there, so the residue is again 1; this is the pole that will furnish the main term $\vartheta(x) \sim x$. The representation is valid only on $\operatorname{Re}(s) > 1$, where both Φ converges and the boundary term decays, and the entire difficulty of the Prime Number Theorem lies in pushing analytic control of this transform onto the line $\operatorname{Re}(s) = 1$.

That push is mediated by a single auxiliary function. Newman's tauberian theorem, the analytic engine of section 5.3, applies to the Laplace transform of a *bounded* function whose transform extends holomorphically across the imaginary axis. The function $\vartheta(e^t)$ is not bounded, but the normalized quantity $\vartheta(e^t)e^{-t}$ is, by Chebyshev's estimates, and subtracting its limiting value 1 produces a function tending to a transform with the required regularity.

Proposition 5.2.4: The Auxiliary Function $g(s)$

Define $f: [0, \infty) \rightarrow \mathbb{R}$ by

$$f(t) = \vartheta(e^t) e^{-t} - 1$$

and let $g(s) = \int_0^\infty f(t) e^{-st} dt$ be its Laplace transform. Then for $\operatorname{Re}(s) > 0$,

$$g(s) = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}$$

Proof:

Write $\sigma = \operatorname{Re}(s)$ and assume $\sigma > 0$. The integrand splits as $f(t)e^{-st} = \vartheta(e^t)e^{-t}e^{-st} - e^{-st} = \vartheta(e^t)e^{-(s+1)t} - e^{-st}$, so over any finite range $[0, T]$,

$$\int_0^T f(t) e^{-st} dt = \int_0^T \vartheta(e^t) e^{-(s+1)t} dt - \int_0^T e^{-st} dt$$

the splitting being legitimate on a bounded interval where each piece is integrable. The two pieces are handled separately, then the limit $T \rightarrow \infty$ is taken.

Step 1: The first integral. Since $\sigma > 0$, the shifted argument satisfies $\operatorname{Re}(s+1) = \sigma + 1 > 1$, so proposition 5.2.3 applies at $s+1$ in place of s :

$$\int_0^\infty \vartheta(e^t) e^{-(s+1)t} dt = \frac{\Phi(s+1)}{s+1}$$

the integral converging absolutely because $\operatorname{Re}(s+1) > 1$. In particular the partial integrals $\int_0^T \vartheta(e^t) e^{-(s+1)t} dt$ converge to $\Phi(s+1)/(s+1)$ as $T \rightarrow \infty$.

Step 2: The second integral. By example 5.2.2, for $\sigma > 0$,

$$\int_0^\infty e^{-st} dt = \frac{1}{s}$$

so $\int_0^T e^{-st} dt \rightarrow 1/s$ as $T \rightarrow \infty$.

Step 3: Combine. Both pieces converge as $T \rightarrow \infty$, so their difference converges and $g(s) = \int_0^\infty f(t) e^{-st} dt$ exists for $\sigma > 0$, with

$$g(s) = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}$$

completing the proof.

The function g packages exactly the data Newman's theorem will need. Its definition is engineered so that the integrand $f(t) = \vartheta(e^t)e^{-t} - 1$ is bounded: Chebyshev's estimates (theorem 3.2.3) give $\vartheta(x) \leq (\log 4)x$ for all $x \geq 1$, while $\vartheta(x) \geq 0$ holds trivially, so $0 \leq \vartheta(x)/x \leq \log 4$ for every $x \geq 1$ and the normalized quantity $\vartheta(e^t)e^{-t} = \vartheta(e^t)/e^t$ stays inside $[0, \log 4]$. Hence $f(t) = \vartheta(e^t)/e^t - 1$ stays inside the bounded range $[-1, \log 4 - 1]$ for all $t \geq 0$; the lower Chebyshev bound $\vartheta(x) \geq c_1 x$, valid for large x , pins $f(t)$ eventually above $c_1 - 1$ as well. Boundedness of f on $[0, \infty)$ is the hypothesis the tauberian theorem of section 5.3 imposes on the function being transformed. The formula $g(s) = \Phi(s+1)/(s+1) - 1/s$ then exhibits g , on $\operatorname{Re}(s) > 0$, as a difference of two functions each carrying a pole at $s = 0$. The shifted prime series $\Phi(s+1)$ has its pole at $s+1 = 1$, that is, at $s = 0$, with residue 1 (proposition 4.3.4), so $\Phi(s+1)/(s+1)$ has a simple pole at $s = 0$ with residue 1; the second term $1/s$ has a simple pole at $s = 0$ with residue 1 as well. The two principal parts cancel: g is holomorphic at $s = 0$, where it would otherwise inherit the singularity of Φ .

This cancellation is the crux. The entire proof of the Prime Number Theorem reduces to showing that g extends holomorphically to the closed half-plane $\operatorname{Re}(s) \geq 0$, and the analytic facts of chapter 4 are precisely what make that extension possible. At $s = 0$ the pole of Φ at the point $s = 1$ is exactly cancelled by the term $1/s$, as computed above; this is where the residue-1 pole of Φ enters, and it is the only point of the line $\operatorname{Re}(s) = 0$ at which $\Phi(s+1)$ is singular. At every other point $s = it$ with $t \neq 0$, the shifted argument $s+1 = 1+it$ lies on the line $\operatorname{Re} = 1$ away from the pole, where Φ is holomorphic because ζ has no zeros there (corollary 4.4.4), the non-vanishing of ζ on $\operatorname{Re}(s) = 1$ established in chapter 4. The two inputs together, the pole of Φ at $s = 1$ cancelling $1/s$ and the non-vanishing of ζ on the rest of the line, give g a holomorphic extension to a neighborhood of

$\operatorname{Re}(s) \geq 0$, the hypothesis that drives section 5.3 and section 5.4.

A numerical look at f makes its boundedness, the structural feature that lets the argument proceed, concrete.

Example 5.2.5: g at a Real Point

The integrand $f(t) = \vartheta(e^t)/e^t - 1$ measures the deviation of $\vartheta(x)/x$ from 1 at $x = e^t$. Tabulating $\vartheta(x)/x - 1$ at a range of x , using the values of $\vartheta(x)$ from example 3.1.3 together with further computation, gives

x	$\vartheta(x)/x$	$\vartheta(x)/x - 1$
10	0.5347	-0.4653
100	0.8373	-0.1627
1000	0.9562	-0.0438
10000	0.9896	-0.0104
100000	0.9969	-0.0031

Every entry in the last column lies in the interval $[-1, \log 4 - 1] \approx [-1, 0.386]$ guaranteed by theorem 3.2.3, since $0 \leq \vartheta(x)/x \leq \log 4$ for all $x \geq 1$; the upper constant $\log 4 \approx 1.386$ caps $\vartheta(x)/x$ for every x , and the lower Chebyshev constant $\log 2 \approx 0.693$ confines $\vartheta(x)/x$ from below once x is large, so $\vartheta(x)/x - 1$ cannot escape a fixed range no matter how large x grows. The table also shows the values drifting toward 0, consistent with the conjectured limit $\vartheta(x)/x \rightarrow 1$; but the proof does not assume this convergence. It assumes only the boundedness visible in the table, that $f(t)$ stays within fixed bounds for all $t \geq 0$. This boundedness is exactly the hypothesis Newman's theorem (section 5.3) will require of f , and the convergence $\vartheta(x)/x \rightarrow 1$ is the conclusion the theorem will deliver, not an input it demands.

5.3 Newman's Tauberian Theorem

The analytic engine of the proof of the Prime Number Theorem is a single theorem of D. J. Newman about Laplace transforms [4], in the streamlined form given by Zagier [15]. The Perron representation of example 2.5.4 expresses the prime-counting data through a contour integral of $-\zeta'/\zeta$, and chapter 4 supplied the analytic input that the integrand has no pole on the line $\operatorname{Re}(s) = 1$ except the one at $s = 1$. What remains is to convert that holomorphy-up-to-the-line into an asymptotic for the counting function, and the conversion is exactly the business of a *Tauberian* theorem: a result that upgrades a statement about the boundary behavior of a transform into a statement about convergence of the function it transforms. Newman's theorem is the cleanest such statement adapted to the situation at hand, and its proof is a short contour argument requiring nothing past Cauchy's integral formula.

The shape of the statement is the following. A bounded function f on $[0, \infty)$ has a Laplace transform $g(s) = \int_0^\infty f(t)e^{-st} dt$ that converges and is holomorphic on the open half-plane $\operatorname{Re}(s) > 0$. Convergence of the *improper integral* $\int_0^\infty f(t) dt$ is the assertion that g extends continuously to the boundary point $s = 0$ with the limit equal to the integral, and this is a strictly stronger demand than boundedness of f alone. Newman's theorem supplies the missing hypothesis in the form most useful for the zeta function: if g extends not just continuously but *holomorphically* across the imaginary axis, then the improper integral converges and equals $g(0)$. Holomorphy up to the line is the Tauberian side condition, and boundedness of f is the standing hypothesis that the application will draw from Chebyshev's bound.

Theorem 5.3.1: Newman's Analytic Theorem

Let $f: [0, \infty) \rightarrow \mathbb{C}$ be bounded and integrable (Riemann or Lebesgue) on every finite interval $[0, T]$, and for $\operatorname{Re}(s) > 0$ define

$$g(s) = \int_0^\infty f(t)e^{-st} dt$$

The integral converges absolutely and g is holomorphic on $\operatorname{Re}(s) > 0$. Suppose g extends to a holomorphic function on an open set containing the closed half-plane $\operatorname{Re}(s) \geq 0$. Then the improper integral $\int_0^\infty f(t) dt$ converges and

$$\int_0^\infty f(t) dt = g(0)$$

Proof:

Write $B = \sup_{t \geq 0} |f(t)|$, finite by hypothesis. For $\operatorname{Re}(s) > 0$ the bound $|f(t)e^{-st}| \leq Be^{-\operatorname{Re}(s)t}$ and $\int_0^\infty e^{-\operatorname{Re}(s)t} dt = 1/\operatorname{Re}(s) < \infty$ show that the defining integral converges absolutely; that g is holomorphic on $\operatorname{Re}(s) > 0$ is the holomorphy of a parameter integral whose integrand is holomorphic in s and dominated locally uniformly, established in [13]. The substance is the boundary statement, proved by a contour integral.

Step 1: The truncated transforms and the goal. For each $T > 0$ set

$$g_T(s) = \int_0^T f(t)e^{-st} dt$$

The integrand is entire in s for each fixed t , and the integral runs over the finite interval $[0, T]$, so g_T is entire (again by holomorphy of a parameter integral, now over a compact interval, [13]). The improper integral $\int_0^\infty f(t) dt$ is by definition $\lim_{T \rightarrow \infty} \int_0^T f(t) dt = \lim_{T \rightarrow \infty} g_T(0)$, so the assertion to be proved is exactly

$$g_T(0) \longrightarrow g(0) \quad (T \rightarrow \infty)$$

The difference $g(0) - g_T(0)$ will be written as a contour integral and estimated.

Step 2: A contour and the Cauchy representation. Fix $R > 0$ for the remainder of the argument. By hypothesis g is holomorphic on an open set $\Omega \supseteq \{\operatorname{Re}(s) \geq 0\}$. The intersection of Ω with the disc $\{|s| \leq R\}$ is an open neighborhood of the compact arc $\{|s| = R, \operatorname{Re}(s) \geq 0\} \cup \{\operatorname{Re}(s) = 0, |s| \leq R\}$, so there is a $\delta = \delta(R) > 0$, which may be taken with $\delta < R$, such that g is holomorphic on a neighborhood of the closed region

$$D = \{s \mid |s| \leq R, \operatorname{Re}(s) \geq -\delta\}$$

Let $C = \partial D$ be its positively oriented boundary, consisting of the major arc $A = \{|s| = R, \operatorname{Re}(s) \geq -\delta\}$ of the circle of radius R together with the vertical segment $\{\operatorname{Re}(s) = -\delta, |\operatorname{Im}(s)| \leq \sqrt{R^2 - \delta^2}\}$. The point $s = 0$ lies in the interior of D , and g_T is entire, so

the function

$$s \mapsto (g(s) - g_T(s)) e^{sT} \left(\frac{1}{s} + \frac{s}{R^2} \right)$$

is holomorphic on a neighborhood of D except for a simple pole at $s = 0$. The factor $1/s + s/R^2$ has residue 1 at $s = 0$ (the term s/R^2 is regular there), and $e^{sT} = 1$ at $s = 0$, so the residue of the displayed function at $s = 0$ is $g(0) - g_T(0)$. Cauchy's residue theorem ([13]) gives

$$g(0) - g_T(0) = \frac{1}{2\pi i} \int_C (g(s) - g_T(s)) e^{sT} \left(\frac{1}{s} + \frac{s}{R^2} \right) ds \quad (5.1)$$

The plan is to split C into the part in $\operatorname{Re}(s) > 0$ and the part in $\operatorname{Re}(s) \leq 0$, and to bound each.

Step 3: The kernel on the circle $|s| = R$. On the circle $|s| = R$ one has $\bar{s} = R^2/s$, hence

$$\frac{1}{s} + \frac{s}{R^2} = \frac{\bar{s}}{R^2} + \frac{s}{R^2} = \frac{s + \bar{s}}{R^2} = \frac{2\operatorname{Re}(s)}{R^2}$$

so that on $|s| = R$,

$$\left| e^{sT} \left(\frac{1}{s} + \frac{s}{R^2} \right) \right| = e^{\operatorname{Re}(s)T} \frac{2|\operatorname{Re}(s)|}{R^2} \quad (5.2)$$

This identity is the reason the second term s/R^2 was inserted; it is recorded for use on both halves of the contour.

Step 4: The right half $C_+ = C \cap \{\operatorname{Re}(s) > 0\}$. Here C_+ is the right portion $\{|s| = R, \operatorname{Re}(s) > 0\}$ of the circle, a semicircle of length πR . For $\operatorname{Re}(s) > 0$ the tail estimate

$$|g(s) - g_T(s)| = \left| \int_T^\infty f(t) e^{-st} dt \right| \leq B \int_T^\infty e^{-\operatorname{Re}(s)t} dt = \frac{B e^{-\operatorname{Re}(s)T}}{\operatorname{Re}(s)}$$

holds, where the first equality uses that the integral defining $g(s)$ converges and exceeds that defining $g_T(s)$ by the tail over $[T, \infty)$. Multiplying by the kernel bound (5.2) and using $|\operatorname{Re}(s)| = \operatorname{Re}(s) > 0$,

$$\left| (g(s) - g_T(s)) e^{sT} \left(\frac{1}{s} + \frac{s}{R^2} \right) \right| \leq \frac{B e^{-\operatorname{Re}(s)T}}{\operatorname{Re}(s)} \cdot e^{\operatorname{Re}(s)T} \frac{2\operatorname{Re}(s)}{R^2} = \frac{2B}{R^2}$$

The integrand on C_+ is bounded by $2B/R^2$ pointwise, so by the standard length estimate ([13]) the contribution of C_+ to (5.1) satisfies

$$\left| \frac{1}{2\pi i} \int_{C_+} (g(s) - g_T(s)) e^{sT} \left(\frac{1}{s} + \frac{s}{R^2} \right) ds \right| \leq \frac{1}{2\pi} \cdot \frac{2B}{R^2} \cdot \pi R = \frac{B}{R}$$

This bound is uniform in T .

Step 5: The left half, the g_T part. Let $C_- = C \cap \{\operatorname{Re}(s) \leq 0\}$ be the remainder of the contour: the left portion of the arc A together with the vertical segment at $\operatorname{Re}(s) = -\delta$. On C_- the

integrand of (5.1) is treated by separating g from g_T , which is legitimate because each piece is integrable on the compact contour. Consider first the g_T piece,

$$I_T^- = \frac{1}{2\pi i} \int_{C_-} g_T(s) e^{sT} \left(\frac{1}{s} + \frac{s}{R^2} \right) ds$$

Since g_T is entire, the integrand $g_T(s) e^{sT} (1/s + s/R^2)$ is holomorphic on the punctured plane $\mathbb{C} \setminus \{0\}$, in particular on the open left half-plane $\{\operatorname{Re}(s) < 0\}$. The contour C_- and the left semicircle

$$C'_- = \{|s| = R, \operatorname{Re}(s) \leq 0\}$$

share the two endpoints $\pm iR$, and the closed region they bound lies entirely in $\{\operatorname{Re}(s) \leq 0\}$, with the only point of nonholomorphy $s = 0$ on its boundary rather than its interior. Cauchy's theorem ([13]) therefore permits replacing C_- by C'_- in I_T^- without changing its value. On C'_- one has $\operatorname{Re}(s) \leq 0$, and for such s the entire transform is bounded by

$$|g_T(s)| = \left| \int_0^T f(t) e^{-st} dt \right| \leq B \int_0^T e^{-\operatorname{Re}(s)t} dt = B \frac{e^{-\operatorname{Re}(s)T} - 1}{-\operatorname{Re}(s)} \leq \frac{B e^{-\operatorname{Re}(s)T}}{|\operatorname{Re}(s)|}$$

using $-\operatorname{Re}(s) = |\operatorname{Re}(s)| > 0$ on the open left semicircle and $e^{-\operatorname{Re}(s)T} - 1 \leq e^{-\operatorname{Re}(s)T}$. Multiplying by the kernel bound (5.2),

$$\left| g_T(s) e^{sT} \left(\frac{1}{s} + \frac{s}{R^2} \right) \right| \leq \frac{B e^{-\operatorname{Re}(s)T}}{|\operatorname{Re}(s)|} \cdot e^{\operatorname{Re}(s)T} \frac{2|\operatorname{Re}(s)|}{R^2} = \frac{2B}{R^2}$$

the factors $e^{\pm \operatorname{Re}(s)T}$ and $|\operatorname{Re}(s)|$ cancelling exactly as on the right half. The semicircle C'_- has length πR , so the length estimate gives

$$|I_T^-| \leq \frac{1}{2\pi} \cdot \frac{2B}{R^2} \cdot \pi R = \frac{B}{R}$$

again uniformly in T .

Step 6: The left half, the g part. The complementary piece is

$$J_T^- = \frac{1}{2\pi i} \int_{C_-} g(s) e^{sT} \left(\frac{1}{s} + \frac{s}{R^2} \right) ds$$

Here R , hence δ and the contour C_- , are fixed, and $T \rightarrow \infty$. The function $g(s) (1/s + s/R^2)$ is continuous on the compact set C_- : the only candidate for a singularity is $s = 0$, but C_- has $\operatorname{Re}(s) \leq 0$ and meets the imaginary axis only at the two points $\pm iR$, where $|s| = R \neq 0$, so $s = 0 \notin C_-$ and the factor $1/s$ is bounded there. Let

$$M = \max_{s \in C_-} \left| g(s) \left(\frac{1}{s} + \frac{s}{R^2} \right) \right| < \infty$$

a constant depending on R and δ but not on T . The remaining factor satisfies $|e^{sT}| = e^{\operatorname{Re}(s)T} \leq 1$ on C_- , since $\operatorname{Re}(s) \leq 0$, so the integrand of J_T^- is dominated by the constant M on the finite-length contour C_- , uniformly in T . For each fixed $s \in C_-$ with $\operatorname{Re}(s) < 0$ one has $e^{\operatorname{Re}(s)T} \rightarrow 0$ as $T \rightarrow \infty$; the exceptional points $\pm iR$, where $\operatorname{Re}(s) = 0$, form a set of measure zero on the

contour. By the dominated convergence theorem ([6]), applied to the arc-length integral with dominating function M ,

$$J_T^- \longrightarrow 0 \quad (T \rightarrow \infty)$$

This is the only one of the three pieces that depends on letting T grow, and it is where holomorphy of g across the line is used: only because g continues to $\operatorname{Re}(s) = -\delta$ can the contour C_- be pushed off the imaginary axis into the region $\operatorname{Re}(s) < 0$ where e^{sT} decays.

Step 7: Assembling the estimate. Splitting (5.1) as $C = C_+ \cup C_-$ and the C_- integral as its g and g_T parts,

$$g(0) - g_T(0) = \underbrace{\frac{1}{2\pi i} \int_{C_+} (\cdots) ds}_{\text{Step 4}} - \underbrace{I_T^-}_{\text{Step 5}} + \underbrace{J_T^-}_{\text{Step 6}}$$

where the sign on I_T^- records that the g_T contribution over C_- enters (5.1) as $\int_{C_-} (g - g_T) = J_T^- - I_T^-$. Steps 4 and 5 bound the first two terms by B/R each, uniformly in T , and Step 6 sends the third to 0. Hence

$$\limsup_{T \rightarrow \infty} |g(0) - g_T(0)| \leq \frac{B}{R} + \frac{B}{R} + 0 = \frac{2B}{R}$$

The left-hand side does not depend on R , while the right-hand side is $2B/R$ for every $R > 0$. Letting $R \rightarrow \infty$ forces $\limsup_{T \rightarrow \infty} |g(0) - g_T(0)| = 0$, so $g_T(0) \rightarrow g(0)$. By Step 1 this is the statement that $\int_0^\infty f(t) dt$ converges with value $g(0)$, completing the proof.

The hypothesis does double duty, and its two halves play distinct roles. Boundedness of f is what makes the tail estimates of Steps 4 and 5 produce the bound $2B/R^2$ on the circle; without a uniform bound on f the kernel could not absorb the growth of $e^{\operatorname{Re}(s)T}$ on the right, and the argument would collapse. Holomorphy of g across the line is the Tauberian content: it is the hypothesis that lets the contour in Step 6 be displaced into $\{\operatorname{Re}(s) < 0\}$, where the exponential e^{sT} decays and the g -integral vanishes in the limit. Strip away the holomorphy and only boundedness of f remains, which guarantees that g is defined and holomorphic on $\operatorname{Re}(s) > 0$ but says nothing about the boundary; the improper integral $\int_0^\infty f(t) dt$ may then diverge by oscillation even though f is bounded, as the example $f(t) = \cos t$ shows, where $g(s) = s/(s^2 + 1)$ has poles at $s = \pm i$ on the imaginary axis and the holomorphy hypothesis fails.

The name *Tauberian* marks the logical direction of the theorem. An *Abelian* theorem deduces the behavior of a transform from the convergence of the function: if $\int_0^\infty f(t) dt$ converges, then $g(s) \rightarrow g(0)$ as $s \rightarrow 0^+$, a statement of continuity that is comparatively soft. A Tauberian theorem reverses the implication, recovering convergence of $\int_0^\infty f(t) dt$ from analytic regularity of g at the boundary, and such reversals are false without a side condition that restrains the function from oscillating, the *Tauberian condition*. In Newman's theorem the side condition is the boundedness of f together with the holomorphic extension of g , and the value $g(0)$ that the holomorphic transform assigns to the boundary point is promoted from a formal or Abel-summed value into the genuine value of the improper integral.

5.3.2 Key Ideas: The Kernel $e^{sT}(1/s + s/R^2)$ The factor $e^{sT}(1/s + s/R^2)$ in (5.1) is engineered so that a single contour integral does three things at once. The term $1/s$ carries a simple pole at $s = 0$ with residue 1; against the holomorphic factor $g(s) - g_T(s)$ it produces the residue $g(0) - g_T(0)$, which is the quantity to be estimated. The factor e^{sT} is the device that lets the truncation $g - g_T$ be controlled: on the right half-plane the tail $g(s) - g_T(s)$ carries a decay $e^{-\operatorname{Re}(s)T}$, and the inserted e^{sT} cancels it, while on the left half-plane the same e^{sT} decays and kills the g -integral as $T \rightarrow \infty$. The role of the added term s/R^2 is to tame the kernel on the circle $|s| = R$: by the identity $1/s + s/R^2 = 2\operatorname{Re}(s)/R^2$ valid on that circle, the modulus of the kernel collapses to $e^{\operatorname{Re}(s)T} (2|\operatorname{Re}(s)|/R^2)$, in which the prefactor $2|\operatorname{Re}(s)|/R^2$ supplies exactly the factor $\operatorname{Re}(s)$ needed to cancel the $1/\operatorname{Re}(s)$ from the tail bound $Be^{-\operatorname{Re}(s)T}/\operatorname{Re}(s)$, and the residual $e^{\pm\operatorname{Re}(s)T}$ cancels the exponential. The net bound $2B/R^2$ on both semicircles is symmetric in the two halves precisely because the kernel modulus depends only on $|\operatorname{Re}(s)|$. Without the s/R^2 term the bare $1/s$ would contribute $1/R$ on the circle with no $\operatorname{Re}(s)$ factor to absorb the $1/\operatorname{Re}(s)$ from the tail, and the right-half estimate would diverge as s approaches the imaginary axis; the corrective term costs nothing at $s = 0$, where it is regular, so the pole and its residue are untouched.

The application in section 5.4 fits the theorem exactly. The function to which Newman's theorem will be applied is built from the Chebyshev theta function ϑ (definition 3.1.2), arranged so that the boundedness hypothesis becomes the assertion that $\vartheta(x)/x$ is bounded, which is the content of Chebyshev's estimates $c_1x \leq \vartheta(x) \leq c_2x$ in theorem 3.2.3. The holomorphic-extension hypothesis becomes the non-vanishing of ζ on $\operatorname{Re}(s) = 1$ from theorem 4.4.2, in the form that $-\zeta'/\zeta$, and with it the relevant transform, continues across the line save for the controlled pole at $s = 1$ (corollary 4.4.4). Newman's theorem then returns the convergence of an improper integral in ϑ , and a short argument converts that convergence into $\vartheta(x) \sim x$, hence into the Prime Number Theorem through the equivalence of theorem 3.1.5. The theorem proved here is thus the one analytic lever that turns the boundary data on ζ into an asymptotic count of primes.

5.4 Proof of the Prime Number Theorem

The pieces assembled in the three preceding sections now combine into the theorem. The auxiliary function g of section 5.2, defined on $\operatorname{Re}(s) > 0$ as the Laplace transform of $f(t) = \vartheta(e^t)e^{-t} - 1$, was computed in proposition 5.2.4 to be $g(s) = \Phi(s+1)/(s+1) - 1/s$. The pole of Φ at $s = 1$ (proposition 4.3.4) cancels against the term $1/s$ at $s = 0$, and the non-vanishing of ζ on the line $\operatorname{Re}(s) = 1$ (corollary 4.4.4, equivalently theorem 4.4.2) removes every other obstruction on the imaginary axis; together they extend g holomorphically across $\operatorname{Re}(s) = 0$. Newman's theorem (theorem 5.3.1) then converts that extension into the convergence of $\int_0^\infty f(t) dt$, which the substitution $x = e^t$ turns into the convergence of $\int_1^\infty (\vartheta(x) - x)/x^2 dx$.

What remains after that convergence is purely real-variable. The integrand $(\vartheta(x) - x)/x^2$ records the normalized deviation of ϑ from its conjectured asymptotic, and the monotonicity of ϑ forbids that deviation from persisting: a function that climbs only upward cannot keep $\vartheta(x)/x$ away from 1 without forcing the integral to diverge. This section carries out both halves, the analytic convergence and the monotonicity argument, and assembles them into the headline result of the chapter.

Theorem 5.4.1: Convergence of the Key Integral

The improper integral

$$\int_1^\infty \frac{\vartheta(x) - x}{x^2} dx$$

converges.

Proof:

The proof applies Newman's theorem (theorem 5.3.1) to the function f of proposition 5.2.4 and then changes variables. The work is in verifying the holomorphic-extension hypothesis, which occupies Steps 2 and 3.

Step 1: The function and its transform. Define $f: [0, \infty) \rightarrow \mathbb{R}$ by

$$f(t) = \vartheta(e^t) e^{-t} - 1$$

The function f is bounded: by Chebyshev's estimates (theorem 3.2.3) one has $0 \leq \vartheta(x) \leq (\log 4)x$ for all $x \geq 1$, so writing $x = e^t$ gives $0 \leq \vartheta(e^t) e^{-t} = \vartheta(x)/x \leq \log 4$, whence $-1 \leq f(t) \leq \log 4 - 1$ for every $t \geq 0$. The function f is also locally integrable, indeed Riemann integrable on every finite interval $[0, T]$, since $\vartheta(e^t)$ is a nondecreasing step function with finitely many jumps on any bounded interval and e^{-t} is continuous. Both hypotheses on the transformed function in theorem 5.3.1 are thus met. Its Laplace transform is, by proposition 5.2.4, the function

$$g(s) = \int_0^\infty f(t) e^{-st} dt = \frac{\Phi(s+1)}{s+1} - \frac{1}{s}, \quad \operatorname{Re}(s) > 0 \quad (5.3)$$

holomorphic on $\operatorname{Re}(s) > 0$. To invoke Newman's theorem it remains to show that g extends holomorphically to an open set containing the closed half-plane $\operatorname{Re}(s) \geq 0$.

Step 2: Holomorphic extension at $s = 0$. Near $s = 0$ the shifted argument $s + 1$ is near 1, where Φ has its only singularity. By proposition 4.3.4 the prime series has a simple pole at $s = 1$ with residue 1, so $\Phi(w) - 1/(w - 1)$ is holomorphic in a neighborhood of $w = 1$; setting $w = s + 1$, the function

$$\Phi(s+1) - \frac{1}{s}$$

is holomorphic in a neighborhood of $s = 0$, since $w - 1 = s$. Call this holomorphic function $h_0(s)$, so that $\Phi(s+1) = 1/s + h_0(s)$ near $s = 0$. Substituting into (5.3),

$$g(s) = \frac{1}{s+1} \left(\frac{1}{s} + h_0(s) \right) - \frac{1}{s} = \frac{1}{s(s+1)} - \frac{1}{s} + \frac{h_0(s)}{s+1}$$

The first two terms combine through the partial-fraction identity $1/(s(s+1)) = 1/s - 1/(s+1)$, which gives

$$\frac{1}{s(s+1)} - \frac{1}{s} = \left(\frac{1}{s} - \frac{1}{s+1} \right) - \frac{1}{s} = -\frac{1}{s+1}$$

so the singular term $1/s$ cancels exactly, leaving

$$g(s) = -\frac{1}{s+1} + \frac{h_0(s)}{s+1} = \frac{h_0(s) - 1}{s+1} \quad (5.4)$$

near $s = 0$. The numerator $h_0(s) - 1$ is holomorphic near $s = 0$, and the denominator $s + 1$ is nonzero there (it equals 1 at $s = 0$), so the right-hand side of (5.4) is holomorphic in a neighborhood of $s = 0$. This is the holomorphic extension of g to the origin.

Step 3: Holomorphic extension at $s = i\tau$ with $\tau \neq 0$. Fix a real $\tau \neq 0$ and consider the point $s = i\tau$ on the imaginary axis. The shifted argument is $s + 1 = 1 + i\tau$, a point on the line $\text{Re} = 1$ distinct from the pole $s = 1$. By proposition 4.3.4 the prime series decomposes as

$$\Phi(w) = -\frac{\zeta'(w)}{\zeta(w)} - R(w)$$

where R is holomorphic on $\text{Re}(w) > 1/2$. The point $w = 1 + i\tau$ lies in this half-plane, so R is holomorphic in a neighborhood of it. The logarithmic derivative $-\zeta'/\zeta$ is holomorphic on the line $\text{Re}(w) = 1$ except for the simple pole at $w = 1$, by corollary 4.4.4, which records the non-vanishing $\zeta(1 + i\tau) \neq 0$ for $\tau \neq 0$ of theorem 4.4.2; since $\tau \neq 0$, the point $1 + i\tau$ is not the pole, so $-\zeta'/\zeta$ is holomorphic in a neighborhood of $w = 1 + i\tau$. Therefore Φ , the difference of two functions holomorphic near $w = 1 + i\tau$, is holomorphic in a neighborhood of that point, hence so is $w \mapsto \Phi(w)/w$ because $w = 1 + i\tau \neq 0$. Translating back by $w = s + 1$, the function $\Phi(s + 1)/(s + 1)$ is holomorphic in a neighborhood of $s = i\tau$. The remaining term $-1/s$ of (5.3) is holomorphic at $s = i\tau$ as well, since $i\tau \neq 0$. Their sum g is thus holomorphic in a neighborhood of every point $i\tau$ with $\tau \neq 0$.

Step 4: Assembling the extension and applying Newman's theorem. The function g is holomorphic on the open half-plane $\text{Re}(s) > 0$ by Step 1. Step 2 gives a neighborhood of $s = 0$ on which g extends holomorphically, and Step 3 gives, for each $\tau \neq 0$, a neighborhood of $i\tau$ on which g extends holomorphically; the extensions agree with g on the overlaps with $\text{Re}(s) > 0$, where they all equal the original transform, so by the identity theorem ([13]) they patch into a single holomorphic function on the union. That union is an open set containing the open half-plane $\text{Re}(s) > 0$ together with every point of the imaginary axis $\text{Re}(s) = 0$, hence an open set containing the closed half-plane $\text{Re}(s) \geq 0$. The hypotheses of theorem 5.3.1 are satisfied: f is bounded and locally integrable, and its transform g extends holomorphically across the line. The theorem concludes that the improper integral $\int_0^\infty f(t) dt$ converges (with value $g(0)$).

Step 5: The change of variables $x = e^t$. The convergence just obtained is

$$\int_0^\infty (\vartheta(e^t) e^{-t} - 1) dt = \lim_{T \rightarrow \infty} \int_0^T (\vartheta(e^t) e^{-t} - 1) dt < \infty$$

On each truncation $\int_0^T$ substitute $x = e^t$, so $t = \log x$, $dt = dx/x$, and $e^{-t} = 1/x$; as t runs over $[0, T]$ the variable x runs over $[1, e^T]$. The substitution is a C^1 diffeomorphism of $[0, T]$ onto $[1, e^T]$, so

$$\int_0^T (\vartheta(e^t) e^{-t} - 1) dt = \int_1^{e^T} \left(\frac{\vartheta(x)}{x} - 1 \right) \frac{dx}{x} = \int_1^{e^T} \frac{\vartheta(x) - x}{x^2} dx$$

the last equality combining the two terms over the common denominator x^2 . As $T \rightarrow \infty$ the upper limit $e^T \rightarrow \infty$ through all values, and the left-hand side converges by Step 4, so the right-hand side converges to the same finite limit:

$$\int_1^\infty \frac{\vartheta(x) - x}{x^2} dx = \int_0^\infty (\vartheta(e^t) e^{-t} - 1) dt < \infty$$

Therefore the improper integral converges, as we sought to show.

The theorem is the analytic payload of the entire chapter delivered in a single real statement. Everything complex-analytic has been spent: the meromorphic continuation of ζ , the pole at $s = 1$, the non-vanishing on the line, Newman's contour argument. What survives is the assertion that $\vartheta(x) - x$, measured against the natural scale x^2 that makes the integrand the logarithmic-mean deviation $(\vartheta(x)/x - 1)/x$, integrates to a finite value. This is already a strong statement about how rarely and by how little $\vartheta(x)$ can stray from x , but it is not yet the asymptotic $\vartheta(x) \sim x$. A finite integral permits the integrand to be large on thin sets, and a priori $\vartheta(x)/x$ could oscillate, dipping below 1 and rising above it infinitely often, so long as the excursions are brief enough that their total weighted contribution stays bounded. Ruling out such oscillation is the role of monotonicity, and it is the one property of ϑ that the analytic machinery never used.

The next result isolates exactly that real-variable principle. It is stated for a general nondecreasing function so that the mechanism, monotonicity against a convergent integral, stands clear of the arithmetic.

Lemma 5.4.2: Monotonicity Forces the Asymptotic

Let $\vartheta: [1, \infty) \rightarrow [0, \infty)$ be nondecreasing, and suppose the integral

$$\int_1^\infty \frac{\vartheta(x) - x}{x^2} dx$$

converges. Then $\vartheta(x) \sim x$ as $x \rightarrow \infty$, that is, $\vartheta(x)/x \rightarrow 1$.

Proof:

The argument is by contradiction, ruling out in turn that $\vartheta(x)/x$ exceeds 1 along a sequence and that it falls below 1 along a sequence. Both directions exploit the same device: over an interval whose endpoints are in a fixed ratio, monotonicity of ϑ bounds the integrand below (respectively above) by a positive (respectively negative) constant independent of where the interval sits, while the Cauchy criterion for the convergent integral forces the integral over any such far-out interval to be small. The Cauchy criterion is the statement that since $\int_1^\infty (\vartheta(t) - t)/t^2 dt$ converges, for every $\varepsilon > 0$ there is an X_0 with

$$\left| \int_a^b \frac{\vartheta(t) - t}{t^2} dt \right| < \varepsilon \quad \text{whenever } X_0 \leq a \leq b \quad (5.5)$$

the integral over $[a, b]$ being the difference of the partial integrals up to b and up to a .

Step 1: The upper excursion cannot occur. Suppose, for contradiction, that there is a constant $\lambda > 1$ and arbitrarily large x with $\vartheta(x) \geq \lambda x$. Fix such an x and consider the interval $[x, \lambda x]$. For t in this interval, $t \leq \lambda x$, and because ϑ is nondecreasing and $t \geq x$,

$$\vartheta(t) \geq \vartheta(x) \geq \lambda x$$

so the numerator satisfies $\vartheta(t) - t \geq \lambda x - t \geq 0$ throughout $[x, \lambda x]$, the last inequality because $t \leq \lambda x$. Hence

$$\int_x^{\lambda x} \frac{\vartheta(t) - t}{t^2} dt \geq \int_x^{\lambda x} \frac{\lambda x - t}{t^2} dt$$

In the integral on the right substitute $t = ux$, so $dt = x du$, $t^2 = u^2 x^2$, and as t runs over $[x, \lambda x]$ the variable u runs over $[1, \lambda]$:

$$\int_x^{\lambda x} \frac{\lambda x - t}{t^2} dt = \int_1^\lambda \frac{\lambda x - ux}{u^2 x^2} x du = \int_1^\lambda \frac{\lambda - u}{u^2} du$$

the factors of x cancelling completely. The resulting quantity

$$c_\lambda^+ := \int_1^\lambda \frac{\lambda - u}{u^2} du$$

is a positive constant depending only on λ , not on x : the integrand $(\lambda - u)/u^2$ is strictly positive for $u \in [1, \lambda)$ and continuous, so its integral over $[1, \lambda]$ is strictly positive. The antiderivative $\int (\lambda - u)/u^2 du = -\lambda/u - \log u$ evaluates it to $c_\lambda^+ = \lambda - 1 - \log \lambda$, which is positive for every $\lambda > 1$. Thus for every one of the arbitrarily large x with $\vartheta(x) \geq \lambda x$,

$$\int_x^{\lambda x} \frac{\vartheta(t) - t}{t^2} dt \geq c_\lambda^+ > 0$$

a fixed positive lower bound. But by the Cauchy criterion (5.5) with $\varepsilon = c_\lambda^+/2$, there is an X_0 beyond which every such integral, over an interval $[x, \lambda x]$ with $x \geq X_0$, has absolute value below $c_\lambda^+/2$. Choosing one of the arbitrarily large x with $x \geq X_0$ produces an integral that is simultaneously $\geq c_\lambda^+$ and $< c_\lambda^+/2$, a contradiction. Hence no such $\lambda > 1$ exists, and $\limsup_{x \rightarrow \infty} \vartheta(x)/x \leq 1$.

Step 2: The lower excursion cannot occur. Suppose, for contradiction, that there is a constant λ with $0 < \lambda < 1$ and arbitrarily large x with $\vartheta(x) \leq \lambda x$. Fix such an x and consider now the interval $[\lambda x, x]$, which lies below x . For t in this interval, $t \geq \lambda x$, and because ϑ is nondecreasing and $t \leq x$,

$$\vartheta(t) \leq \vartheta(x) \leq \lambda x$$

so the numerator satisfies $\vartheta(t) - t \leq \lambda x - t \leq 0$ throughout $[\lambda x, x]$, the last inequality because $t \geq \lambda x$. Hence

$$\int_{\lambda x}^x \frac{\vartheta(t) - t}{t^2} dt \leq \int_{\lambda x}^x \frac{\lambda x - t}{t^2} dt$$

Substituting $t = ux$ as before, with u now running over $[\lambda, 1]$,

$$\int_{\lambda x}^x \frac{\lambda x - t}{t^2} dt = \int_\lambda^1 \frac{\lambda - u}{u^2} du$$

again with all factors of x cancelling. The quantity

$$c_\lambda^- := \int_\lambda^1 \frac{\lambda - u}{u^2} du$$

is a strictly negative constant depending only on λ : on $(\lambda, 1]$ the integrand $(\lambda - u)/u^2$ is strictly negative and continuous, so $c_\lambda^- < 0$. The same antiderivative $-\lambda/u - \log u$ evaluates it to $c_\lambda^- = 1 - \lambda + \log \lambda$, which is negative for every λ with $0 < \lambda < 1$. Thus for every one of the arbitrarily large x with $\vartheta(x) \leq \lambda x$,

$$\int_{\lambda x}^x \frac{\vartheta(t) - t}{t^2} dt \leq c_\lambda^- < 0$$

a fixed negative upper bound. The Cauchy criterion (5.5) with $\varepsilon = |c_\lambda^-|/2$ gives an X_0 beyond which every integral over $[\lambda x, x]$ with $\lambda x \geq X_0$ has absolute value below $|c_\lambda^-|/2$. Selecting an arbitrarily large x with $\lambda x \geq X_0$ yields an integral that is at once $\leq c_\lambda^- = -|c_\lambda^-|$ and greater than $-|c_\lambda^-|/2$, a contradiction. Hence no such $\lambda \in (0, 1)$ exists, and $\liminf_{x \rightarrow \infty} \vartheta(x)/x \geq 1$.

Steps 1 and 2 give $\limsup_{x \rightarrow \infty} \vartheta(x)/x \leq 1 \leq \liminf_{x \rightarrow \infty} \vartheta(x)/x$. Since the liminf never exceeds the limsup, both equal 1, so $\lim_{x \rightarrow \infty} \vartheta(x)/x = 1$, completing the proof.

The lemma turns a statement about an average into a statement about a limit, and monotonicity is the entire reason the conversion is legitimate. The integrand $(\vartheta(t) - t)/t^2$ has a convergent integral, which controls only the cumulative deviation; the integrand could in principle be large at isolated places. Monotonicity forbids isolation. If ϑ overshoots λx at a single point x , it stays at least λx for the entire interval $[x, \lambda x]$, because it cannot come back down, and the overshoot is therefore spread across a window of multiplicative width λ , contributing a fixed positive amount to the integral. The same window appears in both directions, and its scale invariance, the fact that $c_\lambda^\pm$ depends on λ but not on x , is what lets a single excursion at arbitrarily large x contradict the smallness guaranteed far out by the Cauchy criterion. The choice of weight x^2 in the denominator is exactly what produces this invariance: it makes the substitution $t = ux$ scale the integral to a form free of x , so a deviation at height x costs the same regardless of x . Replace ϑ by a function that is not monotone, and the argument collapses, since a brief spike could be undone immediately and the window of forced deviation would vanish.

With both halves in place, the proof of the Prime Number Theorem is immediate. The function ϑ supplied by arithmetic is nondecreasing, and theorem 5.4.1 furnishes exactly the convergent integral the lemma requires.

Theorem 5.4.3: The Prime Number Theorem

As $x \rightarrow \infty$,

$$\vartheta(x) \sim x, \quad \psi(x) \sim x, \quad \pi(x) \sim \frac{x}{\log x}$$

where ϑ and ψ are the Chebyshev functions (definition 3.1.2) and π is the prime counting function (definition 3.1.1).

Proof:

The Chebyshev theta function $\vartheta(x) = \sum_{p \leq x} \log p$ is nondecreasing in x , since increasing x can only admit further primes into the sum, each contributing a nonnegative term $\log p$. By theorem 5.4.1 the integral $\int_1^\infty (\vartheta(x) - x)/x^2 dx$ converges. The two hypotheses of lemma 5.4.2, monotonicity of ϑ and convergence of this integral, are therefore met, and the lemma yields

$$\vartheta(x) \sim x$$

By the equivalence of the three asymptotic forms (theorem 3.1.5), $\vartheta(x) \sim x$ holds if and only if $\psi(x) \sim x$ and if and only if $\pi(x) \sim x/\log x$. Hence all three hold simultaneously, completing the proof.

This is the theorem the chapter was built to prove, and its history measures the distance the argument has travelled. The asymptotic $\pi(x) \sim x/\log x$ was conjectured by Gauss and Legendre

from numerical tables near the end of the eighteenth century, refined by Gauss into the sharper guess $\pi(x) \sim \text{Li}(x)$, and proved only in 1896, independently by Jacques Hadamard and Charles-Jean de la Vallée Poussin. The century of delay separates the two kinds of information the statement contains. The *order* of $\pi(x)$, that it grows like $x/\log x$ up to bounded factors, is elementary: Chebyshev obtained it in 1852 by comparing the central binomial coefficient against its prime factorization, and that argument, reproduced in theorem 3.2.3, places $\vartheta(x)/x$ between $\log 2$ and $\log 4$ using nothing past unique factorization and Legendre's formula. What the elementary method cannot reach is the *constant*. Chebyshev's corridor has width a factor of two, and no sharpening of the binomial comparison closes it to the single value 1. The constant is analytic in origin. It is the residue of the pole of ζ at $s = 1$, transmitted through the pole of $-\zeta'/\zeta$ and of Φ at $s = 1$, each of residue exactly 1 (proposition 4.3.4), and it reaches the counting function as the value $g(0)$ that Newman's theorem assigns at the boundary. The non-vanishing of ζ on the line $\text{Re}(s) = 1$ is what licenses that boundary value to exist, and it is precisely the input that the elementary theory of chapter 3 could not produce. The Prime Number Theorem is, in this sense, the meeting of two centuries of method: the order from Chebyshev's combinatorics, the constant from Riemann's complex analysis of ζ .

5.4.4 Historical Note: Riemann's Memoir and the Elementary Proof The path proved here is the one Riemann charted in his 1859 memoir, which proposed studying $\pi(x)$ through the analytic continuation of $\zeta(s)$ and the location of its zeros, though Riemann himself proved no asymptotic. Hadamard and de la Vallée Poussin completed the program in 1896 with the non-vanishing of ζ on $\text{Re}(s) = 1$ as the technical core. For half a century afterward it was believed that the analytic input was unavoidable, that no proof could avoid the complex zeros of ζ . This was refuted in 1949 by Atle Selberg and Paul Erdős, who gave an *elementary* proof, in the technical sense of avoiding complex analysis, built on Selberg's symmetry formula $\sum_{p \leq x} (\log p)^2 + \sum_{pq \leq x} (\log p)(\log q) = 2x \log x + O(x)$. The elementary proof is longer and less transparent than the analytic one and yields a weaker error term; the analytic method through zero-free regions, taken up in section 5.5, remains the route to quantitative refinements. Newman's 1980 contour argument [4], in Zagier's exposition [15], is the streamlining of the analytic proof followed in this chapter.

The theorem is best felt against the numbers that first suggested it, where the slow approach of $\pi(x)$ to $x/\log x$ and the closer fit of the logarithmic integral are both visible.

Example 5.4.5: The Prime Number Theorem Numerically

Write $\text{Li}(x) = \int_2^x dt/\log t$ for the logarithmic integral, the approximation to $\pi(x)$ that emerged inside the proof of theorem 3.1.5. Comparing $\pi(x)$ against the two approximations $x/\log x$ and $\text{Li}(x)$ at three widely spaced values of x gives

x	$\pi(x)$	$\frac{x}{\log x}$	$\frac{\pi(x)}{x/\log x}$	$\text{Li}(x)$	$\frac{\pi(x)}{\text{Li}(x)}$
10^3	168	144.8	1.1605	176.6	0.9515
10^6	78498	72382.4	1.0845	78626.5	0.9984
10^9	50847534	48254942.4	1.0537	50849233.9	0.99997

The fourth column, $\pi(x)/(x/\log x)$, is the ratio whose limit the theorem asserts to be 1. It does tend to 1, but slowly: even at $x = 10^9$ it is still 1.05, off by five percent, and the convergence is governed by the next-order term, which is of relative size $1/\log x$, equal to $1/\log 10^9 \approx 0.048$ at $x = 10^9$, matching the observed gap. The logarithmic integral $\text{Li}(x)$ is the better approximation

by a wide margin: the last column $\pi(x)/\text{Li}(x)$ reaches 1 far faster, already agreeing to four decimal places at $x = 10^9$. The reason is that $\text{Li}(x) = x/\log x + x/(\log x)^2 + \dots$ captures the secondary term $x/(\log x)^2$ that $x/\log x$ omits, and it is exactly this secondary term that accounts for the persistent excess of $\pi(x)$ over $x/\log x$ seen in the fourth column. Both approximations are asymptotically correct, $\pi(x)/(x/\log x) \rightarrow 1$ and $\pi(x)/\text{Li}(x) \rightarrow 1$, since $\text{Li}(x) \sim x/\log x$; the theorem is the statement that the ratios converge, and the table is the statement that for $x/\log x$ the convergence is unhurried. A consequence read directly from $\pi(x) \sim x/\log x$, made precise in corollary 5.4.6 below, is that the n -th prime p_n satisfies $p_n \sim n \log n$: at $n = 10^4$ the prime is $p_n = 104729$ against $n \log n \approx 92103$, the ratio 1.137 drifting toward 1 at the same logarithmic rate.

The asymptotic for p_n used at the close of the example is the most frequently quoted reformulation of the theorem, and it follows by inverting $\pi(x) \sim x/\log x$. Two further consequences are worth recording in the same breath: the precise sense in which primes thin out, and the inversion itself.

Corollary 5.4.6: Consequences

Let p_n denote the n -th prime. Then as $n \rightarrow \infty$,

$$p_n \sim n \log n$$

Moreover the density of primes near x is asymptotically $1/\log x$: the probability that a randomly chosen integer in $[1, x]$ is prime, namely $\pi(x)/x$, satisfies $\pi(x)/x \sim 1/\log x$.

Proof:

The density statement is a restatement of the theorem: dividing $\pi(x) \sim x/\log x$ from theorem 5.4.3 by x gives $\pi(x)/x \sim 1/\log x$, which is the asserted density. The substance is the asymptotic for p_n , proved by inverting the relation $\pi(x) \sim x/\log x$.

Step 1: A logarithmic comparison. Set $x = p_n$ in the theorem. By definition $\pi(p_n) = n$, the count of primes up to the n -th prime, so theorem 5.4.3 gives

$$n = \pi(p_n) \sim \frac{p_n}{\log p_n}, \quad \text{that is} \quad \frac{n \log p_n}{p_n} \rightarrow 1 \quad (5.6)$$

Taking logarithms of $n \sim p_n/\log p_n$, and using that $\log$ of a quantity tending to 1 tends to 0,

$$\log n = \log p_n - \log \log p_n + o(1)$$

Since $p_n \rightarrow \infty$, both $\log p_n \rightarrow \infty$ and $\log \log p_n \rightarrow \infty$, but the second grows far slower; dividing the last display by $\log p_n$,

$$\frac{\log n}{\log p_n} = 1 - \frac{\log \log p_n}{\log p_n} + \frac{o(1)}{\log p_n} \rightarrow 1$$

because $\log \log p_n / \log p_n \rightarrow 0$. Hence $\log n \sim \log p_n$.

Step 2: Inverting. Rewrite (5.6) as $p_n \sim n \log p_n$. Substituting the equivalence $\log p_n \sim \log n$ from Step 1,

$$p_n \sim n \log p_n \sim n \log n$$

where the replacement of $\log p_n$ by $\log n$ is legitimate because the two are asymptotically equal and n is a common factor: $p_n/(n \log n) = (p_n/(n \log p_n)) \cdot (\log p_n/\log n) \rightarrow 1 \cdot 1 = 1$. Therefore $p_n \sim n \log n$, as we sought to show.

The corollary translates the theorem into the language of the primes as a sequence. The statement $p_n \sim n \log n$ says that the n -th prime sits near $n \log n$, so the primes, though individually unpredictable, are on average spaced $\log n \approx \log p_n$ apart at height p_n : the gap between consecutive primes near x is, on average, about $\log x$. This is the same fact as the density $1/\log x$ viewed from the dual side, counting the spacing between primes rather than their frequency, and the two are reciprocal as one expects. The average gap $\log x$ grows without bound, so the primes do thin out, but only logarithmically, far more slowly than the squares or the powers of 2; between x and $2x$ there are still about $x/\log x$ primes for every large x . The estimate $p_n \sim n \log n$ is the crudest term of a longer expansion, $p_n = n \log n + n \log \log n - n + o(n)$, whose secondary terms come from the same logarithmic-integral correction that made $\text{Li}(x)$ outperform $x/\log x$ in example 5.4.5; the leading term alone is the direct content of the Prime Number Theorem.

5.5 Error Terms and the Zero-Free Region

The proof of the Prime Number Theorem completed in section 5.4 rests on a single qualitative fact: $\zeta(s)$ has no zeros on the line $\text{Re}(s) = 1$ (theorem 4.4.2). That fact is exactly enough to force $\psi(x) \sim x$, equivalently $\vartheta(x) \sim x$ and $\pi(x) \sim x/\log x$ (theorem 5.4.3), but it is silent on the rate of convergence. Non-vanishing on the line says the ratio $\psi(x)/x$ tends to 1; it does not say how fast, and the Tauberian route of Newman's theorem, which trades all quantitative information for a clean limit, gives no error term at all. To measure the speed one needs a quantitative strengthening: not just that ζ avoids the line, but that it avoids an open region to the *left* of the line, a region whose width near a given height controls how far the contour of integration can be pushed past $\text{Re}(s) = 1$ and hence how small the resulting error can be made.

This section proves the de la Vallée Poussin zero-free region, the first such quantitative result, in full: there is a constant $c > 0$ with $\zeta(s) \neq 0$ throughout $\sigma \geq 1 - c/\log(|t| + 2)$. The argument is the quantitative form of the positivity inequality $3 + 4\cos\theta + \cos 2\theta \geq 0$ that proved qualitative non-vanishing in chapter 4, sharpened by a growth bound for ζ near the line. The width of this region then dictates an error term, $\psi(x) = x + O(x \exp(-c\sqrt{\log x}))$, whose derivation by contour optimization is stated with a precise reference. The section closes with the Riemann–von Mangoldt explicit formula, the Riemann Hypothesis, and the modern Korobov–Vinogradov region, each placing the elementary error term in the larger landscape that the functional equation of the next chapter opens up.

The proof of the zero-free region needs one analytic input beyond the structural facts of chapter 4: a bound on the size of ζ and its derivative just to the left of $\text{Re}(s) = 1$. The continuation formula of proposition 1.4.6 supplies it directly.

Lemma 5.5.1: A Growth Bound for ζ near the Line

There is a constant $A > 0$ such that

$$|\zeta(s)| \leq A \log |t| \quad \text{and} \quad |\zeta'(s)| \leq A (\log |t|)^2$$

uniformly for all $s = \sigma + it$ with $\sigma \geq 1 - \frac{1}{\log |t|}$ and $|t| \geq 5$.

Proof:

By the reflection $\zeta(\bar{s}) = \overline{\zeta(s)}$, which holds because ζ is real on the real axis and analytic, both moduli are symmetric under $t \mapsto -t$, so it suffices to treat $t \geq 5$. Write $\sigma = \operatorname{Re}(s)$ throughout. The threshold $t \geq 5$ gives $\log t \geq \log 5 > \frac{3}{2}$, which keeps every point used below inside the half-plane $\operatorname{Re}(s) > 0$ where the continuation formula of proposition 1.4.6 is valid: the lower edge of the region satisfies $1 - 1/\log t \geq 1 - 1/\log 5 > 0$, and the Cauchy circle of Step 3 stays in $\operatorname{Re}(w) > 0$ as well.

Step 1: A truncated continuation formula. proposition 1.4.6 gives, for $\operatorname{Re}(s) > 0$ and $s \neq 1$ and every real $x \geq 1$,

$$\sum_{n \leq x} \frac{1}{n^s} = \frac{x^{1-s}}{1-s} + \zeta(s) + O(x^{-\sigma})$$

with an absolute implied constant. Solving for $\zeta(s)$,

$$\zeta(s) = \sum_{n \leq x} \frac{1}{n^s} - \frac{x^{1-s}}{1-s} + O(x^{-\sigma}) \quad (5.7)$$

The estimate of the three pieces will be carried out at the choice $x = t$, the truncation height matched to the imaginary part.

Step 2: The bound for $|\zeta(s)|$. The argument is carried out uniformly in a width parameter: for each fixed $\kappa \geq 1$ there is a constant $A_0(\kappa)$ with $|\zeta(s)| \leq A_0(\kappa) \log t$ for all $s = \sigma + it$ with $\sigma \geq 1 - \kappa/\log t$, $\sigma > 0$, and $t \geq 2$. The positivity constraint $\sigma > 0$ is what makes the continuation formula available; in the stated region of the lemma ($\kappa = 1$, $t \geq 5$) it holds automatically since $1 - 1/\log t > 0$ there, while in the $\kappa = 3$ application of Step 3 it is checked directly at each point. Fix $\kappa \geq 1$ and such an s , and take $x = t$ in (5.7); the continuation formula applies since $\operatorname{Re}(s) = \sigma > 0$ and $s \neq 1$. First note that for these σ ,

$$t^{1-\sigma} = \exp((1-\sigma) \log t) \leq \exp\left(\frac{\kappa}{\log t} \cdot \log t\right) = e^\kappa$$

since $1 - \sigma \leq \kappa/\log t$; this bound on $t^{1-\sigma}$ uses only the region constraint, not the sign of σ . The three pieces are estimated in turn.

For the Dirichlet sum, write $|n^{-s}| = n^{-\sigma} = n^{-1} n^{1-\sigma}$ and bound the factor $n^{1-\sigma}$ for $1 \leq n \leq x$ by $\max(x^{1-\sigma}, 1)$. For $\sigma \leq 1$ the exponent $1 - \sigma$ is nonnegative, so $n^{1-\sigma} \leq x^{1-\sigma}$; for $\sigma > 1$ it is negative, so $n^{1-\sigma} \leq 1$. In both cases $n^{1-\sigma} \leq \max(x^{1-\sigma}, 1) \leq e^\kappa$, using $x^{1-\sigma} = t^{1-\sigma} \leq e^\kappa$ and

$1 \leq e^\kappa$. Hence $n^{-\sigma} \leq n^{-1}e^\kappa$ for $n \leq x$, and

$$\left| \sum_{n \leq x} \frac{1}{n^s} \right| \leq e^\kappa \sum_{n \leq x} \frac{1}{n} \leq e^\kappa (1 + \log x) = e^\kappa (1 + \log t)$$

the harmonic sum bounded by $1 + \log x$ through comparison with $\int_1^x du/u$.

For the rational term, $|x^{1-s}/(1-s)| = x^{1-\sigma}/|1-s|$, and $|1-s| = |(1-\sigma) - it| \geq |t| = t \geq 2$, so

$$\left| \frac{x^{1-s}}{1-s} \right| = \frac{t^{1-\sigma}}{|1-s|} \leq \frac{e^\kappa}{t} \leq \frac{e^\kappa}{2}$$

For the error term, $x^{-\sigma} = t^{-\sigma} = t^{-1}t^{1-\sigma} \leq t^{-1}e^\kappa \leq e^\kappa/2$. Summing the three contributions,

$$|\zeta(s)| \leq e^\kappa(1 + \log t) + \frac{e^\kappa}{2} + O\left(\frac{e^\kappa}{2}\right) \leq A_0(\kappa) \log t$$

for a suitable constant $A_0(\kappa)$ and all $t \geq 2$, since $\log t$ dominates the constants once $t \geq 2$ and $1 + \log t \leq (1 + 1/\log 2) \log t$ there. This is the asserted bound on $|\zeta|$, valid on $\sigma \geq 1 - \kappa/\log t$ with $\sigma > 0$.

Step 3: The bound for $|\zeta'(s)|$ by a Cauchy estimate. Fix $s = \sigma + it$ with $\sigma \geq 1 - 1/\log t$ and $t \geq 5$, and let Γ be the positively oriented circle of radius $r = 1/(2 \log t)$ centered at s . Since $t \geq 5$ gives $\log t \geq \log 5$, the radius satisfies $r \leq 1/(2 \log 5) < 1$. Each point $w = u + iv$ on Γ satisfies

$$u \geq \sigma - r \geq 1 - \frac{1}{\log t} - \frac{1}{2 \log t} = 1 - \frac{3}{2 \log t}, \quad |v - t| \leq r < 1$$

so $|v| \geq t - 1 \geq 4$ and $|v| \leq t + 1 \leq 2t$, the last because $1 \leq t$. The lower bound on u is positive: $\log t \geq \log 5 > \frac{3}{2}$ gives $1 - 3/(2 \log t) > 0$, so $\operatorname{Re}(w) = u > 0$ and w lies in the half-plane where ζ is defined and the bound of Step 2 is available. From $|v| \leq 2t$ and $t \geq 5$ one gets $\log |v| \leq \log(2t) = \log 2 + \log t \leq 2 \log t$, hence $1/\log t \leq 2/\log |v|$, and therefore

$$u \geq 1 - \frac{3}{2 \log t} \geq 1 - \frac{3}{\log |v|}$$

so w lies in the region of Step 2 with $\kappa = 3$, imaginary part v satisfying $|v| \geq 4 \geq 2$, and $\operatorname{Re}(w) = u > 0$. Step 2 with $\kappa = 3$ therefore applies to every $w \in \Gamma$, giving $|\zeta(w)| \leq A_0(3) \log |v| \leq 2A_0(3) \log t$, the last step using $\log |v| \leq 2 \log t$. Thus $|\zeta(w)| \leq A_1 \log t$ for an absolute constant $A_1 = 2A_0(3)$ and every $w \in \Gamma$. The Cauchy integral formula for the derivative ([13]) gives

$$\zeta'(s) = \frac{1}{2\pi i} \int_{\Gamma} \frac{\zeta(w)}{(w-s)^2} dw$$

the contour Γ lying in $\operatorname{Re}(w) > 0$, $w \neq 1$, where ζ is holomorphic. On Γ one has $|w - s| = r$, so $|(w - s)^{-2}| = r^{-2}$, and the length of Γ is $2\pi r$; the standard length estimate yields

$$|\zeta'(s)| \leq \frac{1}{2\pi} \cdot \frac{\max_{w \in \Gamma} |\zeta(w)|}{r^2} \cdot 2\pi r = \frac{1}{r} \max_{w \in \Gamma} |\zeta(w)| \leq (2 \log t) \cdot A_1 \log t = 2A_1 (\log t)^2$$

Taking $A = \max(A_0(1), 2A_1)$ gives both bounds simultaneously on the stated region, completing the proof.

The bound is crude in its dependence on σ , holding uniformly across the thin strip $1 - 1/\log |t| \leq \sigma$ with no improvement as σ increases past 1, but its dependence on t is what matters: ζ grows no faster than a power of $\log |t|$ near the line. This polynomial-in-log growth is the quantitative substitute for the boundedness that holds deeper inside the half-plane $\operatorname{Re}(s) > 1$, and it is the only size information the zero-free argument consumes. The choice $x = |t|$ in the proof is the balance point: a larger truncation would shrink the error term $O(x^{-\sigma})$ but inflate the Dirichlet sum, and a smaller one would do the reverse, so matching the truncation height to the imaginary part is what produces the single factor of $\log |t|$ rather than a power of $|t|$. The derivative bound costs one further factor of $\log |t|$, the price of the radius $r \asymp 1/\log |t|$ in the Cauchy estimate, the largest scale of radius on which the growth bound for ζ still applies.

Example 5.5.2: The Growth Bound at $s = 1 + it$

On the line itself, $\sigma = 1$, the lemma reads $|\zeta(1 + it)| = O(\log |t|)$ and $|\zeta'(1 + it)| = O((\log |t|)^2)$. The first is the familiar statement that ζ grows slowly along the line $\operatorname{Re}(s) = 1$. At the concrete height $t = 100$ the bound gives $|\zeta(1 + 100i)| \leq A \log 100 = A \cdot 4.605 \dots$, while the true value is $|\zeta(1 + 100i)| \approx 1.63$, comfortably inside the bound for any reasonable constant; the gap reflects the crudeness of the constant A , not of the $\log |t|$ shape. The logarithmic growth is genuine, not an artifact: the partial sums $\sum_{n \leq |t|} n^{-1-it}$ that dominate (5.7) have modulus of order $\log |t|$ for typical t , since the phases $n^{-it} = e^{-it \log n}$ fail to produce full cancellation in the harmonic-weighted sum. The same computation with the second derivative, a Cauchy estimate on a circle of radius $\asymp 1/\log |t|$ applied to the bound for ζ' , would give $|\zeta''(1 + it)| = O((\log |t|)^3)$, each derivative costing one additional logarithm.

The growth bound is the missing quantitative ingredient. The qualitative non-vanishing theorem of chapter 4 ruled out zeros *on* the line by showing that a zero there would drive the positive combination $|\zeta(\sigma)|^3 |\zeta(\sigma + it)|^4 |\zeta(\sigma + 2it)|$ below 1 as $\sigma \rightarrow 1^+$. The quantitative version measures how far a zero must sit from the line for the same combination to remain admissible, and the distance it extracts is exactly $c/\log(|t| + 2)$.

Theorem 5.5.3: The de la Vallée Poussin Zero-Free Region

There is a constant $c > 0$ such that $\zeta(s) \neq 0$ throughout the region

$$\sigma \geq 1 - \frac{c}{\log(|t| + 2)}$$

In particular ζ has no zeros in an open neighborhood of the line $\operatorname{Re}(s) = 1$ whose width at height t shrinks like $1/\log |t|$.

Proof:

The half-plane $\operatorname{Re}(s) > 1$ contains no zeros, by the Euler product (corollary 4.1.2), and a finite portion of the line is handled by continuity, so it suffices to find $c > 0$ such that no zero $\rho = \beta + i\gamma$ has

$$\beta > 1 - \frac{c}{\log(|\gamma| + 2)}$$

for $|\gamma|$ large. By the symmetry $\zeta(\bar{s}) = \overline{\zeta(s)}$, zeros come in conjugate pairs, so it is enough to treat $\gamma \geq \gamma_0$ for a fixed $\gamma_0 \geq 7$, the range $0 \leq \gamma < \gamma_0$ being a compact piece of the closed half-plane $\operatorname{Re}(s) \geq 1$ on which ζ is nonzero (no zeros on $\operatorname{Re}(s) = 1$ by theorem 4.4.2, none in $\operatorname{Re}(s) > 1$)

and hence bounded away from zero on a strip $\sigma \geq 1 - c'$ by continuity and compactness. Fix a zero $\rho = \beta + i\gamma$ with $\gamma \geq \gamma_0$ and write $L = \log(\gamma + 2)$.

Step 1: The positivity inequality. For real $\sigma > 1$ the von Mangoldt series of proposition 4.3.1 converges absolutely, and taking real parts of $-\zeta'(s)/\zeta(s) = \sum_n \Lambda(n)n^{-s}$ at $s = \sigma + i\tau$ gives, since $\Lambda(n) \geq 0$ and $|n^{-i\tau}| = 1$ with $\operatorname{Re}(n^{-i\tau}) = \cos(\tau \log n)$,

$$\operatorname{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + i\tau)\right) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\sigma}} \cos(\tau \log n)$$

Form the combination with weights 3, 4, 1 at the heights $\tau = 0, \gamma, 2\gamma$:

$$3\left(-\frac{\zeta'}{\zeta}(\sigma)\right) + 4\operatorname{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + i\gamma)\right) + \operatorname{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + 2i\gamma)\right) = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\sigma}} (3 + 4\cos\theta_n + \cos 2\theta_n)$$

with $\theta_n = \gamma \log n$. By lemma 4.4.1 every bracket $3 + 4\cos\theta_n + \cos 2\theta_n = 2(1 + \cos\theta_n)^2 \geq 0$, and the coefficients $\Lambda(n)n^{-\sigma}$ are nonnegative, so the entire sum is nonnegative:

$$3\left(-\frac{\zeta'}{\zeta}(\sigma)\right) + 4\operatorname{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + i\gamma)\right) + \operatorname{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + 2i\gamma)\right) \geq 0 \quad (5.8)$$

This is the quantitative engine; the three terms are now bounded above individually, the first from the pole and the latter two from the growth bound, and the inequality is read backward to constrain β .

Step 2: The pole term. By proposition 4.3.1 the function $-\zeta'/\zeta$ has near $s = 1$ the form $-\zeta'(s)/\zeta(s) = 1/(s-1) - h(s)$ with h holomorphic in a fixed neighborhood of $s = 1$, hence bounded on the segment $1 < \sigma \leq 2$. Therefore there is a constant A_2 with

$$-\frac{\zeta'}{\zeta}(\sigma) \leq \frac{1}{\sigma-1} + A_2 \quad (1 < \sigma \leq 2) \quad (5.9)$$

the inequality being equality up to the bounded term $-h(\sigma)$, which contributes at most a constant.

Step 3: The off-line terms from the growth bound. The two remaining terms are controlled by a standard bound on the logarithmic derivative of a function of moderate growth, applied to ζ . The bound states that there is a constant $A_3 > 0$ such that for $1 < \sigma \leq 2$ and any height τ with $|\tau| \geq \gamma_0$,

$$\operatorname{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + i\tau)\right) \leq A_3 \log(|\tau| + 2) - \sum_{\rho'} \operatorname{Re} \frac{1}{\sigma + i\tau - \rho'} \quad (5.10)$$

where the sum runs over the zeros ρ' of ζ with $|\operatorname{Im}\rho' - \tau| \leq 1$, and every term of that sum is nonnegative because $\operatorname{Re}(\sigma + i\tau - \rho') = \sigma - \beta' > 1 - 1 = 0$ for a zero $\rho' = \beta' + i\gamma'$ (which has $\beta' \leq 1$). The inequality (5.10) is the partial-fraction estimate for $-\zeta'/\zeta$ obtained from the Borel–Carathéodory theorem and Jensen’s formula applied to ζ on the disc of radius slightly larger than 1 centered at $2 + i\tau$: Borel–Carathéodory converts the upper bound $|\zeta(s)| \leq A \log |t|$ of lemma 5.5.1 on the boundary circle into a bound on ζ'/ζ in the interior, and Jensen’s formula identifies the zeros ρ' as the only obstruction, contributing the displayed negative partial-fraction

terms; the derivation is standard and is carried out in full in [1, Ch. 12] and [3, Ch. 6], using the complex-analytic tools of [13] (the Borel–Carathéodory theorem and Jensen’s formula), with $\log(|\tau| + 2)$ entering as the logarithm of the growth bound’s M . Two applications of (5.10) are needed.

At height $\tau = 2\gamma$, drop the entire nonnegative sum to obtain the clean upper bound

$$\operatorname{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + 2i\gamma)\right) \leq A_3 \log(2\gamma + 2) \leq A_4 L \quad (5.11)$$

for a constant A_4 , using $\log(2\gamma + 2) \leq 2\log(\gamma + 2) = 2L$ for $\gamma \geq 1$.

At height $\tau = \gamma$, the chosen zero $\rho = \beta + i\gamma$ has $\operatorname{Im}\rho = \gamma = \tau$, so $|\operatorname{Im}\rho - \tau| = 0 \leq 1$ and ρ appears in the sum of (5.10). Keep only its term and drop the rest, all nonnegative:

$$\operatorname{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + i\gamma)\right) \leq A_3 \log(\gamma + 2) - \operatorname{Re}\frac{1}{\sigma + i\gamma - \rho} = A_3 L - \operatorname{Re}\frac{1}{\sigma - \beta}$$

since $\sigma + i\gamma - \rho = \sigma + i\gamma - (\beta + i\gamma) = \sigma - \beta$ is real and positive. Thus

$$\operatorname{Re}\left(-\frac{\zeta'}{\zeta}(\sigma + i\gamma)\right) \leq A_3 L - \frac{1}{\sigma - \beta} \quad (5.12)$$

the term $-1/(\sigma - \beta)$ recording the proximity of $\sigma + i\gamma$ to the zero.

Step 4: Combining the bounds. Substitute (5.9), (5.12), and (5.11) into the positivity inequality (5.8):

$$0 \leq 3\left(\frac{1}{\sigma - 1} + A_2\right) + 4\left(A_3 L - \frac{1}{\sigma - \beta}\right) + A_4 L$$

Collecting the logarithmic terms into a single constant times L , there is a constant $A_5 = 3A_2/L + 4A_3 + A_4 \leq A_6$ (bounded since $L \geq \log(\gamma_0 + 2) \geq 1$) with

$$\frac{4}{\sigma - \beta} \leq \frac{3}{\sigma - 1} + A_6 L \quad (5.13)$$

valid for every $\sigma \in (1, 2]$. This single inequality, true for all such σ , will be made to contradict β being too close to 1 by an appropriate choice of σ .

Step 5: Optimizing σ . Choose

$$\sigma = 1 + \frac{\delta}{L}, \quad \delta > 0 \text{ a constant to be fixed,}$$

which lies in $(1, 2]$ once $\delta \leq L$, true for large γ since $L \rightarrow \infty$. Then $\sigma - 1 = \delta/L$, so $3/(\sigma - 1) = 3L/\delta$, and (5.13) becomes

$$\frac{4}{\sigma - \beta} \leq \frac{3L}{\delta} + A_6 L = \left(\frac{3}{\delta} + A_6\right) L$$

Inverting,

$$\sigma - \beta \geq \frac{4}{(3/\delta + A_6)L} = \frac{4\delta}{(3 + A_6\delta)L}$$

Now $\beta = \sigma - (\sigma - \beta) \leq \sigma - \frac{4\delta}{(3 + A_6\delta)L} = 1 + \frac{\delta}{L} - \frac{4\delta}{(3 + A_6\delta)L}$, that is,

$$\beta \leq 1 - \frac{1}{L} \left(\frac{4\delta}{3 + A_6\delta} - \delta \right) = 1 - \frac{\delta}{L} \cdot \frac{4 - (3 + A_6\delta)}{3 + A_6\delta} = 1 - \frac{\delta}{L} \cdot \frac{1 - A_6\delta}{3 + A_6\delta}$$

The bracketed factor $(1 - A_6\delta)/(3 + A_6\delta)$ is positive precisely when $\delta < 1/A_6$. Fix any such δ , for instance $\delta = 1/(2A_6)$, giving the positive constant

$$c = \frac{\delta(1 - A_6\delta)}{3 + A_6\delta} = \frac{1}{2A_6} \cdot \frac{1/2}{3 + 1/2} = \frac{1}{14A_6} > 0$$

With this δ ,

$$\beta \leq 1 - \frac{c}{L} = 1 - \frac{c}{\log(\gamma + 2)}$$

for every zero $\rho = \beta + i\gamma$ with $\gamma \geq \gamma_0$. Shrinking c if necessary to also cover the compact range $0 \leq \gamma < \gamma_0$, where a fixed zero-free strip $\sigma \geq 1 - c'$ exists by the non-vanishing on the line, the bound $\beta \leq 1 - c/\log(|\gamma| + 2)$ holds for all zeros, which is the assertion that $\zeta(s) \neq 0$ for $\sigma > 1 - c/\log(|t| + 2)$, completing the proof.

The proof is the qualitative non-vanishing argument run with a ruler. In chapter 4 the inequality (5.8) was used only to exclude $\beta = 1$: a zero on the line would make $-4/(\sigma - 1)$ blow up faster than $+3/(\sigma - 1)$ as $\sigma \rightarrow 1^+$, breaking nonnegativity. Here the same competition between the $3/(\sigma - 1)$ from the pole and the $4/(\sigma - \beta)$ from the zero is resolved quantitatively, and the factor $4 > 3$ is what does the work: because the zero is weighted by 4 and the pole by 3, the zero cannot approach the pole's line faster than a definite rate without the weighted sum turning negative. The growth bound enters only through the constant A_3L in (5.10), a term of size $\log \gamma$; it sets the scale $1/L = 1/\log(\gamma + 2)$ of the forbidden region, since balancing the constant against the pole's $1/(\sigma - 1)$ forces the optimal offset $\sigma - 1 \asymp 1/L$, and the zero is then trapped at distance $\asymp 1/L$ from the line. The shape $c/\log(|t| + 2)$ is thus the direct fingerprint of the logarithmic growth of ζ : a slower-growing ζ would widen the region, and the conjectural bound $|\zeta(1 + it)| = O((\log |t|)^\varepsilon)$ would not, by this argument alone, change the log in the denominator, because the method's loss is in the constant, not the shape.

The width of the zero-free region translates directly into an error term for $\psi(x)$. The mechanism is the truncated Perron formula: the contour can be pushed left to the boundary of the region, and the further left it goes, the smaller the factor x^σ on the shifted line, hence the smaller the error.

Theorem 5.5.4: The Prime Number Theorem with Error Term

There is a constant $c > 0$ such that

$$\psi(x) = x + O\left(x \exp(-c\sqrt{\log x})\right)$$

and correspondingly, with $\text{Li}(x) = \int_2^x dt/\log t$,

$$\pi(x) = \text{Li}(x) + O\left(x \exp(-c'\sqrt{\log x})\right)$$

for a constant $c' > 0$.

The full proof is a contour-optimization computation: the truncation height and the depth of the contour shift must be chosen as coupled functions of x , and tracking the resulting error through the curved boundary of the zero-free region is several pages of estimation with no conceptual turning point beyond the choices themselves. The argument is given in full in [1, Ch. 18] and [3, Ch. 6]; only the structure and the origin of the term $\exp(-c\sqrt{\log x})$ are recorded here.

Proof Key ideas:

Apply the truncated Perron formula (theorem 2.5.3) to the von Mangoldt coefficients $a_n = \Lambda(n)$, whose Dirichlet series is $-\zeta'/\zeta$ (example 2.5.4), at a height T to be chosen:

$$\psi(x) = \frac{1}{2\pi i} \int_{c_0-iT}^{c_0+iT} \left(-\frac{\zeta'}{\zeta}(s)\right) \frac{x^s}{s} ds + R(x, T)$$

with $c_0 = 1 + 1/\log x$ and an explicit error $R(x, T)$ from the truncation. *Step 1: shift the contour.* Move the segment of integration left to a contour hugging the boundary $\sigma = 1 - c/\log(|t| + 2)$ of the zero-free region of theorem 5.5.3, closing it off with horizontal segments at $\pm T$. The shift is legitimate because $-\zeta'/\zeta$ is holomorphic in the region swept, save for the single pole at $s = 1$; the residue of $(-\zeta'/\zeta)(s) x^s/s$ there is x , by the residue-1 pole of proposition 4.3.1, and this residue is the main term. *Step 2: bound the shifted integrand.* On and near the boundary contour the size of $-\zeta'/\zeta$ is controlled by the growth bound lemma 5.5.1, which gives $-\zeta'/\zeta = O((\log |t|)^2)$ there, while the factor x^s contributes $x^\sigma = x \cdot x^{-c/\log(|t|+2)} = x \exp(-c \log x / \log(|t|+2))$, exponentially smaller than x by an amount that worsens as $|t|$ grows. *Step 3: optimize T .* The truncation error $R(x, T)$ decreases as T grows, while the contribution of the shifted contour increases with T through the $\log(|t| + 2)$ in the exponent; balancing the two competing errors fixes the optimal height. Setting $\log T \asymp \sqrt{\log x}$ equalizes them, because the dominant saving $\exp(-c \log x / \log T)$ on the contour and the dominant loss from the truncation both become $\exp(-c''\sqrt{\log x})$ at that height. The common value $\exp(-c''\sqrt{\log x})$, times x , is the error term. *Step 4: pass to $\pi(x)$.* The estimate for ψ transfers to ϑ by removing the prime-power contribution (of size $O(\sqrt{x} \log x)$, absorbed into the error), and to $\pi(x)$ by partial summation against $1/\log t$ (theorem 1.4.1), which converts the main term x into $\text{Li}(x) = \int_2^x dt/\log t$ and preserves an error of the same exponential shape with a possibly smaller constant c' . The full execution of these four steps is carried out in [1, Ch. 18] and [3, Ch. 6], completing the proof.

The error term $x \exp(-c\sqrt{\log x})$ is smaller than $x/(\log x)^N$ for every fixed N , since $\exp(-c\sqrt{\log x})$ decays faster than any negative power of $\log x$, yet it is far larger than the power saving $x^{1-\delta}$ that a zero-free region of *fixed* width δ to the left of the line would yield. The square root in the exponent is

the exact arithmetic of the optimization: the contour saving is $\exp(-c \log x / \log T)$ and the truncation loss grows with $\log T$, so the two balance when $(\log x) / \log T \asymp \log T$, that is, $\log T \asymp \sqrt{\log x}$, making each error $\exp(-c\sqrt{\log x})$. A wider zero-free region, one whose boundary recedes from the line more slowly in $|t|$, would permit a deeper contour shift at each height and hence a larger exponent; the shape of the region and the shape of the error term are two readings of the same data. The constant c here is not the same as the constant in the zero-free region, but it is determined by it, and any improvement in the width of the region propagates to an improvement in the error.

The error term and the zero-free region are two faces of a single object, made precise by the explicit formula, which expresses the deviation $\psi(x) - x$ as a sum over the zeros of ζ .

5.5.5 Note: The Explicit Formula The deviation of ψ from its main term is given exactly, not just asymptotically, by the *Riemann-von Mangoldt explicit formula*: for $x > 1$ not a prime power,

$$\psi_0(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log\left(1 - \frac{1}{x^2}\right)$$

where $\psi_0(x) = \frac{1}{2}(\psi(x^-) + \psi(x^+))$ is the normalized Chebyshev function and the sum runs over the nontrivial zeros ρ of ζ , taken in conjugate pairs and in order of increasing $|\operatorname{Im}\rho|$. The main term x is the residue at the pole $s = 1$; each zero $\rho = \beta + i\gamma$ contributes an oscillation $-x^{\rho}/\rho$ of size $x^{\beta}/|\rho|$; the constant $-\log(2\pi)$ is the value at $s = 0$ from the kernel; and the final term collects the trivial zeros at $s = -2, -4, \dots$. Its proof lies beyond the present chapter: it requires the meromorphic continuation of ζ to all of $\mathbb{C}$ through the functional equation (??) and the Hadamard product factorization of the entire function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, which expresses ξ as a product over its zeros and thereby converts the contour integral of $-\zeta'/\zeta$ into the sum over ρ . The full derivation is given in [1, Ch. 17]; see also [3]. The formula foreshadows the role of the functional equation in ???: every term but the first is governed by the symmetry $s \leftrightarrow 1-s$ and the zeros it organizes, and the error term of theorem 5.5.4 is precisely a bound on the zero-sum $\sum_{\rho} x^{\rho}/\rho$ obtained by knowing that all $\beta < 1$ stay a definite distance below 1.

The explicit formula reframes every statement about the error term as a statement about the location of the zeros. The size of $\sum_{\rho} x^{\rho}/\rho$ is governed by $\sup_{\rho} \beta = \sup_{\rho} \operatorname{Re}(\rho)$: the larger the real parts of the zeros, the larger the oscillations x^{β} , and the worse the error. The de la Vallée Poussin region bounds β below 1 by a margin that narrows with height, and the resulting error $x \exp(-c\sqrt{\log x})$ is what that margin yields. The optimal possible control corresponds to the zeros lying as far left as they can, on the line $\operatorname{Re}(s) = 1/2$, which is the content of the Riemann Hypothesis.

Conjecture 5.5.6: The Riemann Hypothesis

Every nontrivial zero ρ of the Riemann zeta function satisfies $\operatorname{Re}(\rho) = \frac{1}{2}$.

The hypothesis is equivalent to the sharpest possible error term in the Prime Number Theorem. If every nontrivial zero has $\beta = \frac{1}{2}$, then each term x^{ρ}/ρ in the explicit formula has size $x^{1/2}/|\rho|$, and summing over zeros with the known density produces the bound below; conversely, that bound forces the zeros onto the line, so the equivalence runs both ways.

Proposition 5.5.7: Riemann Hypothesis and the Error Term

The Riemann Hypothesis holds if and only if

$$\psi(x) = x + O(\sqrt{x} (\log x)^2)$$

and this error term is essentially optimal: the deviation $\psi(x) - x$ is not $o(\sqrt{x})$, so the exponent $\frac{1}{2}$ cannot be lowered.

The equivalence is proved in [3, Ch. 13]; the forward direction reads the bound off the explicit formula under $\beta \equiv \frac{1}{2}$, and the reverse recovers $\sup \beta \leq \frac{1}{2}$ from the bound by a contour argument, while the optimality is a consequence of the oscillation forced by the infinitely many zeros on the line. Unconditionally, the best known zero-free region is wider than de la Vallée Poussin's but still far from the half-plane $\operatorname{Re}(s) > \frac{1}{2}$ that the Riemann Hypothesis asserts: the *Korobov–Vinogradov* region states that $\zeta(s) \neq 0$ for

$$\sigma \geq 1 - \frac{c}{(\log |t|)^{2/3} (\log \log |t|)^{1/3}}$$

established by Vinogradov's method for estimating exponential sums and proved in full in [9, Ch. 6]. By the same contour optimization that produced theorem 5.5.4 from de la Vallée Poussin's region, this wider region yields the best unconditional error term $\psi(x) = x + O(x \exp(-c(\log x)^{3/5} (\log \log x)^{-1/5}))$.

The progression from de la Vallée Poussin to Korobov–Vinogradov to the Riemann Hypothesis is a progression in the width of a zero-free region and, by the explicit formula, in the size of an error term. De la Vallée Poussin's region has width $\asymp 1/\log |t|$ and gives the error $x \exp(-c\sqrt{\log x})$; Korobov–Vinogradov widens the region near the real axis by a power of $\log \log |t|$ and improves the exponent from $\frac{1}{2}$ to $\frac{3}{5}$; the Riemann Hypothesis would push the region all the way to the open half-plane $\operatorname{Re}(s) > \frac{1}{2}$, the widest a zero-free region can be, since zeros are known to lie on $\operatorname{Re}(s) = \frac{1}{2}$ and so the region cannot extend past it, and would replace the exponential saving by the square-root power $x^{1/2}$. Each step trades a better understanding of where the zeros are not for a sharper count of the primes, and the line $\operatorname{Re}(s) = \frac{1}{2}$ marks the limit of what any zero-free region can achieve. The functional equation of the next chapter is what brings that line into view, by exhibiting the symmetry $s \leftrightarrow 1 - s$ under which the critical line $\operatorname{Re}(s) = \frac{1}{2}$ is the axis, and by supplying the continuation and the product factorization that the explicit formula and the Riemann Hypothesis both presuppose.